Krishna's

TEXT BOOK

Physical Optics and Lasers

B.Sc. Second Year | First Paper

(for B.Sc. IInd year students of All Colleges affiliated to Universities in Uttar Pradesh)

As per U.P. UNIFIED Syllabus

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KRISHNA HOUSE, 11, Shivaji Road, Meerut-250 001 (U.P.), India

Jai Shri Radhey Shyam

Dedicated

to

Lord

Krishna

Authors & Publishers

Preface

In year 2011, U.P. State universities adopted **Unified Syllabus** suggested by **M.H.R.D. New Delhi, U.G.C. and U.P. Government.** So a great need was being felt to suit the requirement of **B.Sc. students.**

We are happy to present this book which fulfils the requirements entitled **"Physical Optics and Lasers".**

Efforts are made to make the language of the book. Simple, fluent, lucid and rather comprehensive. Some salient features of the book are as follows:

1. The book has been written strictly according to the unified syllabus of U.P. state universities. The complete syllabus has been divided into **Four Units.**
2. According to the unified paper pattern, each unit has been divided into three sections *i.e.*, **Section A: Descriptive part, Section B: Short part** & **Section C: Very short part.**
3. In section A, the complete syllabus of concerned unit is described in details and after each article, some simple numerical examples are given based upon it, So that the student may understand the concept well.
4. In each unit, Miscellaneous Problems are given which are based on different articles of that unit.
5. At of the end of each unit, **Some Probable Questions** are given which have been or may be asked in the university examination in a different manner.

We are thankful to our near and dear ones for their blessings, good wishes and cooperation due to which the implementation of this project could become possible. It is God who has blessed us to complete this work in time.

We are also thankful to **Mr. S.K. Rastogi**, *M.D.,* **Mr. Sugam Rastogi**, *E.D.*, **Mrs.Kanupriya Rastogi**, *Director* and **entire team** of **Krishna Prakashan Media (P) Ltd.**, for taking keen interest in getting the book published.

In spite of great care, some misprints and omissions might have crept in. We will be grateful to teachers and students who will point out the same and give suggestions for further improvement of the book.

It is hoped that present edition of the book will be found useful and fulfill a genius need.

—Authors

Syllabus

Physical Optics and Lasers

B.Sc. IInd Year Paper-I

U.P. UNIFIED (*w.e.f.* 2012-13)

M.M. – 50

UNIT-I

Interference of a light: The principle of superposition, two-slit interference, coherence requirement for the sources, optical path retardations, lateral shift of fringes, Rayleigh refractometer and other applications. Localised fringes; thin films, applications for precision measurements for displacements.

Haidinger fringes: Fringes of equal inclination. Michelson interferometer, its application for precision determination of wavelength, wavelength difference and the width of spectral lines. Twymann Green interferometer and its uses. Iriensity distribution in multiple beam interference, Tolansky fringes, Fabry-Perrot interferometer and etalon.

UNIT-II

Fresnel diffraction: Fresnel half-period zones, plates, straight edge, rectilinear propagation.

Fraunhoffer diffraction: Diffraction at a slit, half-period zones, phasor diagram and integral calculus methods, the intensity distribution, diffraction at a circular aperture and a circular disc, resolution of images, Rayleigh criterion, resolving power of telescope and microscopic systems, outline of phase contrast microscopy.

Diffraction gratings: Diffraction at N parallel slits, intensity distribution, plane diffraction grating, reflection grating and blazed gratings. Concave grating and different mountings. Resolving power of a grating and comparison with resolving powers of prism and of a Fabry-Perrot etalon.

UNIT - III

Polarization, Double refraction in uniaxial crystals, Nicol prism, polaroids and retardation plates, Babinet's compensator. Analysis of polarised light. Optical activity and Fresnel's explanation, Half shade and Biquartz polarirneters.

Matrix representation of plane polarized waves, matrices for polarizers, retardation plates and rotators, Application to simple systems.

UNIT-IV

Laser system: Purity of a special line, coherence length and coherence time, spatial coherence of a source, Einstein's A and B coefficients, spontaneous and induced emissions, conditions for laser action, population inversion.

Application of Lasers: Pulsed lasers and tunable lasers, spatial coherence and directionality, estimates of beam intensity; temporal coherence and spectral energy density.

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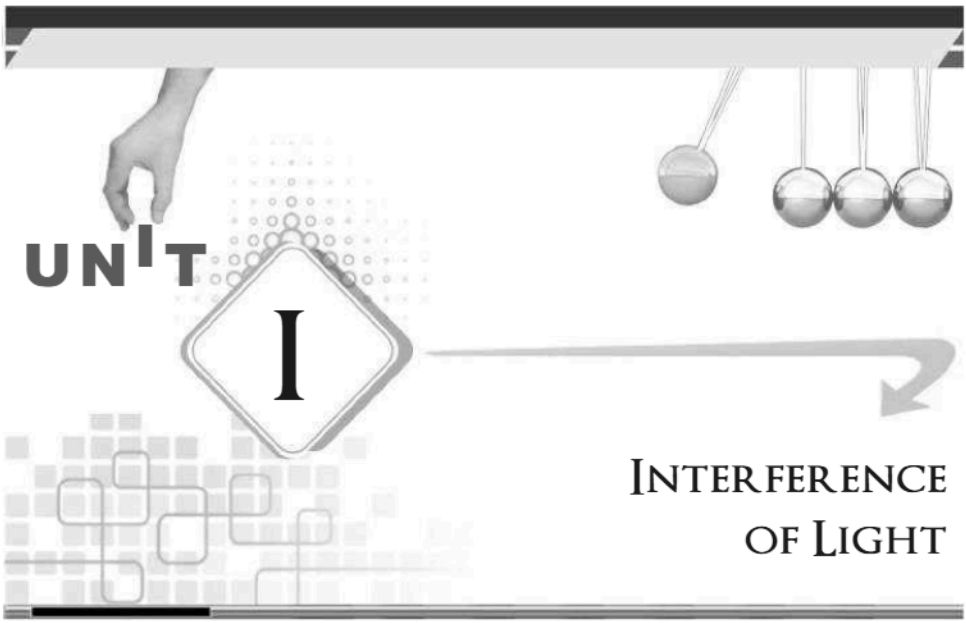
INTERFERENCE OF LIGHT

- The Principle of Superposition
- Interference of Light
- Two Slit Interference
- Fresnel's Biprism and Formation of Fringes
- Conditions for Interference of Light
- Coherent Requirements of the Sources
- Optical Path Retardation
- Laser Light Source
- Applications of Fringes
- Air Wedge Film and its Applications
- Interference in Thin Films of the Cases of Uniform Films
- Newton Rings and its Applications
- Hair diameter, Fringes of Equal Inclination
- Michelson's Interferometer and its Applications
- Twyman and Green Interferometer and its uses
- Intensity Distribution in Multiple Beam Interference
- Fabry Perot Interferometer



UNIT I

INTERFERENCE OF LIGHT

When two or more waves travel in a medium simultaneously in any directions, it is observed that the net displacement of the waves at an instant is the algebraic sum of the displacement due to each wave and it is called *superposition of waves.*

1.1 The Principle of Superposition

It states that, *the displacement at an instant due to a number of waves, meeting simultaneously at a point in a medium, is the vector sum of the individual displacement due to each wave at that point at the same time.*

If $\vec{y}_1, \vec{y}_2, \vec{y}_3$ are the displacements due to individual waves at a particular time and at particular position, then the resultant displacement $\vec{y}$ at the same time at the given position will be,

$$\vec{y} = \vec{y}_1 + \vec{y}_2 + \vec{y}_3 + \ldots$$

It is the general principal of superposition of waves and is applicable to all kinds of wave. as a special case, *when the superposition of waves of the same frequency takes place, then the resultant amplitude or intensity is some where lesser and somewhere greater than the amplitude or intensity due to a single wave. This modification of amplitude or intensity is called interference between the waves.*

1.2 Resultant Intensity due to Superposition of Two Waves

Let S_1 and S_2 are two similar parallel slits which are illuminated by a monochromatic light and are very close to each other. Now, we find the resultant intensity of light at point P on the screen AB placed at a distance D from the sources S_1 and S_2 (Fig. 1.1).

Let a_1 and a_2 are the amplitudes at P due to the waves emitting from S_1 and S_2 respectively. These waves may be represented by,

$$y_1 = a_1 \sin \omega t$$

$$y_2 = a_2 \sin (\omega t + \delta)$$

Where δ is the phase difference between the waves at point P.

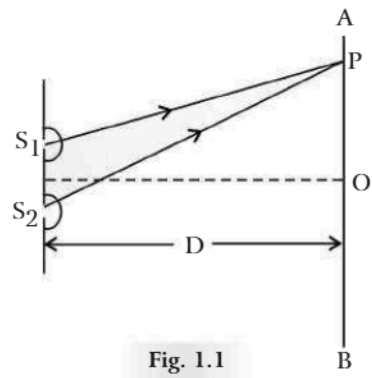


Fig. 1.1

By the principle of superposition, the resultant displacement at P may be given by,

$$y = y_1 + y_2 = a_1 \sin \omega t + a_2 \sin (\omega t + \delta)$$

$$= a_1 \sin \omega t + a_2 \sin \omega t \cos \delta + a_2 \sin \delta \cos \omega t$$

$$= \sin \omega t (a_1 + a_2 \cos \delta) + a_2 \sin \delta \cos \omega t$$

Now, if we take

$$R \cos \theta = a_1 + a_2 \cos \delta \quad ...(1)$$

and

$$R \sin \theta = a_2 \sin \delta \quad ...(2)$$

where R and θ are new constants.

then,

$$y = \sin \omega t \, R \cos \theta + R \sin \theta \cos \omega t$$

or

$$y = R \sin (\omega t + \theta) \quad ...(3)$$

The value of R and θ is given by equation (1) and (2). Using equations (1) and (2) we get,

$$R^2 \cos^2 \theta + R^2 \sin^2 \theta = a_1^2 + a_2^2 \cos^2 \delta + 2a_1 a_2 \cos \delta + a_2^2 \sin^2 \delta$$

or

$$R^2 = a_1^2 + a_2^2 + 2a_1 a_2 \cos \delta \quad ...(4)$$

and

$$\tan \theta = \frac{a_2 \sin \delta}{a_1 + a_2 \cos \delta} \quad ...(5)$$

We know that the intensity at a point is directly proportional to the square of the amplitude of the wave. So, the resultant I at point P is given by,

$$I \propto R^2 \quad \text{or} \quad I = R^2 \qquad \text{(proportionality constant is 1)}$$

So,

$$\boldsymbol{I = a_1^2 + a_2^2 + 2a_1 a_2 \cos \delta} \quad ...(6)$$

1.2.1 Phase Difference and Path Difference

If the path difference between two waves is λ, then the phase difference = 2π

$\therefore$ for a path difference x, the phase difference will be $\frac{2\pi}{\lambda}x$

Here, the path difference = $S_2P - S_1P$

So, the phase difference, $\delta = \frac{2\pi}{\lambda}(S_2P - S_1P)$

1.2.2 Special Cases

1. If the phase difference, $\delta = 0, 2\pi, 4\pi, \ldots, 2n\pi$ or the path difference, $x = 0, \lambda, 2\lambda, \ldots, n\lambda$, then the intensity I is maximum, from equation (6),

$$I = a_1^2 + a_2^2 + 2a_1a_2 = (a_1 + a_2) = 4a^2 \qquad (\text{if } a_1 = a_2 = a)$$

So, *the maximum intensity is greater than the sum of two individual intensities* and this is called *constructive interference.*

2. If the phase difference, $\delta = \pi, 3\pi \ldots (2n+1)\pi$ or the path difference $x = \frac{\lambda}{2}, \frac{3\lambda}{2} \ldots (2n+1)\frac{\lambda}{2}$, then the intensity I will be minimum *i.e.*, $I = a_1^2 + a_2^2 - 2a_1a_2$

or $$I = (a_1 - a_2)^2 = 0 \qquad (\text{if } a_1 = a_2 = a)$$

So, *the minimum intensity is less than the sum of the two individual intensities and this is called destructive interference.*

Thus, as we move on the screen, the path difference between the two waves changes gradually and there is a variation in the intensity of light, being alternately maximum and minimum. *This pattern of alternate maximum (brightness) and minimum (darkness) intensity is known as interference fringes.*

1.3 Interference of Light

When two waves of the same frequency travel in the same direction and have a constant phase difference with each other, the resultant intensity of light is not distributed uniformly in space. The non uniform distribution of the light intensity, due to the superposition of two waves, is called *interference.* At some points the intensity is maximum while at some other points the intensity is minimum. The maximum intensity is called *constructive interference* while the minimum intensity is called *destructive interference.* Generally, when two light waves are made to interfere, we get alternate dark and bright bands. These are called *interference fringes.*

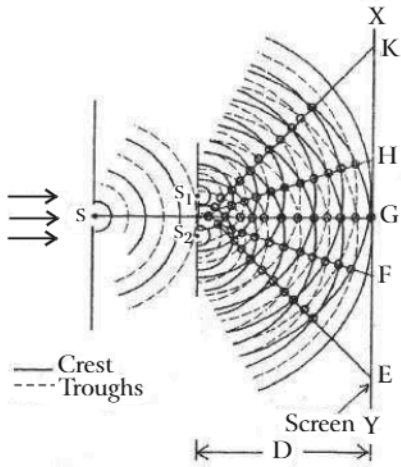


Fig. 1.2

In the year 1802, **young** demonstrated an experiment on interference of light. He passed sunlight through a pinhole S and then through two pinholes S_1 and S_2, making two coherent sources as shown in Fig. 1.2. S_1 and S_2 are equidistant from S and are closed to each other. Let $S_1S_2 = 2d$. Finally light is received on a screen XY which is placed at a distance D from the slits. He observed the alternate dark and bright spots on the screen. These spots can be explained due to interference of light waves.

Wave theory of light states that spherical waves spread out from S and also from S_1 and S_2. According to Huygen's principal, S_1 and S_2 become the centres of secondary wavelets. Their radii increase as they move away from S_1 and S_2, so they get superimposed more and more on each other. At point where a crest (or trough) due to one falls on a crest (or trough) due to other, the resultant amplitude is the sum of the amplitude due to each wave separately. The intensity, which is proportional to the square of the amplitude, at these points is therefore maximum, it is the case of *constructive interference.* At point where a crest due to one falls on a trough due to other, the resultant amplitude is the difference of the amplitude due to separate waves and the resultant intensity is a minimum. This is the case of *destructive interference*.

1.4 Two Slit Interference

Let S be a narrow slit illuminated by monochromatic light and S_1 and S_2 are two parallel narrow slits which are very close to each other and equidistant from S. The light waves from S arrive at S_1 and S_2 in the same phase *i.e.*, these two sources S_1 and S_2 are identical in all respects (Fig. 1.3). Finally, light is received on a screen AB which is placed at a distance D from the slits. One can observed the alternate dark and bright spots on the screen due to interference.

1.4.1 Fringe–Width in Young's Experiment

Let at any instant the waves from S_1 and S_2 will reach at any point P on the screen AB.

Let $S_1S_2 = 2d$, $OR = D$ and $RP = x$

To get the resultant intensity at P, we join S_1P and S_2P. Now, the two waves meeting at P are following the different path *viz.*, S_1P and S_2P. So, we calculate the path difference $S_2P - S_1P$ from the Fig. 1.3. In ΔS_1PR_1,

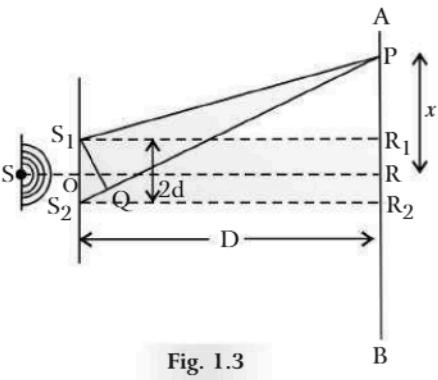


Fig. 1.3

$$(S_1P)^2 = (S_1R_1)^2 + (R_1P)^2$$

$$= (OR)^2 + (R_1P)^2 = D^2 + (x-d)^2$$

or $$S_1P = [D^2 + (x-d)^2]^{1/2} = D\left[1 + \frac{(x-d)^2}{D^2}\right]$$

and similarly in ΔS_2PR_2,

$$(S_2P)^2 = (S_2R_2)^2 + (PR_2)^2$$

$$= (OR)^2 + (PR_2)^2 = D^2 + (x+d)^2$$

or $$S_2P = D\left[1 + \frac{(x+d)^2}{D^2}\right]^{1/2}$$

So, $$S_2P - S_1P = D\left[1 + \frac{(x+d)^2}{D^2}\right]^{1/2} - D\left[1 + \frac{(x-d)^2}{D^2}\right]^{1/2}$$

But, we have $D >> (x+d)$ or $(x-d)$. So, $\frac{(x-d)^2}{D^2}$ and $\frac{(x+d)^2}{D^2}$ is very-very small. Therefore, by using Binomial theorem we have,

$$S_2P - S_1P = D + \frac{1}{2}\frac{(x+d)^2}{D} - D - \frac{1}{2}\frac{(x-d)^2}{D}$$

or $$S_2P - S_1P = \frac{1}{2D}[(x+d)^2 - (x-d)^2] = \frac{2xd}{D}$$

If the path difference *i.e.*, $S_2P - S_1P$ is a whole number multiple of wavelength λ, then the point P is bright or at that point the intensity is maximum. Let the point P be the centre of nth bright fringe, then for bright fringes,

$$S_2P - S_1P = \frac{2x_nd}{D} = n\lambda, \text{ where } n = 0,1,2\ldots$$

or $$x_n = \frac{D}{2d}n\lambda \qquad \ldots(1)$$

This equation gives the distance of the bright fringes from the point R.At R, the path difference is zero and a bright image is formed.

If $$n=1, x_1=\frac{D\lambda}{2d}$$

$$n=2, x_2=\frac{2D\lambda}{2d}$$

$$n=3, x_3=\frac{3D\lambda}{2d}$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$$

and $$x_n=\frac{nD\lambda}{2d}$$

also $$x_{n+1}=\frac{(n+1)D\lambda}{2d}$$

Therefore, the distance between any two consecutive bright image,

$$x_{n+1}-x_n=x_3-x_2=x_2-x_1=\mathbf{\frac{D\lambda}{2d}}$$

It is independent of n. Hence, the distance (spacing) between any two consecutive bright fringes is the same.

Similarly, if the path difference *i.e.,* S_2P-S_1P is an odd number multiple of half wavelength, then the point P is dark or at point P the intensity is minimum. Let the point be the centre nth dark fringe, then for dark fringes,

$$S_2P-S_1P=\frac{2x_nd}{D}=(2n+1)\frac{\lambda}{2}, \text{where } n=0,1,2,\ldots$$

or $$x_n=\frac{(2n+1)\lambda D}{2.2d} \qquad \ldots(2)$$

This equation gives the distance of the dark fringes from the point R.

If $$n=0,\ x_0=\frac{D\lambda}{2.2d}$$

$$n=1, x_1=\frac{3D\lambda}{2.2d}$$

$$n=2, x_2=\frac{5D\lambda}{2.2d}$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$$

and $$x_n=\frac{(2n+1)D\lambda}{2.2d}$$

also $$x_{n+1} = \frac{(2n+3)D\lambda}{2.2d}$$

Therefore, the spacing between any two consecutive dark image,

$$x_{n+1} - x_n = x_2 - x_1 = \frac{\mathbf{D\lambda}}{\mathbf{2d}}$$

Again, it is independent of n. Hence the distance between any two consecutive dark fringes is the same.

Hence, the spacing between any two consecutive bright or dark fringes is same and is called *fringe width.* it is indicated by β.

Thus, $$\beta = \frac{\mathbf{D}}{\mathbf{2d}}\lambda \qquad ...(3)$$

Regarding equation (3) it is worthy to note that:

1. The width of fringe is directly proportional to wavelength of light *i.e.*, $\beta \propto \lambda$.

2. The width fringe is directly proportional to the distance of the screen from the sources *i.e.*, $\beta \propto D$.

3. The width of fringe is inversely proportional to the distance between two sources, *i.e.*, $\beta \propto \frac{1}{2d}$.

   Thus, *if the width of slit increases then fringe width decreases.*

Example 1: A light of wavelength 5100 Å from a narrow slit is incident on a double slit. If the overall separation of 10 fringes on a screen kept 200 cm away is 2 cm, find the slit separation.

Solution: We know that fringe width,

$$\beta = \frac{D}{2d}\lambda$$

Where D is distance of screen from slit, λ is wavelength of light used and $2d$ is the separation between slits.

Given that,

$$10\beta = 2 \text{ cm} \quad \text{or} \quad \beta = 0.2 \text{ cm}, D = 200 \text{ cm}$$

and $$\lambda = 5100 \text{ Å} = 5100 \times 10^{-8} \text{ cm}$$

Putting the values, we get

$$0.2 = \frac{200 \times 51 \times 10^{-6}}{2d} \quad \text{or} \quad 2d = \frac{51 \times 200 \times 10^{-6}}{0.2}$$

or $$2d = 51 \times 10^{-3} = \mathbf{0.051\ cm}$$

Example 2: In young's double slit experiment the separation of the slit is 1.9 mm and the fringes spacing is 0.31 mm at a distance of 1 m from the slits. Calculate the wavelength of light.

Solution: The fringe width, $\beta = \frac{D}{2d}\lambda$

Given that, $D = 1m = 100\ \text{cm},\ 2d = 1.9\ \text{mm} = 0.19\ \text{cm}$

and $\beta = 0.31\ \text{mm} = 0.031\ \text{cm}$

Putting the values, we get

$$0.031 = \frac{100 \times \lambda}{0.19}$$

or $$\lambda = \frac{0.031 \times 0.19}{100} = 589 \times 10^{-7}\ \text{cm} = \mathbf{5890\ \text{Å}}$$

Example 3: Two straight narrow parallel slits 2.0 mm apart illuminated with a monochromatic light of wavelength 5890 Å. Fringes are observed at a distance of 60 cm from the slits. Find the width of the fringes.

Solution: The width of the fringes is given by,

$$\beta = \frac{D}{2d}\lambda$$

Here given that, $D = 60\ \text{cm}, \lambda = 5890\ \text{Å},$ and $2d = 2.0\ \text{mm} = 0.2\ \text{cm}$

Putting the values, we get

$\therefore$ $$\beta = \frac{5890 \times 10^{-8} \times 60}{0.2} = \mathbf{1.76 \times 10^{-2}\ cm = 0.176\ mm}$$

1.5 Fresnel's Biprism

A biprism is a combination of two identical right angled prisms of very small refracting angle (of the order of 30′) joint base to base as shown in Fig. 1.4. In actual practice the biprism is made from a single glass plate by grinding and polishing, so that it is single prism with one of its angle about 179° and the other two about $\frac{1}{2}°(30')$ each.

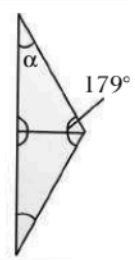


Fig. 1.4

The biprism is used, to obtain two coherent sources to produce sustained interference.

1.5.1 Formation of Fringes

A source of monochromatic light illuminates a narrow vertical slit *S*. The light from *S* is allowed to fall symmetrically on the biprism *ABC,* placed at a small distance from *S* having its refracting edge parallel to the slit. Each half of the biprism **refracts** the incident light in different directions and produces virtual image of *S* as shown in Fig. 1.5. the distance between *S* and biprism *ABC* is so adjusted that the virtual images S_1, S_2, are quite close to each other. Now, these two virtual sources S_1 and S_2 may be supposed to act as coherent sources. Interference fringes of equal width are produced in the overlapping region *EH* of the screen (in which the light reaches from S_1 and S_2 both). Beyond *E* and *H* fringes of large width are produced which are due to diffraction. To observe the fringes, the screen can be replaced by an eyepiece or a low power microscope.

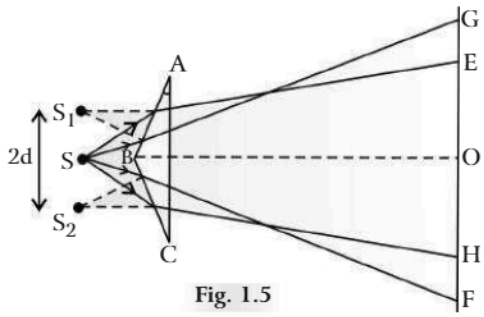


Fig. 1.5

1.5.2 Expression for Fringe Width

Let the distance between S_1 and S_2, the two virtual source be $2d$ and D is the distance of the screen from the sources on which the fringes are obtained. The point *R* on the screen is equidistant from S_1 and S_2. Therefore, the waves from S_1 and S_2 reach *R* in the same phase and reinforce each other. So the point *R* will be the centre of a bright fringe while the illumination (brightness or darkness) at any other point *P* on the screen will depend upon the path difference between the waves from S_1 and S_2.

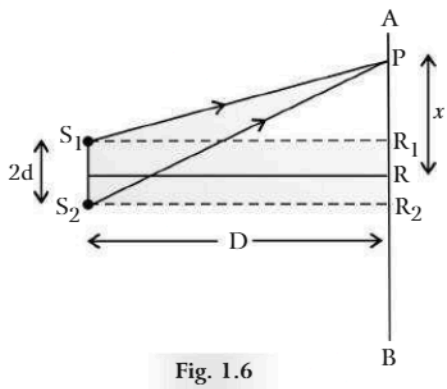


Fig. 1.6

To derive the expression for fringe width joint $S_1 P$ and $S_2 P$ and draw perpendiculars S_1R_1 and S_2R_2 on the screen *AB.* Let $RP = x$, then from the Fig. 1.6.

$$(S_2P)^2 = (S_2R_2)^2 + (PR_2)^2 = D^2 + (x+d)^2$$

or
$$S_2P = [D^2 + (x+d)^2]^{1/2} = D\left[1 + \frac{(x+d)^2}{D^2}\right]^{1/2}$$

$$= D\left[1 + \frac{1}{2}\frac{(x+d)^2}{D^2}\right] \qquad [\because\ x + d << D]$$

and
$$(S_1P)^2 = (S_1R_1)^2 + (PR_1)^2 = D^2 + (x-d)^2$$

or
$$S_1P = D\left[1 + \frac{(x-d)^2}{D^2}\right]^{1/2} = D\left[1 + \frac{1}{2}\frac{(x-d)^2}{D^2}\right]$$

$\therefore$
$$S_2P - S_1P = D + \frac{1}{2}\frac{(x+d)^2}{D} - D - \frac{1}{2}\frac{(x-d)^2}{D} = \frac{2xd}{D}$$

The resultant intensity at a point P is maximum if the path difference between the waves is an integral multiple of wavelength. Let the point P be the centre of nth bright fringe, then

$$\frac{2x_n d}{D} = n\lambda \qquad \text{where} \quad n = 0, 1, \ldots$$

or
$$x_n = \frac{D}{2d} n\lambda \qquad \ldots(1)$$

This equation gives the distance of the nth bright fringe from the point R. At R, the path difference is zero and a bright image is formed.

If $n = 1$, $x_1 = \frac{D}{2d}\lambda$

$n = 2$ $x_2 = \frac{D\,2\lambda}{2d}$

$\vdots \quad \vdots \quad \vdots \quad \vdots$

and
$$x_n = \frac{D\,n\lambda}{2d}$$

Also
$$x_{n+1} = \frac{D\,(n+1)\,\lambda}{2d}$$

Therefore, the distance between any two consecutive bright fringes $= x_{n+1} - x_n$

$$= x_2 - x_1 = \frac{D\lambda}{2d}$$

It is independent of n. hence, the distance (spacing) between any two consecutive bright fringes is the same.

Similarly, if the path difference between the waves *i.e.*, $S_2P - S_1P$ is an odd multiple of half wavelength, then the point P is dark or at point P the intensity is minimum. Let the point P be the centre of nth dark fringe, then

$$\frac{2x_n d}{D} = (2n+1)\frac{\lambda}{2}, \qquad \text{where } n = 0,1,2,\ldots$$

or
$$x_n = \frac{(2n+1)\lambda D}{4d} \qquad \ldots(2)$$

This equation gives the distance of the nth fringe from the point R.

If $n = 0$, $x_0 = \frac{\lambda D}{4d}$

$n = 1$, $x_1 = \frac{3\lambda D}{4d}$

$n = 2$, $x_2 = \frac{5\lambda D}{4d}$

$\vdots \quad \vdots \quad \vdots \quad \vdots$

and
$$x_n = \frac{(2n+1)\lambda D}{4d}$$

also
$$x_{n+1} = \frac{[2(n+1)+1]\lambda D}{4d}$$

Therefore, the spacing between any two consecutive dark fringes,

$$x_{n+1} - x_n = x_2 - x_1 = x_1 - x_0 = \frac{D\lambda}{2d}$$

Again, it is independent of n. hence the distance (spacing) between any two consecutive dark fringes is the same as in the case of bright fringes.

Hence, the spacing between any two consecutive bright or dark fringes is the same and is called *fringe width.* It is represented by β.

Thus,
$$\beta = \frac{D\lambda}{2d}$$

Example 4: A biprism is placed 5 cm away from a slit illuminated by sodium light ($\lambda = 5890$ Å) ***. The width of the fringes obtained on a screen kept at 75 cm from the biprism is*** 9.424×10^{-2} ***cm. What is the distance between two coherent sources ?***

Solution: We have,
$$\beta = \frac{D\lambda}{2d}$$

Where D is the distance of the screen from the source

λ is the wavelength of the light used

β is the fringe width and

$2d$ is the distance between the coherent sources.

Here given that, $\beta = 9.424 \times 10^{-2}$ cm, $D = 75 + 5 = 80$ cm and $\lambda = 5890$Å

Putting the values, we get

$$2d = \frac{80 \times 5890 \times 10^{-8}}{9.424 \times 10^{-2}} = \frac{4.712 \times 10^{-1}}{9.424} \text{ cm} = \mathbf{0.05\ cm}$$

Example 5: Calculate the separation between two coherent sources formed by a biprism whose inclined faces make angle of 1° with its base. The slit source is 20 cm from the biprism. **($\mu = 1.5$)**

Solution: The separation between two coherent slits is given by,

$$2d = 2a(\mu - 1)\alpha$$

Where a is the source to biprism distance, μ is the refractive index of the biprism material and α is the angle of biprism and $2d$ is the separation of slits.

Here given that, $a = 20$ cm, $\mu = 1.5$ and $\alpha = 1° = (\pi/180)$ rad

Putting the values, we get

$$2d = 2 \times 20\,(1.5 - 1)\frac{\pi}{180} = \frac{40 \times 0.5 \times 22}{7 \times 180}$$

$$= \mathbf{0.35\ cm}$$

Example 6: In a Fresnel biprism experiment the fringe width observed is 0.087 mm. What will it become if the slit to biprism distance is reduced to (3/4) the original distance though all remaining things unchanged.

Solution: The separation between the virtual sources is given by,

$$2d = 2a(\mu - 1)\alpha$$

Where a is the source to biprism distance, μ is the refractive index of the biprism material and α is the angle of biprism and $2d$ is the separation of slits.

Here given that, the slit biprism distance a is reduced to (3/4), then by above said formula, the separation $2d$ will also become (3/4) of its original value. Hence the fringe width will be

$$\beta = \frac{D\lambda}{2d} = \frac{D\lambda}{2d \times (3/4)} = \frac{4}{3}\frac{D\lambda}{2d} = \frac{4}{3} \times 0.087$$

$$= \mathbf{0.116\ mm}$$

1.6 Conditions for Interference of Light

If we have two or more sources of light sending waves simultaneously in any region, then according to definition, we must always have the phenomenon of interference. But in practice this interference pattern is not good or visible. To obtain a well defined observable interference pattern, the intensity at the points corresponding constructive and destructive interference must be maintained maximum and minimum (zero) respectively. For obtaining a visible interference following conditions must be fulfilled:

1. *The two interfering waves should be coherent i.e., the waves must be in the same phase or must have a constant phase difference.* If the phase difference is not constant *i.e.*, changes with time, then at any point the constructive interference may exist and a very short time later at this point destructive interference may be. Hence, the same point will appear sometimes bright and sometimes dark or pattern will not be sustain. So, this condition is known as *condition for sustained interference.*

2. *The two waves must have the same wavelength and frequency i.e., sources should be monochromatic.* If the sources are chromatic (have many frequencies), then overlapping of various coloured fringes will produce a confused pattern of fringes. Also, if wavelengths are different, the phase difference between the two will continuously vary. Hence, the intensity of light at the same point will vary with time.

3. *The two waves must have the same amplitudes.* If it is not so, then the minimum amplitude $(a_1 - a_2)$ will not be zero and good contrast between bright and dark fringes will not be observed. So, this condition is known as *condition for good contrast.*

   It may also be noted that the fringe width β is given by,

$$\beta = \frac{D\lambda}{2d}$$

   So, in order to get clearly observable fringes,

   (i) The separation between the light sources $(2d)$ should be very small.

   (ii) The distance D, between sources and screen must be kept quite large.

   Here, it is important to note that according to theories of light, every source emerging the light emits a continuous wave only for a very short time (10^{-8}sec) and after that it emits a fresh wave having different initial phase. The change in phase disturb the interference pattern. So, two independent sources can never maintain a constant phase difference even for a short time and therefore can never be considered to behave as coherent sources.Thus, to obtain sustained interference pattern we take two sources orginated by the same sources as in Young's experiment.

1.7 Requirement of Coherent Sources

Two sources are said to be coherent if they emit light waves of the same frequency, nearly the same amplitude and always maintain a constant phase difference with each other. In actual, it is not possible that two independent sources are coherent. So for experimental purpose, two virtual sources are formed by placing two narrow slits in front of a single source thus, all the random phase changes occurring in the parent sources are repeated in the secondary sources, so that phase remain constant. these are called *coherent sources of light*. Coherence of sources is main the condition of interference.

1.8 Optical Path Retardation

When a thin transparent plate is introduced in the path of one of the two interfering beams, then the optical path of that beam is retarded and it is known as optical path retardation.

1.9 Lateral Shift of Fringes

When a thin transparent plate is introduced in the path of one of the two interfering beams, the entire fringe pattern is shifted towards the beam in the path of which plate is introduced. This shift of entire fringe pattern is known as lateral shift of fringes.

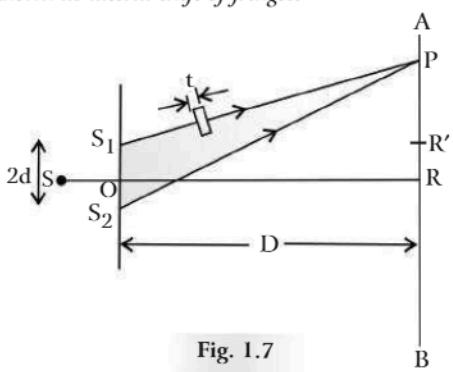


Fig. 1.7

1.9.1 Displacement of Fringes by the Introduction of a Thin Plate of Glass

When a thin transparent plate is introduced in the path of one of the two interfering beams the entire fringe pattern is displaced to a point towards the beams in the path of which the plate is introduced. Let S_1 and S_2 be two coherent monochromatic source of lights which are producing interference fringes on the screen (Fig. 1.7) in such a way that R is the position of the central bright fringe (since the optical path S_1R and S_2R are equal). Let a thin glass plate of thickness, t and refractive index, μ be introduced in the path of one of the two beam (Let S_1). Light from S_1 travel partly in air and partly in the plate while light from S_2 travel wholly in the air. So, the path difference between

the light beams will be changed and also the fringe system. Since the glass plate is placed in the path S_1P, the fringes must shift upward. Let the central fringe shift to the position R' (at this position the optical path S_1R' is equal to S_2R'). Let the nth fringe lies at point P, then

the path difference between S_2P and S_1P

$$= S_2P - [(S_1P - t) + \mu t]$$

$$= S_2P - S_1P - (\mu - 1)t$$

Also, we have $$S_2P - S_1P = \frac{2xd}{D},$$

where $2d = S_1S_2$ and D = Distance of screen from S_1 or S_2 and $RP = x$

$\therefore$ Effective path difference at $P = \frac{2xd}{D} - (\mu - 1)t$

Let the point P be the centre of the nth bright fringe, then the path difference is equal to $n\lambda$.

i.e., $$\frac{2x_n d}{D} - (\mu - 1)t = n\lambda$$

or $$x_n = \frac{D}{2d}\left[n\lambda + (\mu - 1)t\right] \quad ...(1)$$

In the absence of the plate ($t = 0$), the distance of the nth bright fringe from R is $\frac{nD\lambda}{2d}$.

Therefore, the lateral shift of the nth bright fringe due to thin plate of glass in one of the two interfering beam is given by,

$$x_0 = \frac{D}{2d}\left[n\lambda + (\mu - 1)t\right] - \frac{nD\lambda}{2d}$$

$\therefore$ $$\mathbf{x_0 = \frac{D}{2d}(\mu - 1)\,t = \frac{\beta}{\lambda}(\mu - 1)t} \qquad \left[\because \beta = \frac{D}{2d}\lambda\right] \quad ...(2)$$

This expression is independent of n, so that the displacement is the same for all the bright fringes (central or any other). Similarly, it can be determined that the displacement for any dark fringe is also $(D/2d)(\mu - 1)t$. Thus, the entire fringe system is displaced through a distance $(D/2d)(\mu - 1)t$ in the upward direction. Similarly, when the plate is introduced in S_2P beam, then the entire fringe system is displaced through a distance $\frac{D}{2d}(\mu - 1)t$ in the downward direction. In this shifting of fringe, the fringe width remains same. We can prove this as follows. We get the central fringe by putting $n = 0$ in equation (1).

$$x_0 = \frac{D}{2d}(\mu - 1)t$$

Similarly, $$x_1 = \frac{D}{2d}[\lambda + (\mu - 1)t]$$

$$x_2 = \frac{D}{2d}[2\lambda + (\mu - 1)t]$$

$$x_n = \frac{D}{2d}[n\lambda + (\mu - 1)\, t]$$

and $$x_{n+1} = \frac{D}{2d}[(n+1)\,\lambda + (\mu - 1)]$$

The fringe width, $$\beta = x_2 - x_1 = x_3 - x_2 = x_{n+1} - x_n$$

$$= \frac{D}{2d}[n\lambda + \lambda + (\mu - 1)\, t - n\lambda - (\mu - 1)\, t]$$

$$\beta = \frac{\mathbf{D}}{\mathbf{2d}}\lambda \qquad \ldots(3)$$

which is the same as before for all fringes.

The wavelength λ of the light or the thickness of plate t can be determined if μ, t or λ, x_0 and β are known. The displacement x_0 and the fringe width β can be determined experimentally. However, if x_0, β, λ and t are given, the value of μ can be calculated.

Example 7: A thin sheet of plastic of reflective index 1.45 for light of wavelength 5890 Å is inserted normally in the path of one of the interfering beams. The central bright fringe shifts through 5 fringes. Find the thickness of the film.

Solution: We have the shift in fringe,

$$x_0 = \frac{D}{2d}(\mu - 1)\, t = \frac{\beta}{\lambda}(\mu - 1)\, t \qquad \left[\because \beta = \frac{D\lambda}{2d}\right]$$

or $$5\beta = \frac{\beta}{\lambda}(\mu - 1)t \quad \text{or} \quad t = \frac{5\lambda}{\mu - 1}$$

Given that, $\mu = 1.45, \lambda = 5890 Å, x_0 = 5\beta$, where β is the fringe width.

Putting the values, we get

$$t = \frac{5 \times 5890 \times 10^{-8}}{(1.45 - 1)}\ \text{cm} = \mathbf{6.544 \times 10^{-4}\,cm}$$

Example 8: A thin sheet of glass of thickness 6×10^{-4} cm is placed in the path of one of the interfering beams. The central bright band shifts a distance equal to the width of 3 bands. Calculate the refractive index of glass if wavelength of light used is 6×10^{-5} cm.

Solution: The shift in the fringes is given by,

$$x_0 = \frac{D\lambda}{2d}(\mu - 1)\, t \quad x_0 = (\beta/\lambda)(\mu - 1)\, t \qquad \left[\because \beta = \frac{D\lambda}{2d}\right]$$

Given that, $x_0 = 3\beta, \lambda = 6 \times 10^{-5}$ cm and $t = 6 \times 10^{-4}$ cm

Putting the values, we get

$$3\beta = \frac{\beta(\mu - 1)6\times10^{-4}}{6\times10^{-5}}$$

or $$\mu - 1 = 0.3 \quad \text{or} \quad \mu = 1.3$$

1.10 Rayleigh Refractometer

Rayleigh refractometer is shown in Fig. 1.8. Light from a monochromatic source S, after passing through the slit, is incident on the less, L_1. The parallel beam is splitted in two beams by the slits S_1 and S_2.

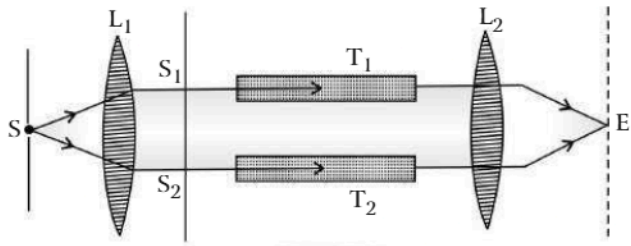


Fig. 1.8

These coherent beams pass through two similar tubes T_1 and T_2. Interference fringes are observed in the focal plane of the lens L_2 with the help of eyepiece. *Rayleigh refractometer is used to determine the refractive index of gases.*

Initially, both the tubes T_1 and T_2 are evacuated and the central white fringe is observed in the field of view of the eyepiece. Now the gas, whose refractive index is to be determined, is introduced in the tube T_1 at known pressure and temperature, then the central white fringe shifts from the field of view as the optical path of this beam increases. The central white fringe is brought back to the centre of the field of view. The number of fringes passed across the field of view is noted. If the length of tube is l and μ is the refractive index of the gas, then the change in the optical path difference,

$$(\mu - 1)\, l = m\lambda$$

where m is number of fringes passed across the field of view. Hence, μ can be calculated.

Example 9: When one of the tube of Rayleigh refractometer is fully evacuated and the other tube is filled with air at atomospheric pressure, 150 fringes cross the field of view. The wavelength of light used is 5000 Å. The tube is 25 cm long. Calculate the refractive index of air at atomospheric pressure.

Solution: For Rayleigh refractometer, we have

$$(\mu - 1)\, l = n\lambda$$

Here given that, $l = 25 \text{ cm} = 0.25 \text{ m}, n = 150$ and $\lambda = 5000 \text{ Å} = 5\times10^{-7} \text{ m}$

putting the values, we get $(\mu - 1)\times0.25 = 150\times5\times10^{-7}$

or $$\mu - 1 = \frac{150 \times 5 \times 10^{-7}}{0.25} = 3 \times 10^{-4}$$

or $$\mu = 1 + 0.0003 = \mathbf{1.0003}$$

1.11 Interference in Thin Films or the Colours of Thin Films

Newton and Hooke observed and developed the interference phenomenon due to multiple reflections from the surface of thin transparent materials.When a thin film of oil spread on the surface of the water or a thin glass plate, is illuminated by a source of light (sun or other source), interference occurs between the light waves reflected from the film the and also in between the light waves transmitted through the film. As a result some brilliant colours are seen in that films. Similar colour phenomenon is also exhibited by soap bubbles.

1.12 Interference due to Reflected Light in Thin Films

Consider, a transparent film of thickness t and refractive index $\mu (>1)$. Let a ray of monochromatic light PQ be incident on the upper surface of the film (Fig. 1.9) PQ ray is partly reflected along QR and partly refracted along QS. At S it is again partly reflected along ST and partly refracted along SV and this process continues throughout the film. Thus, we get a set of several reflected and transmitted rays coming out from the film. Let us first consider the reflected rays only.

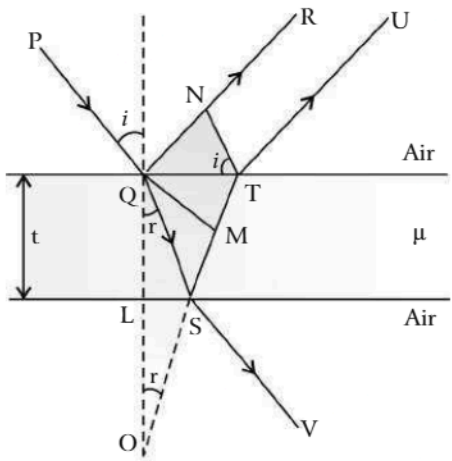


Fig. 1.9

The difference in path between the reflected rays *QR* and *TU* can be calculated. Draw *TN* normal to *QR* and *QM* to *ST*. The angle of incidence is *i* and the angle of refraction is *r*.

The optical path difference between rays *QR* and *TU*,

$$x = \mu(QS + ST) - QN$$

We have that, $$\mu = \frac{\sin i}{\sin r} = \frac{QN}{TM}$$

or $$QN = \mu TM$$

$\therefore$ $$x = \mu(QS + ST) - \mu TM$$

$$= \mu(QS + ST - TM) = \mu(OT - TM)$$

$$= \mu(OM)$$

In the triangle *QOM*,

$$\cos r = OM/QO$$

or $$OM = QO\cos r = (QL + LO)\cos r$$

$$= 2t\cos r \qquad (\because QL = LO = t)$$

$\therefore$ $$x = \mu(OM) = 2\mu t\cos r \qquad ...(1)$$

This path difference is not actual but is it is apparent because when the light is reflected from the surface of an optically denser medium (air medium interface) a phase change of π equivalent to a path difference $\lambda/2$ occurs.

Therefore, the total path difference in this case,

$$x = 2\mu t\cos r - \lambda/2 \qquad ...(2)$$

1.12.1 Condition for Maxima and Minima

1. If the path difference, $x = n\lambda$, where $n = 0,1,2,\ldots$ etc., constructive interference takes place and the film will appear bright.

   $\therefore$ $$2\mu t\cos r - \lambda/2 = n\lambda$$

   or $$\mathbf{2\mu t\cos r = (2n+1)\lambda/2} \qquad ...(3)$$

   It is the condition of maxima.

2. If the path difference, $x = (2n-1)\lambda/2$, where $n = 1,2,\ldots$ etc., destructive interference takes place and the film will appear dark.

   $\therefore$ $$2\mu t\cos r - \lambda/2 = (2n-1)\lambda/2$$

   or $$\mathbf{2\mu t\cos r = n\lambda} \qquad ...(4)$$

   It is condition of minima.

   It should be remembered that the interference, pattern will not be perfect because the intensities of rays will not be the same and also their amplitude are different.

1.13 Interference due to Transmitted Light in Thin Films

Consider a transparent film of thickness t and refractive index μ. A ray of monochromatic light PQ be incident on the upper surface of the film (Fig. 1.10). After refraction it goes along QS. At S it is partly reflected along ST and partly refracted along SV. The ray ST after reflection at X,finally emerges along XY. Here at S and T reflection takes place at the rarer medium (medium-air interface).Therefore, no phase change occurs. Draw SM normal to TX and XN to SV. The optical path difference between SV and XY is given by,

$$x = \mu(ST + TX) - SN$$

and we have that, $\mu = \dfrac{\sin i}{\sin r} = \dfrac{SN}{MX}$

or $$SN = \mu MX$$

from Fig. 1.10,

$$\angle SOT = r \text{ and } OT = ST = TX$$

$$\therefore \quad ST + TX = OT + TX = OX$$

$$\therefore \quad x = \mu OX - \mu MX$$

$$= \mu(OX - MX) = \mu OM$$

In the $\Delta OSM, \cos r = OM/OS$

or $$OM = OS \cos r$$

But $$OS = 2t$$

$$\therefore \quad OM = 2t \cos r$$

and $$x = \mu.OM = 2\mu t \cos r$$

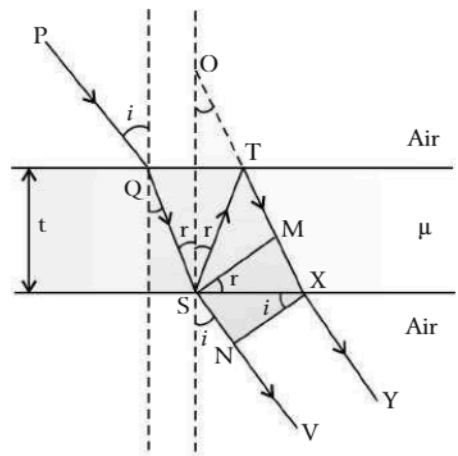


Fig. 1.10

1.13.1 Condition for Maxima and Minima

1. For constructive interference, the path difference must be equal to $n\lambda$ *i.e.*,

$$\mathbf{2\mu t \cos r = n\lambda} \qquad \ldots(1)$$

where $$n = 0, 1, 2, \ldots \text{ etc.}$$

It is the condition of maxima and film will appear bright.

2. For destructive interference, the path difference must be equal to $(2n-1)\,\lambda/2$

i.e., $$\mathbf{2\mu t \cos r = (2n-1)\,1/2} \qquad \ldots(2)$$

where $$n = 1, 2, \ldots, \text{ etc.}$$

It is the condition of minima and film will appear dark.

In the case of transmitted light, the interference fringes obtained are less distinct because the difference in amplitude of respective rays are very large. It is seen from the relations deduced above for the brightness and darkness of the film in the

reflected and transmitted system that the conditions of interference for the two systems are quite opposite to each other. A bright band in the reflected light will appear dark in transmitted light for the same path difference and vice-versa. Thus, the interference pattern of the reflected and transmitted light are **complementary.**

So, when a thin film of oil on water, or a soap bubble, exposed to an extended source of light is observed under reflected light, brilliant colour are seen in the film or the bubble. These colours arise due to the interference of the light waves reflected from the top and bottom surfaces of the film.

Example 10: A parallel beam of sodium light $(\lambda = 5890\text{Å})$ ***is incident on a thin glass plate*** $(\mu = 1.5)$ ***such that the angle of refraction into the plate is*** $60°$. ***Calculate the smallest thickness of the plate which will make it appear dark by reflection.***

Solution: For dark fringes in reflection system the condition is given by,

$$2\mu t \cos r = n\lambda$$

Here given that, $\mu = 1.5, r = 60°, \lambda = 5890\text{Å}$ and $n = 1$ ($\because$ thickness is least)

Putting all values, we get

$$t = \frac{5890 \times 10^{-8} \times 1}{2 \times 1.5 \times \cos 60} = \frac{5890 \times 10^{-8}}{2 \times 1.5 \times 0.5}$$

$$= 3927 \times 10^{-8}\ \text{cm} = 3.92 \times 10^{-5}\ \text{cm}$$

Example 11: Find the thickness of a soap film $(\mu = 1.33)$ ***which gives constructive second order interference of reflected yellow light*** $(\lambda = 6000\ \text{Å})$ ***if incidence is normal.***

Solution: For constructive interference in reflected light the condition is given by,

$$2\mu\, t \cos r = (2n + 1)\,\lambda/2$$

Here given that, $\mu = 1.33, r = 0$ (normal incidence), $n = 2$ and $\lambda = 6000$ Å

Putting the values, we get $t = \dfrac{(2 \times 2 + 1) \times 6000 \times 10^{-8}\ \text{Å}}{2 \times 2 \times 1.33 \times \cos 0} = \dfrac{3 \times 10^{-4}}{5.32}\ \text{cm}$

$$= 5.7 \times 10^{-5}\ \text{cm}$$

Example 12: White light falls normally upon a film of soapy water whose thickness is 5×10^{-5} **cm** ***and refractive index is*** 1.33. ***What wavelength in the visible region will be reflected more strongly.***

Solution: For more strong reflection (*i.e.,* constructive interference)

$$2\mu t \cos r = (2n + 1)\,\lambda/2,\ n = 0, 1, 2, \ldots$$

Here given that, $\mu = 1.33, t = 5 \times 10^{-5}$ cm and $r = 0$ (normal incidence)

Putting values, we get $(2n+1)\lambda/2 = 2\times1.33\times5\times10^{-5}\cos 0$

or $$\lambda = \frac{2.66\times10^{-4}}{(2n+1)}\text{ cm} = \frac{26600}{(2n+1)}\text{ Å}$$

Putting $n = 0,1,2,\ldots$ we get

$$\lambda = 26600\text{ Å}, 8867\text{ Å}, 5320\text{ Å}, 3800\text{ Å}$$

These are the reflected wavelengths. But out of these only **5320 Å** is in visible region. So the film will appear **greenish** in reflected light.

1.14 Newton's Ring

When a plano-convex lens of large focal length is placed on a plane glass plate (Fig.1.11), a thin film of air is formed between the lower surface of the lens and the upper surface of the plate. The thickness of the air film is very small at the point of contact and gradually increases from the centre outwards. If a monochromatic light is allowed to fall normally on this film, a set of alternate dark and bright fringes will be seen in the film. The fringes are concentric circles with their centre dark. These circles or rings are called **Newton's rings.**

Newton's rings are formed as a result of interference between the light waves reflected from the upper and lower surfaces of the air film. Fig.1.11 shows that *AB* is incident ray and ray 1 and 2 will be interfering rays which forms the **Newton's rings.**

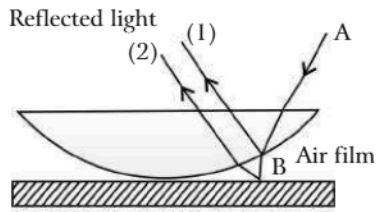


Fig. 1.11

1.14.1 Newton's Ring by Reflected Light

Suppose the radius of curvature of the lens is R and the thickness of the air film is t at a distance of $OQ = r$, from the point of contact O (Fig. 1.12). The effective path difference between the interfereing rays is,

$$2\mu t\cos r + \lambda/2$$

where μ is the refractive index of the film (air). For the bright fringes (rings) or for the maximum intensity,

$$2\mu t\cos r + \lambda/2 = n\lambda$$

or $$2\mu t\cos r = (2n-1)\lambda/2 \qquad ...(1)$$

for air $\mu = 1$ and for normal incidence $r = 0$

So, $$\cos r = \cos 0 = 1$$

$\therefore$ $$\mathbf{2t = (2n-1)\lambda/2} \qquad ...(2)$$

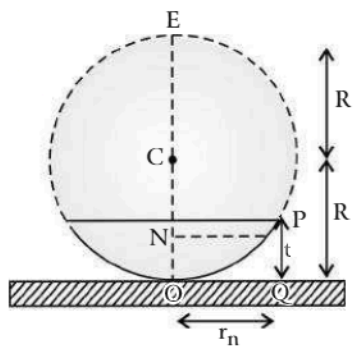


Fig. 1.12

For the dark rings (fringes) or for the minimum intensity,

$$2\mu t \cos r + \lambda/2 = (2n+1)\lambda/2$$

or $$2\mu t \cos r = n\lambda$$

again for $\mu = 1$, and $\cos r = 1$

$$\therefore \quad \mathbf{2t = n\lambda} \qquad ...(3)$$

It is clear that a bright or dark fringe of any order n depends upon the thickness of the air film. Since t is constant along a circle with its centre at the point of contact, the fringes, are in the form of concentric circles and these are known as the *fringes of constant thickness.* These fringes also known as ***localized fringes*** because these are seen in the thin film area.

1.14.2 Diameter of Bright Rings

Let r_n be the radius of a Newton's ring corresponding to a point P where the film thickness is t. From the Fig. 1.12,

$$PN^2 = ON \times NE = ON \times (2R - ON)$$

$$= PQ \times (2R - PQ) = t\,(2R - t) = 2Rt - t^2$$

$$= 2Rt \quad (\because\ t \text{ is very less in comparison to } R, \text{ so } t^2 \text{ can be neglected})$$

$$\therefore \quad r_n^2 = 2Rt$$

or $$2t = r_n^2/R \qquad ...(4)$$

Substituting the value of t in eq. (2), we get

For bright rings, $$r_n^2/R = (2n-1)\,\lambda/2$$

or $$r_n^2 = \frac{(2n-1)}{2}\lambda R$$

or $$r_n = \sqrt{\frac{(2n-1)}{2}\lambda R} \qquad ...(5)$$

and $D_n = 2r_n$, where D is the diameter of the nth bright ring.

So, from the equation (5), $$D_n^2 = 2\,(2n-1)\,\lambda R$$

or $$D_n = \sqrt{(2\lambda R)}\,\sqrt{(2n-1)}$$

or $$D_n \propto \sqrt{(2n-1)} \qquad \text{(if } \lambda \text{ and } R \text{ is constant)}$$

Since n is an integer, $(2n-1)$ is an odd number. Thus, the *diameters of bright rings are proportional to the square roots of the odd natural number i.e.,*

$$D_1 = \sqrt{1}, D_2 = \sqrt{3}, D_3 = \sqrt{5}$$

The separation between successive rings are in there ratio 0.732, 0.504, 0.410 *i.e., the separation decreases as the order increases.*

1.14.3 Diameters of Dark Rings

From equation (3) and (4), $r_n^2 / R = n\lambda$

or $$r_n = \sqrt{(nR\lambda)}$$

If D is the diameter of the dark ring, then

$$D_n = 2r_n = 2\sqrt{(nR\lambda)} = \sqrt{(4nR\lambda)}$$

or $$D_n \propto \sqrt{(n)} \qquad [\because R, \lambda \text{ and } 4 \text{ are constants}]$$

Thus, *the diameters of dark rings are proportional to the square root of natural numbers.*

$$D_1 = \sqrt{1},\ D_2 = \sqrt{2},\ D_3 = \sqrt{3} \ldots$$

Here, again the *fringe width decreases as the order of the fringe increases.*

Example 13: Newton's rings are observed normally in reflected light of wavelength 5.9×10^{-5} ***cm. The diameter of the*** 10^{th} ***dark ring is 0.50 cm. Find the radius of curvature of the lens and the thickness of the film.***

Solution: The diameter of nth dark rings is given by,

$$D_n = \sqrt{(4nR\lambda)}$$

or $$R = \frac{D_n^2}{4n\lambda}$$

Here given that, $D_n = 0.50 \text{ cm},\ n = 10,\ \lambda = 5.9 \times 10^{-5} \text{ cm}$

$\therefore$ $$R = \frac{(0.50)^2}{4 \times 10 \times 5.9 \times 10^{-5}} = 105.9 \approx \mathbf{106 \text{ cm}}$$

If t is the thickness of the film corresponding to a ring diameter D, then we have

$$2t = D^2 / 4R$$

or $$t = D^2 / 8R$$

Putting the value, we get $$t = \frac{(0.50)^2}{8 \times 106} = \mathbf{3 \times 10^{-4} \text{ cm}}$$

Example 14: A thin equi-convex lens of focal length 4 m and reflective index 1.50 rests on and is in contact with an optically flat glass plate. Using light of wavelength 5460Å. Newton's ring are viewed normally by reflection. What is the diameter of the 5 th ***bright ring?***

Solution: As we have, $$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Here given that, $R_1 = R, R_2 = -R$ (since lens if equi-convex)

and given that, $\mu = 1.50,\ f = 4 \text{ m} = 400 \text{ cm}$

$$\therefore \quad \frac{1}{400} = (1.5-1)\left(\frac{1}{R}+\frac{1}{R}\right) = 0.5\times\frac{2}{R}$$

$$\therefore \quad R = 400 \text{ cm}$$

The diameter of the nth bright ring is given by,

$$D_n = \sqrt{2\,(2n-1)\lambda R}$$

Here given that, $n = 5, \lambda = 5460\times10^{-8}$ cm, $R = 400$ cm

Putting the values, we get

$$\therefore \quad D_n = \sqrt{(2\times9\times5460\times10^{-8}\times400)} = \mathbf{0.627cm.}$$

1.15 Determination of the Wavelength of Light Using Newton's Ring

The experimental arrangement for determining the wavelength of used light by Newton's ring is shown in Fig. 1.13. S is a source of sodium light. A plano convex lens L of large focal length is placed on a plane glass P in such way that the curved surface of the lens touches the glass plate. A parallel beam of light is reflected by the glass plate B which is held inclined at an angle of 45° to the horizontal. The reflected light from B falls normally on the air film enclosed between L and P. Interference occurs between the rays reflected from the upper and lower surfaces of the film. Interference rings (Newton's ring) are viewed through B by the travelling microscope M focused on the air film.

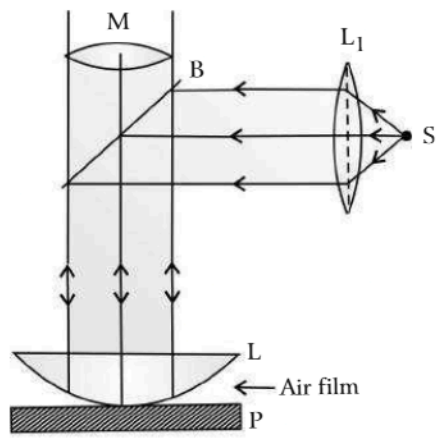


Fig. 1.13

1.15.1 Theory and Formula

In this arrangement the effective path difference between the interfering rays is $2\mu t\cos r + \lambda/2$. Here $\mu = 1$ (since film is of air) and $r = 0$ (since light is incident normally). Hence, the effective path difference is written simply

$$2t + \lambda/2.$$

The condition for the nth bright fringe is,

$$2t + \lambda/2 = n\lambda$$

or

$$2t = (2n-1)\,\lambda/2 \qquad \ldots(1)$$

Let R be the radius of curvature of the lens. If r_n be the radius of nth bright ring at which the film thickness is t, then we have

$$2t = r_n^2/R \qquad ...(2)$$

from equation (1) and (2), $r_n^2/R = (2n-1)\lambda/2$

or $$r_n^2 = (2n-1)\lambda R/2 \qquad ...(3)$$

If D_n is the diameter of the nth bright ring *i.e.*, $D_n = 2r_n$, then

$$D_n^2 = 2(2n-1)\lambda R \qquad ...(4)$$

For $(n+P)$ th ring the diameter D_{n+P} is given by,

$$D_{n+P}^2 = 2[2(n+P)-1]\lambda R \qquad ...(5)$$

Subtracting equation (4) from equation (5), we get

$$D_{n+P}^2 - D_n^2 = 2[2(n+P)-1]\lambda R - 2(2n-1)\lambda R$$
$$= 2\lambda R[2n+2P-1-2n+1]$$
$$= 4P\lambda R$$

or $$\lambda = \frac{\mathbf{D_{n+P}^2 - D_n^2}}{\mathbf{4PR}}$$

Similarly, for dark rings,

$$r_n^2 = n\lambda R$$

or $$D_n^2 = 4n\lambda R$$

and $$D_{n+P}^2 = 4(n+P)\lambda R$$

or $$D_{n+P}^2 - D_n^2 = 4\lambda R[n+P-n] = 4P\lambda R$$

and $$\lambda = \frac{\mathbf{D_{n+P}^2 - D_n^2}}{\mathbf{4PR}}$$

Measuring the diameters of the respective rings and the radius of curvature of the lens R, the wavelength λ can be determined.

1.15.2 Method

As described above, we observe the Newton's rings in microscope and focus on the bright or dark rings. To measure the diameter of the rings, we set the cross wire once on the left and once on the right of each ring. The difference of both readings will be the diameter of the ring. Now we plot a graph between (diameter)^2, D^2 and number of ring. By taking slope the line (Fig. 1.14) we get the average value of $\frac{D_{n+P}^2 - D_n^2}{P}$

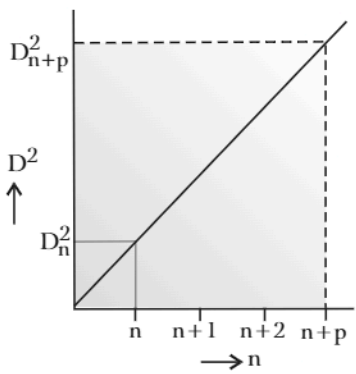


Fig. 1.14

To find out the radius of curvature R of the lens we use Boy's method or spherometer method. Substituting the value of $\frac{D_{n+P}^2 - D_n^2}{P}$ and R in the formula, the wavelength of light can be determined.

1.16 Determination of Refractive Index of a Liquid

We introduce the given liquid between the lens and glass whose refractive index is to be determined (Fig. 1.15). If the liquid is rarer than glass, a phase change of π will occur at reflection from the lower surface of the liquid film but if denser than glass, the phase change of π will occur at reflection from the upper surface of the film. In either case it corresponds to the path difference is λ/2. Therefore, the effective path difference between interfering rays is,

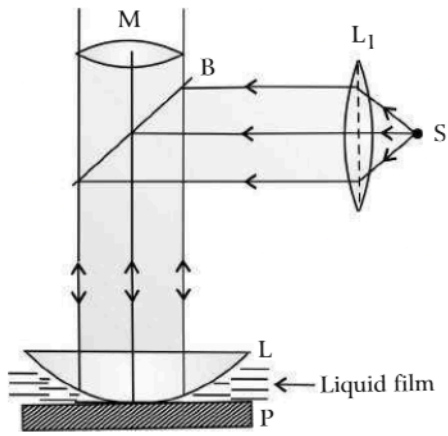


Fig. 1.15

$$= 2\mu t \cos r + \lambda/2$$

or
$$= 2\mu t + \lambda/2$$

(since light falls normally on the film)

The condition for the bright fringe is,

$$2\mu t + \lambda/2 = n\lambda$$

or
$$2t = \frac{(2n-1)\lambda}{2\mu} \quad \ldots(1)$$

If r_n is the radius of nth bright ring and R is the radius of curvature of the lens L, then we have

$$2t = r_n^2/R \quad \ldots(2)$$

From equation (1) and (2), we have

$$\frac{r_n^2}{R} = \frac{(2n-1)\lambda}{2\mu}$$

or
$$r_n^2 = \frac{(2n-1)\lambda R}{2\mu}$$

or
$$D_n^2 = 2(2n-1)\frac{\lambda R}{\mu} \quad \ldots(3)$$

where $D_n = 2r_n$ *i.e.*, diameter of the nth ring.

For the $(n+P)$th ring, diameter D_{n+P} is given by,

$$D_{n+P}^2 = 2[2(n+P)-1]\frac{\lambda R}{\mu} \quad \ldots(4)$$

Subtracting equation (3) from (4), we get

$$(D_{n+P}^2 - D_n^2)_{liquid} = \frac{2\lambda R}{\mu}[2n+2P-1-2n+1] = \frac{4P\lambda R}{\mu} \quad \ldots(5)$$

For air film, $\mu = 1$, then $(D_{n+P}^2 - D_n^2)_{air} = 4P\lambda R \quad \ldots(6)$

From equation (5) and (6), we get $\mu = \dfrac{(D^2_{n+p} - D^2_n)_{air}}{(D^2_{n+P} - D^2_n)_{liquid}}$

Hence, μ of the liquid can be determined by measuring the values of $D^2_{n+P} - D^2_n$ for the same ring in air and in the given liquid.

Example 15: In a Newton's ring experiment the diameter of the 15^{th} ring was found to be 0.590 cm and that of the 5^{th} ring was 0.336 cm. If the radius of curvature of the plano concave lens is 100 cm, calculate the wavelength of light used.

Solution: We have that, $\lambda = \dfrac{D^2_{n+P} - D^2_n}{4PR}$

Here given that, $D_{n+P} = D_{15} = 0.590 \text{ cm}, D_n = D_5 = 0.336 \text{ cm}$

$P = 10$ and $R = 100 \text{ cm}$

Putting the values, we get $\lambda = \dfrac{(0.590)^2 - (0.336)^2}{4 \times 10 \times 100} = 5882 \times 10^{-8} \text{ cm} = 5882 \text{ Å}$

Example 16: In a Newton's ring experiment the diameter of third dark ring is 3.2 mm. Find the radius of curvature of the lens if λ of light used = 5890×10^{-8} cm

Solution: The diameter of dark ring is given by,

$$D_n^2 = 4n\lambda R$$

Here given that, $D_n = D_3 = 3.2 \text{ mm} = 0.32 \text{ cm}, \ n = 3$ and $\lambda = 5890 \times 10^{-8} \text{ cm}$

Putting the values, we get

$$R = \frac{(0.32)^2}{4 \times 3 \times 5890 \times 10^{-8}} = 145 \text{ cm}.$$

Example 17: In a newton's ring experiment the diameter of the 5^{th} ring was 0.336 cm and the diameter of the 15^{th} ring was 0.590 cm. Find the radius of curvature of the plano convex lens if the wavelength of light used is 5890 Å.

Solution: We have that, $\lambda = \dfrac{D^2_{n+P} - D^2_n}{4PR}$

or $R = \dfrac{D^2_{n+P} - D^2_n}{4P\lambda}$

Here given that, $D_{n+P} = D_{15} = 0.590 \text{ cm}, D_n = D_5 = 0.336 \text{ cm}$

$P = 10$ and $\lambda = 5890 \times 10^{-8} \text{ cm}$

Putting the values, we get

$\therefore$ $R = \dfrac{(0.590)^2 - (0.336)^2}{4 \times 10 \times 5890 \times 10^{-8}} = 99.83 \text{ cm}.$

1.17 Haidinger Fringes

As we have $2\mu t \cos r = n\lambda$

If t is large, a very small change in r will change the path difference by one wavelength. In this case the ray must pass through a plate as a parallel beam and must be received by the eye on the telescope focused for infinity. The interference patterns are known as *fringes of equal inclination.* These are different from Newton's rings. These fringes of equal inclination were first observed by Haidinger and afterwards studied by Lummer and Mascart. From an extended source *S*, light rays fall on the plate. The rays striking at the same angle and refracted at the same angle form a parallel beam and are viewed through the telescope focused for infinity (Fig. 1.16). The pattern is a series of concentric circles whose centre is the principal focus of the objective of the telescope.

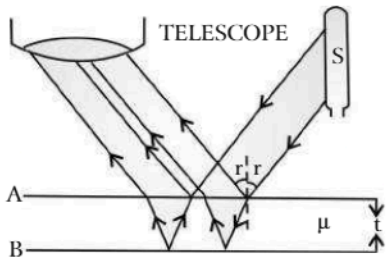


Fig. 1.16

1.18 Michelson's Interferometer

Michelson Interferometer is an excellent device to observe and study the interference fringes of various shapes which have a number of applications in optics.

1.18.1 Construction

It consists of two highly polished mirrors M_1 and M_2 and two plane glass plates *A* and *B*. The plane mirrors M_1 and M_2 are silvered on their front surfaces and mounted vertically on two arms at right angle to each other (Fig. 1.17). There are three tilt screwas at mirrors to adjust the planes of mirrors about vertical and horizontal axes. The mirror M_1 is mounted on a carriage *C* which can be moved forward or backward by a screw *E*. The plates *A* and *B* are mounted exactly parallel to each other and inclined at 45° to M_1 and M_2. The surface of glass plate *A* towards *B* is partially silvered.

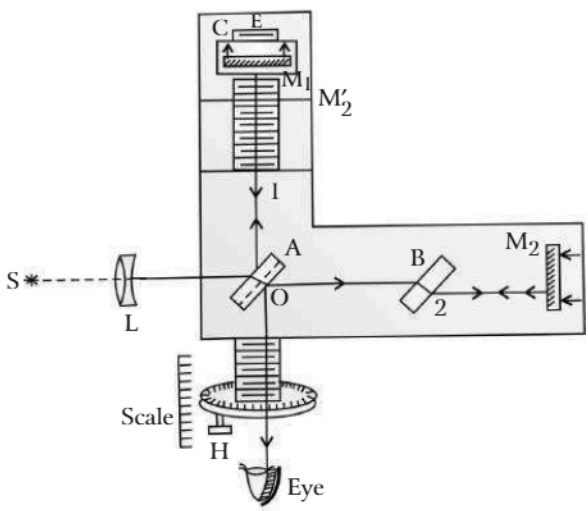


Fig. 1.17

1.18.2 Working

The light from a monochromatic sources S is made incident on the glass plate A through a lens L. The plate A is inclined at an angle of 45° and is partially silvered, so that the light incident on it is partly reflected towards mirror M_1 and partly transmitted towards mirror M_2. These two beams (reflected and transmitted) travel along two mutually perpendicular paths and are reflected back by the mirror M_1 and M_2. These two beams return to the plate A. The beam reflected back by M_1 is transmitted through the glass plate A and the beam reflected back by M_2 is reflected by the glass plate A towards the eye (Fig. 1.17) through a telescope T.

Since, the rays entering the telescope are derived from the same incident ray, they are coherent and hence interference fringes can be seen in the telescope. After partial reflection and transmission at O, the ray 1 travels through the glass plate A twice, while the ray 2 does not do so even once. Thus, to equalise the path of transmitted beam with reflected beam we introduced plate B, of the same thickness, parallel to A. So, plate B is called the **compensating plate.** The form of fringes depends on the inclination of M_1 and M_2 mirrors. Let M_2' be the image of M_2 formed by reflection at the half silvered surface of A, so that $OM_2' = OM_2$. The interference fringes may be regarded to be formed by light reflection from the surfaces of M_1 and M_2'. Thus, the arrangement is equivalent to an air film enclosed between the reflecting surfaces M_1 and M_2'.

1.18.3 Circular Fringes

If the mirrors M_1 and M_2 are perfectly perpendicular, the observer's eye will see the images of the mirrors M_1 and M_2 through A. Thus, an air film $M_1 M_2'$ of uniform thickness will be formed and we obtain circular fringes localised at infinite, which can be explained as:

Now, M_1 and M_2' are the parallel reflecting surfaces (Fig. 1.18). The actual source S will be replaced by its virtual image S' due to reflection in the partially silvered surface. S' forms two virtual image sources S_1 and S_2 in mirrors M_1 and M_2'. Light from a point such as P on the source appears to an eye E to come from the corresponding points P_1 and P_2 on S_1 and S_2 respectively. If the thickness of $M_1 M_2'$ is d and the distance between sources S_1 and $S_2 = 2d$, then the path difference between the two beams will be $2d \cos\theta$. According to usual law of interference, we have

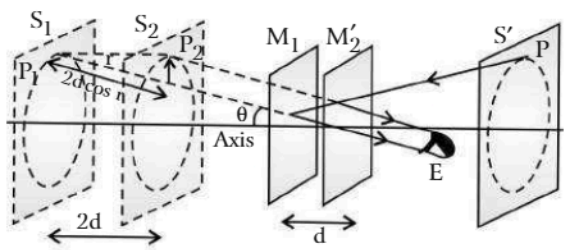


Fig. 1.18

$$2d\cos\theta = n\lambda \qquad \text{(for bright fringes)}$$

and $$2d\cos\theta = (2n+1)\lambda/2 \qquad \text{(for dark fringes)}$$

Since, the waves coming from different points of sources make different angles θ with the axis and the value of θ goes on increasing, as we move away from axis. So, in the telescope alternate dark and bright fringes are observed. Further it is important that the locus of points(lying on S_1 or S_2) having the constant angle θ with the axis of telescope, is a circle and consequently fringes will be circular. Since the interfering rays are parallel, these fringes are formed at infinity.

The centre of the rings will be dark or bright according to

$$\left.\begin{aligned} 2d &= n\lambda \quad \text{(bright)} \\ &= (2n+1)\lambda/2 \quad \text{(dark)} \end{aligned}\right\} \qquad (\because\ \theta = 0 \text{ or } \cos\theta = 1)$$

And if $d = 0$, then whatever may be the value of θ, the path difference will be zero and consequently the whole view in telescope will appear bright.

1.18.4 Straight Fringes

When the mirrors M_1 and M_2 are not exactly perpendicular *i.e.,* the mirror M_1 and virtual mirror M_2' are inclined, the air film enclosed is wedge-shaped. Since light is incident on the film at different angles so straight fringes with convexity towards the thin edge are obtained. They are alternate dark and bright straight line fringes if monochromatic light is used (Fig. 1.19) This type of fringes formed near the film are not visible for large path differences.

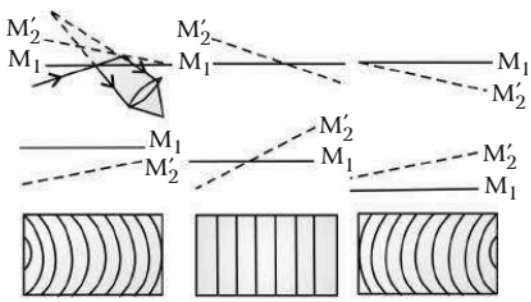


Fig. 1.19

1.18.5 White Light Fringes

When the mirrors M_1 and M_2 are not exactly perpendicular and monochromatic light is replaced by white light, then the different colours overlap on one another and only the first few coloured localised fringes are visible. The central fringe is dark and the other fringes are coloured. White light fringes are useful for the determination of zero path difference, especially in the standardization of the metre.

1.18.6 Applications of Michelson Interferometer

Michelson interferometer can be used to determine:

1. The wavelength of a monochromatic source
2. The difference between the two neighboring wavelengths
3. The refractive index of a thin transparent film.

1.19 Determination of the Wavelength of Monochromatic Light Using Michelson's Interferometer

The mirrors M_1 and M_2 of Michelson's interferometer are so adjusted that circular fringes are visible in the field of view. If M_1 and M_2 are equidistant from the glass plate A, the field of view will be perfectly dark. The mirror M_2 is kept fixed and the mirror M_1 is moved with the help of screw so that fringes with a bright spot at the centre of field of view are visible. If d is the distance through which M_1 is moved, n is the number of fringes that cross the centre of field of view *i.e.*, order of the spot obtained and λ is the wavelength of monochromatic light used, then we have

$$2d\cos\theta = n\lambda$$

But at the centre $\theta = 0$ or $\cos r = 1$,

So, $2d = n\lambda$

or $d = (n\lambda/2)$

From this we can say that for one fringe shift, the mirror M_1 moves through a distance equal to half the wavelength. Hence, λ can be determined by,

$$\lambda = \frac{\mathbf{2d}}{\mathbf{n}}$$

Example 18: In moving one mirror in a Michelson interferometer through a distance of 0.1474 mm, 500 fringes cross the centre of the field of view. What is the wavelength of light?

Solution: In Michelson interferometer we know that,

$$\lambda = \frac{2d}{n}$$

Given that, $n = 500$ and

$$d = 0.1474 \text{ mm} = 0.01474 \text{ cm}$$

Putting the value, we get $\lambda = \dfrac{2d}{n} = \dfrac{2\times 0.01474}{500} = 5896\times 10^{-8}\text{ cm} = \mathbf{5896\ \text{Å}}$

Example 19: Fringes of equal inclination are observed in a Michelson interferometer. As one of the mirrors is moved back by 1 mm, 3663, fringes move out from the centre of the pattern. Calculate λ.

Solution: In Michelson interferometer we have that,

$$\lambda = \frac{2d}{n}$$

$$n = 3663 \text{ and } d = 1\,\text{mm} = 0.1\,\text{cm}$$

putting the values, we get

$$\lambda = \frac{2 \times 0.1}{3663} = 5459 \times 10^{-8} \text{ cm}$$

$$\lambda = \mathbf{5459\,\text{Å}}.$$

1.20 Determination of Difference in Two Neighbouring Wavelength Width of Spectral Lines

When the source of light has two wavelengths very close to each other (like sodium D lines D_1 and D_2), each wavelength produces its own system of fringes. Suppose, the wavelength of D_1 line is λ_1 and the wavelength of D_2 line is λ_2 and $\lambda_1 > \lambda_2$ (slightly). Each spectral line will give its fringes in the interferometer.

By adjusting mirror M_1 of the Michelson interferometer, a position is found when the fringes are very bright. In this position, the bright fringes due to D_1 coincides with the bright fringes due to D_2 line. When the mirror is moved further, both the set of fringes get separated because their wavelengths are different. When the mirror M_1 is moved through a certain distance, the bright fringes due to one wavelength will be seen. By moving the mirror M_1 again a position is reached when bright fringe of D_1 falls on the bright fringe of D_2 and the fringe are again distinct. This is possible when nth order fringes due to wavelength λ_1 coincides with $(n+1)th$ order fringes due to wavelength. λ_2 Let n_1 and n_2 are the number of fringes that cross the centre of field of view, when mirror M_1 is displaced through a distance d, between two consecutive positions of bright fringes.

$\therefore$ $$2d = n_1\lambda_1 = n_2\lambda_2$$

Here, $$n_2 = n+1 \text{ and } n_1 = n$$

$\therefore$ $$2d = n\lambda_1 = (n+1)\lambda_2 \qquad \ldots(1)$$

or $$n\lambda_1 = (n+1)\lambda_2$$

or $$n = \frac{\lambda_2}{\lambda_1 - \lambda_2} \qquad \ldots(2)$$

and from equation (1) we have,

$$2d = n\lambda_1 = \frac{\lambda_1\lambda_2}{\lambda_1 - \lambda_2} \qquad \text{[by using (2)]}$$

or $$\lambda_1 - \lambda_2 = \frac{\lambda_1\lambda_2}{2d} \qquad \ldots(3)$$

since λ_1 and λ_2 are very close together, so $\lambda_1\lambda_2 \approx \lambda^2$ where λ is the mean of λ_1 and λ_2

$\therefore$ $$\lambda_1 - \lambda_2 = \frac{\lambda^2}{\mathbf{2d}} \qquad \ldots(4)$$

Thus, if we measure the distance moved by M_1 between two consecutive positions of bright at the centre of the field of view, and the mean wavelength is known, then we can determine the difference between two neighbouring wavelengths or width of spectral lines.

***Example 20:** In an experiment with a Michelson's interferometer, the distance travelled by the mirror for two consecutive position of maximum distinctions was* **0.2945 mm.** *If the mean wavelength for two lines of sodium is* **5893 Å,** *calculate the difference between them.*

Solution: We have that

$$\Delta\lambda = \lambda_1 - \lambda_2 = \frac{\lambda^2}{2d}$$

Here given that, mean wavelength, $\lambda = 5893 \times 10^{-8}$ cm and $d = 0.2945$ mm $= 0.02945$cm

putting the values, we get

$$\Delta\lambda = \frac{(5893 \times 10^{-8})^2}{2 \times 0.02945} \approx 6 \times 10^{-8} \text{ cm} \approx \mathbf{6\,\mathring{A}}$$

***Example 21:** Calculate the distance between the two successive positions of a movable mirror of a Michelson's interferometer giving distinct fringes in the case of sodium having lines of wavelength* **5890 Å** *and* **5896 Å.**

Solution: We have,

$$d = \frac{\lambda_1 \lambda_2}{2(\lambda_1 - \lambda_2)}$$

Given that,

$$\lambda_1 = 5896 \text{ Å} = 5896 \times 10^{-8} \text{ cm}$$

$$\lambda_2 = 5890 \text{ Å} = 5890 \times 10^{-8} \text{ cm}$$

$$\lambda_1 - \lambda_2 = 6\text{Å} = 6 \times 10^{-8} \text{ cm}$$

Putting the values, we get

$$d = \frac{5896 \times 10^{-8} \times 5890 \times 10^{-8}}{2 \times 6 \times 10^{-8}}$$

or

$$\mathbf{d = 0.02894\,cm = 0.2894\,mm}$$

1.21 Determination of Refractive Index of Thin Transparent Film

Michelson interferometer is adjusted to produce straight white light fringes and the cross wire is adjusted on one straight fringe. When a thin transparent film is introduced in the path of one of interfering beam (beam going towards M_1), a path difference of $2(\mu - 1)t$ is introduced between the interfering beams. The fringes are therefore shifted. The mirror M_1 is then moved till the fringes are again seen. If the displacement of M_1 is d, then

$$2d = 2(\mu - 1)t$$

where μ is the refractive index of the film and t is the thickness of the film.

or

$$d = (\mu - 1)t$$

Thus, measuring d and t we can determine μ.

Example 22: Straight fringes are observed in a Michelson interferometer with light of wavelength 4800 Å. In one of the paths, a liquid film of gradually changing thickness is introduced and 500 fringes cross the field of view. If the refractive index of the liquid is 1.40, calculate the change in the thickness of film.

Solution: We have **that,** $2d = 2(\mu - 1)t = n\lambda$

or $$t = \frac{n\lambda}{2(\mu - 1)}$$

Here given that, $\lambda = 4800\text{Å} = 4800 \times 10^{-10}$ m; $n = 500$ and $\mu = 1.40$

Putting the values, we get $$t = \frac{500 \times 4800 \times 10^{-10}}{2(1.4 - 1)}$$

$$t = 3 \times 10^{-4} \text{ m}$$

$$\mathbf{t = 0.3 \text{ mm}}$$

Example 23: A thin plate of refractive index 1.5 displaces 10 fringes when it is introduced in one of the arms of the Michelson interferometer. Calculate the thickness of the plate. Given that: $\lambda = 6000\ \theta°$

Solution: **We have that,** $2d = 2(\mu - 1)t = n\lambda$

or $$t = \frac{n\lambda}{2(\mu - 1)}$$

Here given that, $n = 10, \lambda = 6000\ \mathring{\text{A}} = 60 \times 10^{-7}$ m and $\mu = 1.5$

Putting the value, we get $$t = \frac{10 \times 6 \times 10^{-7}}{2(1.5 - 1)}$$

$$\mathbf{t = 6 \times 10^{-6} m}$$

$$\mathbf{= 6 \times 10^{-3} mm.}$$

1.22 Twyman and Green Interferometer

F. Twyman and A. Green in 1916 designed an interferometer, known after their names, for testing the optical homogeneity of prisms, lenses and glass plates. It is also used for testing the ruling of plane reflection grating for the absence of *ghosts*

Twyman and Green interferometer, is shown in Fig. 1.20. It is a modification of Michelson interferometer in which extended source si replaced by a monochromatic point source S at the focus of a well corrected lens L_1 and the interfering wave fronts are focussed at E by another lens L_2. The compensating plate is not essential

since it is invariably used with monochromatic light. The point source is realised in practice by focussing mono chromatic light from an extended source by an auxiliary lens on a small hole in a screen placed at *S*. (Mercury light is commonly employed while a Wratten filter in the eye-piece isolates the green line). The interferometer is illuminated with strictly parallel light. The single plane wavefront *W*, emerging from the lens L_1, after partial reflection at the lightly silvered surface of the plate *P*, gives rise to two plane wavefronts W_1 and W_2 which are respectively reflected at the plane mirrors M_1 and M_2 at normal incidence. Finally, the reflected plane wavefronts, by partial reflection and refraction at *P*, give rise to superposed plane wavefronts, proceeding towards the second well corrected lens, L_2 and thus they are focussed at *E*, where the observer's eye is situated. When M_1 and M_2 are exactly perpendicular so that the lightly silvered side of *P*, exactly bisects the angle between them, the superposed wavefronts are exactly parallel. Therefore the phase difference between the superposed disturbances is the same at every point in the field of view. The field is, therefore, of uniform colour or intensity, the value depending on the extent to which wavefronts reinforce each other.

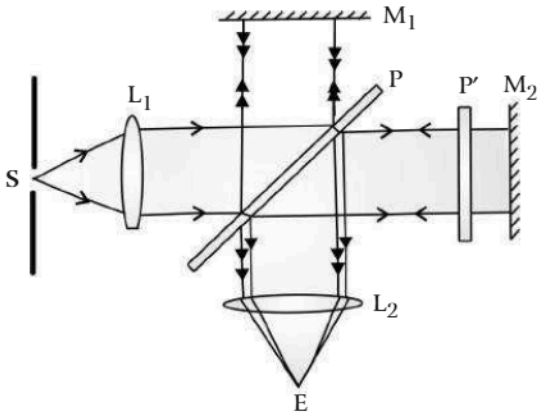


Fig. 1.20

Now a test plate *P′* is introduced. If it is uniform in thickness and structure, the field of view remains of uniform intensity. But if there are irregularities in *P′*, fringes show up around those irregularities and one can correct them (by polishing etc.,) while viewing the improvements. If *P′* is convex lens, then M_2 is replaced by a convex mirror of radius of curvature equal to the focal length of the lens. If mirrors are to be tested and corrected, *P′* is an ideal lens and M_2 is the mirror under test. In effect the arrangement is used to make the test plates, lenses and mirrors accurate to better then $\lambda/10$ or so in precision.

This interferometer cannot be used to test lenses and mirrors of large aperture because both the glass plate *P* and the comparison mirror M_1 must be as large as the prism under test and to construct large accurately plane surfaces is very laborious process. Bates interferometer has been designed for testing large mirrors.

1.23 Multiple Beam Interference

When the interference pattern is obtained due to the superimposition of many coherent beams together then it is known as **multiple beam interference.** To understand it, let us consider a plane and parallel sided highly reflecting plate of refractive index μ placed in air. Let *t* be the thickness of the plate. The incident ray of light is partly reflected and partly transmitted at different points *i.e., A, B, C, B, D....* on both surfaces of the plate.

It results into a number of parallel reflected rays and a number of parallel transmitted rays as shown in Fig. 1.21. If the reflected or transmitted rays are seen collectively by a converging lens, then they interfere constructively or destructively at a point in the focal place of lens. Thus, an interference, pattern is obtained in the focal plane of lens due to the rays from different point of source.

Let r and t be the fraction of amplitudes of reflected and transmitted wave while travelling from air into plate and r' and t' while travelling from plate into air. Therefore, the amplitudes of successive reflected and transmitted waves will be as shown in Fig. 1.21.

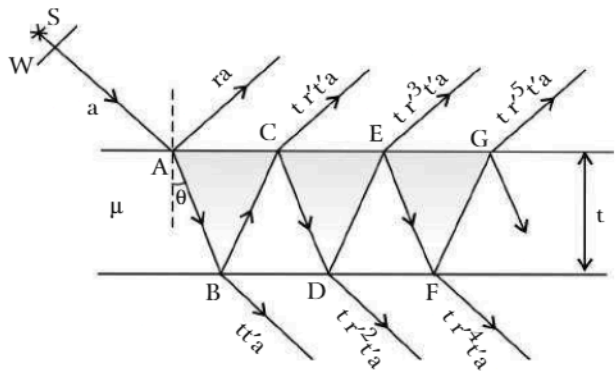


Fig. 1.21

1.24 Intensity Distribution in Interference Pattern of Reflected Waves

As different reflected or transmitted rays traverse different optical paths, so there will a phase difference (say δ) between two successive reflected (or transmitted) rays. If θ is the angel between normal and transmitted wave as shown in Fig. 1.21, then path difference between two successive reflected (or transmitted) rays $= 2\mu t \cos\theta$ (using Snell's law), then the phase difference will be,

$$\frac{2\pi}{\lambda}(2\mu t \cos\theta)$$

From the method of complex representation of sinusoidal functions, the complex amplitude of successive reflected waves will be,

$$ra, tr' t' ae^{-i\delta}, tr'^3 t' ae^{-2i\delta}, tr'^5 t' ae^{-3i\delta} \ldots..$$

Let Z_{ref} be the complex amplitude of the resultant reflected wave, then

$$Z_{ref} = a[r + tt' r' e^{-i\delta}(1 + r'^2 e^{-i\delta} + r'^4 e^{-2i\delta} + \ldots.)]$$

If the plate is long enough, the number of reflected waves will be infinite. Then we may write,

$$Z_{ref} = a\left\{r + tt' r' e^{-i\delta}\left(\frac{1}{1 - r'^2 e^{-i\delta}}\right)\right\}$$

And from Stoke's treatment, we have

$$r = -r', \text{ and } tt' = 1 - r^2 = 1 - r'^2$$

So,
$$Z_{\text{ref}} = a\left\{-r' + \frac{(1-r'^2)\, r'\, e^{-i\delta}}{(1-r'^2\, e^{-i\delta})}\right\} = a\left\{\frac{r'\,(e^{-i\delta}-1)}{1-r'^2\, e^{-i\delta}}\right\}$$

But also, we have that the intensity $\propto a^2$, and square of the absolute value of a complex number is equal to the product of the number and its complex Conjugate. So, the reflected intensity will be,

$$I_{\text{ref}} = K\, Z_{\text{ref.}}\, \overline{Z}_{\text{ref}}$$

where K is some constant and $\overline{Z}_{\text{ref}}$ is complex conjugate of $Z_{\text{ref.}}$ Thus,

$$I_{\text{ref}} = K\left[\frac{ar'\,(e^{-i\delta}-1)}{1-r'^2\, e^{-i\delta}}\right]\left[\frac{ar'\,(e^{i\delta}-1)}{1-r'^2\, e^{i\delta}}\right]$$

$$= K\, a^2\left[\frac{r'^2\,\{2-(e^{i\delta}+e^{-i\delta})\}}{1-r'^2\,(e^{i\delta}+e^{-i\delta})+r'^4}\right]$$

But we have that, $e^{i\delta} + e^{-i\delta} = 2\cos\delta$ and $Ka^2 = I =$ intensity of incident light.

$\therefore$
$$I_{\text{ref}} = I\left[\frac{2r'^2\,(1-\cos\delta)}{1+r'^4-2r'^2\cos\delta}\right]$$

or
$$I_{\text{ref}} = I\left[\frac{4r'^2\sin^2(\delta/2)}{(1-r'^2)^2+4r'^2\sin^2(\delta/2)}\right] \qquad [\because \cos\delta = 1-2\sin^2(\delta/2)]$$

or
$$I_{\text{ref}} = \frac{I}{1+\dfrac{(1-r'^2)^2}{4r'^2\sin^2(\delta/2)}}$$

But $r'^2 = r^2 = R$, the reflectivity of the plane surfaces,

$\therefore$
$$\mathbf{I_{ref} = \frac{I}{1+\dfrac{(1-R)^2}{4R\sin^2(\delta/2)}}}$$

This is the expression for reflected intensity and is known as Airy's formula.

A plot of $\frac{I_{\text{ref}}}{I}$ vs δ for different values of reflectively R is shown in Fig. 1.22. From the curves it is obvious that minima appear as dark rings on bright background in the interference pattern. Higher the reflectivity, darker are the minima rings.

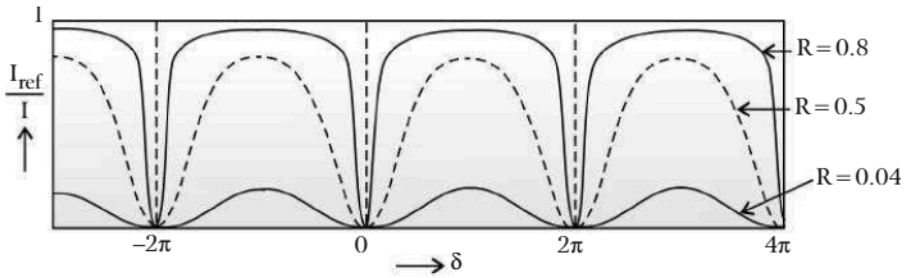


Fig. 1.22

1.25 Intensity Distribution in Interference Pattern of Transmitted Waves

Although the expression for transmitted intensity can be obtained by proceeding in a similar way but we derive it here using the fact that the sum of the reflected and transmitted intensities is equal to incident intensity *i.e.,*

$$I_{ref} + I_{trans} = I$$

or
$$I_{trans} = I - I_{ref} = I - \frac{I}{1 + \frac{(1-R)^2}{4R\sin^2(\delta/2)}}$$

or
$$\mathbf{I_{trans} = \frac{I}{1 + \frac{4R\sin^2(\delta/2)}{1-R^2}}}$$

A plot of I_{trans} vs δ for different values of reflectivity R is shown in Fig. 1.23. From the curves it is obvious that the maxima appear as bright rings on a dark background in transmitted interference pattern. Higher the value of R, brighter are the maximarings.

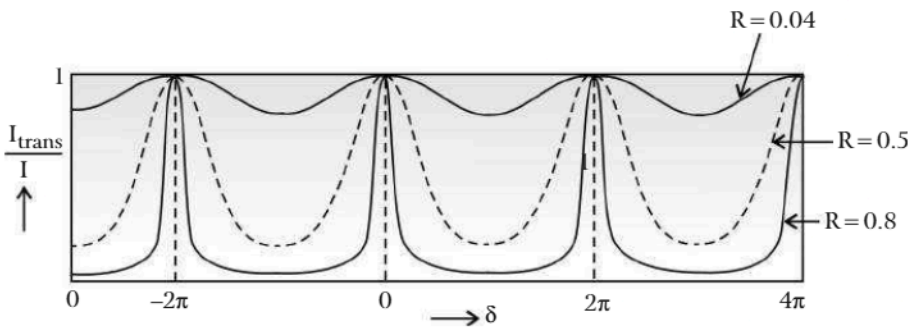


Fig. 1.23

1.26 Febry-Perot Interferometer

It is a high resolving power instrument which is based on the multiple beam interference. Because of its high resolving power, it is commonly used for the *study of hyperfine structure of the spectral lines.*

1.26.1 Construction

A Fabry-Perot interferometer consists of two plane parallel glass plates A and B separated by a distance. The inner surfaces of the plates are silvered. S_1 is a broad source of monochromatic light and L_1 is a convex lens which makes the ray parallel. Here, the incident rays on glass plates gets a large number of internal reflections successively at the two silvered surfaces as shown in Fig. 1.24. At each reflection a small fractional part of the light is transmitted also. Thus, each incident ray produces a group of coherent and parallel transmitted rays with a constant path difference between any two successive rays. A second convex lens L_2 brings these rays together at a point P in its focal plane where they interfere. Hence, the rays from all points of the source produce an interference pattern on a screen S_2 placed in the focal plane of L_2. In this system, one of the two plates is kept fixed while the other can be moved vary the separation of the two plates. This configuration of this system is called a **Fabry-Perot interferometer.** Sometimes both the plates are kept at a fixed separation. The system with fixed spacing is known as **Fabry-perot etalon.**

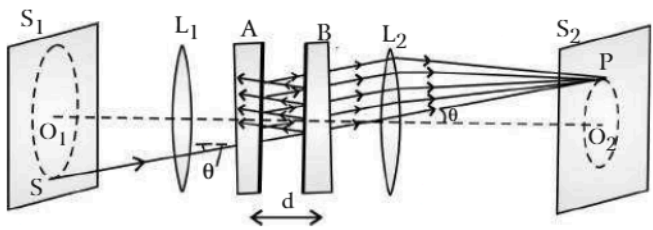


Fig. 1.24

If separation between two silvered surfaces is d and θ is the inclination of a particular ray with the normal to the plates, then the difference between any two successive transmitted rays corresponding to the incident ray is $2d\cos\theta$. So, the condition for maximum intensity is given by,

$$2d\cos\theta = n\lambda$$

where n is an integer which represents the order of interference and λ is the wavelength of light. The locus of points in the source which give rays of a constant inclination θ, is a circle. Therefore, an extended source will give the interference pattern of bright concentric rings on a dark background.

Intensity Distribution: Let P_1 and P_2 represent the internal reflecting surfaces of parallel plates A and B. Let SA_1 be a plane wave incident on the plate at an angle θ with the normal. The wave under goes multiple reflections at the two surfaces P_1 and P_2 and consequent transmissions as shown in Fig. 1.25. Let T and R by the fractions of the

incident light intensity which represent transmitted and reflected wave intensity respectively. The fractional transmitted and reflected amplitude will be $\sqrt{T}$ and $\sqrt{R}$, and at each reflection the amplitude becomes $\sqrt{R}$ times, at each transmission the amplitude becomes $\sqrt{T}$ times. As the amplitude of the incident wave is 'a', the amplitude of the wave A_1B_1 is $\sqrt{T}$ and that of B_1T_1 is $(\sqrt{T} \times \sqrt{T}) = aT$. Similarly, the amplitude of the wave B_2T_2 is aTR. In a similar way the amplitudes of the waves B_3T_3 and B_4T_4.... etc., are aR^2T, aR^3T.... etc. respectively. All these transmitted waves (derived from the same incident wave SA_1) are coherent and capable of producing interference.

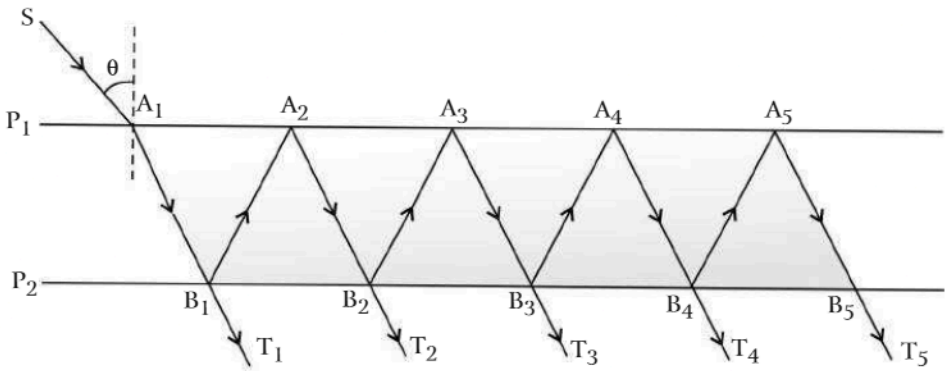


Fig. 1.25

The phase difference δ between any two consecutive rays reaching a point on the screen is given by,

$$\delta = \frac{2\pi}{\lambda} \times \text{Path difference} = \frac{2\pi}{\lambda}(2d\cos\theta) \qquad ...(1)$$

Let the incident wave be represented by,

$$Y = ae^{i\omega t},$$

then the waves reaching a point on the screen will be,

$$y_1 = aTe^{i\omega t},$$

$$y_2 = aTR\, e^{i(\omega t - \delta)}$$

$$y_3 = aTR^2 e^{i(\omega t - 2\delta)} \ \ldots \text{etc.}$$

By the principle of superposition, the resultant amplitude is given by,

$$A = aT + aTRe^{-i\delta} + aTR^2e^{-2i\delta} + ...$$

$$= aT\,[1 + Re^{-i\delta} + R^2e^{-2i\delta} + R^3e^{-3i\delta} + ...]$$

$$= aT\left(\frac{1}{1 - Re^{-i\delta}}\right)$$

Thus, the complex conjugate of A,

$$A^* = aT\left(\frac{1}{1 - Re^{i\delta}}\right)$$

Hence, the resultant intensity, $I = AA^*$

$$= \frac{a^2T^2}{(1-Re^{-i\delta})(1-Re^{i\delta})}$$

$$= \frac{a^2T^2}{1+R^2-R(e^{i\delta}+e^{-i\delta})} = \frac{a^2T^2}{1+R^2-2R\cos\delta}$$

$$= \frac{a^2T^2}{(1-R)^2+4R\sin^2(\delta/2)} \qquad [\because \cos\delta = (1-2\sin^2\delta/2)]$$

$$= \frac{a^2T^2}{(1-R)^2}\left[\frac{1}{1+\frac{4R}{1-R^2}\sin^2(\delta/2)}\right] \qquad ...(2)$$

or $$\mathbf{I = \frac{a^2T^2}{(1-R)^2[1+F\sin^2(\delta/2)]}} \qquad ...(3)$$

where $F = \frac{4R}{(1-R)^2}$ is called the **coefficient of fitness or coefficient of sharpness.**

1.26.2 Conditions for Maximum and Minimum Intensities

For Maximum Intensity

The intensity will be a maximum, when $\sin^2\delta/2 = 0$

or $$\delta = 2n\pi, \quad \text{where } n = 0,1,2,\ldots \text{ etc.}$$

Thus, the maximum intensity will be,

$$\mathbf{I_{max} = \frac{a^2T^2}{(1-R)^2}} \qquad ...(4)$$

For Minimum Intensity

The intensity will be a minimum, when $\sin^2(\delta/2) = 1$

i.e., $$(\delta/2) = (2n+1)\pi/2$$

or $$\delta = (2n+1)\pi, \text{ where } n = 0,1,2,\ldots \text{ etc.}$$

Thus, the minimum intensity will be,

$$I_{min} = \frac{a^2T^2}{(1-R)^2\left[1+\frac{4R}{(1-R)^2}.1\right]} = \mathbf{\frac{a^2T^2}{(1+R)^2}} \qquad ...(5)$$

The expression of intensity in equation (3) can be written in terms of equation (4) as,

$$I = \frac{I_{max}}{\left[1+F\sin^2\frac{\delta}{2}\right]} \qquad ...(6)$$

This is the intensity expression for the F.P. fringes.

or
$$\frac{I}{I_{max}} = \frac{1}{1+\frac{4R}{(1-R)^2}\sin^2\frac{\delta}{2}} \quad ...(7)$$

1.26.3 Sharpness of the Fringes

If I versus δ is plotted for different values of R the reflectivity of plate then a set of curves is obtained as shown in Fig. 1.26. They show that larger the value of R, more

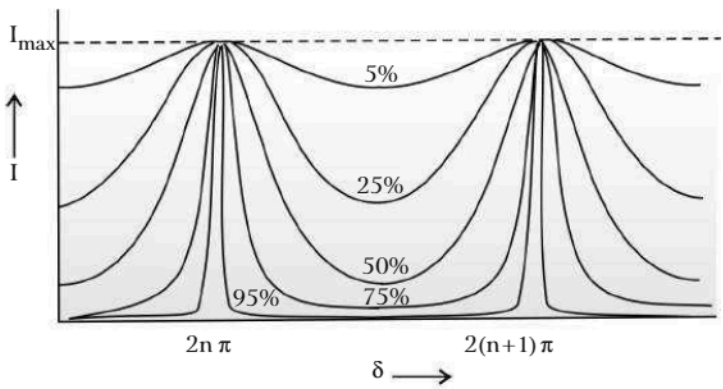


Fig. 1.26

abrupt in the fall of intensity on either side of a maximum. *i.e., higher the reflective of the plates, sharper are interference fringes or brighter is the fringe pattern.* A measurement of the sharpness of a fringe is the "half fringe width" which is defined as the *total width of a fringe at points where the intensity has fallen to half of its maximum intensity* (shown in Fig. 1.27).

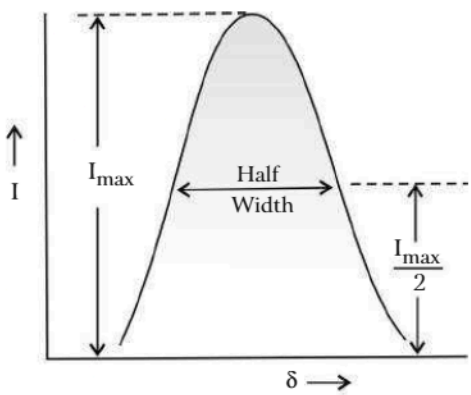


Fig. 1.27

We have
$$\frac{I}{I_{max}} = \frac{1}{1+\frac{4R}{(1-R)^2}\sin^2\frac{\delta}{2}} \quad ...(8)$$

From the definition of half fringe width,

$$\frac{I}{I_{max}} = \frac{1}{2}$$

Therefore,

$$\frac{1}{2} = \frac{1}{1 + \frac{4R}{(1-R)^2}\sin^2\frac{\delta}{2}}$$

or $$\frac{4R}{(1-R)^2}\sin^2\frac{\delta}{2} = 1$$

or $$\sin^2\frac{\delta}{2} = \frac{(1-R)^2}{4R} \quad \ldots(9)$$

$$\delta = 2\sin^{-1}\frac{(1-R)}{2\sqrt{R}}$$

If $R = 1, \delta = 0,$

It is clear from equation (9) that the value of δ at half width decreases with increase in the value of R.

If $R = 0.90$, then $\delta = 0.211$ radian

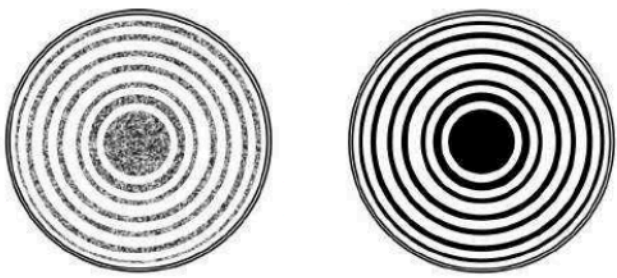
In case of Michelson interferometer, the intensity of the fringe system is given by,

$$I = I_{max}\cos^2\frac{\delta}{2}$$

At half width, $$\frac{I}{I_{max}} = \frac{1}{2}, \cos^2\frac{\delta}{2} = \frac{1}{2}$$

or $$\delta = 2\cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = 1.57 \text{ radian}$$

It shows that the fringes obtained with a Fabry-Perot interferometer are comparatively much sharper then those obtained with Michelson interferometer.



(a) Fringes obtained by Michelson interferometer

(b) Fringes obtained with F.P. interferometer

Fig. 1.28

Example 24: Two F.P. interferometer have equal separation, but their coefficients of reflections are 0.64 and 0.81. Determine the relative widths of maxima in two cases.

Solution: We have the width of maxima,

$$\sin^2\frac{\delta}{2}=\frac{(1-R)^2}{4R}=\frac{1}{F}$$

or $$\sin\frac{\delta}{2}=\frac{1}{\sqrt{F}}$$

For large R, $\frac{1}{\sqrt{F}}$ will be very small *i.e.*, $\sin\frac{\delta}{2}\approx\frac{\delta}{2}$

$\therefore$ $$\frac{\delta}{2}=\frac{1}{\sqrt{F}}$$

For $$R=0.64,\quad F_1=\frac{4R}{(1-R)^2}=\frac{4\times0.64}{(1-0.64)^2}=19.6$$

For $$R=0.81\quad F_2=\frac{4\times0.81}{(1-0.81)^2}\approx90$$

So, $$\frac{\delta_1}{\delta_2}=\sqrt{\frac{F_2}{F_2}}=\sqrt{\frac{90}{19.6}}\approx2.14$$

So, the relative widths of maxima is 2.14. Thus, it shows that the half width with $R=0.64$ is greater than the half of width with $R=0.81$.

Example 25: Two Fabry-Perot interferometers have equal plate separations but the coefficients of intensity reflection are 0.80 and 0.90. Deduce the relative width of the maxima in the two cases.

Solution: The intensity distribution in *F-P* fringes is given by,

$$I=\frac{I_{max}}{1+F\sin^2(\delta/2)}$$

where $F=\frac{4R}{(1-R)^2}$, and R is the reflecting power of the plates,

if $$I=\frac{I_{max}}{2}\ \text{(For half width)}$$

So $$\frac{I}{I_{max}}=\frac{1}{1+F\sin^2(\delta/2)}$$

or $$\frac{1}{2}=\frac{1}{1+F\sin^2(\delta/2)}$$

or $$\sin\frac{\delta}{2}=\frac{1}{\sqrt{F}}$$

For large R, $\frac{1}{\sqrt{F}}$ will be so small $\Rightarrow \sin\frac{\delta}{2} = \frac{\delta}{2}$

$\therefore \quad \frac{\delta}{2} = \frac{1}{\sqrt{F}}$

For $R = 0.80$, $F_1 = \frac{4R}{(1-R)^2} = \frac{4\times 0.8}{0.04} = 80$

and For $R = 0.90$, $F_2 = \frac{4\times 0.9}{0.01} = 360$

$\therefore \quad \frac{\delta_1}{\delta_2} = \sqrt{\frac{F_2}{F_1}} = \sqrt{\frac{360}{80}} = \sqrt{\frac{9}{2}} = 2.12$

Miscellaneous Problems

Problem 1: Find the ratio of intensity of the centre of a bright fringe to the intensity at a point quarter of the distance between two fringes from the centre.

Solution: Since coherent sources forming fringes always have same amplitude *i.e.*, $a_1 = a_2 = a$. So, the resultant intensity at any point is given by,

$$I = a_1^2 + a_2^2 + 2a_1a_2\cos\delta$$
$$= a^2 + a^2 + 2a^2\cos\delta = 2a^2(1+\cos\delta)$$

where δ is phase difference between the interfering waves.

At the centre of a bright fringes, $\delta = 0$ or $\cos\delta = 1$

So, $$I_C = 2a^2(1+1) = 4a^2$$

The phase difference between successive fringes is 2π. Thus, at a point distant one quarter of the distance between two fringes, the phase difference will be $\delta = 2\pi/4 = \pi/2$. Hence, the intensity at this point I_1 will be given by,

$$I_1 = 2a^2(1+\cos\pi/2) = 2a^2$$

Therefore, the required ratio $= \frac{I_C}{I_1} = \frac{4a^2}{2a^2} = 2$

Problem 2: Two coherent sources of monochromatic light of wavelength 6000 Å produce an interference pattern on a screen kept a distance of 1 m from them. The distance between two consecutive bright fringes on the screen is 0.5 mm. Find the distance between the two coherent sources.

Solution: We have the fringe width, $\beta = \frac{D\lambda}{2d}$ or $2d = \frac{D\lambda}{\beta}$

where $2d$ is the separation between coherent sources.

Here given that, $D = 1\,\text{m} = 100\,\text{cm}, \lambda = 6000 \times 10^{-8}\,\text{cm}$ and $\beta = 0.5\,\text{mm} = 0.05\,\text{cm}$

Putting all the values, we get

$$2d = \frac{6000 \times 10^{-8} \times 100}{0.05} = 0.12\,\text{cm} = \mathbf{1.2\ mm}$$

Problem 3: Light of wavelength 5500 Å from a narrow slit is incident on a double slit. The overall separation of 5 fringes on a screen 200 cm away is 1 cm. Calculate (i) the fringe width (ii) the slit separation.

Solution: (i) The fringe width, $\beta = \dfrac{\text{Overall separation}}{\text{Number of fringes}} = \dfrac{1}{5}\,\text{cm} = \mathbf{0.2\,cm}$

(ii) We have that, $\beta = \dfrac{D\lambda}{2d}$ or $2d = \dfrac{D\lambda}{\beta}$

Here given that, $D = 200\,\text{cm}, \lambda = 5500 \times 10^{-8}$ cm and $\beta = 0.2$ cm

$\therefore$ $$2d = \frac{200 \times 5500 \times 10^{-8}}{0.2} = \mathbf{0.055\,cm}$$

Problem 4: Two coherent sources are 0.18 mm apart and the fringes are observed on a screen 80 cm away. It is found that with a certain monochromatic source of light, the fourth bright fringes is situated at a distance of 10.8 mm from the central fringe. Calculate the wavelength of light.

Solution: We have that, $\beta = \dfrac{D\lambda}{2d}$ or $\lambda = \dfrac{\beta 2d}{D}$

Given that, $4\beta = 10.8\,\text{mm} = 1.08\,\text{cm}, D = 80\,\text{cm},$ and $2d = 0.18\,\text{mm} = 0.018\,\text{cm}$

$\therefore$ $$\lambda = \frac{1.08 \times 0.018}{4 \times 80} = 6075 \times 10^{-8}\,\text{cm} = \mathbf{6075\,Å}$$

Problem 5: A light source emits light of two wavelengths $\lambda_1 = 4300$ Å and $\lambda_2 = 5100$ Å. The source is used in a double slit interference experiment. The distance between the sources and the screen is 1.5 m and the distance between the slits is 0.025 mm. Calculate the separation between the third order bright fringes due to these two wavelengths.

Solution: The distance of the nth bright fringe from the central fringe is given by,

$$x_n = \frac{n\lambda D}{2d}$$

Let for $\lambda = \lambda_1$, the third fringe be at x_{31} and for $\lambda = \lambda_2$, the third fringe at λ_{32} from the central fringe.

$\therefore$ $$x_{31} = \frac{3\lambda_1 D}{2d} \quad \text{and} \quad \therefore \quad x_{32} = \frac{3\lambda_2 D}{2d}$$

The separation between the third order bright fringes due to these two wave lengths will be,

$$x_{32} - x_{31} = \frac{3\lambda_2 D}{2d} - \frac{3\lambda_1 D}{2d} = \frac{3D}{2d}[\lambda_2 - \lambda_1]$$

$$= \frac{3 \times 150}{0.0025}[5100 \times 10^{-8} - 4300 \times 10^{-8}]\,\text{cm}$$

$$= \mathbf{1.44\,cm}$$

Problem 6: ***In a two slit interference pattern with $\lambda = 6000$ Å the zero order and the 10^{th} order maxima fall at 12.34 mm and 14.73 mm respectively. If λ is changed to 5000 Å, deduce the position of the zero order and twentieth order fringes, the other arrangement remaining the same.***

Solution: We have that, $\beta = \dfrac{D\lambda}{2d}$

Given that, $10\beta = 14.73 - 12.34$ mm

or $\beta = 2.39/10 = 0.239$ mm $= 0.0239$ cm for $\lambda = 6000$ Å

$\therefore$ $\dfrac{(\beta)_{6000}}{(\beta)_{5000}} = \dfrac{6000\,\text{Å}}{5000\,\text{Å}} = \dfrac{6}{5}$ [$\because$ D and $2d$ remain same]

or $(\beta)_{5000}\,\text{Å} = 5/6 \times 0.0239 = 0.0199$ cm $= .199$ mm

Thus, with $\lambda = 5000$ Å, **the zero order fringe will be at 12.34 mm and the twentieth order fringe will be at $12.34 + 0.199 \times 20 = 16.32$ mm**

Problem 7: ***Calculate the separation between the coherent sources formed by a biprism whose inclined faces make angle of 2° with its base. The slit source distance is 10 cm from the biprism.*** **($\mu = 1.50$).**

Solution: We have that, $2d = \mathbf{2a(\mu - 1)\alpha}$

Here given that, $a = 10$ cm, $\mu = 1.50$ and $\alpha = 2° = 2\pi/180 = (\pi/90)$ radian

$\therefore$ $2d = 2 \times 10 \times (1.50 - 1)(\pi/90) = \pi/9 = \mathbf{0.349\,cm}$

Problem 8: ***The inclined faces of a biprism of refractive index 1.50 makes an angle of 2° with the base. A slit illuminated by a monochromatic light is placed at a distance of 10 cm from the biprism. If the distance between two dark fringes observed at a distance of 1 m from the biprism is 0.18 mm, find the wavelength of light used.***

Solution: We have that, $\lambda = \dfrac{\beta 2d}{D}$

Here given that, $\beta = 0.18\,\text{mm} = 0.18 \times 10^{-3}\,\text{m}$

and $D = 10 + 100 = 110\,\text{cm} = 1.1\,\text{m}$

Also, we have, $2d = 2a(\mu - 1)\alpha$

$a = 10 \text{ cm} = 0.1 \text{ m}, \alpha = (\pi/90) \text{ rad, and } \mu = 1.5$

$\therefore \quad 2d = 2 \times 0.1(1.5 - 1)(\pi/90) = 3.49 \times 10^{-3} \text{ m}$

Therefore, $\lambda = \dfrac{0.18 \times 10^{-3} \times 3.49 \times 10^{-3}}{1.1} = 5.710 \times 10^{-7} \text{ m}$

$= \mathbf{5710 \text{ Å}}$

Problem 9: In a biprism experiment with sodium light, bands of width 0.0195 cm are observed at 100 cm from the slit on introducing a convex lense, 30 cm away from the slit. Two images of the slit are seen 0.7 cm apart at 100 cm distance from the slit. Calculate the wavelength of sodium light.

Solution: The fringe width is given by, $\beta = \dfrac{D\lambda}{2d}$ or $\lambda = \dfrac{\beta . 2d}{D}$

Here given that, $\beta = 0.0195 \text{ cm}$ and $D = 100 \text{ cm}$

And for a convex lens, $I/O = v/u$ and $v + u = 100 \text{ cm}$

Given that, $u = 30 \text{ cm}$, so $v = 70 \text{ cm}$

So, $0.7/O = 70/30$ or $O = 0.30 \text{ cm}$

i.e., the distance between the two coherent sources, $2d = O = 0.30 \text{ cm}$

$\therefore \quad \lambda = \dfrac{0.0195 \times 0.30}{100} = 5850 \times 10^{-8} \text{ cm}$

$= \mathbf{5850 \text{ Å}}$

Problem 10: In a biprism experiment, the eyepiece was placed at a distance of 120 cm from the source. The distance between two virtual sources was found to be 0.075 cm. Find the wavelength of light used if the eyepiece has to be moved through a distance 1.888 cm for 20 fringes to cross the field of view.

Solution: The fringe width, $\beta = \dfrac{\text{Total travellled distance}}{\text{Number of fringes}} = \dfrac{l}{n}$

$\therefore \quad \beta = \dfrac{1.888}{20} = 0.0944 \text{ cm}$

and also given that, $D = 120 \text{ cm}, 2d = 0.075 \text{ cm}$

$\therefore \quad \lambda = \dfrac{\beta 2d}{D} = \dfrac{0.0944 \times 0.075}{120} = 5.9 \times 10^{-5} \text{ cm}$

$= 5900 \times 10^{-8} \text{ cm} = \mathbf{5900 \text{ Å}}$

***Problem 11:** In an experiment with Fresnel's biprism, fringes for light of wavelength 5×10^{-5} cm are observed 0.2 mm apart at a distance of 175 cm from the prism. The prism is made of glass of refractive index 1.50 and it is at a distance of 25 cm from the illuminated slit. Calculate the angle at the vertex of the biprism.*

Solution: We have that, $\beta=\dfrac{D\lambda}{2d}$ or $2d=\dfrac{D\lambda}{\beta}$

Given that, $D=175+25=200\text{ cm},\ \lambda=5\times10^{-5}\text{ cm}$

So, $2d=\dfrac{200\times5\times10^{-5}}{2\times10^{-2}}=0.5\text{ cm}$

but, we also have $2d=2a(\mu-1)\alpha$ or $\alpha=\dfrac{2d}{2a(\mu-1)}$

Here given that, $a=25\text{ cm},\ \mu=1.5$ and $2d=0.5\text{ cm}$

So, $\alpha=\dfrac{0.5}{2\times25\times(1.5-1)}=\dfrac{0.5}{50\times0.5}$

$=0.02$ radian

The vertex angle $\theta=(\pi-2\alpha)$ radian $=(\pi-0.04)$ radian $=\mathbf{177°42'}$

***Problem 12:** A biprism is placed at a distance of 5 cm in front of a narrow slit, illuminated by sodium light ($\lambda=5890$ Å) and the distance between the virtual sources is found to be 0.05 cm. Find the width of the fringes observed in an eyepiece placed at a distance of 75 cm from the biprism.*

Solution: The fringe width is given by, $\beta=\dfrac{D\lambda}{2d}$

Here given that, $D=5+75=80\text{ cm}, 2d=0.05\text{ cm}$

and $\lambda=5890\times10^{-8}\text{ cm}$

$\therefore$ $\beta=\dfrac{80\times5890\times10^{-8}}{0.05}=\mathbf{9.424\times10^{-2}\text{ cm}}$

***Problem 13:** Interference fringes are observed with a biprism of refracting angle 1° and refractive index 1.5 on a screen 80 cm away from it. If the distance between the source and the biprism is 10 cm, calculate the fringe width when the wavelength of light used is 5900 Å.*

Solution: The fringe width is given by, $\beta=\dfrac{D\lambda}{2d}$

Here given that, $D=80+10=90\text{ cm},\ \lambda=5900\text{ Å}$

and $2d = 2a(\mu - 1)\alpha = 2 \times 10\,(1.5 - 1)\,(\pi/180)$ [$\because\ \alpha = 1° = \pi/180$ rad]

$= 1.74 \times 10^{-1}$ cm

$\therefore\quad \beta = \dfrac{90 \times 5900 \times 10^{-8}}{1.74 \times 10^{-1}} = 3.05 \times 10^{-2}$ cm

$= \mathbf{0.030\ cm}$

Problem 14: ***The distance between the slit and biprism and between biprism and screen is 50 cm each. The obtuse angle of biprism is 179° and its $\mu = 1.5$. If the width of fringes is 0.0135 cm, calculate the λ of light used.***

Solution: **The angle of biprism** $= \dfrac{180° - 179°}{2} = 0.5° = \dfrac{0.5 \times \pi}{180}$ rad.

Here given that, $a = 50$ cm, $\mu = 1.5$, $D = 50 + 50 = 100$ cm and $\beta = 0.0135$ cm

$\therefore\quad 2d = 2a(\mu - 1)\alpha = 2 \times 50\,(1.5 - 1) \times \dfrac{0.5 \times \pi}{180} = \dfrac{25\,\pi}{180}$

The wavelength of light is given by,

$$\lambda = \frac{\beta 2d}{D} = \frac{0.0135 \times 25 \times 22}{7 \times 180 \times 100}\ \text{cm}$$

or $\lambda = 5890 \times 10^{-8}$ cm $= \mathbf{5890\ Å}$

Problem 15: ***In any interference arrangement at a certain point, we get 100th maximum for $\lambda = 6000$ Å. What will be order of maximum at that point for $\lambda = 5000$ Å and 7000 Å?***

Solution: Given that, 100th maximum at a certain point for $\lambda = 6000$ Å

Then we have, $\beta_1 = \dfrac{D\lambda_1}{2d}$ ...(1)

$\beta_2 = \dfrac{D\lambda_2}{2d}$ ...(2)

[$\because\ D$ and $2d$ remain same]

and also $100\,\beta_1 = n\beta_2$

or $n = 100\,\dfrac{\beta_1}{\beta_2} = 100\,\dfrac{\lambda_1}{\lambda_2}$ [by using eq. (1) and (2)]

$\therefore\quad n = 100 \times \dfrac{6000}{5000} = \mathbf{120}$ for $\lambda = 5000$ Å

Similarly, for light of $\lambda_2 = 7000$ Å,

$$m = 100 \times \frac{6000}{7000} = \mathbf{85.7}$$

***Problem 16:** In a biprism experiment, the positions of the slit and eyepiece were 1 and 100 cms respectively. The distance between two images for the two positions of lens was 0.3 and 1.2 mm. The width of 10 fringes was 9.72 mm. Find the wavelength of the light used.*

Solution: We have that, $\beta = \dfrac{D\lambda}{2d}$ or $\lambda = \dfrac{\beta 2d}{D}$

Here given that, $\beta = 9.72/10 = 0.972 \text{ mm} = 0.0972 \text{ cm}, D = 100 - 1 = 99 \text{ cm}$

and also we have, $2d = \sqrt{(d_1 d_2)} = \sqrt{(0.3 \times 1.2)} = 0.6 \text{ mm} = 0.06 \text{ cm}$

$\therefore$ $$\lambda = \frac{0.0972 \times 0.06}{99} = \frac{648}{11} \times 10^{-6} \approx 59 \times 10^{-6}$$

$$= 5900 \times 10^{-8} \text{ cm} = \mathbf{5900 \text{ Å}}$$

***Problem 17:** A Fresnel's biprism arrangement is set with sodium light ($\lambda = 6000$ Å) and 50 fringes are observed in its field of view. If λ is changed to 5000 Å, how many fringes will appear in the field of view?*

Solution: We have that, $\beta_s = \dfrac{D\lambda_s}{2d}$ and $\beta_o = \dfrac{D\lambda_o}{2d}$

where λ_s = wavelength of sodium light and λ_o = wavelength of other light

Since, D and $2d$ are same. So,

$$\frac{\beta_s}{\beta_o} = \frac{\lambda_s}{\lambda_o}$$

Given that, width of the view $= 50\beta_s$ for sodium light.

Let n be the number of fringes observed by using other light, then width of the view $= n\beta_o$

or $$50\beta_s = n\beta_o \text{ or } n = 50\frac{\beta_s}{\beta_o}$$

or $$n = 50\frac{\lambda_s}{\lambda_o} = 50 \times \frac{6000}{5000} = \mathbf{60}$$

***Problem 18:** The inclined faces of a biprism of glass ($\mu = 1.5$) make an angle of 2° with the base. The slit is at 10 cm from the biprism and is illuminated by light of wavelength 5500 Å. Find the fringe width at a distance of one meter from the biprism.*

Solution: The fringe width is given by, $\beta = \dfrac{D\lambda}{2d}$

Here given that, $D = 10 + 100 = 110 \text{ cm}, \lambda = 5500 \text{ Å}$

and the separation between the two virtual sources, $2d = 2a(\mu - 1)\alpha$

Given that, $a = 10\text{ cm},\ \mu = 1.5, \alpha = 2° = \dfrac{2\times\pi}{180}$

$\therefore$ $$\beta = \frac{110\times5500\times10^{-8}\times7\times180}{2\times10\times(1.5-1)\times2\times22}\text{ cm} = \frac{7.623}{880\times0.5}\text{ cm} = \mathbf{0.0173\ cm}$$

Problem 19: Interference fringes are produced by Fresnel's biprism in the focal plane of an eyepiece 200 cm away from the slit. The two images of the slit that are formed for each of the two positions of a convex lens placed between the biprism and eyepiece are found to be separated by 4.5 mm and 2.9 mm respectively. If the width of the interference fringes be 0.326 mm, find the wavelength of the light used.

Solution: The wavelength of light is given by,

$$\lambda = \frac{\beta 2d}{D}$$

Here given that, $\beta = 0.326\text{ mm} = 0.0326\text{ cm}, D = 200\text{ cm}$ and $d_1 = 4.5\text{ mm}, d_2 = 2.9\text{ mm}$,

The separation between two virtual sources will be,

$$2d = \sqrt{(d_1 d_2)} = \sqrt{(4.5\times2.9)} = 3.6\text{ mm} = 0.36\text{ cm}$$

putting all the values, we get $\lambda = \dfrac{0.0326\times0.36}{200} = \mathbf{5860\ Å}$

Problem 20: In a biprism experiment, the distance between the slit and the screen is 180.00 cm. The biprism is 60.0 cm away from the slit and its refractive index is 1.52. When a source of wavelength 5890 Å is used, the fringe-width is found to be 0.010 cm. Find the angle between the two refracting surfaces of the biprism.

Solution: The fringe width is given by,

$$\beta = \frac{D\lambda}{2d} \quad \text{or} \quad 2d = \frac{D\lambda}{\beta}$$

Here given that, $\beta = 0.010\text{ cm}, D = 180\text{ cm}$ and $\lambda = 5890Å$

$\therefore$ $$2d = \frac{180\times5890\times10^{-8}}{0.010} = 1.061\text{ cm}$$

But we have the separation between two virtual sources.

$$2d = 2a(\mu-1)\alpha \quad \text{or} \quad \alpha = \frac{2d}{2a(\mu-1)}$$

Here given that, $a = 60\text{ cm}, \mu = 1.52$ and $2d = 1.061\text{ cm}$

$\therefore$ $$\alpha = \frac{1.061}{2\times60\times(1.52-1)} = \frac{1.061}{120\times0.52} = 0.0170\text{ rad.}$$

$$= 0.017\times\frac{180}{\pi} = 0.97° \approx \mathbf{1°}$$

Problem 21: *When a plate of glass of thickness* 3.4×10^{-4} *cm is placed in the path of one of the interfering beams in a biprism arrangement, it is found that the central bright fringe shifts through a distance equal to the width of four fringes. Find the refractive index of the glass plate, wavelength of light used is* 5.46×10^{-5} *cm.*

Solution: The shift in fringes is given by,

$$x_0 = \frac{D}{2d}(\mu - 1)t$$

Here given that, $t = 3.4\times10^{-4}\,\text{cm}, \lambda = 5.46\times10^{-5}\,\text{cm}$ and $x_0 = 4\beta$

But we have, $$\beta = \frac{D\lambda}{2d} \Rightarrow x_0 = 4\times\frac{D\lambda}{2d}$$

$\therefore$ $$4\times\frac{D\lambda}{2d} = \frac{D}{2d}(\mu - 1)t$$

or $$(\mu - 1) = \frac{4\lambda}{t} = \frac{4\times5.46\times10^{-5}}{3.4\times10^{-4}} = 0.64$$

or $$\mu = 1 + 0.64 = \mathbf{1.64}$$

Problem 22: *Fresnel's fringes are produced with homogeneous light of wavelength* 6×10^{-5} *cm. A thin glass plate* $(\mu = 1.50)$ *is introduced in the path of one of the interfering beams. The central bright band is shifted to the position previously occupied by the* 5*th bright band. Find the thickness of the plate.*

Solution: The shift in fringe is given by,

$$x_0 = \frac{D}{2d}(\mu - 1)t$$

If $$x_0 = n\beta \quad \text{and} \quad \beta = \frac{D\lambda}{2d}$$

then, $$\frac{nD\lambda}{2d} = \frac{D}{2d}(\mu - 1)t$$

or $$n\lambda = (\mu - 1)t$$

Here given that, $n = 5, \lambda = 6\times10^{-5}$ cm, $\mu = 1.50$

$\therefore$ $$t = \frac{5\times6\times10^{-5}}{(1.5 - 1)} = \mathbf{6\times10^{-4}\,cm}$$

Problem 23: *Fresnel's biprism fringes are observed with white light. When a thin transparent sheet covers one-half of prism, the central fringe shifts sideways by* 14.97 *mm. With the same geometry, the fringe-width with mercury green light* (5461) *comes to* 0.274 *mm. Deduce the thickness of the sheet, assuming the refractive index of its material* 1.58.

Solution: The shift in fringes is given by,

$$x_0 = \frac{D}{2d}(\mu - 1)t$$

and the fringe width,

$$\beta = \frac{D\lambda}{2d} \quad \text{or} \quad \frac{D}{2d} = \frac{\beta}{\lambda}$$

$\therefore$ $$x_0 = (\beta/\lambda)(\mu - 1)t$$

or $$t = \frac{x_0 \lambda}{\beta(\mu - 1)}$$

Here given that, $x_0 = 14.97$ mm, $\lambda = 5461$ Å, $\beta = 0.274$ mm and $\mu = 1.58$

Putting all the values, we get

$$t = \frac{5461 \times 10^{-8} \times 1.497}{(1.58 - 1) \times 0.0274} = \mathbf{5.14 \times 10^{-3}\ cm}$$

Problem 24: On introducing a thin sheet of mica (thickness 12×10^{-5} cm) in the path of one of the interfering beams in a biprism arrangement, the central fringe is shifted through a distance equal to the spacing between successing bright fringe. Calculate the refractive index of mica. ($\lambda = 6 \times 10^{-5}$ cm).

Solution: The shift in fringes is given by,

$$x_0 = \frac{D}{2d}(\mu - 1)t = \frac{\beta}{\lambda}(\mu - 1)t \qquad \left[\because \beta = \frac{D\lambda}{2d}\right]$$

or $$(\mu - 1) = \frac{x_0 \lambda}{\beta t}$$

Here given that, $x_0 = \beta, \lambda = 6 \times 10^5$ cm and $t = 12 \times 10^{-5}$ cm

$\therefore$ $$(\mu - 1) = \frac{\beta \times 6 \times 10^{-5}}{\beta \times 12 \times 10^{-5}} = 0.5 \quad \text{or} \quad \mu = \mathbf{1.5}$$

Problem 25: A thin sheet of mica ($\mu = 1.6$) of 7 micron thickness introduction in the path of one of the interfering beams in a biprism arrangement shifts the central fringe to a position normally occupied by the 7^{th} bright fringe from the centre. Find the wavelength of light used. (1 micron = 10^{-4} cm).

Solution: We have that, $$x_0 = \frac{D}{2d}(\mu - 1)t = \frac{\beta}{\lambda}(\mu - 1)t$$

or $$\lambda = (\beta/x_0)(\mu - 1)t$$

Here given that, $x_0 = 7\beta, \mu = 1.6$ and $t = 7 \times 10^{-4}$ cm

putting all the values, we get

$$\lambda = \frac{\beta}{7\beta}(1.6 - 1) \times 7 \times 10^{-4} = \mathbf{6000\ Å}$$

Problem 26: In a double-slit arrangement fringes are produced using light of wavelength 4800 Å. One slit is covered by a thin plate of glass of refractive index 1.4 and the other slit by another plate of glass of the same thickness but of refractive index 1.7. On doing so the central bright fringes shifts to the position originally occupied by the fifth bright fringe from the centre. Find the thickness of the glass plates.

Solution: On introducing a thin sheet of glass in one interfering beam, the shift in fringe is given by,

$$x_{01} = \frac{D}{2d}(\mu - 1)\, t = \frac{\beta}{\lambda}(\mu - 1)\, t$$

If another sheet of same thickness t but refractive index μ' is introduced in other beam, the shift in the fringe will be,

$$x_{02} = \frac{\beta}{\lambda}(\mu' - 1)\, t$$

The resultant shift in the fringe ,

$$x_0 = x_{01} - x_{02} = (\beta/\lambda)(\mu - \mu')\, t$$

or
$$t = \frac{x_0 \lambda}{\beta(\mu - \mu')}$$

Here given that, $x_0 = 5\beta,\ \lambda = 4.8 \times 10^{-5}$ cm, $\mu = 1.7$ and $\mu' = 1.4$

putting all the values, we get $t = \dfrac{5\beta \times 4.8 \times 10^{-5}}{\beta(1.7 - 1.4)} = \mathbf{8 \times 10^{-4}\ cm}$

Problem 27: A light wave having λ_s 5890 Å and 4000 Å is incident normally on a thin air film of $t = 0.2945 \times 10^{-6}$ m. What is the colour shown in reflection by the film?

Solution: For constructive interference in the reflected light,

$$2\mu t \cos r = (2n + 1)\lambda/2$$

Here given that, $\mu = 1$ (air film), $t = 2.945 \times 10^{-7}$ m, $r = 0$ (normal incidence)

So, $$2 \times 1 \times 2.945 \times 10^{-7} \cos 0 = (2n + 1)\lambda/2$$

or
$$\lambda = \frac{11.78 \times 10^{-7}}{2n + 1}\ \text{m} = \frac{11780}{2n + 1}\ \text{Å}$$

Where $n = 0, 1, 2, \ldots$

$\therefore$ $\lambda = 11780$ Å, 3927 Å, 2356 Å .... etc. By putting $n = 0, 1, 2, \ldots$ etc.

Only these λ_s will appear bright in the film after reflection. Since only **3927 Å** is in visible region so, corresponding it *i.e.*, **violet colour** will appear in reflection by the film.

Problem 28: A soap film, 5×10^{-5} cm thick is viewed at an angle of 30° to the normal. Find the wavelength of visible light which will appear in the reflected light from the film ($\mu = 1.5$).

Solution: For constructive interference in the reflected light,

$$2\mu t\cos r=(2n+1)\frac{\lambda}{2} \quad \text{or} \quad (2n+1)\lambda=2\times 2\mu t\cos r$$

Here given that, $\mu=1.5, t=5\times10^{-5}$ cm and $i=30°$

Also we have, $\frac{\sin i}{\sin r}=\mu$ so, $\sin r=\frac{\sin i}{\mu}=\frac{\sin 30°}{1.5}$

$$\sin r=\frac{0.5}{1.5}=\frac{1}{3}, \cos r=\sqrt{(1-\sin^2 r)}=\sqrt{(1-1/9)}$$

$\therefore$ $$\cos r=\sqrt{\frac{8}{9}}=\frac{2\sqrt{2}}{3}$$

So, $$(2n+1)\lambda=2\times2\times1.5\times5\times10^{-5}\times\frac{2\sqrt{2}}{3}$$

or $$\lambda=\frac{20\times1.41\times10^{-5}}{(2n+1)}=\frac{28.2\times10^{-5}}{2n+1}$$

For $n=1$ $\lambda=9.4\times10^{-5}$ cm $=9400$ Å

$n=2$ $\lambda=5.64\times10^{-5}$ cm $=5640$ Å

$n=3$ $\lambda=4.02\times10^{-5}$ cm $=4020$ Å

These all wavelengths will appear in the reflected light but 9400 Å and 4020 Å lie in invisible region. So, only **5640 Å** will appear in the beam reflected by film.

Problem 29: A glass slab **($\mu=1.5$)** ***is to be coated with a film of a transparent material*** **($\mu=1.25$)**. ***What should be the thickness of the film so that yellow light*** **($\lambda=6000$ Å)** ***incident normally on the film gives destructive interference.***

Solution: In this case, the reflections at both the upper and lower surfaces of material film take place from rarer to denser medium. Hence, there will be zero net phase difference between two reflecting interfering beams. Therefore, the destructive interference will occur when,

$$2\mu t\cos r=(2n+1)\frac{\lambda}{2} \quad \text{where } n=0,1,2,\ldots$$

Here given that, $\mu=1.25$, $r=0$ (normal incidence) and $\lambda=6000$ Å

Putting the values, we get

$$2\times1.25\times t\cos0=(2n+1)\frac{6000}{2}$$

or $$t=(2n+1)1200 \text{ Å}$$

Putting $n=0,1,2,\ldots.$

$t=$ **1200 Å, 3600 Å, 6000 Å, … etc.**

Problem 30: Two adjacent positions of complete destructive interference are obtained by a uniform thin film of oil when light of wavelength 5×10^{-5} ***cm and*** 7×10^{-5} ***cm falls normally on it. Find the thickness of film if refractive index of the oil is*** 1.3.

Solution: The condition of destructive interference in the reflected light from a film,

$$2\mu t\cos r = n\lambda$$

Here given that, $\mu = 1.3, r = 0$ (normal incidence)

So, $2\times1.3\times t\times\cos 0 = n\lambda$

or $2.6\,t = n\lambda$

Now, let for the wavelength 7×10^{-5} cm the order be n, then for the wavelength 5×10^{-5} cm, the order will be $(n+1)$

So, $2.6\times t = n\times7\times10^{-5}$ ...(1)

and $2.6\times t = (n+1)5\times10^{-5}$ ...(2)

$\therefore$ $7\times10^{-5}n = 5\times10^{-5}(n+1)$ or $7n = 5n+5$

or $2n = 5, n = 5/2 = 2.5 \approx 3$

Putting in eq. (1), we get

$$t = \frac{3\times7\times10^{-5}}{2.6} = \mathbf{8.07\times10^{-5}\ cm}$$

Problem 31: White light is reflected from an oil film of thickness 0.01. ***mm and*** $\mu = 1.4$ ***at an angle of*** 45° ***to the vertical. Calculate the number of dark bands seen in reflected light between*** 4000 Å ***and*** 5000 Å.

Solution: The condition for dark band in the reflected light is given by,

$$2\mu t\cos r = n\lambda$$

Here given that, $\mu = 1.4, t = 0.01\,\text{mm} = 1.0\times10^{-3}\,\text{cm}$, and angle of incidence, $i = 45°$,

Also we have, $\dfrac{\sin i}{\sin r} = \mu$

So, $1.4 = \dfrac{\sin 45}{\sin r}, \sin r = \dfrac{1}{\sqrt{2}\times1.4}$

or $\cos r = \sqrt{1-\sin^2 r} = \sqrt{1-\dfrac{1}{2\times1.4\times1.4}} = 0.863$

$\therefore$ $2\times1.4\times1\times10^{-3}\times0.863 = n\lambda$

or $n\lambda = 2.42\times10^{-3}$ cm

If $\lambda = 4000$ Å, then

$$n = \frac{2.42 \times 10^{-3}}{4000 \times 10^{-8}} = \frac{242}{4} \approx 60$$

and if $\lambda = 5000$ Å, then $$n = \frac{2.42 \times 10^{-3}}{5000 \times 10^{-8}} = \frac{242}{5} \approx 48$$

Therefore, number of dark bands observed between $\lambda = 4000$ Å and 5000 Å will be $\mathbf{60 - 48 = 12}$

Problem 32: Calculate the thickness of the thinnest film $(\mu = 1.4)$ ***in which interference of violet component*** $(\lambda = 4000$ Å$)$ ***of incident light can take place by reflection.***

Solution: For constructive interference of light reflected from a film, the condition is given by,

$$2\mu t \cos r = (2n + 1)\frac{\lambda}{2}, n = 0, 1, 2, \ldots.$$

For thinnest film $n = 0$, and taking normal incidence *i.e.*, $r = 0$

$\therefore$ $$2\mu t \cos 0 = \frac{\lambda}{2} \quad \text{or} \quad t = \frac{\lambda}{4\mu}$$

Here given that, $\lambda = 4000$ Å and $\mu = 1.4$

$\therefore$ $$t = \frac{4000 \times 10^{-8}}{4 \times 1.4} \text{ cm} = \mathbf{7.14 \times 10^{-6} cm}$$

Problem 33: A thin film of transparent material $(\mu = 1.25)$ ***is coated on glass*** $(\mu = 1.50)$. ***White light falls normally on the film. In the reflected light, complete destruction is observed at*** 6000 ***and constructive interference at*** 7000 Å. ***Find the thickness of the film.***

Solution: In this case the complete destruction condition is,

$$2\mu t \cos r = (2n+1)\frac{\lambda}{2} = (2n+1)\frac{6000}{2} \text{ Å} = (2n+1)3000 \qquad \ldots(1)$$

and complete constructive interference condition is,

$$2\mu t \cos r = n\lambda = n7000 \text{ Å} \qquad \ldots(2)$$

or $(2n+1)3000 = 7000\, n$ [From equation (1)]

or $n = 3$

Given that, $\mu = 1.25$ (material), $r = 0$ and $n = 3$

$\therefore$ Putting in eq. (1) we get, $$t = \frac{7 \times 6000 \times 10^{-8}}{2 \times 2 \times 1.25} \text{ cm} = \mathbf{8.4 \times 10^{-5} cm}$$

Problem 34: A parallel beam of sodium light (5890 Å) strikes a film of oil ($\mu = 1.46$) floating on water ($\mu = 1.33$). When viewed at an angle of 30° from the normal, the eighth dark band is seen. Find the thickness of the film.

Solution: In this case reflection takes place at denser to rare, so there will be a phase change of π at the upper surface only. So, the condition for dark band is given by,

$$2\mu t \cos r = n\lambda \qquad ...(1)$$

Here given that, $\mu = 1.46, n = 8, \lambda = 5890$ Å, and $i = 30°$

We have $$\frac{\sin i}{\sin r} = \mu, \sin r = \frac{\sin i}{\mu} = \frac{\sin 30}{1.46}$$

or $$\sin r = \frac{0.5}{1.46} = 0.342$$

and $$\cos r = \sqrt{(1 - \sin^2 r)} = \sqrt{(1 - 0.117)} = \sqrt{(0.882)} = 0.939$$

So, from equation (1), $$t = \frac{8 \times 5890 \times 10^{-8}}{2 \times 1.46 \times 0.939} = \frac{4.71 \times 10^{-4}}{2.74} = \mathbf{1.71 \times 10^{-4}\,cm}$$

Problem 35: A thin film of oil ($\mu = 1.2$) is formed on the surface of water ($\mu = 4/3$). What is the thickness of the oil at the first bright fringe for $\lambda = 4800$ Å? At the third dark fringe ?

Solution: In this case the condition of constructive interference is given by,

$$2\mu t \cos r = n\lambda$$

Here given that, $\mu = 1.2, r = 0$ (normal incidence), $n = 1$, and $\lambda = 4800$ Å

$\therefore$ $$t = \frac{1 \times 4800 \times 10^{-8}}{2 \times 1.2} \text{ cm} = \mathbf{2.0 \times 10^{-5}\,cm}$$

And for destructive interference,

$$2\mu t \cos r = (2n + 1)\frac{\lambda}{2} \text{ where } n = 0, 1, 2 \ldots$$

Here given that, $n = 2$ (third dark fringe)

or $$t = \frac{5 \times 4800 \times 10^{-8}}{2 \times 2 \times 1.2} \text{ cm} = \mathbf{5 \times 10^{-5}\,cm}$$

Problem 36: A soap film of $\mu = 1.33$ is illuminated with white light incident at an angle of 45°. The light refracted by it is examined and two bright fringe are focused corresponding to wavelengths 6.1×10^{-5} cm. Find the thickness of the film.

Solution: In this case bright fringes are examined in the refracted or transmitted light. The condition of constructive interference in transmitted light is given by,

$$2\mu t \cos r = n\lambda$$

Here, $\mu = 1.33, \lambda_1 = 6.1\times10^{-5}$ cm and $\lambda_2 = 6.0\times10^{-5}$ cm, $i = 45°$

$\because$ $\dfrac{\sin i}{\sin r} = \mu$ $\quad\therefore\quad \sin r = \dfrac{\sin i}{\mu} = \dfrac{\sin 45°}{1.33}$

$$\sin r = \frac{1}{1.41\times1.33} = 0.534$$

$$\cos r = \sqrt{(1-\sin^2 r)} = \sqrt{(0.712)} = 0.8443$$

Putting the values, we get

$$2\times1.33\times t\times0.843 = n\times6.1\times10^{-5} \qquad ...(1)$$

and $$2\times1.33\times t\times0.843 = (n+1) = 6.0\times10^{-5} \qquad ...(2)$$

From equation (1) and (2),

$$6.1n = 6.0\,(n+1) = 6n+6$$

$$0.1n = 6.0 \text{ or } n = 60$$

From equation (1),

$$t = \frac{60\times6.1\times10^{-5}}{0.843\times2.66} = \mathbf{1.63\times\ 10^{-3}cm}$$

Problem 37: In a Newton's ring arrangement with lens of radius of curvature 100 cm and diameter (aperture) 2 cm, how many rings will be observed if λ of light is 5×10^{-5} cm. If arrangement is immersed in water, then? μ for water is 1.3.

Solution: Let the total number of fringes be n, then the diameter of nth ring D_n will be the diameter of the lens. We have diameter of nth ring.

$$D_n^2 = 4n\lambda R$$

Here given that, $\quad D_n = 2 \text{ cm}, \lambda = 5\times10^{-5} \text{ cm}, R = 100 \text{ cm}$

$\therefore$ $$n = \frac{D_n^2}{4\lambda R} = \frac{(2)^2}{4\times5\times10^{-5}\times100} = \frac{10^3}{5} = \mathbf{200}$$

Now, if arrangement is immersed in water, then diameter of nth ring is given by,

$$D_n^2 = 4n\lambda R/\mu$$

or $$n = \frac{D_n^2\mu}{4\lambda R} = \frac{(2)^2\times1.3}{4\times5\times10^{-5}\times100} = \mathbf{260}$$

Problem 38: A Newton's arrangement is used with a source of light emitting two wavelengths $\lambda_1 = 6000$ Å and $\lambda_2 = 4500$ Å. It is found that nth dark ring due to λ_1 coincides with $(n+1)$th dark ring for λ_2. If radius of curvature of the lens is 90 cm, find the diameter of nth dark ring for λ_1.

[Meerut 2007]

Solution: Let D be the diameter of nth dark ring of λ_1 and $(n+1)$th dark ring for λ_2 then we have

$$D^2 = 4n\lambda_1 R \quad \text{and} \quad D^2 = 4(n+1)\lambda_2 R$$

Here, $\lambda_1 = 6000 \times 10^{-8}$ cm, $\lambda_2 = 4500 \times 10^{-8}$ cm, and $R = 90$ cm

$$4n \times 6000 \times 10^{-8} \times 90 = 4(n+1)\, 4500 \times 10^{-8} \times 90$$

or $$6n = 4.5\,(n+1) = 4.5n + 4.5$$

or $$n = 3$$

So, $$D^2 = 4 \times 3 \times 6000 \times 10^{-8} \times 90 = 648 \times 10^{-4}$$

or $$D = \mathbf{0.254\ cm}$$

Problem 39: A convex lense of radius 350 cm placed on a flat plate and illuminated by monochromatic light gives the 6^{th} bright ring of diameter 0.68 cm. Calculate the wavelenegth of light used.

Solution: The diameter of nth bright ring is given by,

$$D_n^2 = 2(2n-1)\lambda R$$

or $$\lambda = \frac{D_n^2}{2(2n-1)R}$$

Here given that, $$D_n = 0.68\ \text{cm}, n = 6, R = 350\ \text{cm}$$

$\therefore$ $$\lambda = \frac{(0.68)^2}{2 \times 11 \times 350} = \frac{0.462}{7700} = 6.0 \times 10^{-5}\ \text{cm}$$

$$= \mathbf{6000\ Å}$$

Problem 40: The diameter of 5^{th} dark rings is 9 mm in Newton's ring experiment when sodium light (λ = 5890 Å) is used. The light passes through the air film at an angle of 30° to the normal. Find the radius of the lens.

Solution: The conditon for the nth dark ring seen in Newton's ring experiment when light falls at an angle,

$$2t \cos r = n\lambda$$

where r is the angle of refraction in the film.

Also, we have $$2t = \frac{D_n^2}{4R}$$

where D_n is the diameter of the nth ring corresponding to film thickness t. So, we can write,

$$\frac{D_n^2}{4R} = \frac{n\lambda}{\cos r} \qquad (\because \ \mu = 1 \text{ for air}$$

write)

or $$D_n^2 = \frac{4nR\lambda}{\cos r}$$

or $$R = \frac{D_n^2 \cos r}{4n\lambda}$$

Here given that, $D_n = D_5 = 9 \text{ mm} = 0.9 \text{ cm}, r = 30^\circ$ or $\cos r = 0.866, n = 5$

and $\lambda = 5890 \times 10^{-8}$ cm

$\therefore$ $$R = \frac{(0.9)^2 \times 0.866}{4 \times 5 \times 5890 \times 10^{-8}} = \mathbf{590 \text{ cm}}$$

Problem 41: Newton's rings are formed by reflection in the air film between a plane glass plate and a lens of radius 100 cm. If the diameter of 3rd bright ring is 0.181 cm and that of 13th bright ring is 0.501 cm, calculate the wavelength of the light used.

Solution: We have that,

$$D_{n+P}^2 - D_n^2 = 4P\lambda R$$

$$\lambda = \frac{D_{n+P}^2 - D_n^2}{4PR}$$

Here given that, $D_{n+P} = D_{13} = 0.501 \text{ cm}, D_n = D_3 = 0.181 \text{cm}, P = 10$ and $R = 100$ cm

putting the values, we get

$$\lambda = \frac{(0.501)^2 - (0.181)^2}{4 \times 10 \times 100} = \frac{0.218}{4000} = \mathbf{5450 \text{ Å}}$$

Problem 42: Newton's rings are formed by the light reflected normally from a liquid film between lens and plate. If the diameter of nth and $(n+10)$th ring is 2.18 and 4.51 mm, calculate μ of the liquid. Given that, R = 90 cm and λ = 5893 Å.

Solution: In liquid film we have,

$$[D_{n+P}^2 - D_n^2] = \frac{4PR\lambda}{\mu}$$

or $$\mu = \frac{4PR\lambda}{D_{n+P}^2 - D_n^2}$$

Here given that, $D_{n+P} = D_{n+10} = 4.51\,\text{mm} = 0.451\,\text{cm}, D_n = 2.18\,\text{mm} = 0.218\,\text{cm}$

and $P = 10, R = 90\,\text{cm}$ and $\lambda = 5893 \times 10^{-8}\,\text{cm}$

Putting the values, we get

$$\mu = \frac{4 \times 10 \times 90 \times 5893 \times 10^{-8}}{(0.451)^2 - (0.218)^2} = \frac{0.212}{0.155}$$

$$= \mathbf{1.36}$$

Problem 43: In a Newton's ring arrangement, if a drop of water ($\mu = 4/3$) ***be placed between the lens and the plate, the diameter of the*** $\mathbf{10^{th}}$ ***dark ring is found to be*** **0.6** ***cm. Obtain the radius of curvature of the face of the lens in contact with the plate. The wavelength of light used is*** **6000 Å.**

Solution: The diameter of *n*th ring in the presence of liquid is given by,

$$D_n^2 = \frac{4n\lambda R}{\mu} \quad \text{or} \quad R = \frac{D_n^2 \mu}{4n\lambda}$$

Here given that, $\mu = 4/3, D_{10} = 0.6$ cm, $n = 10$ and $\lambda = 6000$ Å

Putting the values, we get, $R = \dfrac{(0.6)^2 \times 4}{3 \times 4 \times 10 \times 6000 \times 10^{-8}}$

$$= 2 \times 10^2 = \mathbf{200\,cm}$$

Problem 44: In a Newton's ring experiment the diameter of the $\mathbf{10^{th}}$ ***bright ring changes from*** **1.40** ***cm to*** **1.27** ***when a liquids is introduced between the plate and the lens. Calculate the refractive index of the liquid.***

[Meerut 2006 B]

Solution: The diameter of *n*th bright ring in the presence of liquid between plate and lens is given by

$$(D_n^2)\,\text{liquid} = \frac{2(2n-1)\lambda R}{\mu}$$

The diameter of *n*th bright ring in the air film,

$$(D_n^2)_{\text{air}} = 2(2n-1)\lambda R$$

$\therefore$ $$\mu = \frac{(D_n)^2_{\text{air}}}{(D_n)^2_{\text{liquid}}} = \frac{(1.40)^2}{(1.27)^2} = \mathbf{1.215}$$

Problem 45: If in a Newtion's ring experiment, the air in the inter space is replaced by the liquid of reflractive index 1.33, in what proportion would the diameter of the ring change ?

Solution: The refractive index of the liquid is given by,

$$\mu = \frac{(D_n)^2_{\text{air}}}{(D_n)^2_{\text{liquid}}}$$

Where $(D_n)_{\text{air}}$ is the diameter of nth ring (bright or dark) in air film.

and $(D_n)_{\text{liquid}}$ is the diameter of nth ring (bright or dark) in liquid film.

or $$\frac{(D_n)_{\text{air}}}{(D_n)_{\text{liquid}}} = \sqrt{\mu}$$

or $$\frac{(D_n)_{\text{liquid}}}{(D_n)_{\text{air}}} = \sqrt{\frac{1}{\mu}} = \sqrt{\frac{1}{1.33}} = 0.867$$

The rings are contracted to 0.867 of their original diameter.

Problem 46: In an arrangement for observing Newton's ring with two different media between the glass surfaces, the nth rings have diameters as 10:7. Find the ratio of the refractive indices of the two media.

Solution: We have that,

$$\frac{(D_n)^2_{\text{air}}}{(D_n)^2_{\text{liquid}}} = \mu$$

So, $$\frac{(D_n)^2_{\text{air}}}{(D_n)^2_{\text{liquid}_1}} = \mu_1 \quad \text{and} \quad \frac{(D_n)^2_{\text{air}}}{(D_n)_{\text{liquid}_2}} = \mu_2$$

or $$\frac{(D_n)^2_{\text{liquid}_2}}{(D_n)^2_{\text{liquid}_1}} = \frac{\mu_1}{\mu_2}$$

Here given that, $$\frac{(D_n)_{\text{liquid}_1}}{(D_n)_{\text{liquid}_2}} = 10:7 \quad \text{or} \quad \frac{(D_n)_{\text{liquid}_2}}{(D_n)_{\text{liquid}_1}} = 7:10$$

So, $$\frac{(7)^2}{(10)^2} = \frac{\mu_1}{\mu_2}$$

or $$\frac{\mu_1}{\mu_2} = \mathbf{49:100}.$$

Problem 47: Newton's rings are proudced by means of a plate and a lense of radius of curvature 5 meters and diameter 2.0 cm. How many bright rings are observed? How many bright ring will be observed if the arrangement is immersed in water **(μ = 1.33).** ***Given that,*** **λ = 5890 Å.**

Solution: The diameter of the nth bright ring in air film,

$$(D_n)^2 = 2(2n-1)\lambda R$$

Since the diameter of lens is 2.0 cm so, the last bright ring's diameter will be 2.0 cm.

or $(D_n)_{\max} = 2.0 \text{ cm}$

$\therefore$ $(2.0)^2 = 2(2n-1)\times 5890\times 10^{-8}\times 500$

$$4 = (2n-1)\times 0.0589$$

or $$(2n-1) = \frac{4}{0.0589} = 67.9$$

or $2n = 68.9$ or $n \approx \mathbf{34.5}$

Now, the diameter of *n*th bright ring in the presence of liquid film,

$$(D_n)^2 = \frac{2(2n-1)\lambda R}{\mu}$$

Again we have, $(D_n)_{\max} = 2.0 \text{ cm}$

$\therefore$ $$(2.0)^2 = \frac{2(2n-1)\times 5890\times 10^{-8}\times 500}{1.33}$$

$$= \frac{2(2n-1)\times 0.0589}{1.33} = (2n-1)\times 0.044$$

or $$(2n-1) = \frac{4}{0.044} = 90.90 \quad \text{or } 2n = 91.90$$

or $n = \mathbf{45.9}.$

Problem 48: In a Michelson's interferometer **200** ***fringes cross the field of view when the movable mirror is displaced through*** **0.0589** ***mm. Calculate the wavelength of monochromatic light used.***

Solution: In Michelson's interferometer, the condition for central bright spot

$$2d = n\lambda \quad \text{or} \quad \lambda = \frac{2d}{n}$$

Here given that, $n = 200, d = 0.0589 \text{ mm} = 5.89\times 10^{-3} \text{ cm}$

putting the values, we get

$$\lambda = \frac{2\times 5.89\times 10^{-3}}{200} = 5890\times 10^{-8} \text{ cm}$$

$$= \mathbf{5890 \text{ Å}}$$

Problem 49: Calculate the distance through which the mirror of the Michelson interferometer has to be displaced between two consecutive positions of maximum distinctness of D_1 and D_2 lines of sodium. Wavelength of D_2 line $= 5890$ Å and $D_1 = 5896$ Å.

Solution: In Michelson's interferometer, we have

$$2d = \frac{\lambda_1 \lambda_2}{\lambda_1 - \lambda_2}$$

Here given that, $\lambda_1 = 5896 \times 10^{-10}$ m, $\lambda_2 = 5890 \times 10^{-10}$ m

putting the values, we get

$$2d = \frac{5896 \times 5890 \times 10^{-20}}{6 \times 10^{-10}} = 5.79 \times 10^{-4} \text{ m}$$

or $$d = \mathbf{2.895 \times 10^{-4} m}$$

Problem 50: How far must the movable mirror of a Michelson's interferometer be displaced for 2500 fringes of the red cadmium light ($\lambda = 6438$ Å) to cross the field of view ?

Solution: In Michelson's interferometer, we have

$$2d = n\lambda$$

Here given that, $n = 2500, \lambda = 6438 \text{ Å} = 6438 \times 10^{-10}$ m

putting theses values, we get

$$2d = 2500 \times 6438 \times 10^{-10} = 1.60 \times 10^{-3} \text{ m}$$

$$d = \mathbf{0.804 \text{ mm}}$$

Problem 51: When a thin film of a transparent material of refractive index 1.45 for $\lambda = 5890$ Å is inserted in one of the arms of a Michelson's interferometer, a shift of 65 circular fringes is observed. Calculate the thickness of the film.

Solution: When a film of thickness t and refractive index μ is inserted in the path of one of the interfering beams of Michelson's interferometer, an additional path difference of $2(\mu - 1)t$ is introduced. If n is the number of fringes shifted, then we have

$$2(\mu - 1)t = n\lambda$$

Here given that, $n = 65, \lambda = 5890 \times 10^{-10}$ m and $\mu = 1.45$

putting these values, we get

$$2(1.45-1)t = 65\times5890\times10^{-10}$$

or $$t = \frac{3.828\times10^{-5}}{0.90} = 4.25\times10^{-5}\,\text{m} = \mathbf{0.0425\ mm}$$

Problem 52: Light having wavelength 6000 Å and 5000 Å is incident on a Fabry-Perot etalon. It is observed that bright fringe is obtained for both colours when the light passes through the etalon in the direction $\cos^{-1}(0.7)$ ***to the normal and that a coincidence is next obtained in the direction*** $\cos^{-1}(0.8)$ ***to the normal. Determine the thickness of the air film within the etalon.***

Solution: If the fringes of λ_1 and λ_2 coincide in a direction θ_1, then order will be different (Let m and n)

So, $$2d\cos\theta_1 = m\lambda_1 = n\lambda_2 \qquad ...(1)$$

where d is the thickness of the air film within the etalon.

Now, in the direction θ_2 fringe again coincide. Suppose in this position the order of λ_1 has been increases by p and that of λ_2 by $(p+1)$, then

$$2d\cos\theta_2 = (m+p)\lambda_1 = (n+p+1)\lambda_2 \qquad ...(2)$$

Subtracting equation (1) from equation (2), we get

$$2d(\cos\theta_2 - \cos\theta_1) = p\lambda_1 = (p+1)\lambda_2 \qquad ...(3)$$

or $$p\lambda_1 = (p+1)\lambda_2$$

or $$p = \frac{\lambda_2}{\lambda_1 - \lambda_2}$$

putting this value in equation (3), we get

$$2d(\cos\theta_2 - \cos\theta_1) = \frac{\lambda_1\lambda_2}{\lambda_1 - \lambda_2}$$

or $$d = \frac{\lambda_1\lambda_2}{2(\lambda_1-\lambda_2)(\cos\theta_2 - \cos\theta_1)}$$

Here given that, $\lambda_1 = 6000\ \text{Å}, \lambda_2 = 5000\ \text{Å}, \cos\theta_2 = (0.8), \cos\theta_1 = (0.7)$

putting these values, we get

$$d = \frac{6000\times10^{-10}\times5000\times10^{-10}}{2\times1000\times10^{-10}\times(0.8-0.7)} = \mathbf{1.5\times10^{-3}\,cm}$$

Problem 53: White light is incident normally on a Fabry-Perot etalon with a plate separation of 4×10^{-4} cm. Calculate the wavelengths for which there are interference maxima in the transmitted beam in the range 4000 to 4500 Å.

Solution: In Fabry-Perot etalon, for maxima, we have

$$2d = n\lambda$$

Here given that, $d = 4\times10^{-4}\,\text{cm} = 4\times10^{-6}\,\text{m}$

$\therefore$ $n\lambda = 2\times4\times10^{-6}$

or $$\lambda = \frac{8\times10^{-6}}{n}\,\text{m}$$

But the value of λ lies in the range of 4000Å to 4500 Å. For that the value of n should be 18, 19 and 20.

(i) When $n = 18$, then

$$\lambda = \frac{8\times10^{-6}}{18} = \mathbf{4444.4\text{Å}}$$

(ii) When $n = 19$, then

$$\lambda = \frac{8\times10^{-6}}{19} = \mathbf{4210.5\text{Å}}$$

(iii) When $n = 20$, then

$$\lambda = \frac{8\times10^{-6}}{20} = \mathbf{4000\,\text{Å}}$$

Problem 54: In a spectrometer, one half of the slit is illuminated with mercury light and the second half is illuminated with light through a Fabry-Perot etalon. In the wavelength range of 4000 Å to 5000 Å, 20 fringes are observed in the continuous spectrum. Calculate the plate separation of F-P etalon.

Solution: In a Fabry-Perot etalon, we have

$$2d = n\lambda$$

If n is the order for 5000 Å, then $(n+20)$ will be the order for wavelength 4000 Å.

So, $$2d = n\times5000\times10^{-10} \quad \ldots(1)$$

and $$2d = (n+20)\times4000\times10^{-10} \quad \ldots(2)$$

From equation (1) and equation (2),

$$n = 80$$

putting in equation (1), we get

$$d = \frac{80\times5000\times10^{-10}}{2} = 2\times10^{-5}\ \text{m} = \mathbf{0.02\ mm}$$

Problem 55: Calculate the frequency bandwidth for white light (range 4×10^{14} Hz to 7.5×10^{14} Hz). Also find the coherence time and coherence length.

Solution: Here given that, $n_1 = 4\times10^{14}\text{Hz}$ and $n_2 = 7.5\times10^{14}\text{Hz}$

then the frequency band width,

$$\Delta n = n_2 - n_1 = \mathbf{3.5\times10^{14}Hz}$$

Now coherence time, $$\Delta t = \frac{1}{\Delta n} = \frac{1}{3.5\times10^{14}} = \mathbf{2.86\times10^{-15}sec}$$

and coherence length, $$\Delta l = c\times\Delta t = 3\times10^{8}\times2.86\times10^{-15} = \mathbf{8.58\times10^{-7}\,m}$$

Problem 56: A quasi monochromatic source emits radiations of mean wavelength λ = 5461Å and has a band width $\Delta n = 10^9$ Hz. calculate (i) coherence time (ii) coherence length (iii) frequency stability $\left(\frac{\Delta n}{n}\right)$.

Solution: Given that, $\lambda = 5461\text{Å} = 5461\times10^{-10}$ m and $\Delta n = 10^9$ Hz

(i) Then, coherence time, $$\Delta t = \frac{1}{\Delta n} = \frac{1}{10^9} = 10^{-9}\text{ sec}$$

(ii) Coherence length, $$\Delta l = c\times\Delta t = 3\times10^{8}\times10^{-9} = \mathbf{0.3\,m}$$

(iii) Frequency stability $$= \frac{\Delta n}{n}$$

or $$\frac{\Delta n}{n} = \frac{\Delta n\times\lambda}{c} = \frac{10^9\times5461\times10^{-10}}{3\times10^{8}} \qquad \left(\because n = \frac{c}{\lambda}\right)$$

$$= \mathbf{1.82\times10^{-6}}.$$

Exercise

(A) Descriptive Type Questions

1. Deduce an expression for the intensity at a point in the region of superposition of two waves of same periods and wavelengths.

2. What is interference of light? Explain constructive and destructive interference on the basis of wave theory of light? **[Meerut 2003B, 02B, 02]**

3. Describe the formation of interference fringes using two parallel slits in Young's experiment. Obtain an expression for the fringe width. What is the effect of increasing the width of the slit? **[Meerut 2012, 11B, 11, 10]**

4. Obtain an expression for the fringe width by Young's experiment. **[Meerut 2011]**

5. Explain the formation of interference fringes by means of a Fresnel's biprism when a monochromatic source of light is used and derive an expression for the Fringe width. **[Meerut 2001B, 01]**

6. State and explain conditions for the interference of light.

7. Explain why we require the coherent sources to get interference pattern? **[Meerut 2009B, 03B]**

8. Explain why two independent sources can not produce observable interference pattern. **[Meerut 2007]**

9. Define the term optical path retardation and lateral shift of fringes. Calculate the displacement of the fringes when a thin plate of a glass is introduced in the path of one of the interfering beams. **[Meerut 2009B, 03]**

10. Describe the Rayleigh refractometer. Also explain its applications.

11. Discuss the interference in thin films due to multiple reflection of light and show that with monochromatic light the interference patterns of the reflected and the transmitted light are complementary. **[Meerut 2004B, 03]**

12. Discuss the phenomenon of interference of light due to thin films and find the condition of maxima and minima. Show that the interference pattern of reflected and transmitted monochromatic light are complementary. **[Meerut 2006]**

13. Describe and explain the formation of Newton's ring in reflected monochromatic light. Prove that in reflected light (i) Diameter of bright rings are proportional to the square roots of odd numbers, and (ii) Diameter of dark rings are proportional to the square roots of natural numbers. **[Meerut 2009B, 09, 08B, 07B, 07, 06B, 03B, 01B, 00]**

14. Explain how Newton's ring method can be used for determining the wavelength of monochromatic light?
[Meerut 2011B, 11, 10, 09B, 08B, 08, 07B, 07, 05B, 05, 02]

15. Describe Newton's ring method for measuring the wavelength of monochromatic light and give the necessary theory.
[Meerut 2002]

16. Explain how we determine the refractive index of a liquid using Newton's ring?
[Meerut 2011B, 10, 09, 08, 06B, 01B]

17. Describe Newton's ring (interference) method for determining the refractive index of a transparent liquid. Derive the formula used.
[Meerut 2001B]

18. What do you mean by Haidinger fringes ? Explain fringes of equal inclination, Why these called Haidinger fringes ?

19. Describe and explain the formation of Newton's rings in the reflected monochromatic light. How can these be used to determine the refractive index of a transparent liquid? Derive the formula used.
[Meerut 2007B, 06B]

20. Calculate the displacement of the fringes when a thin transparent sheet is introduced in the path of one of interfering beam in biprism. Show how this is used to find out the thickness of mica sheet.
[Meerut 2007]

21. Describe the construction and working of a Michelson's Interferometer and explain how the interferometer may be used to obtain (i) circular fringes (ii) straight fringes (iii) white light fringes ?
[Meerut 2009B, 08B, 08, 06, 05, 03, 02B, 01, 00B]

22. Explain how the Michelson's interferometer is used to determine the wavelength of monochromatic light?
[Meerut 2009B, 08, 00B]

23. Explain the precision determination of difference in two neighboring wave length by Michelson's interferometer.
[Meerut 2008B, 05, 03, 02, 01B, 00B]

24. How can we determine the refractive index of a thin transparent film using Michelson interferometer ?
[Meerut 2006]

25. Describe Twyman and Green Interferometer.

26. What do you understand by multiple beam interference ? Describe the intensity distribution in reflected and transmitted light in a plane parallel plate.

27. Describe the construction and working of Febry Perot interferometer. Show that the intensity distribution in multiple beam interference is given by

$$I = I_{\max} \frac{1}{1 + F \sin^2 (\delta/2)}$$

where, $\delta = \frac{2\pi}{\lambda}(2\alpha \cos\theta)$ and $F = \frac{4R}{(1-R)^2}$.

All the symbols have their usual meaning. **[Meerut 2012, 11, 10, 09B, 07B, 06B, 05B, 04B, 03B, 02B, 01B]**

28. Explain the principle of Fabry-Perot interferometer. Obtain an expression for the intensity distribution in the transmitted light and discuss the sharpness of the fringes obtained. **[Meerut 2004B]**

29. Explain the construction and working of Fabry-Perot interferometer and deduce an expression for the intensity distribution in the fringes. Explain the effect of increasing reflectivity on the intensity of fringes. **[Meerut 2006B]**

30. Compare the fringes of Fabry-Perot interferometer with those of Michelson interferometer. **[Meerut 2005B]**

31. Explain the perfect blackness of the central spot in Newton's ring system. Can you obtain a bright centre? **[Meerut 2001]**

(B) Short Answer Type Questions

1. If the intensity ratio of two coherent sources is α, then prove that in the interference pattern

$$\frac{I_{max} - I_{min}}{I_{max} + I_{min}} = \frac{2\sqrt{\alpha}}{2+\alpha}$$

[Meerut 2005B, 04B, 02]

2. Two coherent sources whose intensity ratio is 81:1 produce interference fringes. Deduce the ratio of maximum to minimum intensity of the fringe system.

3. Two coherent beams of wavelength 5000 Å reaching a point would individually produce intensities 1.44 and 4.00 units. If they reach together, the intensity is 0.90 units. Calculate the lowest phase difference with which the beams reach that point.

4. Show that in a two slit interference pattern the intensity at a point is given by

$$I = A + B\cos^2\left(\frac{Kx}{2}\right)$$

Where A, B and K are constants of the set up and x is the linear distance of the point from central fringe. **[Meerut 2007B]**

5. Light from a narrow slit passes through two parallel slits 0.2 cm apart. Find the wavelength of the light if the width of interference fringes on a screen one meter distant from the two slits is 0.295 mm.

6. In Fresnel's biprism arrangement show that the distance between two virtual coherent sources is $2a(\mu - 1)\alpha$, where a is the distance between source and biprism, α is the angle of biprism and μ is the refractive index of the material of the biprism.

7. What will happen to biprism fringes if (i) angle of biprism is increased (ii) the width of slit is increased continuously?

8. When white light is used in biprism arrangement the central fringe is achromatic and it is surrounded by a few coloured fringes. Why, explain it?

9. How would you locate the central fringe of zero order ? **[Meerut 2008B, 07]**

10. Explain the method of finding wavelength of a monochromatic light with the help of biprism.

11. Describe displacement method to determine the distance between two virtual sources in a biprism.

12. Determine the thickness of a thin sheet of transparent material by using Fresnel's biprism. **[Meerut 2009B]**

13. A thick film shows no colours in reflected white light, explain why?

14. An extended source is necessary to observe colours in thin films, explain why? **[Meerut 2010]**

15. An oil film ($\mu = 1.47, t = 1.2 \times 10^{-7}$ m) rests on a pool of water. If light strikes the film at an angle of 60°, what is the wavelength reflected in the first order.

16. Newton's rings are formed by reflected light of wavelength 5895 Å with a liquid between the glass plate and plano convex lens. If the diameter of the 5th bright ring is 3 mm, calculate the refractive index of the liquid. The radius of curvature of the lens is 100 cm.

17. In a Newton's ring experiment the diameter of the 15^{th} ring was found to be 0.530 cm and that of 5^{th} ring was 0.306 cm when liquid introduced between plate and lens. If the radius of curvature of the lens is 100 cm, calculate the refractive index of the liquid. The λ of light used is 5895 Å.

18. Explain clearly, when central spot in Newton's ring experiment is dark as seen by reflection? Is it possible to see the central bright spot in reflected light rings? **[Meerut 2009B, 01]**

19. Can Newton's ring be seen in the transmitted light ? How they are different with the rings seen in reflected light ? **[Meerut 2005]**
20. What will be the effect on Newton's ring if white light is used instead of monochromatic light ?
21. Why the Fabry-Perot interferometer is better than Michelson interferometer ? **[Meerut 2004B, 03B]**
22. How can we use Fabry-Perot interferometer for the measurement of wavelength of monochromatic light ?
23. Explain the use of Fabry-Perot interferometer in the measurement of difference in two close wavelengths.
24. A Fabry-Perot interferometer is used to determine the small difference in wavelengths of the two components of a radiation of average wavelength 5893 Å. It is found that at a given point near the centre the two sets of fringes for a given distance between the two plates, and again when the distance is increased by 0.289 mm. Find the difference in wavelength.
25. Explain interference filter.
26. Describe Lummer-Gehrcke Plate.
27. Define coherence time and coherence length. **[Meerut 2012]**
28. Define temporal and spatial coherence. **[Meerut 2012]**

(C) Very Short Answer Type Questions

1. Define interference. **[Meerut 2011B, 10]**
2. Define constructive interference. **[Meerut 2012]**
3. Write conditions for the interference of light.
4. What do you mean by interference fringe pattern?
5. Define biprism.
6. What are coherent sources of light? **[Meerut 2011, 08B, 07B, 06]**
7. What are achromatic fringes? **[Meerut 2011, 10, 09, 08B, 07]**
8. What do your mean by Michelson's interferometer?
9. Write the principle of Fabry-Perot interferometer.
10. How the Fabry-Perot interferometers is better than Michelson's interferometer.
11. Write the difference between F.P. interferometer and F.P. etalon.

(D) Multiple Choice Questions

1. Constructive interference occurs when the phase difference is:
   (a) 0 (b) 90°
   (c) 180° (d) None
2. Destructive interference occurs only when the phase difference is:
   (a) 0 (b) 90°
   (c) 180° (d) None
3. In Young's double slit arrangement, the fringe width, $\beta = \frac{D\lambda}{2d}$. In this formula, the separation between the sources is represented by:
   (a) D (b) d
   (c) $2d$ (d) None
4. When a thin transparent plate is introduced in the path of one interfering beams, the fringe width will be:
   (a) Increase (b) Remain same
   (c) Decrease (d) None of the above
5. In Fresnel's biprism, if the angle (α) of the biprism is increased, then fringe will be:
   (a) Clearly visible (b) Not visible
   (c) Remain same (d) None of the above
6. Newton's rings are formed as a result of:
   (a) Interference (b) Diffraction
   (c) Polarisation (d) Reflection
7. In Newton's rings the diameter of bright rings is directly proportional to the square root of the:
   (a) Natural number (b) Odd natural number
   (c) Even natural number (d) None of the above
8. In Newton's ring experiment, what happens if white light is used instead of monochromatic light:
   (a) First few rings are clear and remaining cannot be viewed
   (b) All rings can be viewed clearly
   (c) No rings can be viewed clearly
   (d) No rings can be viewed at all
9. Interference filter is based on the principle of:
   (a) Michelson's interferometer (b) Fabry-Perot interferometer
   (c) Reflection (d) None of the above

10. The principle of Fabry-Perot interferometer is:
    (a) Dual beam interference (b) Single beam interference
    (c) Multiple beam interference (d) All above
11. In Michelson's interferometer, to get straight fringes, the position of Mirrors M_1 and M_2 is kept:
    (a) Not exactly perpendicular (b) Exactly perpendicular
    (c) Arbitrary in any direction (d) Not necessary to note

(E) Fill in the Blank

1. At constructive interference the intensity is ..... .
2. At destructive interference the intensity is ..... .
3. The width of fringes obtained by Young's double slit arrangement is ..... to the distance of the screen from the two sources.
4. When the width of slit is gradually increased, the contrast between the bright and dark fringes becomes ..... .
5. In Newton's ring, the diameter of dark rings is directly proportional to ..... of natural number.
6. The central spot in Newton's ring experiment is ..... as seen by reflection.
7. Write the formula for the fringe width (β) of a fringe pattern.
8. In Fabry-Perot interferometer, F is known as ..... .
9. In Fabry-Perot interferometer, the value of F in terms of reflection is given by ..... .

(F) True/False

1. The modification in the resultant intensity of the waves is called interference.
2. The maximum intensity point is called destructive interference point.
3. If the phase difference between two sources is zero, then sources are said to be coherent.
4. The width of fringe obtained by Young's double slit arrangement is directly proportional to wavelength of light.
5. The width of fringe obtained by Young's experiment is directly proportional to the distance between the two sources.
6. For interference, the monochromatic light is not necessary.
7. To get Newton's ring plano-concave lens of large focal length is necessary.
8. The fringe width of bright fringes and dark fringes is same.
9. For sustained interference, the wave must have a constant phase difference.

10. In Michelson's interferometer, circular fringes are obtained when mirrors M_1 and M_2 are perfectly perpendicular.

11. In Michelson's interferometer, the formula for determining the refractive index of a transparent thin film is, $2d = (\mu - 1)t$, where all symbols have their usual meanings.

12. In Fabry-Perot interferometer, the coefficient of sharpness is given by,

$$F = \frac{R}{(1-R)^2}$$

where R is the amplitude of the fraction of reflection.

13. Michelson interferometer gives sharp fringes in comparison to the Fabry-Perot interferometer.

Answers

(B) Short Answer Type Questions

(Hints & Solutions)

1. Let I_1 and I_2 be the intensities of two coherent sources and a_1, a_2 be the amplitudes of the waves produced by them. Since $I \propto a^2$ so, we may write,

$$\frac{a_1}{a_2} = \sqrt{\frac{I_1}{I_2}} \text{ but given that, } \frac{I_1}{I_2} = \alpha$$

So, $$\frac{a_1}{a_2} = \sqrt{\alpha} \qquad ...(1)$$

The maximum and minimum intensities in the interference pattern are given by,

$$I_{max} = (a_1 + a_2)^2 \text{ and } I_{min} = (a_1 - a_2)^2$$

$\therefore$ $$\frac{I_{max} - I_{min}}{I_{max} + I_{min}} = \frac{(a_1 + a_2)^2 - (a_1 - a_2)^2}{(a_1 + a_2)^2 + (a_1 - a_2)^2}$$

$$= \frac{4a_1 a_2}{2a_1^2 + 2a_2^2} = \frac{2a_1 a_2}{a_1^2 + a_2^2} = \frac{2a_1/a_2}{1 + \frac{a_1^2}{a_2^2}} = \frac{2\sqrt{\alpha}}{1+\alpha} \quad \text{[by using equation (1)]}$$

2. Let the intensities of two coherent sources be I_1 and I_2 and the amplitude be a_1 and a_2 respectively.

Given that, $$I_1/I_2 = 81/1$$

We have that, the intensity is proportional to the square of the amplitude.

So, $$\frac{I_1}{I_2} = \frac{a_1^2}{a_2^2} \quad \text{or} \quad \frac{a_1}{a_2} = \sqrt{\frac{I_1}{I_2}} = \sqrt{\frac{81}{1}} = 9$$

or $$a_1 = 9a_2$$

Now, the maximum intensity of bright fringe *i.e.*

$$I_{max} = (a_1 + a_2)^2 = (9a_2 + a_2)^2 = 100\, a_2^2$$

Similarly, $$I_{min} = (a_1 - a_2)^2 = (9a_2 - a_2)^2 = 64\, a_2^2$$

So, the ratio of I_{max} to I_{min} *i.e.*, $\dfrac{I_{max}}{I_{min}} = \dfrac{100\, a_2^2}{64\, a_2^2} = \dfrac{\mathbf{25}}{\mathbf{16}}$

3. The resultant intensity at a point due to two coherent sources of amplitude a_1 and a_2 with a phase difference δ will be,

$$I = a_1^2 + a_2^2 + 2a_1a_2 \cos\delta$$

or $$= I_1 + I_2 + 2\sqrt{(I_1 I_2)} \cos\delta$$

Given that, $$I_1 = 1.44, I_2 = 4.00 \text{ and } I = 0.90$$

$\therefore$ $$0.90 = 1.44 + 4.00 + 2\sqrt{(1.44 \times 4.00)} \cos\delta$$

$$= 5.44 + 2 \times 2.4 \cos\delta$$

or $$\cos\delta = \frac{0.90 - 5.44}{4.8} = -0.9458$$

or $$\cos(180 - \delta) = 0.9458$$

or $$180° - \delta = 19°$$

or $$\delta = 180° - 19° = \mathbf{161°}.$$

6. For a prism of very small refracting angle α, the deviation δ produced in ray by the prism is given by,

$$\delta = (\mu - 1)\alpha \qquad ...(1)$$

where μ is the refractive index of the material of the prism. From fig. 1.29, it is clear that the distance $2d$ between the sources S_1 and S_2 is given by,

$$2d = 2\delta \times a,$$

where a is the distance between the slit and biprism.

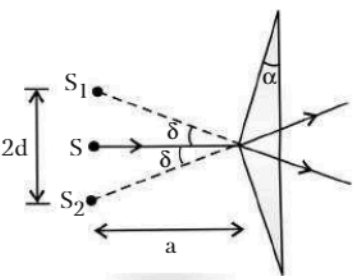


Fig. 1.29

Using eq. (1), we get

$$2d = 2(\mu - 1)\alpha \times a = \mathbf{2}\boldsymbol{a}(\mu - \mathbf{1})\alpha.$$

7. (i) If the angle of the biprism (α) is increased, the distance $2d$, between the virtual sources would also increase, since $2d = 2a(\mu - 1)\alpha$. Due to this, the fringe width, $\beta = \dfrac{D\lambda}{2d}$ would decrease or the fringes will not be separately visible and may disappear ultimately.

(ii) When width of the slit is gradually increased the two virtual source slits are correspondingly widened; due to this the fringe width would decrease accordingly. So the contrast between the bright and dark fringes becomes poorer and poorer. Ultimately, the fringes disappear having a uniform illumination everywhere.

8. Since white light is a mixture of several light waves of different wavelength and different colours, the fringe width, $\beta = \frac{D\lambda}{2d}$, will be different for different colour (as λ is different and also $2d = 2a(\mu - 1)\alpha, \mu$ is different for different colour). Hence with white light the pattern of different colours overlapping with each other will be obtained. In any fringe system the central fringe is defined as a fringe for which the path difference and the phase difference between the interfering waves is zero for all wavelengths. Therefore, all different coloured interfering waves produce a bright fringe at the centre. The superposition of different colours makes the central fringe white and it is known as zero order fringe.

As we move on either side of the centre, the path difference gradually increases from zero, then fringe width for different colour will be different and we get a few coloured fringes in haphazard way on both sides of the central fringe. If we move further away from the centre, the path difference becomes large. Thus, the large number of wavelengths will produce maximum intensity at a given point, and the other wavelength will produce minimum intensity at the same point. Hence, at each point we get uniform illumination.

9. All the bright fringes obtained by monochromatic light are of the same colour. It is very difficult to locate the central or zero order fringe. If the monochromatic light be replaced by white light, the zero order fringe is white while the rest are coloured. Hence, we can easily adjust the cross wire of the eyepiece on the zero order fringe. Now, white light is replaced by the given monochromatic light. The cross wire will still be on the zero order for central fringe. In this way the central fringe is located.

10. Fresnel's biprism can be used to determine the wavelength of a given source of monochromatic light as follow:

This experiment consists of a fine vertical slit *S*, biprism and a micrometer eyepiece. The slit *S* is adjusted close to a source of light and the refracting edge of biprism is also set parallel to the slit *S*. The slit and the biprism are adjusted in the stands on an optical bench. The micrometer eyepiece is placed on the optical bench at some distance from the prism to see the fringes in its focal plane.

For obtaining sharp fringes the following adjustments are necessary.

(i) The optical bench is levelled in the horizontal plane.

(ii) The slit, the biprism and the eyepiece are adjusted to the same height.

(iii) The slit is made narrow and vertical and illuminated by a monochromatic light.

(iv) The biprism is moved at right angles to the optical bench until two equally bright virtual slit images S_1 and S_2 are observed on looking through the biprism along the axis of the bench. The biprism is now rotated in its plane. By doing so the frings appear in eyepiece though they may not be very distinct.

(v) Rotate the biprism (with the help of tangent screw) till the fringes seen in eyepiece, become more distinct. At this position, the edge of the biprism become exactly parallel to the slit.

(vi) The eyepiece is moved at right angles to the length of the bench until the overlapping region comes in the field of view.

(vii) Again adjust the width of slit to make the fringes more distinct.

(viii) Finally, the line joining the slit and the edge of the biprism is made parallel to the length of the bench, the lateral shift between the fringes and the cross wire is completely removed. For this, the cross wire is set on a fringe in the centre and the eyepiece is moved along the bench away from the biprism. If the line joining the slit to the edge is not parallel to the bench, the fringe system shifts laterally. The biprism is moved laterally until the cross wire is again set on the same fringe. Now, the eyepiece is moved toward the biprism when the fringe system shifts again but in opposite direction. The biprism is moved laterally until the cross wire is again set on the same fringe. This process is repeated until the lateral shift completely disappears. The wavelength λ of the light used is determined by using the formula,

$$\beta = \frac{D\lambda}{2d}$$

or
$$\lambda = \frac{\beta 2d}{D} \qquad ...(1)$$

where β is the fringe width, $2d$ is the distance between two coherent sources S_1 and S_2 and D is the distance between slit and eyepiece.

Now, we measure these quantities in the following way:

(a) Read the distance D between slit and eyepiece directly on the optical bench.

(b) After obtaining the fringes, the vertical cross wire of the eyepiece is set on a bright fringe on one side of the interference pattern. The dreading of the micrometer screw is taken. Then the eyepiece is moved laterally so that vertical cross wire coincides with successive bright fringes and again corresponding reading is noted. From these reading the fringe width β can be determined.

(c) To measure $2d$, we use the displacement method.

By putting all these values, we can determine wavelength of light used.

11. **Displacement method to determine the distance between two virtual sources:** A convex lens of short focal length is placed in between the biprism and the eyepiece [Fig. 1.30(a)] without disturbing their position in such a way that the image of the virtual sources S_1 and S_2 are seen in the field of view of the eyepiece. Suppose the lens is in position L_1 and the distance between the images of S_1 and S_2 be d_1. In this case,

$$\frac{d_1}{2d} = \frac{v}{u} \qquad \text{....(1)}$$

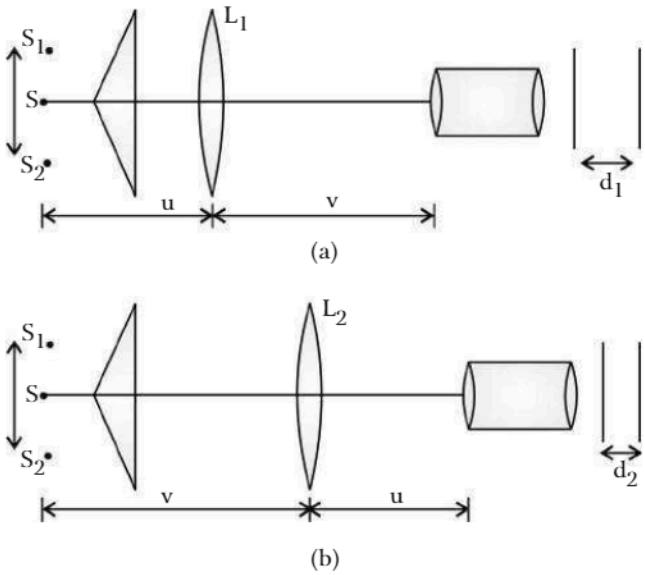


Fig. 1.30

Now, move the lens towards the eye-piece and bring it to some other position L_2, so that the images of S_1 and S_2 are seen clearly in the field of view of eyepiece [Fig.1.30 (b)]. Again measure the distance d_2 between the images of S_1 and S_2.

Here, $$\frac{d_2}{2d} = \frac{u}{v} \qquad \text{....(2)}$$

Multiplying equation (1) and (2), we get $\frac{d_1}{2d} \times \frac{d_2}{2d} = \frac{v}{u} \times \frac{u}{v} = 1$

or $$4d^2 = d_1 d_2$$

or $$2d = \sqrt{d_1 d_2}$$

So, by measuring d_1 and d_2, we can determine the distance between the virtual sources S_1 and S_2 in a biprism.

12. **Determination of the thickness of thin sheet:** The biprism can be used to determine the thickness of a given thin sheet. Suppose S_1 and S_2 are two virtual coherent sources of a biprism. The point R is equidistant from S_1 and S_2. When a thin sheet G of thickness t and refractive index μ is introduced in the path of one of the beams (Fig. 1.31) the fringes shifts to position R'.

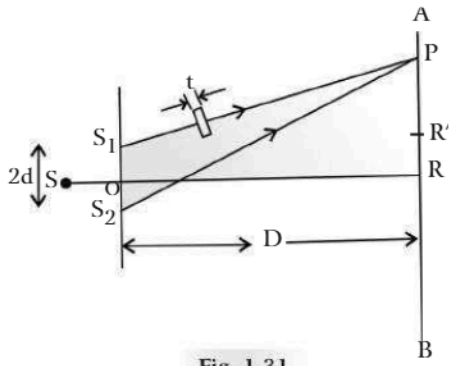


Fig. 1.31

Let the point P be the centre of the nth bright fringe, then path different between S_1P and S_2P,

$$= S_2P - [(S_1P - t) + \mu t]$$

$$= [S_2P - S_1P - (\mu - 1)t]$$

Also, we have $S_2P - S_1P = \frac{2xd}{D}$, where $2d = S_1S_2$

D = Distance of screen from S_1, S_2 and $RP = x$

$\therefore$ Effective path difference at $P = \frac{2xd}{D} - (\mu - 1)t$

For constructive interference *i.e.*, for bright fringe this path difference must be equal to $n\lambda$ *i.e.*,

$$\frac{2xd}{D} - (\mu - 1)t = n\lambda$$

In the absence of the film ($t = 0$), the distance of the nth bright fringe from R is $\frac{nD\lambda}{2d}$. Therefore, the shift of the nth bright fringe due to thin sheet is given by,

$$x_0 = \frac{D}{2d}[n\lambda + (\mu - 1)t] - \frac{nD\lambda}{2d}$$

or $$x_0 = \frac{D}{2d}(\mu - 1)t \quad \text{or} \quad \boldsymbol{t = \frac{2dx_0}{D(\mu - 1)}}$$

Thus, measuring x_0, the distance through which the central fringe is shifted, D, $2d$ and μ, the thickness of the transparent thin sheet can be determined.

13. When the thickness of the film is large as compared to the wavelength of light, the path difference at any point of the film will be large. Then, the same point will have maximum intensity for a large number of wavelengths, and minimum intensity for another large number of wavelengths. In such a case interference fringes cannot be seen. Hence, the resultant effect at a point will be the sum of all colours, *i.e.*, white.

14. When a thin film is illuminated by an extended source, the rays from different points of the sources are reflected from different parts of the film so as to enter the eye placed in a fixed position. Hence, we can see the entire film simultaneously. While, when a thin film is illuminated by a point source then, for different incident rays, different pairs of interfering rays are obtained. All these pairs cannot be received by the eye. The rays only from a small portion of the film can enter the eye. Hence, the entire film cannot be seen by the eye placed in a fixed position.

15. For constructive interference of light reflected,

$$2\mu t \cos r = (2n+1)\,\lambda/2$$

Here given that, $\mu = 1.47, t = 1.2\times10^{-7}\,\text{m}, i = 60°, n = 0$ (first order)

Also, we have $\dfrac{\sin i}{\sin r} = \mu$ or $\sin 60°/\sin r = 1.47$

or $$\sin r = \frac{\sin 60°}{1.47} = \frac{\sqrt{3}}{2\times1.47} = 0.589$$

or $$r = \sin^{-1}(0.589) = 36°$$

Putting all values, we get $\lambda = 2\times2\times1.47\times\cos 36\times1.2\times10^{-7} = 7.056\times0.81\times10^{-7}\,\text{m}$

$$= 5.7\times10^{-7}\,\text{m} = \mathbf{5700\ Å}.$$

16. In the presence of liquid, the diameter of nth bright ring is given by,

$$D_n^2 = \frac{2(2n-1)\lambda R}{\mu}$$

or $$\mu = \frac{2(2n-1)\lambda R}{D_n^2}$$

Here given that, $D_n = D_5 = 3\text{ mm} = 0.3\text{ cm}, n = 5, \lambda = 5895\times10^{-8}\,\text{cm}$ and $R = 100\text{ cm}$

Putting the values, we get $\mu = \dfrac{2(2\times5-1)\times5895\times10^{-8}\times100}{(0.3)^2} = \mathbf{1.178}$

17. We have that, $D_{n+P}^2 - D_n^2 = \dfrac{4PR\lambda}{\mu}$

or $$\mu = \frac{4PR\lambda}{D_{n+P}^2 - D_n^2}$$

Here given that, $D_{n+P} = D_{15} = 0.530\text{ cm}, D_n = D_5 = 0.306\text{ cm}$

$P = 10, R = 100\text{ cm}$, and $\lambda = 5895\times10^{-8}\text{ cm}$

Putting the values, we get $\mu = \dfrac{4\times10\times100\times5895\times10^{-8}}{(0.530)^2 - (0.306)^2} = \mathbf{1.26}$

18. Newton's ring in reflected light are formed by the interference between the ray (1) reflected from the upper surface of the air film, and the rays (2) or (3) reflected from the lower surface of the film [Fig. 1.32]. At the point of contact the thickness of the air film is almost zero and hence, no path difference exists between the interfering rays, but the ray (2) reflected from the lower surface of the film suffers a phase change of π. Thus, the two interfering waves of the centre are opposite in phase and destroy each other if the amplitude of (2) is equal to that of (1). Hence, destructive interference is produced and the centre of the ring system is perfectly black as shown in fig 1.33.

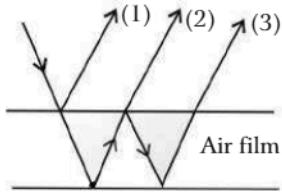


Fig. 1.32

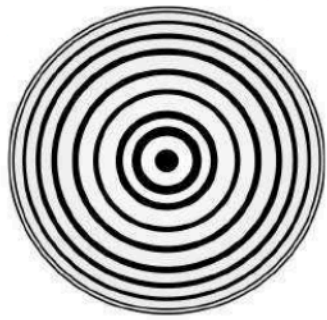


Fig. 1.33

To obtain the Newton's ring with bright centre, we take the lens and plate of different refractive indices (μ_1 and μ_3) materials and introduce a liquid between them whose refractive index μ_2 is intermediate to μ_1 and μ_3 (fig. 1.34). Therefore,

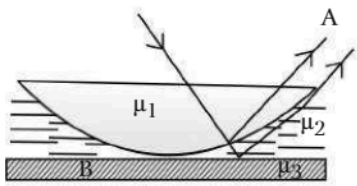


Fig. 1.34

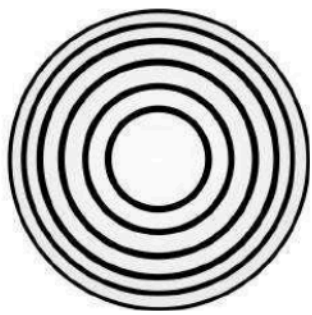


Fig. 1.35

the reflection at both the upper and lower surfaces of the film take place under similar conditions *i.e.*, in going from a rarer to a denser medium ($\mu_1 < \mu_2 < \mu_3$) or denser to rarer medium. Thus, there is a phase change of π at both reflections. Hence, the relative phase difference between the interfering rays at the point of contact is zero or 2π and the central spot appears bright as shown in fig. 1.35.

19. Newton's rings can be seen in transmitted (refracted) light as well as reflected light. [Fig. 1.36] shows the interfering reflected rays and interfering transmitted rays. The reflected and transmitted rays interfere in the similar way and produce Newton's ring. The rings observed in reflected light are exactly complementary to those seen in transmitted light *i.e.*, corresponding to every dark or bright ring in reflected light there is a bright or dark ring in transmitted light. The reason for this

is, the ray 1 reflected from the upper surface of the air film does not suffer any phase change while the ray 2 reflected from the lower surface of the air film suffers a phase change of π. Thus, the two interfering reflected rays have a phase difference of π. On the other hand the ray 1′ is transmitted directly through the air film while the ray 2′ suffers two internal reflections and hence a phase change of 2π occurs before coming out. Thus, the tow interfering transmitted rays have a phase difference of 2π or zero. Hence, the condition for constructive interference in reflected light is the condition of destructive interference in the transmitted light and vice-versa.

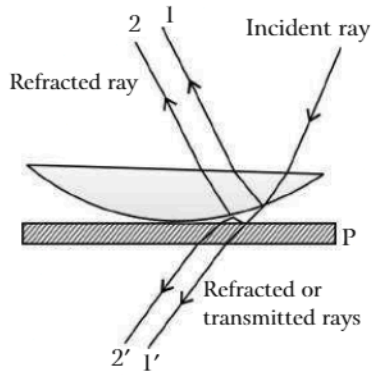


Fig. 1.36

The ring in transmitted light are much poorer in contrast than that in reflected light. It is due to the amplitude ratio of ray 2′ and 1′. Both are very much different so, the contrast of maxima and minima is very poor.

20. With monochromatic light, Newton's rings are alternately dark and bright. The diameter of the ring depends upon the wavelength of light used. With white light only a few coloured rings are visible. It is because the white light is composed of a number of wavelengths (colours). Each produces its own ring system having a different spacing. Only the first few rings are clear and after that due to overlapping of the rings of different colours, the rings can not be viewed.

21. The Fabry-Perot interferometer is better than Michelson interferometer in two respects.

(i) The Fabry-Perot fringes formed by multiple beam interference are much sharper than the Michelson fringes. At half width *i.e.*,

$$I/I_{\text{max.}} = 1/2$$

Then
$$\delta = 2\sin^{-1}\frac{1-R}{2\sqrt{R}}$$

for $R = 0.98$, $\delta = 0.22$ radian

while in the case of Michelson fringes, at half width *i.e.*,

$$I/I_{\text{max.}} = 1/2,\ \delta = 2\cos^{-1}(1/\sqrt{2}) = 1.57 \text{ radian} \qquad [I = I_{\max}\cos^2\delta/2]$$

(ii) When the light consists of two or more close wave lengths (such as D_1 and D_2 lines of sodium), then in a F.P. interferometer each wavelength produces its own pattern, and the rings of one pattern are clearly separated from the corresponding rings of the other pattern. In Michelson interferometer separate patterns are not obtained. So, F.P. is very suitable for the study of the fine structure of spectral lines.

22. **Determination of wavelength of monochromatic light:** In Fabry-Perot interferometer, the condition for maximum is,

$$2d\cos\theta = n\lambda$$

where n is the order of interference fringe and λ is the wave length of light used. At the centre of the fringe system, the angle of incidence, θ is equal to zero *i.e.*, $\cos\theta = 1$, therefore,

$$2d = n_0\lambda \qquad ...(1)$$

where n_0 is the order of interference fringe at the centre.

For this purpose the reflecting surfaces of glass plates A and B are made exactly parallel. The movable plate is moved and at d_1 separation, there is a bright fringe at the centre and corresponding to another separation d_2 there is again a bright fringe at the centre. If n_1 and n_2 are the corresponding orders at the centre, then from eq. (1), we have

$$2d_1 = n_1\lambda$$

$$2d_2 = n_2\lambda$$

or $$2(d_2 - d_1) = (n_2 - n_1)\lambda = N\lambda$$

where $N = (n_2 - n_1)$ is the number of fringes that cross the centre of the field when the plate separation is changed from d_1 to d_2.

$\therefore$ $$\lambda = \frac{2(d_2 - d_1)}{N}$$

So, by taking d_1, d_2 and N, λ can be determined.

23. **Measurement of difference in wavelengths:** If a source emits two close wavelengths λ_1 and λ_2, then two set of fringes are observed in Fabry-Perot interferometer. Initially the plates are brought close to each other. Let for the plate separation d_1, the rings due to λ_1 and λ_2 partially coincide, then

$$2d_1\cos\theta = m\lambda_1 = n\lambda_2 \qquad ...(1)$$

where m and n are the order of fringes of fringes of respective wavelength.

Thus, $$\frac{\lambda_1}{\lambda_2} = \frac{n}{m}$$

Now, the separation between the plate is gradually increased (let d_2). The rings now separate out and at one stage bright rings of λ_1 coincide with the mid points of dark rings of λ_2 and vice-versa. Let in this position the order of fringes due to λ_1 has been increased by p and that of λ_2 by $p+1$, then we have,

$$2d_2\cos\theta = (m + p)\lambda_1 = (n + p + 1)\lambda_2 \qquad ...(2)$$

Subtracting equation (2) from equation (1), we get

$$2(d_2 - d_1)\cos\theta = p\lambda_1 = (p+1)\lambda_2$$

At the centre of the fringes system, $\cos\theta = 1$

$$\therefore \quad 2(d_2 - d_1) = p\lambda_1 = (p+1)\lambda_2 \qquad \ldots(3)$$

or $$p(\lambda_1 - \lambda_2) = \lambda_2$$

$$\therefore \quad p = \lambda_2 / \lambda_1 - \lambda_2$$

putting this value p in equation (3), we get

$$2(d_2 - d_1) = \frac{\lambda_1 \lambda_2}{\lambda_1 - \lambda_2}$$

or $$\lambda_1 - \lambda_2 = \frac{\lambda_1 \lambda_2}{2(d_2 - d_1)} = \frac{(\lambda_{mean})^2}{2(d_2 - d_1)}$$

or $$\Delta\lambda = \frac{(\lambda^2_{\mathbf{mean}})^2}{\mathbf{2(d_2 - d_1)}} \qquad \ldots(4)$$

where λ_{mean} is the mean of λ_1 and λ_2, and $(d_2 - d_1)$ is the distance traversed by the movable plate between two consecutive positions of maximum distinctness.

24. We have that,

$$\Delta\lambda = \lambda_1 - \lambda_2 = \frac{(\lambda_{mean})^2}{2(d_2 - d_1)}$$

Here, λ_{mean}= average wavelength of λ_1 and λ_2. So, λ_{mean}=5893 Å $= 5893 \times 10^{-10}$ m and $d_2 - d_1$ = distance traversed by the movable plate between two consecutive position of maximum intensity $= 0.289$ mm $= 0.289 \times 10^{-3}$ m

So, by putting these values, we get

$$\lambda_1 - \lambda_2 = \Delta\lambda = \frac{(5893 \times 10^{-10})^2}{2 \times 0.289 \times 10^{-3}} = \mathbf{6\,Å}.$$

25. Interference Filter: An interference filter in based on the principle of Fabry-Perot interferometer. It consists of an optical system transmits nearly monochrmatic beam of light (covering a small range of 50 Å).

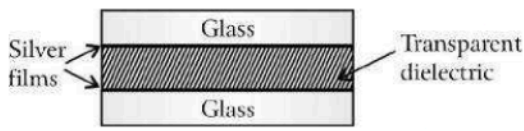


Fig. 1.37

In interference filter there are two glass plates. On the surface of these glass plates semi-transparent silver films are deposited by evaporation method. A thin transparent dielectric *e.g.* magnesium fluoride is placed between the two glass plates (Fig.1.37). When a beam of light is incident normally on the filter, multiple reflections take place within the film. The interference maxima for the transmitted beam will be governed by $2\mu t = n\lambda$. Here, μ is the refractive index of the dielectric, t its thickness, and n is a whole number. If $\mu t = \lambda$, n will be equal to 2. For the value of n=1, the maximum occurs for a wavelength of 2λ. Here, λ and 2λ represent a wide separation in the visible region.

Now, if the thickness of the dielectric is reduced, the transmitted wavelengths are more widely spaced. For an optical thickness (μt) of the dielectric film of 5000 Å, the transmitted wavelengths for n=1,2,3 etc., are 10000 Å, 5000 Å, 3333 Å. These three wavelengths are widely spaced. Only 5000 Å lies in the visible region. If there are two maxima in the visible region, one of them can be eliminated by using a coloured glass filter. This may be the protecting glass of the dielectric itself.

Interference filters are better as compared to the coloured glass filters because in interference filters light is not absorbed and hence, there is no overheating. Interference filters are used in spectroscopic work *i.e.*, to study the spectra in a narrow range of wavelengths.

26. **Lummer-Gehrcke Plate:** It is a high resolving power instrument employed for the study of fine structure of spectral lines and the Zeeman effect. It consists of an optically plane-parallel plate of glass or quartz about 10-15 cm long, 1-2 cm wide, and a few mm thick. A totally reflecting small prim P is cemented at one end of the plate as shown in fig. 1.38.

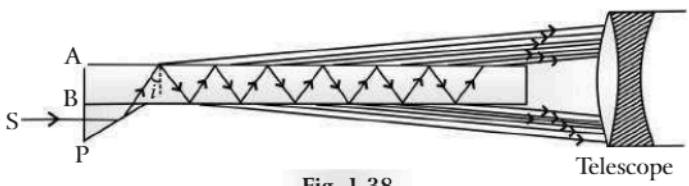


Fig. 1.38

A beam of light from an extended source enters into the plate through the prism P and incident on the lower surface of the plate at an angle slightly smaller than the critical angle. Therefore, a small fraction of the light emerges out, while the rest is internally reflected. This happens many times successively at the faces A and B of the plate. Thus a group of parallel rays leaves each surface of the plate near grazing emergence, having a constant path difference between any two successive rays. They are received by the telescope and interference fringes are viewed in the filed of view.

27. **Coherence time and Coherence length:** A monoc-hromatic of light gives a single frequency (wavelength) is an ideal definition and it is true only theoretically. In practical form, the best source of light emits radiations of a finite range of wavelength, though the range is very narrow. In fig. 1.39 the wave represents a single frequency sinusoidal wave. *The time interval over which the phase is constant is called the coherence time* and it is represented by Δt or τ. For an ideal monochromatic wave,

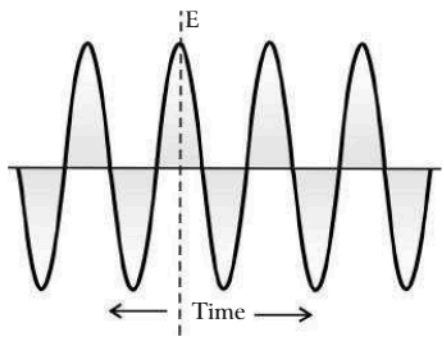


Fig. 1.39

coherence time Δt is infinity because the phase remains constant throughout. Corresponding, for this ideal monochromatic wave, the frequency range Δn is given by,

$$\Delta n = \frac{1}{\Delta t}$$

If $\Delta t = \infty$ then, $\Delta n = 1/\infty = 0$

But in practice, the coherence time Δt exists. The coherence length is also known a spatial interval and is given by,

$$\Delta l = c\Delta t$$

The length over which the phase of the wave remains constant is known as coherence length or spatial interval.

28. **Temporal Coherence:** The temporal coherence is the phase relationship between the radiated wave at a single point at different times. *The average time interval for which the wave remains in same phase is known as coherence time or temporal coherence.*

Spatial Coherence: The spatial coherence is the phase relationship between two radiated waves at different points in the spaces. *The average length for which the waves remain in same phase is known as coherence length or spatial coherence.*

(D) Multiple Choice Questions

1.	(a)	2.	(c)	3.	(c)	4.	(b)	5.	(b)	6.	(a)	7.	(b)
8.	(a)	9.	(b)	10.	(c)	11.	(a)						

(E) Fill in the Blank

1.	maximum	2.	minimum	3.	directly proportional
4.	poorer	5.	square root	6.	dark
7.	$\beta = \frac{D\lambda}{2d}$	8.	coefficient of sharpness or fitness	9.	$F = \frac{4R}{(1-R)^2}$

(F) True/False

1.	T	2.	F	3.	F	4.	T	5.	F	6.	F	7.	F
8.	T	9.	T	10.	T	11.	F	12.	F	13.	F		

❑❑❑

DIFFRACTION OF LIGHT

- Diffraction of Light
- Fresnel's Assumptions
- Rectilinear Propagation of Light
- Zone Plate
- Diffraction at a Single Slit
- Fraunhofer Diffraction at Circular Slit
- The Phase and Intensity of Fraunhofer Diffraction at a Slit
- Integral Calculus Method of Fraunhofer Diffraction at a S
- Fraunhofer Diffraction at Circular Aperture
- Resolving Power of Optical Instrument or Resolution of Images
- Rayleigh's Criterion for Resolution
- Resolving Power of Telescope
- Resolving Power of Microscope
- Phase Contrast Microscope

- N-Parallel Slits or Diffraction Grating
- Diffraction at N-Parallel Slits and Intensity Distribution
- Plane Reflection Grating
- Blazed Grating
- Concave Grating
- Different Methods of Mounting of Concave Grating
- Resolving Power of a Grating
- Experimental Determination of the Resolving Power of a Grating
- Resolving Power of a Prism

UNIT II

DIFFRACTION OF LIGHT

2.1 Diffraction of Light

It is found that the path of sunlight entering a dark room through a small window is straight. Similarly, when an opaque obstacle is placed in the path of light a sharp shadow is cast on the screen. It also indicates that the light travels in straight line. But when a light wave from a source passes through a narrow slit [Fig. 2.1], then we expect the region *AB* of the screen *SS′* to be illuminated and the remaining portion (known as the geometrical shadow) to be absolutely dark. But we found that the light intensity in the region *AB* is not uniform and light spreads to some extent in the region of geometrical shadow also. It is also found that if the width of the slit is made smaller, larger amount of light reach the geometrical shadow*. *The phenomenon of bending of light around corners of obstacles or the spreading of light waves into the geometrical shadow of an object is called diffraction, and the intensity distribution on the screen is known as the diffraction pattern.* In light wave the bending is small due to its shorter wave length (10^{-5} cm) while in the sound wave it is large as its wave length is large.

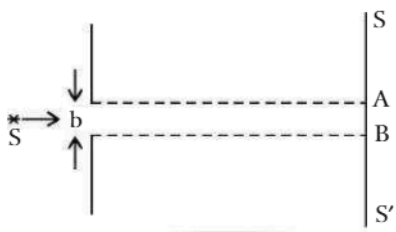


Fig. 2.1

*** The illumination in the geometrical shadow of an obstacle is not commonly observed because the light sources not point sources and the obstacles used are of very large size compared to the wavelength of light.**

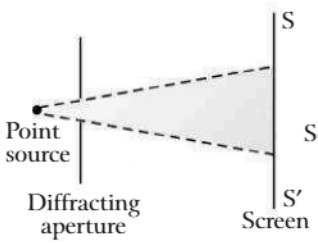
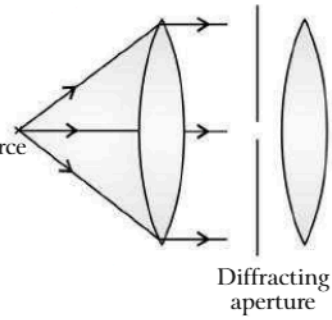


Fig. 2.2 (a)



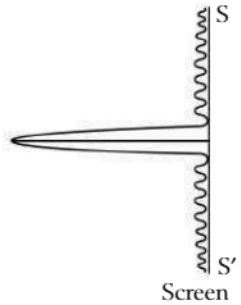


Fig. 2.2 (b)

The diffraction phenomena in light are usually of two types (i) **Fresnel diffraction** (ii) **Fraunhofer diffraction.** *In Fresnel's diffraction, the source of light and the screen are at a finite distance from the obstacle (diffracting aperture) and the wavefronts are cylindrical or spherical* [Fig. 2.2(a)]. *On the other hand in Fraunhofer's diffraction, the source and the screen are at infinite distance from the obstacle and the wavefronts is plane* [Fig. 2.2(b)].

2.2 Fresnel's Half Period Zones

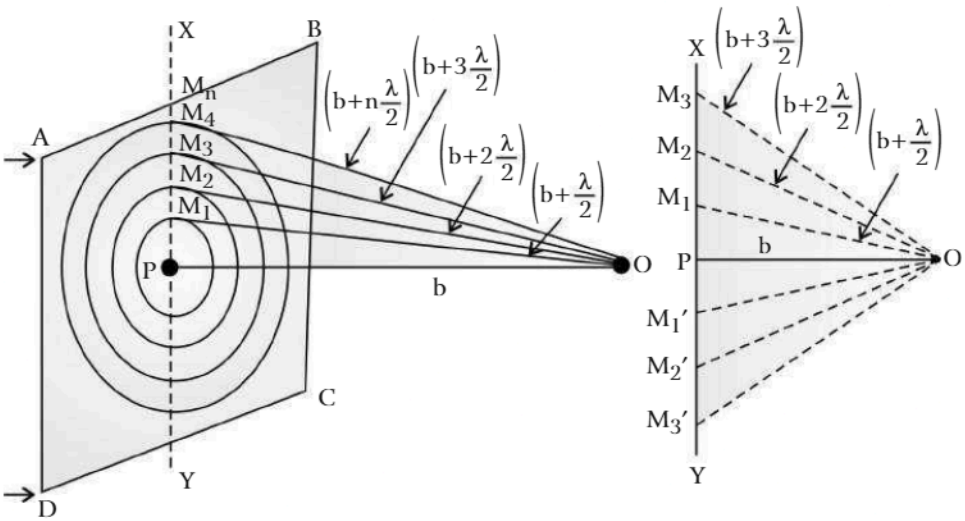
We have that, the light waves traveling in space produces vibration in the hypothetical ether. The continuous locus of ether particles which are vibrating in the same phase at any instant is called the '*wavefront*'. According to Huygen's theory, each point on wavefront sends out the secondary wavelets. Fresnel assumed that these wavelets interfere and produce a resultant intensity of light at any point. In order to calculate the resultant intensity at a point due to a wavefront, Fresnel again assumed that *the wavefront is divided into a large number of small zones. These zones are known as Fresnel's half period zones*. Diffraction means departure from rectilinear propagation and such departure are observable only on close examination. Althought in general, light propagaties rectilinearly.

2.2.1 Construction of Half Period Zones

Consider a plane wavefront *ABCD* perpendicular to the plane of paper. Let the monochromatic light of wavelength λ be moving towards a point *O*, at which the effect of (resultant amplitude) the entire wavefront is to be found. According to Huygen's principle of wave propagation, each point on the wavefront *ABCD* becomes a centre of secondary disturbance. All the points on the wavefront *ABCD* are in the same phase, but the secondary waves originating from *ABCD* will reach *O* with different phases. To determine the resultant effect at *O*, Fresnel subdivided the wavefront *ABCD* into a number of zones.

For this, a perpendicular *PO* is drawn from *O* on the wavefront. The point *P* is called the **pole** of the wavefront with respect to *O*. Let $PO = b$. Let us draw a series of concentric spheres with *O* as centre and radii equal to $b+\frac{\lambda}{2}, b+\frac{2\lambda}{2}, b+\frac{3\lambda}{2}$... etc. It will

cut the circular areas of radii PM_1, PM_2, PM_3 ... etc., on the wavefront $ABCD$ as shown in Fig. 2.3 (a). The area enclosed between P and M_1, M_1 and M_2, M_2 and M_3 etc. are known as *half period zones*. Each zone differs from its neighbour by a phase difference of π or a path difference of $\lambda/2$. The area enclosed by the first circle of radius PM_1 is called the first half period zone. The area enclosed by the annular strip M_1M_2 is known as second half period zone and so on. Thus, the annular area $(n-1)$th circle and nth circle is the nth half period zone.



(a) (b)

Fig. 2.3

2.2.2 Area of Zone

The area of the nth zone $= \pi PM_n^2 - \pi PM_{n-1}^2$

$$= \pi[(OM_n^2 - PO^2) - (OM_{n-1}^2 - PO^2)]$$

$$= \pi[OM_n^2 - OM_{n-1}^2]$$

$$= \pi\left[\left(b + \frac{n\lambda}{2}\right)^2 - \left(b + \frac{(n-1)\lambda}{2}\right)^2\right]$$

$$= \pi\left[b^2 + \frac{n^2\lambda^2}{4} + bn\lambda - b^2 - \frac{(n-1)^2\lambda^2}{4} - b\lambda(n-1)\right]$$

$$= \pi\left[b\lambda + \frac{\lambda^2}{4}(n^2 - n^2 - 1 + 2n)\right]$$

$$= \pi\left[b\lambda + \frac{\lambda^2}{4}(2n-1)\right] \quad \ldots(1)$$

$$\approx \pi b\lambda \qquad (\because b >> \lambda \text{ so } \lambda^2 \text{ term is negligible})$$

Thus, the area of each half period zone ($\pi b\lambda$) is approximately same and independent of the order of zone.

2.2.3 Intensity at an External Point due to Spherical Wavefront

Now, to find the resultant effect (amplitude) of the whole wavefront at O [Fig. 2.3 (b)], we find the resultant effect due to all zones as the whole wavefront is subdivided into zones. So, the amplitude at O due to a zone is:

1. Directly proportional to the area of the zone.
2. Inversely proportional to the average distance of the zone from O*, *i.e.*, greater the distance of the zone, the smaller is the amplitude reaching at O.
3. Directly proportional to the obliquity factor $(1+\cos\theta)$ where θ is the angle between the normal to the zone and the line joining the zone to O.

   Therefore, the amplitude at O due to nth zone,

$$A_n \propto \frac{\pi\left[b\lambda + \frac{\lambda^2}{4}(2n-1)\right]}{\left[b + (2n-1)\frac{\lambda}{4}\right]}(1+\cos\theta)$$

$$\propto \pi\lambda(1+\cos\theta)$$

   The total amplitude at O due to all the zones also depend upon the relative phase of various zones.

Let $A_1, A_2, A_3 \ldots A_n$ represent the amplitudes of ether particles at O due to first, second, third, ..., nth half period zones respectively. They are in descending order of magnitude (as obliquity factor increases). Since, the average distance of O from two consecutive zones differ by $\lambda/2$. So, the waves from two consecutive zones reach to O in opposite phase. Therefore, the difference in phase for wavelets coming from P and M_1 is π radians. Hence, if the amplitude due to first zone is positive, then due to second zone is negative and due to the third is positive and so on. So, the resultant amplitude at O due to entire wavefront is given by,

$$A = A_1 - A_2 + A_3 - A_4 + \ldots (-1)^{n-1} A_n \qquad \ldots(2)$$

Since, the amplitudes $A_1, A_2 \ldots$ gradually decrease in magnitude. So, A_2 is slightly smaller than A_1 but slightly greater than A_3. Hence, approximately, we have

$$A_2 = \frac{A_1 + A_3}{2} \quad \text{and} \quad A_4 = \frac{A_3 + A_5}{2} \text{ and so on.}$$

*** The average distance of the nth zone from O,**

$$= \frac{OM_n + OM_{n-1}}{2}$$

$$= \frac{\left(b + \frac{n\lambda}{2}\right) + \left[b + \frac{(n-1)\lambda}{2}\right]}{2} = b + (2n-1)(\lambda/4)$$

Thus, the resultant amplitude at O, at any instant may also be written as

$$A = \frac{A_1}{2} + \left(\frac{A_1}{2} - A_2 + \frac{A_3}{2}\right) + \left(\frac{A_3}{2} - A_4 + \frac{A_5}{2}\right) + \ldots\ldots$$

The last term will be $\frac{A_n}{2}$ or $\frac{A_{n-1}}{2} - A_n$ according to n is odd or even.

If n is odd, then $$A = \frac{A_1}{2} + \frac{A_n}{2}$$

and if n is even, then $$A = \frac{A_1}{2} + \frac{A_{n-1}}{2} - A_n$$

If the whole wavefront $ABCD$ is unobstructed, a large number of half period zones can be constructed. Since, the amplitude gradually decreases. So, A_n and A_{n-1} tends to zero.

$\therefore$ $$A = \frac{A_1}{2} \text{ for all } n$$

Thus, the amplitude due to whole wavefront at a point in front of it, is just half of that due to the first half period zone acting alone. Also,

$$I \propto A^2, \quad \text{So,} \quad \boldsymbol{I \propto \frac{A_1^2}{4}}$$

Thus, *the intensity due to whole wavefront at O is only one fourth of that due to the first half period zone alone. Hence, only half the area of the first half period zone is effective in producing the illumination at the point O.*

2.2.4 Intensity at an External Point due to Cylindrical Wavefront

Consider a long and narrow slit $S'S''$ held parallel to the plane of the paper. Let it be illuminated by a monochromatic source of light of wavelength λ [Fig. 2.4]. The wavefront emanating from the slit will be cylindrical with the slit as the axis of the cylinder. O is the point at which the effect of the wavefront is to be find.

To determine the effect of this wavefront at O, let it be divided into elementary half period strips with respect to the point O.

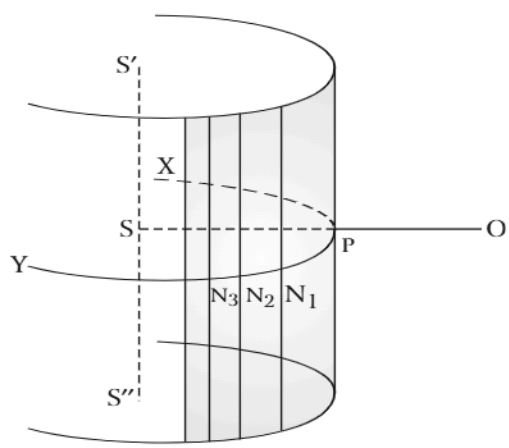


Fig. 2.4

Let XY represents a transverse cross section of the wavefront. Draw a normal OP on the surface of the cylindrical wavefront as shown in Fig. 2.5(a). Then P is the pole of the wavefront with respect to O on the screen. Let $SP = a$, $PO = b$. With O as centre, and $(OP + \lambda/2)$, $\left(OP + \frac{2\lambda}{2}\right), \left(OP + \frac{3\lambda}{2}\right)$... as radii describe the arcs cutting the wave surface in M_1, M_2, M_3 etc., and $N_1 N_2, N_3$ .... etc., respectively. The lines through M_1, N_1, M_2, N_2, ... etc., are drawn parallel to the slit. This gives us the strips of width PM_1, M_1M_2, M_2M_3 .... and PN_1, N_1N_2, N_2N_3 .... each parallel to the slit. These strips are called first, second, third...., half period strips respectively. When viewed from point O they appear as shown in Fig 2.5(b). Thus, the entire wavefront is divided into two exactly similar halves on the two sides of P. It is evident from this figure that the width of each succeeding strip is less than that of proceeding one. Initially, it decreases rapidly and slowly later on. Since, the length of these strips remain constant, so that the areas of the half period strips are proportional to their widths and are in decreasing order. The effects of these strips at O are also in decreasing order and alternately in the same phase.

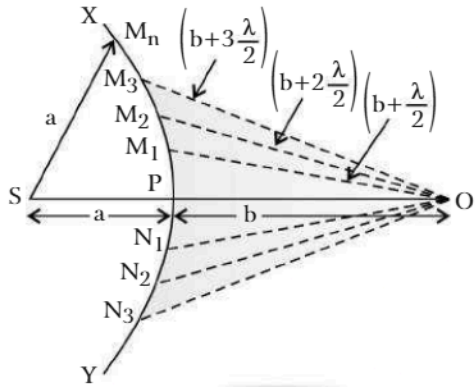


Fig. 2.5 (a)

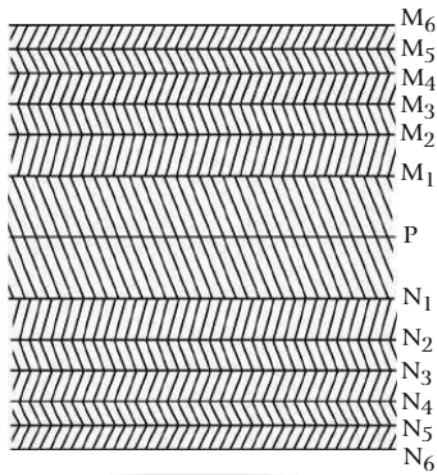


Fig. 2.5 (b)

Let A_1, A_2, A_3 .... etc., be numerical value of the amplitudes at O due to first, second, third .... etc., half period strips on either side of the pole. The resultant amplitude at O due to either half of the wavefront will be,

$$A = A_1 - A_2 + A_3 - A_4 + A_5 \ldots\ldots (-1)^n A_n$$

The alternate terms are with opposite signs because the waves from the consecutive strips differ in path by $\lambda/2$ and hence reach at O in the opposite phase. The series can be written as,

$$A = \frac{A_1}{2} + \left(\frac{A_1}{2} - A_2 + \frac{A_3}{2}\right) + \left(\frac{A_3}{2} - A_4 + \frac{A_5}{2}\right) + \ldots\ldots$$

$$= \frac{A_1}{2}, \text{ since the expressions within the bracket reduce to zero.}$$

Thus, the resultant amplitude due to the entire cylindrical wavefront,

$$A = \frac{A_1}{2} + \frac{A_1}{2} = A_1$$

and the resultant intensity I at O due to entire cylindrical wavefront will be $\boldsymbol{A_1^2}$.

2.3 Rectilinear Propagation of Light

If an opaque obstacle (or an aperture) whose size is very 'large' compared to the wavelength of light, be placed between a source of light and a screen, a distinct shadow (or an illuminated region) is obtained on the screen. This shows that the light travels from the source to the screen in straight lines. This is called 'rectilinear propagation of light'. Let us explain it on wave theory.

Let a plane wavefront of monochromatic light be incident normally on a rectangular aperture *ABCD* Fig. 2.6. Let a screen be placed parallel to the aperture at some distance from it. If the law of rectilinear propagation were strictly true, we would obtain uniform illumination inside the full-line rectangle and complete darkness outside this rectangle called geometrical shadow region.

Let P_1 be a point well-inside the full-line rectangle and O_1 the corresponding pole. O_1 is at a large distance (compared with the wavelength of light) from the edge of the aperture. Therefore if half-period zones are drawn around O_1, a large number of zones can be drawn before the edge of the aperture intersects them. Thus, practically all the effective zones are exposed. Hence, the resultant amplitude at P_1 is equal to one-half that due to the first zone. The neighbouring points are similarly and equally illuminated.

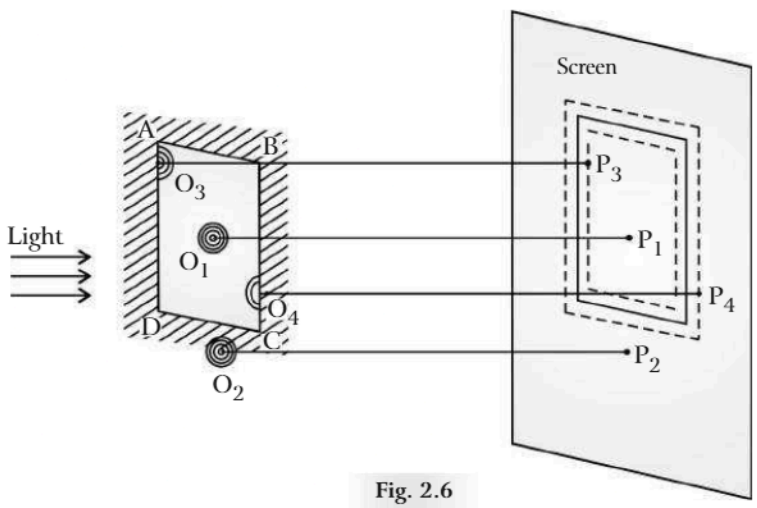


Fig. 2.6

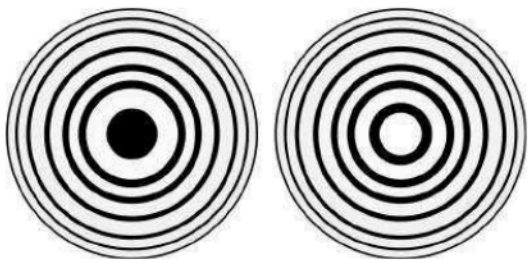
For a point P_2 well inside the geometrical shadow, the pole is at O_2. Thus all the effective half-period zones are cut off, and the resultant amplitude is zero. Hence there is complete darkness at P_2.

For points lying between the dotted rectangles, such as P_3 and P_4, the situation is different. Far the point P_3, the pole is O_3 just inside the aperture. Clearly, the effective area of the zones that intersect the edge *diminishes rapidly*. Similarly, for the point P_4, the pole is at O_4 just outside the aperture so that the effective area of the zones *increases rapidly*. In such cases, the above theory does not apply so that the intensity at P_3 and P_4 cannot be decided.

Thus there is almost uniform illumination at all points inside the inner dotted rectangle and complete darkness at all points outside the outer dotted rectangle. At points which lie between the two dotted rectangles, the elementary theory gives no information, but certainly at these points there is neither uniform illumination nor complete darkness. Since the wavelength of light is very small, the two dotted rectangles lie very close to the full-line rectangle. Thus, the wave theory accounts for the rectilinear propagation of light, but certainly the rectilinear propagation is only "approximate".

2.4 Zone Plate

A zone plate is a specially constructed screen such that light is obstructed from every alternate half period zones. It can be designed so as to cut off light due to the even numbered zones or that due to the odd numbered zones. It is constructed by drawing a series of concentric circles on white paper with radii proportional to the square root of the natural numbers. The odd numbered zones *i.e.*, alternate zones are pained with black ink and a highly reduced photograph of this drawing [Fig. 2.7(a)] is taken on a plane glass plate. When its negative is developed, its odd zones are transparent to light while even zones will stop the light. The negative thus obtained is the zone plate [Fig. 2.7 (b).] This plate behaves like a convex lens.



(a) Zone plate (b)

Fig. 2.7

2.4.1 Theory of the Zone Plate

Consider a point sources S is emitting spherical waves of wavelength λ. Let XY represents the zone plate perpendicular to the plane of the paper. If P is the position of the screen for a bright image. Let a is the distance of the source from the zone plate and b is the distance of the screen from the plate then $OM_1, OM_2, OM_3 \ldots\ldots OM_n$ $(r_1, r_2, r_3, \ldots r_n)$ etc., are the radii of the Ist, 2nd, 3rd and nth half period zones respectively [Fig. 2.7 (c)]. The position of the screen is such that from one zone to the next, there is a path difference of λ/2 *i.e.*,

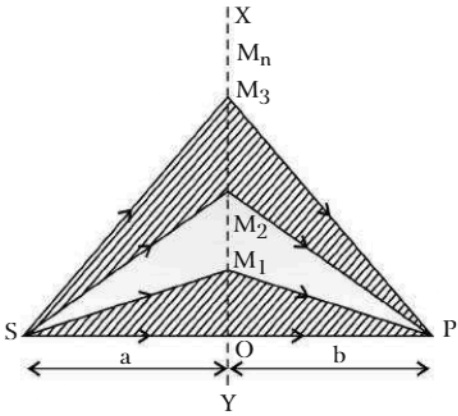


Fig. 2.7 (c)

$$SO + OP = a + b$$

$$SM_1 + M_1P = a + b + \lambda/2$$

$$SM_2 + M_2P = a + b + 2\lambda/2$$

$$SM_3 + M_3P = a + b + 3\lambda/2$$

and $$SM_n + M_nP = a + b + n\lambda/2 \quad \ldots(1)$$

The area of the first circle is the area of the first half period zone, the area between the second and the first circle is the area of second half period zone and so on. The area between the nth and the $(n-1)$th circle is the area of nth zone.

Now, to calculate the radius r_n of the nth circle, we have

$$SM_n = \sqrt{SO^2 + OM_n^2} = (a^2 + r_n^2)^{1/2} \qquad \text{[From fig. 2.7 (c)]}$$

$$= a\left(1 + \frac{r_n^2}{a^2}\right)^{\frac{1}{2}}$$

$$= a\left(1 + \frac{1}{2}\frac{r_n^2}{a^2}\right) \qquad \text{(approximation)}$$

$$= a + \frac{r_n^2}{2a} \quad \ldots(2)$$

Similarly,

$$M_nP = b + \frac{r_n^2}{2b} \quad \ldots(3)$$

Substituting the values of SM_n and M_nP in equation (1), we get

$$a + \frac{r_n^2}{2a} + b + \frac{r_n^2}{2b} = a + b + \frac{n\lambda}{2}$$

or $$\frac{r_n^2}{2}\left(\frac{1}{a}+\frac{1}{b}\right)=\frac{n\lambda}{2} \quad ...(4)$$

or $$r_n^2=\frac{ab\lambda}{a+b}.n$$

or $$r_n=\sqrt{\frac{ab\lambda}{a+b}}.\sqrt{n}$$

or $$r_n \propto \sqrt{n} \qquad \left[\because \sqrt{\frac{ab\lambda}{a+b}} \text{ is a constant}\right]$$

i.e., the radii of circle are proportional to the square root of natural numbers. Thus, if alternate zones are made opaque, the plate will serve as a zone plate.

2.4.2 Action Like a Convex Lens

Now, the amplitude due to a zone at P depends on the area of zone, on the average distance of the zone from P and on the obliquity of the zone.

The area of the $n$the zone $=\pi r_n^2-\pi r_{n-1}^2=\frac{\pi\, a\, b\, n\lambda}{a+b}-\frac{\pi\, ab\,(n-1)\lambda}{a+b}=\frac{\pi\, a\, b\,\lambda}{a+b}$,

This area of the nth zone is independent of n. Hence, the area of each zone is same. But the average distance of the zone from P and the obliquity increases as the order increases. Hence, the amplitude at P due to a zone decreases as the order of zone increases. Also we have that the waves from successive transparent zones (which are alternate) differ in path by λ. Thus, if A_1, A_2, A_3 etc. are the amplitudes due to first, second and third, zones respectively, then, the resultant amplitude will be,

$$A=A_1+A_3+A_5+\ldots. \quad (\because \text{even numbers of elements are opaque})$$

which is many times greater than the amplitude due to all the zones *i.e.*, $A_1/2$. Hence, the point O is sufficiently bright and may be called as the image of S. Thus, the zone plate focuses the light from S to O. It thus behave like a convex lens.

2.4.3 Focal Length of Zone Plate

The relation between the respective distance of the object and the image, *i.e.*, 'a' and 'b' given by equation (4). So, it can be written as,

$$\frac{1}{a}+\frac{1}{b}=\frac{n\lambda}{r_n^2}$$

On comparing with lens formula $\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$, $[\because u \text{ is negative in convex less}]$

we get, $$\frac{1}{f_n}=\frac{n\lambda}{r_n^2} \quad \text{or} \quad \boldsymbol{f_n=\frac{r_n^2}{n\lambda}}$$

This result gives the focal length of the zone plate. Thus, zone plate behaves like a lens (convex) with focal length $\frac{r_n^2}{n\lambda}$. Since, the focal length of zone plate depend upon the wavelength of light. So, a zone plate has severe chromatic aberration.

2.4.4 Multiple Foci of Zone Plate

A zone plate has a number of foci between point O and P [Fig. 2.7 (c)]. Corresponding to each focus point each zone can be divided into an odd number of half period elements. For an object at infinity ($a=\infty$), the radius of nth circle will be given by

$$r_n^2 = bn\lambda$$

So, the area of nth zone

$$= \pi r_n^2 - \pi r_{n-1}^2$$

$$= \pi\, bn\lambda - \pi\lambda b\,(n-1)\lambda$$

$$= \pi\, b\lambda$$

The area of zone depends on the distance between zone plate and screen. If the screen is moved nearer to zone plate, the area of half period elements decreases and more half period zones can be constructed on each zone. For a particular position P_m of the screen (image), $(2m-1)$ half period elements can be present on each zone, then focal length of zone plate is given by,

$$f_m = \frac{r_n^2}{(2m-1)\,n\lambda}$$

where r_n is the radius of nth zone of wavefront and λ is wavelength of light used.

Putting $m=1,2,3,\ldots$ etc., different (positions of image or foci) can be obtained. The position of these foci are given by,

$$f_1 = \frac{r_n^2}{n\lambda},\ f_2 = \frac{r_n^2}{3\,n\lambda},\ f_3 = \frac{r_n^2}{5\,n\lambda}.\ \text{and so on.}$$

It means for first image point P_1 one half period element, for 2nd image point P_2 three half period elements and for third image point P_3,5 half period elements can be constructed on each zone. With the increase in the focal length, the brightness of image decreases. So, the image points P_1, P_2, P_3, P_4, P_5 are of diminishing intensity.

Thus, a zone plate has multiple foci, corresponding to the focal lengths,

$$\frac{r_n^2}{n\lambda}, \frac{r_n^2}{3\,n\lambda}, \frac{r_n^2}{5\,n\lambda} \ldots \text{etc.}$$

according to number of half period elements that each zone of plate contains.

Example 1: What is the radius of the first zone in a zone plate of focal length 20 cm for light of wavelength 5000 Å ?

Solution: The focal length of a zone in a zone plate,

$$f = \frac{r_n^2}{n\lambda}$$

Here given that, $n=1, \lambda = 5000\ \text{Å} = 5000\times10^{-10}$ m and $f_1 = 20\ \text{cm} = 20\times10^{-2}$ m

Putting all the values, we get $20\times10^{-2}=\dfrac{r_1^2}{1\times5000\times10^{-10}}$

or $r_1^2=100000\times10^{-12}=10\times10^{-8}\,\text{m}$

$r_1=3.16\times10^{-4}\,\text{m}=\mathbf{0.0316\ cm}$

Example 2: The diameter of the central zone of a zone plate is 2.3 mm. If a point source of light ($\lambda=5890\,\text{Å}$) is placed at a distance of 6m from it calculate the position of the first image.

Solution: The focal length of a zone plate is,

$$f=\frac{r_n^2}{n\lambda}=\frac{r_1^2}{\lambda}$$

Given that, $2r_1=2.3$ mm or $r_1=1.15\text{ mm}=1.15\times10^{-3}\text{m}$, and $\lambda=5890\times10^{-10}\text{m}$

So, $f=\dfrac{(1.15\times10^{-3})^2}{5890\times10^{-10}}=2.25\text{ m}$

Also, we have $\dfrac{1}{v}-\dfrac{1}{u}=\dfrac{1}{f}$ (for a lens)

Here given that, $f=+2.25\text{ m}$, $u=-6\text{ m}$

$$\frac{1}{v}=\frac{1}{2.25}-\frac{1}{6}=\frac{6-2.25}{6\times2.25}=\mathbf{0.27m.}$$

$$\mathbf{v=27\ cm}$$

The first image will be formed at a distance of 27 cm.

Example 3: A zone plate is found to give series of images of a point source on its axis. If the strongest and the second strongest images are at distance of 0.30 m and 0.060 m respectively from the zone plate (both on the same side remote from the source), Calculate (i) the distance between the source and the zone plate. (ii) The principal focal length and radius of the first zone. (Assume $\lambda=5000\,\text{Å}$)

Solution: The focal length of a zone plate is,

$$f_m=\frac{r_n^2}{(2m-1)\,n\lambda}$$

Here given that, $n=1$ ($\because$ we are considering first zone)

$m=1$ (for stronger image)

$m=2$ (for second strongest image)

and $\lambda=5\times10^{-7}\,\text{m}$

Putting the values, we get $f_1=\dfrac{r_1^2}{\lambda}$ ...(1)

and $$f_2 = \frac{r_1^2}{3\lambda} \quad ...(2)$$

Also, we have $$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Again, given that, $v_1 = 0.30$ m and $v_2 = 0.06$ m

So, $$\frac{1}{0.30} - \frac{1}{u} = \frac{\lambda}{r_1^2} \quad ...(3)$$

and $$\frac{1}{0.06} - \frac{1}{u} = \frac{3\lambda}{r_1^2} \quad ...(4)$$

Multiplying equation (3) by 3 and subtracting it from equation (4), we get

$$\frac{1}{0.06} - \frac{3}{0.30} = \frac{1}{u} - \frac{3}{u} = -\frac{2}{u}$$

$$\frac{1}{0.06} - \frac{1}{0.10} = -\frac{2}{u}$$

or $$u = -\mathbf{0.3\ m}$$

Negative sign shows that lens is convex *i.e.*, point source is on opposite side of the images.

Now, from equation (3), $$\frac{1}{0.30} + \frac{1}{0.30} = \frac{\lambda}{r_1^2} = \frac{5\times10^{-7}}{r_1^2}$$

or $$r_1 = \mathbf{2.74\times 10^{-4}\ m}$$

and from equation (1), $$f_1 = \frac{r_1^2}{\lambda} = \frac{(2.74\times10^{-4})^2}{5\times10^{-7}} = \mathbf{0.15\ m}$$

2.5 Diffraction at a Straight Edge

Let *S* be a narrow slit illuminated by a source of monochromatic light of wavelength λ. The slit is perpendicular to the plane of the paper. A sharp and straight edge *BC* of an opaque obstacle (razor blade) is placed parallel to the slit, [Fig. 2.8]. Let *MN* be a screen parallel to *BC*. *XY* is the incident cylindrical wavefront. *O* is point on the screen which is joined to *S*, through *B* and *SBO* is perpendicular to the screen. Now, let us determine the distribution of light intensity on the screen.

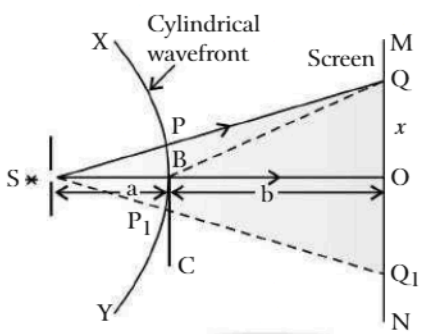


Fig. 2.8

If there were no diffraction of light at the straight edge, we would have obtained uniform illumination above *O* and complete darkness below *O i.e.*, geometrical shadow. But in actual practice, we observe,

1. Alternate dark and bright bands of unequal width with poor contrast and parallel to the straight edge just above *O*.
2. Rapid fall in intensity and complete darkness in geometrical shadow is obtained at a short distance below *O*.

   This is the complete diffraction pattern due to a the straight edge and can be explained on the basis of the wave theory.

2.5.1 Intensity in the Geometrical Shadow

For the point *O* on the screen in Fig. 2.8, the pole of the wavefront is at *B*. Clearly, *O* receives light from the entire upper half wavefront *BX*, the lower half portion *BY* being completely obstructed. Hence, the resultant amplitude at *O* is $A_1/2$ and the resultant intensity will be $A_1^2/4$.

Now, if we move below *O*, into the geometrical shadow *i.e.*, point Q_1, P_1 will be the corresponding pole. Thus, the entire lower half wavefront P_1Y and the portion P_1B of the upper half is obstructed. Only the part *BX* of the upper-half exposes Q_1. If P_1B cuts off only the first half period strip of the upper half wavefront, the amplitude at Q_1 will be $A_2/2$. As Q_1 point moves further into the geometrical shadow, then second, third, fourth ....etc., half period strips are successively cut off and the amplitude at Q_1 will be $\frac{A_3}{2}, \frac{A_4}{2}, \frac{A_5}{2}$ respectively. As the intensity is proportional to the square of the amplitude, the intensity will be $\left(\frac{A_2}{2}\right)^2, \left(\frac{A_3}{2}\right)^2, \left(\frac{A_4}{2}\right)^2, \left(\frac{A_5}{2}\right)^2$, etc., respectively. As A_1, A_2, A_3 .... etc., are in descending order of magnitude, therefore as the point moves into the geometrical shadow the intensity of light falls off rapidly and soon diminish to zero.

2.5.2 Explanation of Diffraction Bands

Now, we move upwards to the point *Q* distance *x* from the point *O*, the corresponding pole *P* of the wavefront also moves up. Then the point *Q* receives light from the entire upper half of the wavefront *PX* and from those half period strips of the lower half which are contained in *BP*. If *BP* contains first half period strip, then the amplitude at *Q* will be,

$$\frac{A_1}{2} + A_1 \quad \text{(maximum)}$$

As point *Q* moves up from *O*, *BP* contains second, third .... etc., half period strips of the lower half wavefront, Then, the respective amplitude at *Q* will be,

$$(A_1/2) + A_1 - A_2 \quad \text{(minimum)}$$

$$(A_1/2) + A_1 - A_2 + A_3 \quad \text{(maximum)}$$

$$(A_1/2) + A_1 - A_2 + A_3 - A_4 \quad \text{(minimum)}$$

$$(A_1/2) + A_1 - A_2 + A_3 - A_4 + A_5 \quad \text{(maximum)}$$

and so on. Thus, the point Q is a maximum or minimum according as BP contains an odd or even number of half period strips.

From the above discussion, it is clear that if we move away from point O in the illuminated region maxima and minima are obtained alternately. The amplitude and hence the intensity of minima are comparable to those of maxima. Hence, the bands have poor contrast. This gives what are known as **diffraction bands**. When point Q is at a sufficient distance from O, the entire half wavefront and a large number of half period strips of the lower half wavefront are unobstructed. Then the resultant amplitude at Q will be,

$$\frac{A_1}{2}+\frac{A_1}{2}=A_1$$

and the resultant intensity is A_1^2. Hence, at a sufficient distance from O the diffraction bands merge into uniform illumination.

2.5.3 Positions and Width of Maximum and Minimum Intensity

To find the positions of the diffraction bands and their width join SQ cutting the wavefront at P. Also join BQ. The number of half period strips contained in PB is equal to the number of half wavelengths in the path difference $(BQ-PQ)$. Thus,

$$BQ-PQ=(2n+1)(\lambda/2) \quad \text{(for maxima)}$$

$$BQ-PQ=2n\frac{\lambda}{2} \quad \text{(for minima)}$$

where $n=0,1,2,\ldots$

Let $SB=a$ and $BO=b, OQ=x$, then we have

$$BQ=(BO^2+OQ^2)^{\frac{1}{2}}=(b^2+x^2)^{1/2}$$

$$=b\left(1+\frac{x^2}{b^2}\right)^{\frac{1}{2}}=b+\frac{x^2}{2b} \quad \text{(approx.)}$$

and

$$PQ=SQ-SP$$

$$=[(a+b)^2+x^2]^{1/2}-a$$

$$=(a+b)\left[1+\frac{x^2}{(a+b)^2}\right]^{1/2}-a$$

$$=(a+b)+\frac{x^2}{2(a+b)}-a \quad \text{[using binomial theorem]}$$

$$=b+\frac{x^2}{2(a+b)}$$

Hence, the path difference, $BQ-PQ=\left[b+\frac{x^2}{2b}\right]-\left[b+\frac{x^2}{2(a+b)}\right]$

$$= \frac{x^2}{2}\left[\frac{1}{b} - \frac{1}{(a+b)}\right] = \frac{x^2}{2}\left[\frac{a}{b(a+b)}\right]$$

$$= \frac{ax^2}{2b(a+b)}$$

Also, we have the condition for maxima,

$$BQ - PQ = (2n+1)(\lambda/2)$$

$$\therefore \quad \frac{ax^2}{2b(a+b)} = (2n+1)\frac{\lambda}{2}$$

or
$$x = \sqrt{\frac{b(a+b)(2n+1)\lambda}{a}} \qquad ...(1)$$

where x is the distance of nth maxima from O.

or
$$x_n = k\sqrt{(2n+1)}$$

where $k = \sqrt{\frac{b(a+b)\lambda}{a}}$ is a constant. This expression gives the position of nth maxima in the diffraction pattern.

The distances of the successive maxima from O are,

$$x_1 = k$$
$$x_2 = k\sqrt{3} \qquad \text{(putting } n = 0,1,2,3 \ldots\ldots\text{)}$$
$$x_3 = k\sqrt{5}$$
$$\ldots\ldots\ldots$$

The separations between successive maxima are,

$$x_2 - x_1 = k(\sqrt{3} - 1) = 0.73\,k$$
$$x_3 - x_2 = k(\sqrt{5} - \sqrt{3}) = 0.50\,k$$
$$x_4 - x_3 = k(\sqrt{7} - \sqrt{5}) = 0.43\,k$$

and so on.

which are in decreasing order.

Similarly, we have the condition for minima,

$$BQ - PQ = 2n(\lambda/2)$$

$$\therefore \quad \frac{ax^2}{2b(a+b)} = 2n.\frac{\lambda}{2}$$

or
$$x = \sqrt{\frac{2b(a+b)n\lambda}{a}}$$

where x is the distance of n th minima from O.

$\therefore$
$$x_n = k\sqrt{2n}$$

where $k = \sqrt{\frac{b(a+b)\lambda}{a}}$ is a constant. This expression gives the position of nth minima in the diffraction pattern.

The distances of the successive minima from O are,

$$x_1 = k\sqrt{2}$$
$$x_2 = k\sqrt{4} \qquad \text{(Putting } n = 1,2,3\ldots.\text{)}$$
$$x_3 = k\sqrt{6}$$

The separations between successive minima are,

$$x_2 - x_1 = k(\sqrt{4} - \sqrt{2}) = 0.58k$$
$$x_3 - x_2 = k(\sqrt{6} - \sqrt{4}) = 0.449k$$

and so on.

which are also in decreasing order.

Thus, as we move far above O on the screen, the fringes come closer and closer.

The intensity distribution on the screen is shown in Fig. 2.9. In the geometrical shadow the intensity falls off rapidly and then complete darkness, while outside the geometrical shadow there is a system of bright and dark bands till uniform illumination results as describe above.

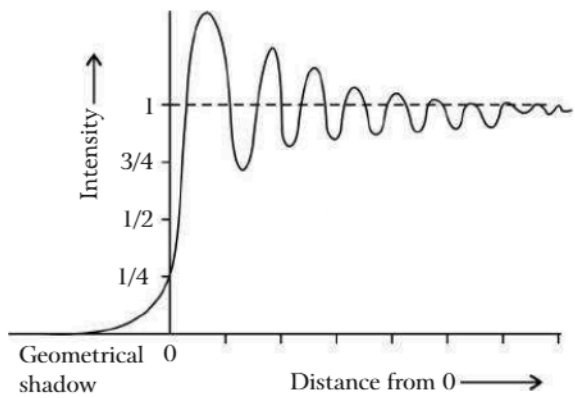


Fig. 2.9

Example 4: A narrow slit illuminated by light of wavelength 6000 Å is placed at a distance of 10cm, from a straight edge. If the screen is at a distance of 100 cm from the straight edge at which diffraction bands are obtained, calculate the distance between the first and second dark band.

Solution: The distance of the nth dark band (nth minima) outside the geometrical shadow is given by,

$$x_n = \sqrt{\frac{2b(a+b)n\lambda}{a}}$$

Here given that, $a = 10$ cm, $b = 100$ cm, and $\lambda = 6000 \times 10^{-8}$ cm $= 6 \times 10^{-5}$ cm

$$\therefore \qquad x_n = \sqrt{\frac{2 \times 100\,(10+100)\,6 \times 10^{-5}\, n}{10}} = 0.363\sqrt{n}$$

and
$$x_1 = 0.363 \text{ cm}$$
$$x_2 = 0.513 \text{ cm}$$

Hence, the distance between the first and second dark band

$$x_2 - x_1 = 0.513 - 0.363 = \mathbf{0.150 \text{ cm}}$$

***Example 5:** In an experiment for observing diffraction pattern due to a straight edge, the distance between the slit source ($\lambda = 6000\,\text{Å}$) and the straight edge is 6 meters and that between the straight edge and eyepiece is 4 meters. Calculate the position of first three maxima and their separation.*

Solution: The distance of the nth bright or maxima band outside the geometrical shadow is given by,

$$x_n = \sqrt{\frac{b(a+b)(2n+1)\lambda}{a}}$$

Given that, $a = 600$ cm, $b = 400$ cm and $\lambda = 6 \times 10^{-5}$ cm

Putting these values we get, $x_n = \sqrt{\frac{400 \times 1000}{600}(2n+1) \times 6 \times 10^{-5}} = 0.2\sqrt{2n+1}$

Here, for getting positions of three maxima, $n = 0,1,2$.

So, the position of first maxima, $x_1 = 0.2\sqrt{(2 \times 0 + 1)} = 0.2 \times \sqrt{1} = \mathbf{0.2\ cm}$

the position of second maxima, $x_2 = 0.2\sqrt{(2 \times 1 + 1)} = 0.2 \times \sqrt{3} = \mathbf{0.346\,cm}$

and the position of third maxima, $x_3 = 0.2\sqrt{(2 \times 2 + 1)} = 0.2 \times \sqrt{5} = \mathbf{0.447\,cm}$

Now, the separation between second and first maxima,

$$x_2 - x_1 = 0.346 - 0.200 = \mathbf{0.146\ cm}$$

and the separation between third and second maxima,

$$x_3 - x_2 = 0.447 - 0.346 = \mathbf{0.101\ cm}$$

2.6 Fraunhofer Diffraction at a Single Slit

In Fraunhofer's diffraction, the source and the screen are at infinite distance from the obstacle and the wavefront is plane.

Let a parallel beam of monochromatic light of wavelength λ be incident normally upon a narrow slit AB of width e where it gets diffracted [Fig. 2.10]. If a converging lens L is placed in the path of the diffracted beam, a real image of the diffraction pattern is formed on the screen MN in the focal plane of the lens.

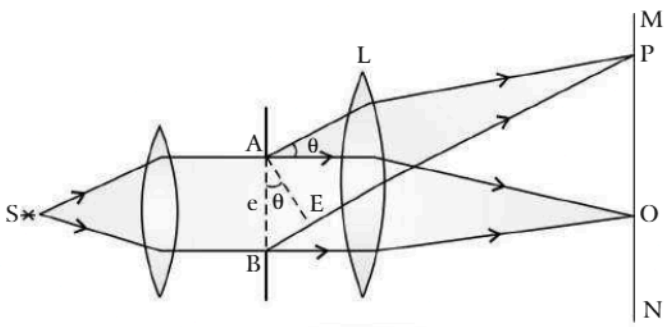


Fig. 2.10

According to Huygen's theory each point of AB sends secondary wavelets in all directions which are in the same phase. They reinforce each other and produce brightness at O, on the axis of the slit. The intensity of the diffracted beam will be different in different directions. Let us find out the resultant intensity at any point P above O on the screen, where set of rays is diffracted through an angle θ, Drop a perpendicular AE from A to the diffracted beam. AE constitutes the diffracted wavefront and BE is the path difference between the rays from the edges A and B of slit AB. The path difference between the wavelets from A to B in the direction θ,

$$BE = AB\sin\theta = e\sin\theta$$

The corresponding phase difference $= (2\pi/\lambda) \times$ path difference

$$= (2\pi/\lambda)(e\sin\theta)$$

Now, consider the width AB of the slit divided into n equal parts. Each part (strip) forms an elementary source. The amplitude of vibration at P due to the waves from each part will be the same, (let equal to a) and the phase difference between the waves from any two consecutive parts is,

$$\frac{1}{n}\left(\frac{2\pi}{\lambda}e\sin\theta\right) = d \text{ (let)}$$

Hence, the resultant amplitude* at P is given by,

$$A = \frac{a\sin(nd/2)}{\sin(d/2)}$$

$$= \frac{a\sin\left(\frac{\pi e\sin\theta}{\lambda}\right)}{\sin\left(\frac{\pi e\sin\theta}{n\lambda}\right)}$$

Let $\frac{\pi e\sin\theta}{\lambda} = \alpha$, then $\quad A = \frac{a\sin\alpha}{\sin\alpha/n} = \frac{a\sin\alpha}{\alpha/n} \quad \left[\because \frac{\alpha}{n} \text{ is very small}\right]$

$$= \frac{na\sin\alpha}{\alpha}$$

If $n \to \infty, a \to 0$, but the product na remains finite. Let $na = A_0$, then

$$A = \frac{A_0\sin\alpha}{\alpha}$$

So, the resultant intensity at P will be,

$$I \propto A^2 \text{ or } I = A_0^2\left(\frac{\sin\alpha}{\alpha}\right)^2 \quad \text{(where proportionality factor is 1)} \quad ...(1)$$

*** The resultant amplitude due to a slit divided into n equal parts, at infinite distance (point P) from the slit is given by,**

$$A = \frac{a\sin(nd/2)}{\sin(d/2)}$$

where, a = amplitude of each slit, d = The phase difference between two successive slit

---

Since, the magnitude of intensity at any point in the focal plane of the lens is a function of α and hence of θ. So, we obtain a series of maxima and minima.

Condition for minima: From equation (1) it is clear that the intensity is zero, when

$$\sin\alpha = 0 \qquad [\text{but } \alpha \neq 0 \quad \because \frac{\sin\alpha}{\alpha} = 1, \text{when } \alpha = 0]$$

or $$\alpha = \pm n\pi$$

where n is integer except zero.

But, we know that, $$\alpha = \frac{\pi e \sin\theta}{\lambda}$$

So, the positions of **minima** are, given by,

$$\frac{\pi e \sin\theta}{\lambda} = \pm n\pi$$

or $e\sin\theta = \pm n\lambda = \pm\lambda, \pm 2\lambda, \pm 3\lambda$ and so on.

This equation gives the directions of first, second, third, ... and so on minima.

Condition for maxima: To locate the positions of maxima of intensity in the diffracting pattern, let us differentiate I with respect to α and equate to zero *i.e.*,

$$\frac{dI}{d\alpha} = 0$$

$$\frac{d}{d\alpha}\left[A_0^2\left(\frac{\sin\alpha}{\alpha}\right)^2\right] = 0$$

$$A_0^2\left[\frac{\alpha^2\, 2\sin\alpha\cos\alpha - \sin^2\alpha.2\alpha}{\alpha^4}\right] = 0$$

or $$\alpha^2 \sin\alpha\cos\alpha - \alpha\sin^2\alpha = 0$$

or $$\alpha\sin\alpha[\alpha\cos\alpha - \sin\alpha] = 0$$

then, either $$\alpha\sin\alpha = 0$$

or $$\alpha\cos\alpha = \sin\alpha$$

or $$\alpha = \tan\alpha$$

This equation is solved graphically by plotting the curves

$$Y = \alpha \qquad ...(2)$$

and $$Y = \tan\alpha \qquad ...(3)$$

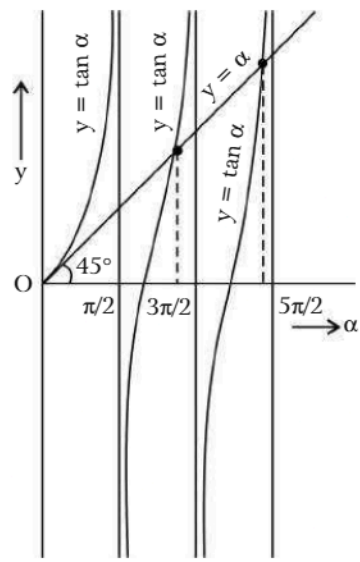


Fig. 2.11

The points of intersection of these two curves gives the roots of equation $\alpha = \tan\alpha$. Equation (2) and (3) are shown in Fig. 2.11. The first equation is the straight line

equation making an angle 45° with the axis, while the second is a discontinuous curve of infinite number of branches with asymptotes at intervals $\alpha = \pi$. The intersection points of both curves are at

$$\alpha = 0, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$$

or in general, $\alpha = (2n+1)(\pi/2)$

Substituting these values of α in the equation (1), we get

$$\underset{\alpha \to 0}{I_0} = A_0^2 \left(\frac{\sin 0}{0}\right)^2 = A_0^2$$

$$\underset{\alpha \to 3\pi/2}{I_1} = A_0^2 \left[\frac{\sin 3\pi/2}{3\pi/2}\right]^2 = \frac{4}{9\pi^2} A_0^2 = \frac{A_0^2}{22}$$

$$\underset{\alpha \to 5\pi/2}{I_2} = A_0^2 \left[\frac{\sin 5\pi/2}{5\pi/2}\right]^2 = \frac{4}{25\pi^2} A_0^2 = \frac{A_0^2}{61}$$

and so on.

Thus, the intensities of the successive maxima are in the ratio,

$$1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2} : \frac{4}{49\pi^2} : \ldots\ldots$$

or $$1 : \frac{1}{22} : \frac{1}{61} : \frac{1}{121} : \ldots\ldots$$

Clearly, most of the incident light is concentrated in the principal maxima which occurs in the direction given by $\alpha = 0$

or $$\frac{\pi e \sin\alpha}{\lambda} = 0$$

$\Rightarrow$ $$\theta = 0$$

i.e., in the direction of the incident light.

Thus, the diffraction pattern consists of a bright principal maxima in the direction of the incident light, having alternately minima and other weak maxima on either side of it, as shown in Fig. 2.12. The minima lie at $\alpha = \pm\pi$, $\pm 2\pi, \pm 3\pi \ldots\ldots$. The weak maxima do not fall exactly mid way between two minima, but are displaced towards the centre of the pattern by a certain amount which decreases with increasing order.

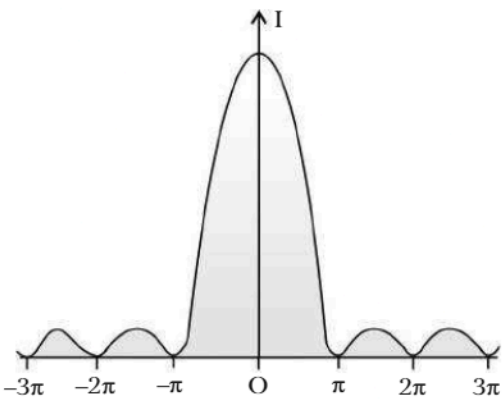


Fig. 2.12

Example 6: In Fraunhofer diffraction due to a narrow slit, a screen is placed 2 m away from the lens to obtain the pattern in its focal plane. Find the slit width if the first minima lies 5 mm on either side of the central maxima when plane waves of wavelength 5×10^{-5} cm are incident on the slit.

Solution: For Fraunhofer diffraction at a narrow slit, the phase difference for the minima is given by,

$$e\sin\theta = n\lambda \text{ where } e \text{ is the width of the slit.}$$

Here given that, $n=1, \lambda = 5\times10^{-5}$ cm $f = 200$ cm.

and linear separation between the central maxima and the first minima $= 0.5$ cm

Also, we have that,

$$\sin\theta = \frac{\text{linear separation between the central maxima and the first minima}}{\text{distance of screen from the lense}}$$

$$\therefore \quad \sin\theta = \frac{0.5}{200}$$

Hence, from the above equation, $e = \dfrac{n\lambda}{\sin\theta} = \dfrac{1\times5\times10^{-5}\times200}{0.5}$

$$= 2\times10^{-2}\text{ cm} = 0.02\text{ cm} = \mathbf{0.2\ mm}$$

Example 7: In Fraunhofer diffraction due to a narrow slit, a screen is placed 2 m away from lens to obtain the pattern. If the slit width is 0.2 mm and the first minima lie 5 mm on either side of the central maxima, find the wave length of light used.

Solution: In the Fraunhofer diffraction at a narrow single slit, the minima is given by,

$$e\sin\theta = n\lambda$$

where e is width of the slit and n is any positive or negative integer.

For first minima, $n=1, e = 0.2$ mm $= 0.02$ cm

So, $$e\sin\theta = \lambda$$

Also, we have that,

$$\sin\theta = \frac{\text{linear separation between the central maxima and the first minima}}{\text{distance of screen from the lense}}$$

$$\therefore \quad \sin\theta = \frac{0.5}{200}$$

So, $$\lambda = \frac{0.02\times0.5}{200} = 5\times10^{-5}\text{ cm} = \mathbf{5000\ \text{Å}}$$

Example 8: A screen is placed 2 m away from a narrow slit which is illuminated with light of wavelength 6000 Å. If the first minimum lies 5 mm on either side of the central maxima, calculate the slit width.

Solution: In Fraunhofer diffraction we have,

$$e\sin\theta = n\lambda$$

Here given that, $n=1, \lambda=6000\ \text{Å}$ and $\sin\theta=\dfrac{0.5}{200}$

So, $$e=\frac{n\lambda}{\sin\theta}=\frac{1\times6000\times10^{-8}\times200}{0.5}$$

or, $$e=2.4\times10^{-2}\ \text{cm}=\mathbf{0.24\ mm}$$

2.7 The Phasor Diagram Method for Fraunhofer Diffraction at a Slit

Consider the case of the addition of two vibrations

$$y_1=a_1\cos\omega t \quad \text{and} \quad y_2=a_2\cos(\omega t+\delta)$$

For that, draw a line AB proportional to a_1 and draw another line BC starting form B proportional to a_2 in a direction making anti clock wise angle δ (phase difference) with AB. Join AC Fig. 2.13. The length AC represents the amplitude of the resultant vibration and the angle $CAB=\phi$ represents its phase relative to vibration y_1.

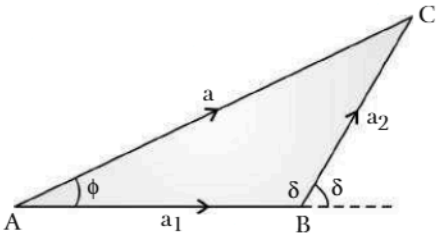


Fig. 2.13

Fig. 2.13 is called *Phasor diagram*, because the angles of BC and AC (*w.r* to AB) represent phase differences, not in the directions of vibration.

Fig.2.14 shows the addition of several vibration of same frequency in phasor diagram method.

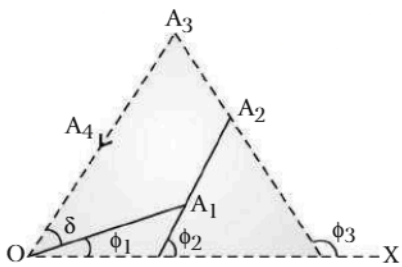


Fig. 2.14

The addition of the secondary wavelets from different parts of slit AB [Fig 2.15] as they reach P can be obtained by the phasor diagram method.

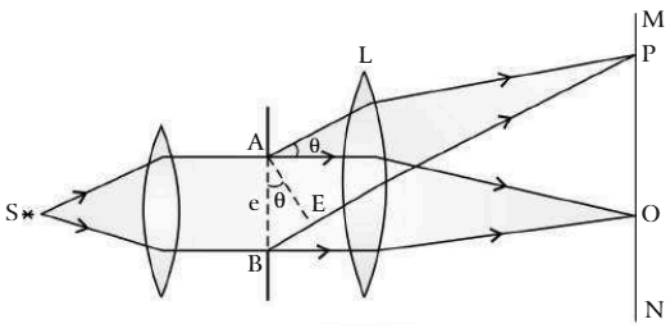
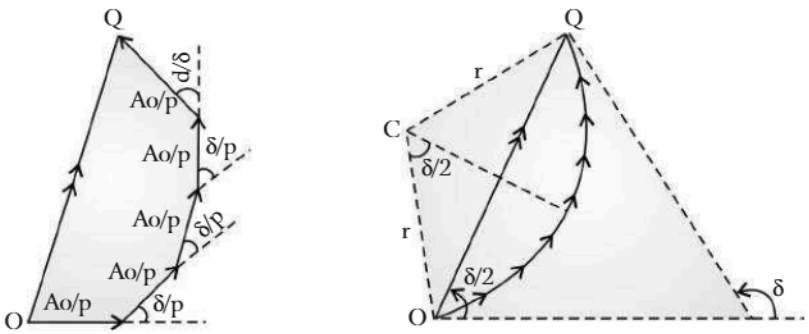


Fig. 2.15

Let us divide slit AB into p equal parts. Then the amplitude contributed by each element is A_0/p, if A_0 is that if the entire slit sends waves in the same phase (as at O). Also , the phase difference between waves for successive elements is δ/p, where

$$\delta = \frac{2\pi}{\lambda}(a\sin\theta) \qquad ...(1)$$

Fig. 2.16(a) shows the phasor diagram to add p disturbances of amplitude A_0/p each and successive phase disturbances δ/p. Fig. 2.16(b) is the result as p tends to infinity; the phasor diagram becomes a smooth circular *arc OQ* with δ as shown.



(a) Stepped addition of wavelets.

(b) Continuous addition of wavelets.

Fig. 2.16

The resultant amplitude A_0 at the observation point P (Fig. 2.15) given by the *chord OQ* in Fig 2.16(b). We thus have

$$\frac{A_\theta}{A_o} = \frac{Chord\ OQ}{Arc\ OQ} = \frac{2r\sin(\delta/2)}{(2r.\delta/2)} = \frac{\sin(\delta/2)}{(\delta/2)} \qquad ...(2)$$

For intensities, we then get

$$\frac{I_\theta}{I_o} = \left[\frac{\sin(\delta/2)}{(\delta/2)}\right]^2 \quad \text{where} \quad \delta = \frac{2\pi\, a\sin\theta}{\lambda} \qquad ...(3)$$

$$= \left(\frac{\sin p}{p}\right)^2 \quad \text{where } p = \frac{\pi\, a\sin\theta}{\lambda} \qquad ...(4)$$

How the phasor diagram changes with increasing θ (hence increasing δ) is shown in Fig.2.17. For $\delta = 0$ (observation point O in [Fig. 2.15] the phasor curve is a straight line of length A_0. For $\delta = \pi$ it becomes a semicircle, for $\delta = 2\pi$ a circle, for $\delta = 3\pi$ it goes round one circle and a half, for $\delta = 4\pi$ two full circles. Thus $\delta = 2\pi, 4\pi, 6\pi$... correspond to chord $OQ = 0$, which means zero amplitude (hence zero intensity). Also, $\delta = 3\pi, 5\pi$ ... correspond to (nearly) secondary maxima, the main maximum being at $\delta = 0$.

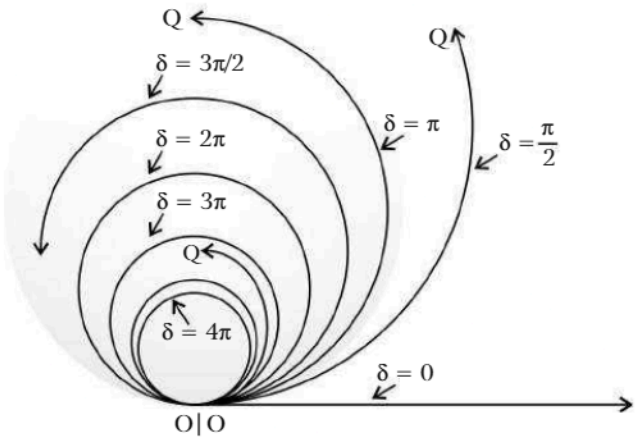


Fig. 2.17

2.8 Integral Calculus Method for Fraunhofer Diffraction at a Slit

The addition of the secondary wavelets reaching P from different parts of slit AB [Fig. 2.15] can also be done by integration. Fig. 2.18 may be used for this method.

Let the disturbance caused at P by the wavelet *from unit width* of the slit of O be

$$y_o = A\cos\omega t$$

Then the wavelet from width dx at C when it reaches P has amplitudes Adx and phase $\omega t + (2\pi/\lambda)\, x \sin\theta$. Calling the disturbance dy we have,

$$dy = Adx\cos\left\{\omega t + \frac{2\pi\, x\sin\theta}{\lambda}\right\} \qquad ...(1)$$

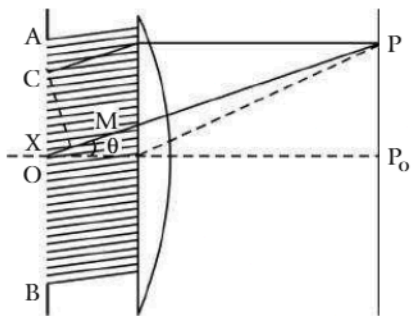


Fig. 2.18

For the total disturbance at the point of observation at angle θ, we get

$$y = \int_{-a/2}^{+a/2} A\cos\left\{\omega t + \frac{2\pi\, x\sin\theta}{\lambda}\right\}.dx$$

$$= \left[A.\frac{\sin(\pi a\sin\theta/\lambda)}{\pi(\sin\theta/\lambda)}\right]\cos\omega t \qquad ...(2)$$

Note: $$y = A\cos\omega t\int_{-a/2}^{+a/2}\cos\frac{2\pi x\sin\theta}{\lambda}\,dx - A\sin\omega t\int_{-a/2}^{+a/2}\sin\frac{2\pi x\sin\theta}{\lambda}\,d\theta$$

The second integral is zero.

The quantity in brackets in Eq.(2) gives the amplitude A_θ of the resultant disturbance. For $\theta = 0$ it becomes Aa, which we call A_o. Thus the result is

$$\frac{A_\theta}{A_0} = \frac{\sin[\pi a(\sin\theta/\lambda)]}{[\pi a(\sin\theta/\lambda)]} = \frac{\sin[\delta/2]}{[\delta/2]}$$

with $\delta = 2\pi a(\sin\theta)/\lambda$. This is the same as in phasor diagram method.

For intensities, we get $$\frac{I_\theta}{I_0} = \left(\frac{\sin p}{p}\right)^2 \quad ...(3)$$

where $p = \dfrac{\pi a \sin\theta}{\lambda}$

The Minima and Maxima: The angular positions of the minima and maxima can be deduced and the same results may be found as we get from the phasor diagram. One may also use Eq. (3) with $(dI_\theta/dp) = 0$ to get their positions. The results are

$$\left.\begin{aligned} \sin p &= 0 \quad (\text{minima, excluding } p = 0) \\ \sin p &= 0 \quad (\text{maxima, including } p = 0) \end{aligned}\right\} \quad ...(4)$$

The first of these eq. gives

$$\text{minima at} \quad a\sin\theta = n\lambda,\ n = \pm 1, \pm 2 \ldots \quad ...(5)$$

Thus, the maximum fall at $p = 0$, and *close to* $p = \pm(n + 1/2)\pi$. The intensities at the maxima, deduced from Eq. (3) are as follows:

For $a\sin\theta = 0$. $$\frac{I_\theta}{I_o} = \left(\frac{\sin p}{p}\right)^2 = 1$$

For $a\sin\theta \sim (n + 1/2)\,\lambda$. $$\frac{I_\theta}{I_o} = \left(\frac{\sin p}{p}\right)^2 \sim \frac{1}{[\pi(n + 1/2)]^2}$$

The relative intensities of successive maxima are, therefore, nearly

$$1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2} : \frac{4}{49\pi^2} \quad ...(6)$$

This means that the first secondary maxima on either side of the central maximum have only about 4% the height of the central maximum. We also recall that the central maximum has double the width of the secondary maxima. Thus, by far a very large amount of light falls in the central diffraction maximum. Hence for most purposes we may just discuss the *spread* of the central maximum and neglect the secondary maxima.

2.9 Fraunhofer Diffraction at a Circular Aperture

Consider a circular aperture *AB* of diameter *d* perpendicular to the plane of paper illuminated by a parallel beam of monochromatic light of wavelength λ. The light, in passing through the aperture, suffers diffraction. If a lens *L* is placed in the path of the diffracted beam very close to aperture a real image of the diffracted pattern is obtained on the screen *RT*, placed perpendicular to the plane of the paper, in the focal plane of the lens and *P* is a point on the screen. [Fig. 2.19] *CP* is perpendicular to the screen.

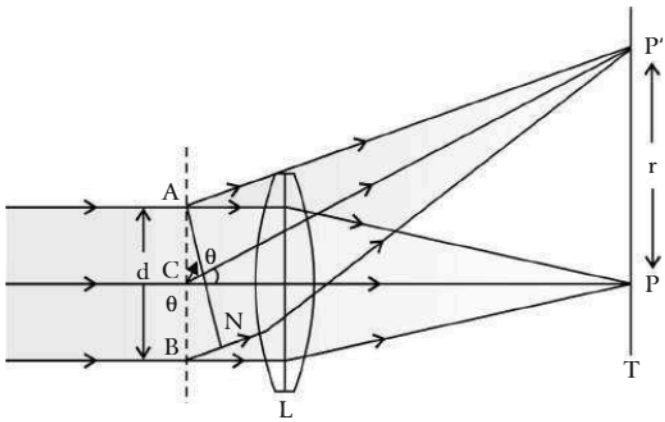


Fig. 2.19: A plane wavefront is incident on the circular aperture.

The secondary waves travelling in the direction *CP* come to focus at *P* and a bright image is formed as the secondary waves, equidistant from *C* in the upper and lower half travel the same distance before reaching *P* and reinforce each other.

Now consider the secondary waves travelling in a direction inclined at an angle θ with the direction *CP*. All these secondary waves meet at *P′* on the screen. Let the distance *PP′* be *x*. The path difference between the secondary waves emanating from the points *B* and *A* (extremities of a diameter) is *BN*.

From the Δ*ABN*,

$$BN = d\sin\theta$$

The point P_1 will be of minimum intensity if this path difference is equal to integral multiples of λ *i.e.*,

$$d\sin\theta_n = n\lambda \qquad ...(1)$$

The point P_1 will be of maximum intensity if the path difference is equal to odd multiples of $\frac{\lambda}{2}$ *i.e.*,

$$d\sin\theta_n = \frac{(2n+1)\lambda}{2} \qquad ...(2)$$

where $n = 1,2,3\ldots\ldots$

As the circular aperture is symmetrical about the axis, the resultant pattern may be obtained by rotating Fig about the same axis. The point *P* thus will trace out a circular ring of uniform intensity, minimum or maximum depending upon the conditions given above. Thus the diffraction pattern due to circular aperture consists of a central bright disc called the **Airy's disc** surrounded by alternate dark and bright concentric rings called Airy's ring. The intensity of dark rings is zero and that of bright rings decreases gradually outwards from *P*.

Further, if the collecting lens is very near the aperture and the screen is at a very large distance from the lens, then

$$\sin\theta = \theta = \frac{r}{f} \qquad ...(3)$$

Also for the first secondary minimum

$$d\sin\theta = \lambda$$

or
$$\sin\theta = \theta = \frac{\lambda}{d} \qquad ...(4)$$

Hence,
$$\frac{r}{f} = \frac{\lambda}{d}$$

or
$$R = \frac{f\lambda}{d} \qquad ...(5)$$

where *r* is the radius of Airy's disc

But actually the radius of the first dark ring is slightly more than that the given by eq.(5). According to Airy, it is given by,

$$r = \frac{1.22}{d} f\lambda \qquad ...(6)$$

The intensity distribution of the bright and dark rings is similar to the one given for a rectangular slit. It is clear from eq.(5) that the radius of the central disc decreases with increase of diameter of the aperture.

The diffraction of a plane wave from a narrow circular aperture is of great importance, on account of its application to resolving power of telescope and microscope.

Example 9: In Fraunhofer diffraction pattern due to a single slit, the screen is at a distance of 100 cm from the slit and the slit is illuminated by monochromatic light of wavelength 5893 Å. The width of the slit is 0.1 mm. Calculate the separation between the central maximum and the first secondary minimum.

Solution: For a rectangular slit,

$$x = \frac{f\lambda}{d}$$

Here given that, $f = 100$ cm, $\lambda = 5893$ Å

$$= 5893 \times 10^{-9} \text{ cm},$$

$$d = 0.1 \text{ mm} = 0.01 \text{ cm}, x = ?$$

$$\therefore \quad x = \frac{100 \times 5893 \times 10^{-8}}{0.01} = \mathbf{0.5893 \text{ cm}}$$

2.10 Resolving Power of an Optical Instrument or Resolution of Images

When two objects are very near together they may appear as one and it may be difficult for the eye to see them separate if the angle subtended by them at the eye is less than one minute, which is the resolving limit of the normal eye. The separation of such close objects is termed as resolution. So, the *ability of an optical instrument* (such as telescope, microscope etc.) *to resolve the images of two close objects is known as resolving power of that instrument*. In the case of a prism or a grating, the term resolving power is referred to the ability of resolving two close spectral lines so that the two lines can be viewed as separate lines. The resolving power of an instrument is given by $\lambda/d\lambda$ where λ is wavelength of any line and $d\lambda$ is separation between their wavelengths.

2.11 Rayleigh's Criterion of Resolution

To express the resolving power of an optical instrument as a numerical value, Lord Rayleigh proposed an arbitrary criterion. According to it, *two spectral lines of equal intensities are said to be resolved by an optical instrument when the central maximum of one coincide with the first minimum of other and vice-versa.*

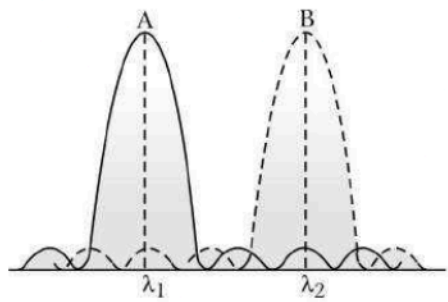


Fig. 2.20 (a)

In Fig. 2.20(a). A and B are the central maxima of the diffraction patterns of two spectral line of wavelength λ_1 and λ_2. The difference in the angle of diffraction is large. Due to the large separation between their central maxima two images can be seen as separate ones. So, the two spectral lines are distinctly separate.

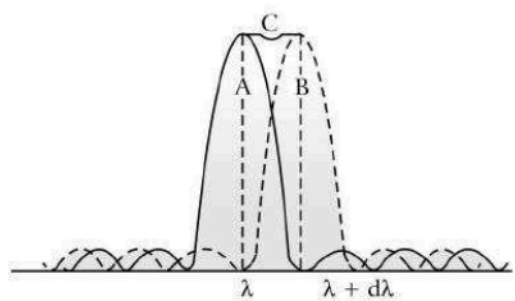


Fig. 2.20 (b)

If however, difference in wave lengths is smaller, the central maxima of the two spectral lines are closer. In fig. 2.20 (b) the difference in wavelengths of the spectral lines is such that the central maxima of one falls exactly over the first minimum of the other and the resultant intensity curve shows a double humped curve with a clear *dip* in the centre of the two central maxima. According to Rayleigh these two spectral lines are said to be just resolved.

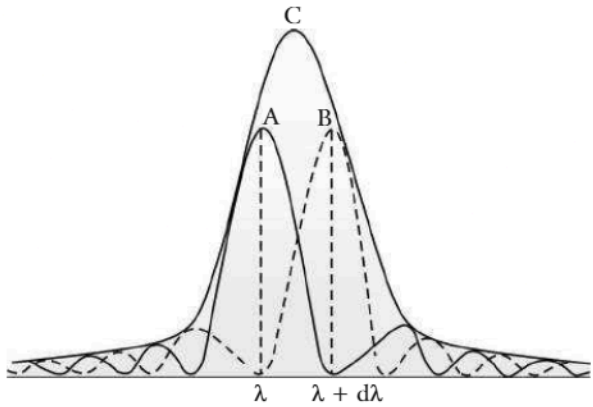


Fig. 2.20 (c)

If however, the difference in the wavelengths is so small that the central maxima of the spectral lines come closer and there is sufficient overlapping of the intensity curve as shown in fig. 2.20 (c). The resultant intensity curve shows only one maximum in the centre without any dip. Under these conditions the two spectral lines are not resolved.

2.12 Resolving Power of a Telescope

The main function of any telescope is not only to magnify an object but also to enable to observe finer details in (not observed with the naked eye). The resolving power of a telescope is defined as *the angle subtended at the objective by two distant point objects* (*stars*) *which are just resolved when seen through the telescope.*

Consider the incident ray of light coming from two neighbouring point of a distance object. The image of each point object is a Fraunhofer diffraction pattern and lies in the focal plane of the telescope objective *AB* as shown in fig.2.21. Let P_1 and P_2 be the positions of the central maxima of the two images. According to Rayleigh's criterion, these two images are said to be resolved if the central maxima of one coincides with the first minimum of the other and vice-versa.

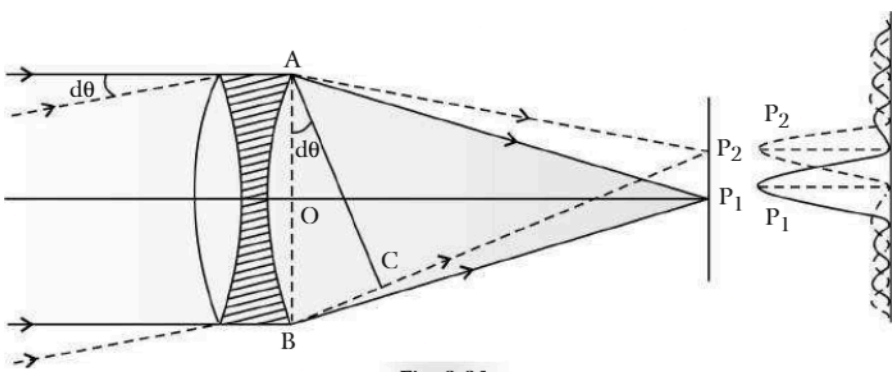


Fig. 2.21

The path difference between the secondary waves travelling in the direction AP_1 and BP_1 is zero and hence, they reinforce one another at P_1. Thus, P_1 will be the position of the central maximum of the first image. The secondary waves travelling in the direction AP_2 and BP_2 will meet at P_2 on the screen. The path difference will be equal to,

$$BP_2 - AP_2 = \lambda \qquad (\lambda \text{ is the wavelength of light})$$

But from the Fig. 2.21,

$$BP_2 - AP_2 = BC = AB d\theta \qquad [\because \text{The angel is small } \sin\theta = d\theta]$$

$$= Dd\theta \quad [AB = D = \text{Diameter of objective lens of telescope}]$$

$$\therefore \qquad \lambda = Dd\theta$$

or
$$d\theta = \lambda / D \qquad \ldots(1)$$

This equation is correct for rectangular apertures. For circular apertures, Airy showed that 1.22λ may be used instead of λ.

So,
$$d\theta = \frac{1.22\,\lambda}{D} \text{ radian}$$

This expression gives the minimum angle between the two point sources (stars) resoluble by a telescope and is known as *the limit of resolution of a telescope*. The reciprocal of $d\theta$ measures the resolving power. If the value of angle is smaller, the resolving power will be higher.

Thus, it is clear that resolving power of a telescope depends upon the diameter of the objective and the wavelength of light used. Telescope with large diameter of the objective has high resolving power.

Example 10: Calculate the aperture of the objective of a telescope which may be used to resolve stars separated by 4.88×10^{-6} ***radian for light of wavelength*** **6000 Å.**

Solution: The limit of resolution of a telescope is given by,

$$d\theta = \frac{1.22\,\lambda}{D}$$ where all symbols have their usual meanings.

Here given that, $d\theta = 4.88\times10^{-6}$ rad and $\lambda = 6000\times10^{-8}$ cm

$$\therefore \qquad D = \frac{1.22\times6000\times10^{-8}}{4.88\times10^{-6}} = \mathbf{15\ cm}$$

Example 11: Find the separation of two points on the moon that can be resolved by a telescope of diameter 500 cm. The distance of the moon is 3.8×10^{8} ***m. The eye is most sensitive to light of wavelength*** **5500 Å**

Solution: The limit of resolution of a telescope is given by,

$$d\theta = \frac{1.22\,\lambda}{D}$$

Here given that, $\lambda = 5500 \times 10^{-8}$ cm and $D = 500$ cm

$$\therefore \quad d\theta = \frac{1.22 \times 5500 \times 10^{-8}}{500} = 1.342 \times 10^{-7} \text{ rad.}$$

Let the separation between the two points be x then,

$$d\theta = x/R$$

where R is distance of the moon $= 3.8 \times 10^{10}$ cm

$$\therefore \quad x = Rd\theta$$
$$= 3.8 \times 10^{-10} \times 1.342 \times 10^{-7} = 5.0996 \times 10^{3} \text{ cm}$$
$$= \mathbf{50.996\ m}$$

2.13 Resolving Power of a Microscope

In the case of a telescope, the smallest permissible angular separation between two distant objects at an unknown distance, determines the limit of resolution when the images appear just resolved. But, in the case of a microscope, the object is very near the objective of the microscope (just beyond the focus of the objective) and the objects subtend a large angle at the objective. The limit of resolution of a microscope is determined by *the least permissible linear distance between the two objects so that the two images appear just resolved.*

In Fig. 2.22, MN is the aperture of the objective of a microscope and A and B are two object points at a distance d apart. A' and B' are the corresponding Fraunhofer diffraction pattern of the two images. A' is the position of the central maximum of A and B' is the position of the central maximum of B. A' and B' are surrounded by alternate dark and bright diffraction rings. The two images are said to be just resolved if the position of the central maximum B' also corresponds to the first minimum of the image of A'.

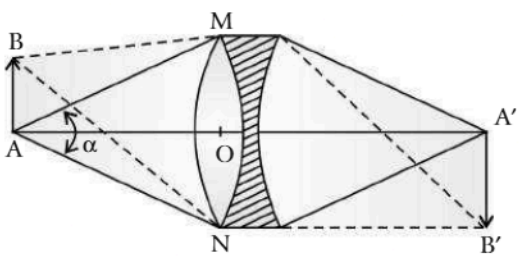


Fig. 2.22

The path difference between the extreme rays from the point B and reaching A' is given by

$$(BN + NA') - (BM + MA')$$

But

$$NA' = MA'$$

$\therefore$ Path difference $= BN - BM$

In Fig. 2.23, AD is perpendicular to DM and AC is perpendicular to BN.

$\therefore$ $BN - BM = (BC + CN) - (DM - DB)$

But $CN = AN = AM = DM$

$\therefore$ Path difference $= BC + DB$

From the ΔACB and ΔADB

$$BC = AB \sin\alpha = d \sin\alpha$$

and $$DB = AB \sin\alpha = d \sin\alpha$$

Path difference $= 2d \sin\alpha$

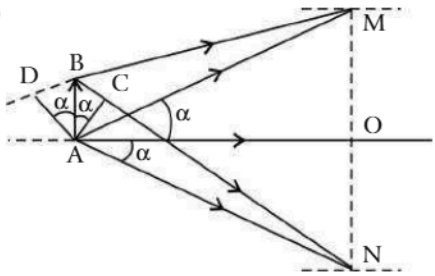


Fig. 2.23

If this path difference $2d \sin\alpha = 1.22\lambda$, then, A' corresponds to the first minimum of the image B' and the two images appear just resolved.

$\therefore$ $$2d \sin\alpha = 1.22\,\lambda$$

or $$d = \frac{1.22\,\lambda}{2 \sin\alpha} \quad \ldots(1)$$

Equation (1) derived above is based on the assumption that the object points A and B are self-luminous. But actually, the objects viewed with a microscope are not self-luminous but are illuminated with light from a condenser. It is found that the resolving power depends on the mode of illumination. According to Abbe, the least distance between two just resolvable object points is given by

$$d = \frac{\lambda_0}{2\mu \sin\alpha} \quad \ldots(2)$$

where λ_0 is the wavelength of light through vacuum and μ is the refractive index of the medium between the object and the objective. The space between the object and the objective is filled with oil (cedar wood oil) in microscopes of high resolving power. This has two advantages. Firstly the loss of light by reflection at the first lens surface is decreased and secondly the resolving power of the microscope is increased. The expression $\mu \sin\alpha$ in equation (2) is called the **numerical aperture** of the objective of the microscope and is a characteristic of the particular objective used. The highest value of numerical aperture obtained in practice is about 1.6. Taking the effective wavelength of white light as 5500 Å and $\mu \sin\alpha = 1.6$,

$$d = \frac{5500 \times 10^{-8}}{2 \times 1.6} = 1.72 \times 10^{-5} \text{ cm}$$

where d is the linear distance between two just resolvable object points. It is clear from equation (2), that decrease of λ_0 and increase of numerical aperture of the objective decreases the value of d and hence the resolving power of the microscope is increased.

An oil immersion objective has higher numerical aperture than an ordinary objective. The resolving power of a microscope can be considerably increased by decreasing the value of λ_0. Thus, by using ultraviolet light and quartz lenses, the resolving power of the microscope can be increased further. In this case the image is photographed. Such a microscope is called an **ultra microscope.**

Example 12: Sodium light of wavelength 5890 Å is used to view an object under a microscope. The aperture of the objective has a diameter of 0.9 cm

(i) Calculate the limiting angle of resolution

(ii) Using visible light, what is the maximum limit of resolution for this microscope.

Solution: (i) Limiting angle of resolution of a microscope is given by,

$$\theta_m = 1.22\left(\frac{\lambda}{d}\right)$$

Here given that, $\lambda = 5890\ \text{Å} = 5.89\times10^{-7}\text{m}$

$$d = 0.9\ \text{cm} = 9\times10^{-3}\text{m}$$

$\therefore$ $$\theta_m = 1.22\left[\frac{5.89\times10^{-7}}{9\times10^{-3}}\right]$$

$$= \mathbf{7.98\times 10^{-5}\ radian}$$

It means that any two points on the object subtending an angle less than 7.98×10^{-5} radian at the objective cannot be distinguished in the image.

(ii) The wavelength of violet light in the visible spectrum is 4000 Å. The maximum limit of resolution corresponds to the smallest angle.

Here $\lambda = 4000\ \text{Å} = 4\times10^{-7}\ \text{m}$

$$\theta_m = 1.22\left(\frac{\lambda}{d}\right)$$

$$\theta_m = 1.22\left[\frac{4\times10^{-7}}{9\times10^{-3}}\right]$$

$$\theta_m = \mathbf{5.42\times 10^{-5}\ radian}$$

2.14 Phase Contrast Microscope

When a beam of light passes through a *transparent object*, phase difference is produced but no change in amplitude takes place. The eye can distinguish only changes in intensity but not changes in phase. In case one wishes to see a small transparent object, *say unstained bacteria*. it is necessary to magnify it and also to convert differences in phase into differences in intensity. *Zernike* in 1935 developed *a microscope in which phase variation in passing through different parts of the specimen are converted into intensity variations in the image. This is called the phase contrast microscope*.

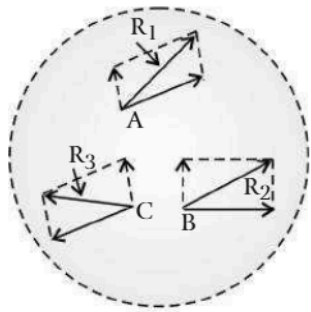


Fig. 2.24

Consider a beam of light passing through a transparent plate of varying thickness. The amplitude vector at the points A,B,C has the same magnitude but is in different directions. The intensity is the same at all points but there is difference in phase between the vectors [Fig.2.24]. This difference in phase cannot be seen by the eye. However, if a constant phase difference is introduced with the help of a phase (represented by the dotted vector), the resultant amplitudes at the points A,B, and C are R_1,R_2 and R_3 respectively. Their magnitudes are different and hence the intensities are different and can be seen by the eye. Thus, by introducing the phase plate, the object slide can be seen by the phase contrast with the help of the phase plate. Phase changes due to varying thickness are converted to amplitudes of varying magnitude at different points.

A phase contrast microscope consists of a transparent object slide placed in between the condensing lens and the objective. An annular aperture is placed at the first focal plane of the condensing lens and the phase plate is kept at the second focal plane of the objective. The phase plate has an annular depression exactly conjugate to the annular aperture.

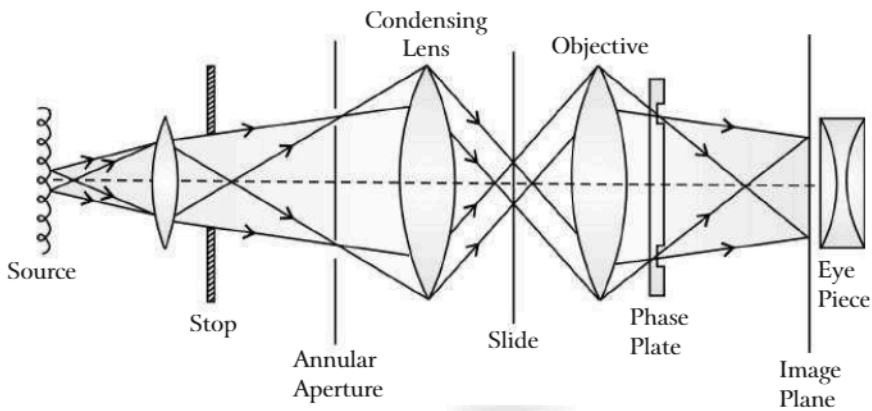


Fig. 2.25

In the Fig. 2.25, the path of zero order light is shown. It is clear that the light passes through the thinner part of the phase plate and consequently its phase is advanced in comparison to the light diffracted through the thicker part of the plate. If t is the thickness of the depression and μ is the refractive index of the phase plate, the advance in phase of the zero order light is $\frac{2\pi}{\lambda}(\mu-1)t$. For maximum efficiency, the image of the annular aperture should coincide exactly with the depression in the phase plate. The final image is formed at the first focal plane of the eyepiece.

2.15 N-Parallel Slits or Diffraction Grating

A diffraction grating is an extremely useful device. A diffraction grating is an arrangement consisting of a large number of N-parallel slits of equal width. The slits are equidistant separated by opaque space. It is made by ruling a large number of fine, equidistant and parallel lines on an optically plane glass plate with a diamond point. The rulings scatter the light and are effectively opaque while the unruled parts transmit light and act as slits. Let AB be the section of a plane transmission grating placed perpendicular to the plane of the paper [Fig 2.26]. Let e be the width of each transparent space and d the width of each opaque portion. Then $(e + d)$ is the grating element. Two points separated by the distance $(e + d)$ in two consecutive slits are known as corresponding points. If the spacing between the lines is of the order of the wavelength of light, then an appreciable diffraction of the light is produced.

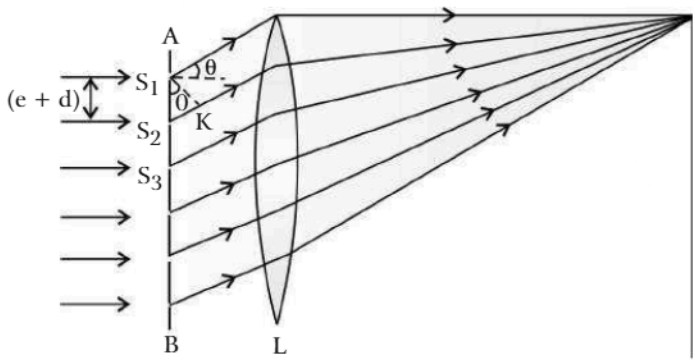


Fig. 2.26

2.16 Diffraction at N-Parallel Slits and Intensity Distribution

Let a parallel beam of monochromatic light of wavelength λ be incident normally on the N slits. By Huygen's principle, all the points in each slit send out secondary wavelets in all directions. By the theory of Fraunhofer diffraction at a single slit, the wavelets from all points in a slit diffracted in direction θ are equivalent to a single wave of amplitude $\frac{A_0 \sin\alpha}{\alpha}$, starting from the middle point of the slit, where $\alpha = \frac{\pi}{\lambda} e \sin\theta$

Thus, if N be the total number of slits, the diffracted rays from all the slits are equivalent to N parallel rays, one from the middle points $S_1, S_2, S_3 \ldots$ of each slit. Let S_1K be perpendicular to S_2K. Then, the path difference between the rays from the slit S_1 and S_2 is

$$S_2K = S_1S_2 \sin\theta = (e + d)\sin\theta$$

and the corresponding phase difference $= \frac{2\pi}{\lambda}(e + d)\sin\theta$

Let this phase difference $(\pi/\lambda)(e+d)\sin\theta = 2\beta$ and the resultant amplitude of vibration from each slit is $A_0 \dfrac{\sin\alpha}{\alpha}$, where $\alpha = (\pi/\lambda)e\sin\theta$. Extending the application of these results to N-slits, the problem reduces to finding the resultant amplitude of N vibration in a direction θ, each of amplitude $A_0 \dfrac{\sin\alpha}{\alpha}$, but the phase increases in arithmetical progression, the common phase difference being $(2\pi/\lambda)(e+d)\sin\theta$. Therefore by the process of vector addition, the resultant amplitude due to all slits in the direction θ is,

$$A = \frac{A_0 \sin\alpha}{\alpha} \times \frac{\sin\left[N\dfrac{\pi}{\lambda}(e+d)\sin\theta\right]}{\sin\left[\dfrac{\pi}{\lambda}(e+d)\sin\theta\right]}$$

$$= \frac{A_0 \sin\alpha}{\alpha} \cdot \frac{\sin N\beta}{\sin\beta}$$

Hence, the intensity is given by, $$I = A^2 = \frac{A_0^2 \sin^2\alpha}{\alpha^2} \cdot \frac{\sin^2 N\beta}{\sin^2\beta} \quad \ldots(1)$$

The first factor of equation (1), $\dfrac{A_0 \sin^2\alpha}{\alpha^2}$ corresponds to the intensity of diffraction pattern due to a single slit, while the second factor $= \dfrac{\sin^2 N\beta}{\sin^2\beta}$ gives the distribution of intensity due to all the slits (interference pattern).

2.16.1 Position of Principal Maxima

Principal maxima are obtained for maximum value of second term $\dfrac{\sin^2 N\beta}{\sin^2\beta}$ For its maximum value

$$\sin\beta = 0,$$

$$\Rightarrow \qquad \beta = \pm n\pi \qquad \text{where } n = 1,2,3,\ldots$$

Then, $$\frac{\sin N\beta}{\sin\beta} = \frac{0}{0} \quad i.e., \text{indeterminate.}$$

To find its value, we use the usual method of differentiating the numerator and the denominator.

Thus, $$\lim_{\beta \to \pm n\pi} \frac{\sin N\beta}{\sin\beta} = \lim_{\beta \to \pm n\pi} \frac{N\cos N\beta}{\cos\beta} = \frac{N\cos N(\pm n\pi)}{\cos(\pm n\pi)} = \pm N$$

Then the intensity, $$I = \frac{A_0^2 \sin^2\alpha}{\alpha^2} N^2 \qquad \text{[Using equation (1)]}$$

So, the intensity is proportional to N^2 and it is maximum. The maxima are very intense and are called **principal maxima**. Their positions correspond to,

$$\beta = \pm n\pi$$

So, $$(\pi/\lambda)(e+d)\sin\theta = \pm n\pi$$

or $$(e+d)\sin\theta = \pm n\lambda \qquad ...(2)$$

where $n = 0, 1, 2, \ldots$

For $n = 0$, we get the zero order maximum.

For $n = \pm 1, \pm 2, \pm 3 \ldots$. we get the first, second and third order principal maxima respectively. The ± indicate that there are two principal maxima for each order lying on either side of the zero order maxima.

2.16.2 Position of Minima

A series of minima occur when $\sin N\beta = 0$, provided that $\sin\beta \neq 0$, then

$$\frac{\sin N\beta}{\sin\beta} = 0$$

From equation (1), we have $$I = \frac{A_0^2 \sin^2\alpha}{\alpha^2}.0 = 0$$

which is a minimum. The position of minima correspond to

$$\sin N\beta = 0 \Rightarrow \pm \sin m\pi$$

or $$N\beta = \pm m\pi$$

or $$\frac{N\pi}{\lambda}(e+d)\sin\theta = \pm m\pi$$

or $$N(e+d)\sin\theta = \pm m\lambda \qquad ...(3)$$

where m has all the integral values except $0, N, 2N, \ldots . nN$, because for these values of $\beta = 0$, which gives principal maxima. It is clear from above that, $m = 0$ gives a principal maximum, and $m = 1, 2, 3 \ldots . (N-1)$ gives minima and then $m = N$ gives again a principal maxima. Thus, *there are* $(N-1)$ *minima between two consecutive principal maxima*.

2.16.3 Position of Secondary Maxima

As there are $(N-1)$ minima between two consecutive principal maxima, there must be $(N-2)$ other maxima between two principal maxima. These are called *secondary maxima*. These maxima are less intense than the principal maxima. Their positions are obtained by differentiating equation (1) with respect to β and equating to zero. Thus,

$$\frac{dI}{d\beta} = \frac{A_0^2 \sin^2\alpha}{\alpha^2} 2\left[\frac{\sin N\beta}{\sin\beta}\right]\frac{N\cos N\beta\sin\beta - \sin N\beta\cos\beta}{\sin^2\beta} = 0$$

or $$N\cos N\beta\sin\beta - \sin N\beta\cos\beta = 0$$

or $$N \cos N\beta \sin\beta = \sin N\beta \cos\beta$$

or $$N \cot N\beta = \cot\beta$$

or $$\tan N\beta = N \tan\beta \quad \ldots(4)$$

Hence, to find their intensity, the fraction $\frac{\sin^2 N\beta}{\sin^2 \beta}$ has to be determined subjected to the condition

$$\tan N\beta = N \tan\beta$$

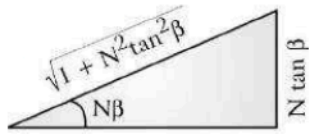


Fig. 2.27

For this we use the triangle shown in Fig. 2.27. From the figure,

$$\sin N\beta = \frac{N \tan\beta}{\sqrt{1+N^2 \tan^2 \beta}}$$

$\therefore$ $$\frac{\sin^2 N\beta}{\sin^2 \beta} = \frac{N^2 \tan^2 \beta}{(1+N^2 \tan^2 \beta)\sin^2 \beta} = \frac{N^2}{\cos^2 \beta + N^2 \sin^2 \beta}$$

$$= \frac{N^2}{1+(N^2-1)\sin^2 \beta}$$

It shows that the intensity of secondary maxima is proportional to $\frac{N^2}{1+(N^2-1)\sin^2 \beta}$, while that of principal maxima is proportional to N^2. Further the intensity ratio of the secondary maxima and the principal maxima *i.e.*,

$$\frac{\text{intensity of secondary maxima}}{\text{intensity of principal maxima}} = \frac{N^2}{[1+(N^2-1)\sin^2]\beta . N^2} = \frac{1}{1+(N^2-1)\sin^2 \beta}$$

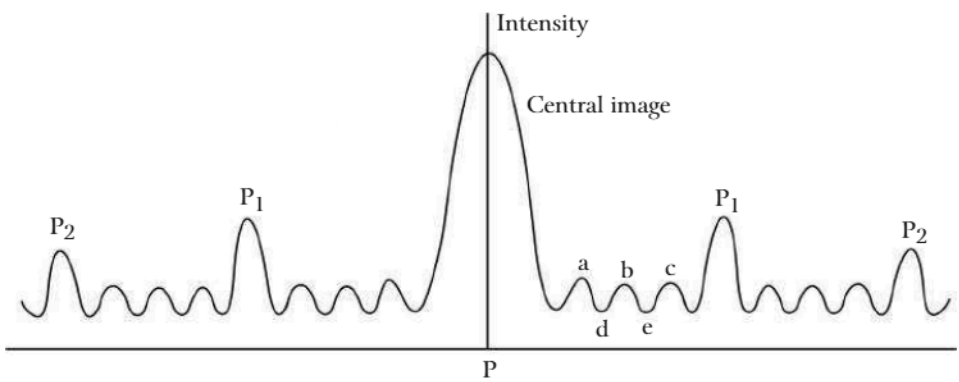


Fig. 2.28

So, for large value of N, these secondary maxima are weak and invisible. But when N is small they may be observed. Fig. 2.28 represents the intensity distribution P corresponds to the position of the central maxima and P_1, P_2 ...., on either side of P represents Ist, 2nd...., principal maxima a, $b,c,\ldots$, are secondary maxima and $d,e\ldots$, are the minima.

Example 13: A parallel beam of monochromatic light is allowed to be incident normally on a plane grating having 4250 lines per cm and a second order spectral lines is observed to be deviated through 30°. Calculate the wavelength of the spectral line.

Solution: The diffraction condition for a diffraction grating,

$$(e+d)\sin\theta = n\lambda$$

Here given that, $e+d = \dfrac{1}{4250}, \theta = 30°$ and $n=2$

Putting all the values, we get, $\dfrac{\sin 30°}{4250} = 2\lambda$

or $$\lambda = \frac{1}{2\times 2\times 4250} = 5.882\times 10^{-5}\ cm = 5882\ \text{Å}$$

Example 14: What is the highest order spectrum which may be seen with monochromatic light of wavelength 6000 Å by means of a diffracting grating with 5000 lines/cm.

Solution: The diffraction condition for diffraction grating,

$$(e+d)\sin\theta = n\lambda$$

The maximum possible value of $\sin\theta = 1$

So, $$(e+d) = n\lambda$$

or $$n = \frac{(e+d)}{\lambda}$$

Here given that, $e+d = \dfrac{1}{5000}$ lines/cm and $\lambda = 6000$ Å

Putting all the values, we get $$n = \frac{1}{5000\times 6000\times 10^{-8}} = \frac{10^8}{3\times 10^7} = \frac{10}{3} = 3.33$$

So the highest order of spectrum which can be seen is **3**.

2.17 Plane Reflection Grating

Consider an optically plane surface with its surface made reflecting by silvering or aluminizing. If a large number (N) of parallel straight lines are scratched on it, each of width b and leaving equal widths a clear between them, we get a plane *reflection* grating.

The reflection grating is used to obtain spectra just like the plane transmission grating, the theory and action being the same. Reflection grating has the particular advantage that it can be used also in the ultra-violet and infrared regions where most materials become opaque. Hence it is very widely used.

2.18 Blazed Grating

Higher order spectra are desirable for good dispersion while, the intensity is maximum in the zero order and falls as n increases Blazed grating is designed to shift the intensity maximum towards a desired high order spectrum. Grooves with inclined reflecting faces are made in the grating surface [Fig. 2.29]. The inclination γ is called the *blaze angle*.

If light is incident along normal AO to the grating surface BB' the *central diffraction* maximum now falls not along OA, but at an angle 2γ from OA. The spectrum seen at angle β from OA corresponds to

$$e \sin \beta = n\lambda$$

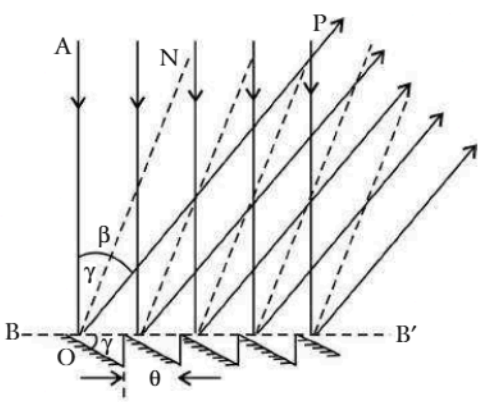


Fig. 2.29

The important thing is that If β is in the neighbourhood of 2γ, then the spectrum has large intensity, despite n being high. If one desires to choose a grating blazed for a given order n for a given region λ then one calculates β and sets $\gamma = 1/2\,\beta$ as the blaze angle.

2.19 Concave Grating

Use of a plane transmission grating requires two lenses, *viz.*, the collimator lens and the telescope objective. The collimating lens gives a parallel beam of light incident on the grating surface and the telescope objective focuses the diffracted beam. If both lenses are not perfectly achromatic, then the formed spectrum will be more complex due to chromatic aberration present in the lenses. Rowland developed the ***concave reflection grating***, which dispenses the use of both lenses. A concave grating consists of a polished spherical concave surface of a metal, like speculum (32% tin and 68% copper). Polished

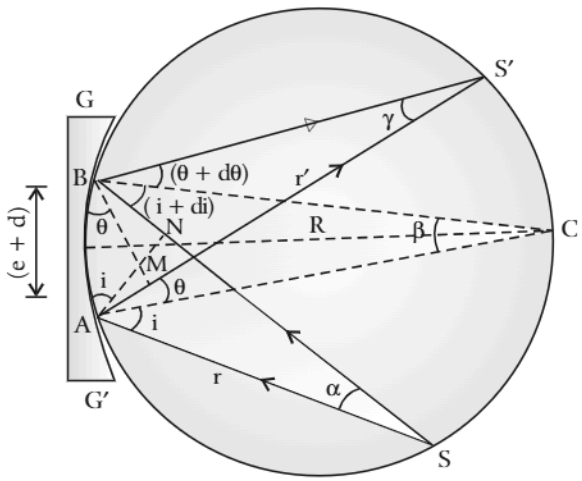


Fig. 2.30

concave surface is ruled with equally spaced fine parallel lines. When light falls on concave grating, then after diffraction, it is automatically focussed without the use of lenses. In such a way, the effect of chromatic aberration is completely eliminated.

2.19.1 Theory of Concave Grating

Let GG' be a concave grating having centre of curvature at C [Fig. 2.30]. Let A and B be two corresponding points of the grating so that $AB = (e + d)$ is the grating element. Let S be a narrow slit perpendicular to the plane of paper and illuminated by monochromatic light of wavelength λ. Let SA and SB be two rays incident on the grating at angles i and $i + di$ respectively. AS' and BS' are the corresponding diffracted rays with angles of diffraction θ and $\theta + d\theta$ respectively.

Let AN and BM be two arcs drawn with S and S' as centres, and SA and $S'B$ as radii respectively. These arcs will be almost straight lines. The path difference between the rays SBS' and SAS'

$$= (SB + BS') - (SA + AS')$$

$$= (SB - SA) - (AS' - BS')$$

$$= BN - AM$$

$$= AB \sin i - AB \sin \theta \qquad \text{[from the Fig. 2.30]}$$

$$= AB (\sin i - \sin \theta)$$

$$= (e + d)(\sin i - \sin \theta) \qquad [\because\ AB = (e + d)]$$

For nth order spectrum, a maximum at S' occurs only when

$$(e + d)(\sin i - \sin \theta) = n\lambda \qquad \ldots(1)$$

In order to form a diffracted image at S', the path difference between any two consecutive rays should be constant *i.e.*,

$$(e + d)(\sin i - \sin \theta) = \text{constant}$$

On differentiating equation (1), we get

$$\cos i\, di - \cos \theta\, d\theta = 0 \qquad \ldots(2)$$

But, from [Fig. 2.30] we have

$$\alpha + i = \beta + i + di \qquad \therefore \quad di = \alpha - \beta$$

$$\beta + \theta = \gamma + \theta + d\theta \qquad \therefore \quad d\theta = \beta - \gamma$$

Putting these values in equation (2), we get

$$\cos i(\alpha - \beta) - \cos \theta(\beta - \gamma) = 0 \qquad \ldots(3)$$

Let $SA = r, AS' = r'$ and radius of curvature of the grating be R, then

$$r.\alpha = AN = (e + d) \cos i$$

$$R\beta = AB = (e + d)$$

$$r'\gamma = BM = (e + d) \cos \theta$$

$$\therefore \qquad \alpha = \frac{(e + d) \cos i}{r}, \qquad \beta = \frac{e + d}{R} \quad \text{and} \quad \gamma = \frac{(e + d) \cos \theta}{r'}$$

Putting these values in equation (3), we get

$$\cos i \left[\frac{\cos i}{r} - \frac{1}{R}\right] - \cos \theta \left[\frac{1}{R} - \frac{\cos \theta}{r'}\right] = 0 \qquad \ldots(4)$$

It is the general equation for the position of S' which shows that,

if, $\quad r = R \cos i \quad$ then $\quad r' = R \cos \theta$

i.e., if S lies at the circumference of a circle of radius R, then S' also lies on the same circle. Thus, we can say that if the slit and the concave grating are placed at the circumference of a circle whose diameter is equal to the radius of curvature of the grating, then the spectra are focussed on the circumference of the same circle.

2.20 Different Methods of Mounting of Concave Grating

To use a concave grating, the slit, grating and eyepiece (photographic plate) can be placed in different positions known as mounting. For concave grating, there are mainly three type of mounting.

1. Rowland Mounting
2. Paschen-Runge Monting
3. Eagle Mounting.

2.20.1 Rowland Mounting

Rowland used a concave grating to observe the diffracted spectrum. The arrangement made by him is known as *Rowland mounting*. In this mounting slit, grating and photographic plate are placed at the corners of a right angle triangle [Fig. 2.31]. *SX* and *SY* are two horizontal iron rails placed perpendicular to each other in a horizontal plane. Grating *G* and plate *P* are mounted at the ends of a rod of length *R* which is equal to the radius of curvature of grating. Arrangement is such that *G* and *P* can move along *Y* and *X* axes respectively simultaneously. *G*,*G*′ and *P*,*P*′ represent two position of grating and plate along *SY* and *SX* respectively. The slit *S* is at the intersection of the rails *SX* and *SY* which remains fixed always. Thus *G*,*P* and *S* always lie on circle, called Rowland circle. When the white light from the slit falls on the grating, it gets diffracted and the photo of spectrum is obtained on the photographic plate *P*. When rod is moved, then *G* and *P* get new positions *G*′ and *P*′ (but always remains on opposite ends of diameter of Rowland circle) and spectra of different orders are obtained. We have that condition for maxima at *P*,

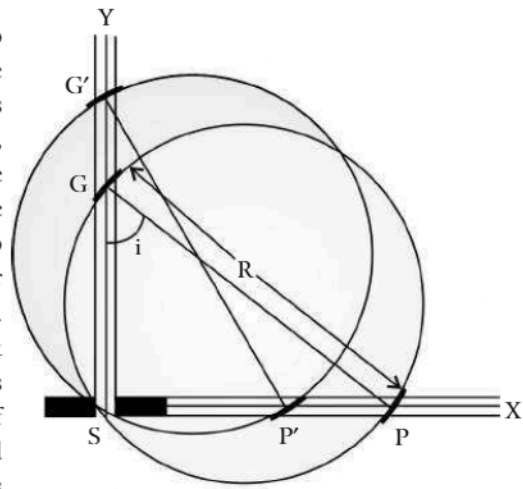


Fig. 2.31

$$(e+d)(\sin i - \sin\theta) = n\lambda$$

Since, grating surface is always parallel to *P* thus, angle of diffraction is zero always and angle of incidence $i = \angle SGP$

$$\therefore \quad (e+d)\sin i = n\lambda$$

But,
$$\sin i = \sin SGP = \frac{SP}{GP}$$

$$\therefore \quad (e+d) = \frac{SP}{GP} = n\lambda$$

$$\because \quad (e+d) \text{ and } GP \text{ are constant}$$

$$\therefore \quad SP \propto n\lambda$$

This shows that for a given wavelength λ, *SP* is proportional to the order *n*. Hence, as *P* moves towards *S* along *SP*, the first, second, third order maxima occur at exactly equal intervals.

Also for a fixed order,

$$SP \propto \lambda$$

If, a scale is attached with *SP*, it gives the direct value of λ. Thus, with this type of mounting it is possible to get wavelength of spectral lines directly.

Advantage

1. Normal spectrum can be observed at 10 to 15° from the normal to the plate on either side of it.
2. By changing the position of grating and plate, photographs of different orders can be obtained.

Disadvantage

1. Focussing is difficult.
2. A big dark room is required.
3. It is expensive and involves a considerable amount of mechanism.
4. It shows some astigmatism.

2.20.2 Paschen-Runge Mounting

Grating of large radius of curvature is mounted by this method. In this mounting grating *G* and slit *S* are fixed in suitable positions along the Rowland circle so as to give a desired angle of incidence, [Fig.2.32]. The spectra of different orders are imaged on the circumference of the circle. In Fig. 2.32, A_1B_1 is the first order spectrum, A_2B_2 and A_3B_3 are the second order and third order spectrum respectively. So, with this mounting several orders of the spectrum can be photographed simultaneously. The photographic plate is held to a circular steel frame along the Rowland circle.

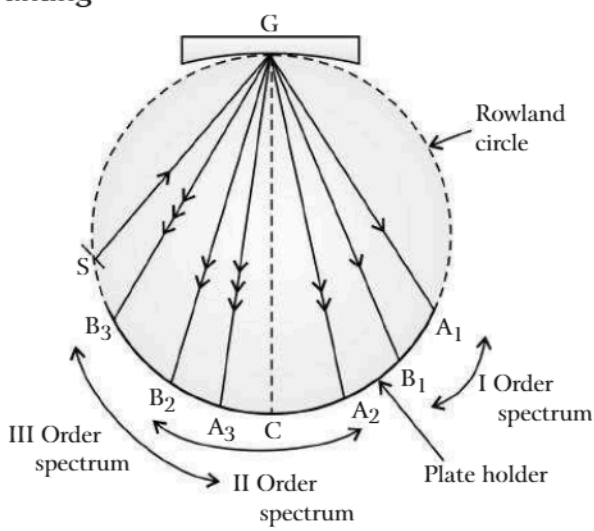


Fig. 2.32

Advantages

1. The arrangement is rigidly fixed.
2. Using a number of plates (or a large plate) different order spectrum can be photographed simultaneously by the same arrangement.

3. Two or more slits can be placed at different points on the Rowland circle to work at different angle of incidence.

Disadvantages

1. The spectrum is normal only in the region near the grating normal.
2. Except the normal spectrum, there is a strong astigmatism.
3. A big dark room is required.
4. It is difficult to avoid temperature variation of the room which may cause shift in the spectral line.

2.20.3 Eagle Mounting

In this mounting, the slit S and the photographic plate P are mounted very close together at one end of a rigid bar and on the other end of bar concave grating is mounted [Fig. 2.33]. In addition to translational motion the grating can be rotated also with the help of a screw. All the three lie on the Rowland circle. Thus, only that part of the spectrum is obtained which is diffracted back, nearly along the incident path. For going from one spectral region to the other, grating G is moved along the rigid bar and the new position of grating (G') is such that the Rowland circle still pass through S. The photographic plate P is also tilted to P' so that it lies on the new Rowland circle.

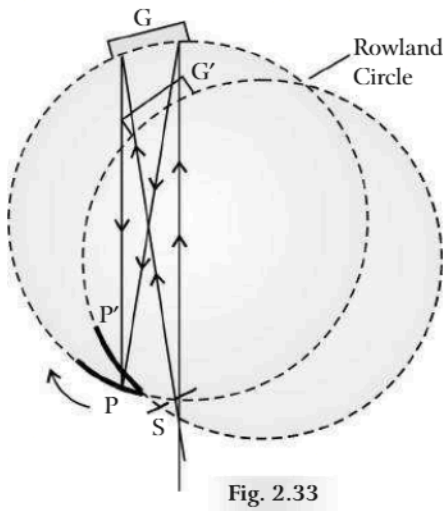


Fig. 2.33

Advantages

1. This mounting is most compact and requires much less space than other mountings. It is therefore used to study the ultraviolet spectrum (because less volume has to be evacuated).
2. Temperature variation can be controlled.
3. Higher order spectrum can be obtained.
4. Astigmatism is negligible.

Disadvantages

1. The spectrum is not normal.
2. It is not focussed equally for all the orders.
3. It involves both translation and rotation of the grating, together with rotation of the plate, which makes the adjustment difficult.

Example 15: A Rowland grating has 6000 lines to a centimetre. Calculate the angular separation of two mercury lines of wavelengths 5770 Å and 5461 Å in the first order spectrum.

Solution: For a Rowland grating that the condition of maximum is,

$$(e+d)\sin i = n\lambda$$

For the first wavelength, λ_1

$$(e+d)\sin i_1 = n\lambda_1 \qquad ...(1)$$

Similarly, for the second wavelength, λ_2

$$(e+d)\sin i_2 = n\lambda_2 \qquad ...(2)$$

Given that, $n=1$ (spectrum is of first order)

and $(e+d)=\frac{1}{6000}, \lambda_1 = 5770\times10^{-8}$ cm and $\lambda_2 = 5461\times10^{-8}$ cm

From equation (1), we have $\sin i_1 = \frac{n\lambda_1}{e+d} \approx \frac{1\times5770\times10^{-8}}{1/6000} = 0.3462$ or $i_1 = 20°12'$

From equation (2), we have $\sin i_2 = \frac{n\lambda_2}{e+d} = \frac{1\times5461\times10^{-8}}{1/6000} = 0.3277$ or $i_2 = 19°6'$

$\therefore$ Angular separation $= i_1 - i_2 = 20°12' - 19°6'$

$= 1°6' = \mathbf{1.9\times10^{-2}}$ **rad.**

2.21 Resolving Power of a Grating

The resolving power of a grating is defined as *its ability to show two neighbouring lines in a spectrum as separate*. It is measured by $\lambda/d\lambda$, where λ is the wavelength of a spectral line and $d\lambda$ is the least difference in the wavelengths of two neighbouring spectral lines which can be just resolved.

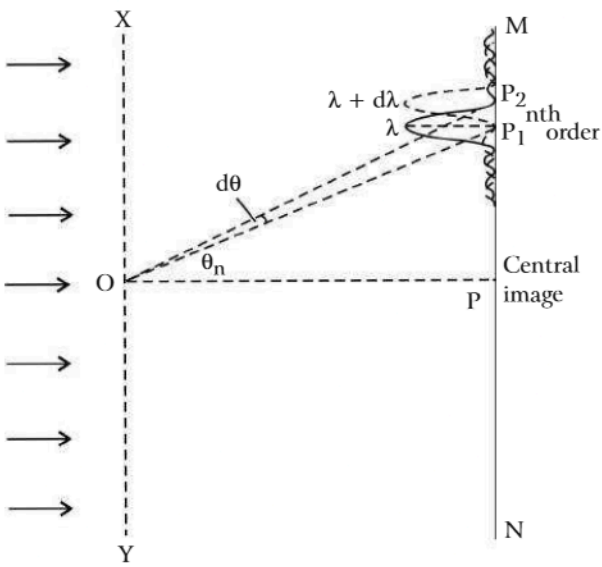


Fig. 2.34

Let a parallel beam of light of two wavelengths λ and $(\lambda + d\lambda)$ be incident normally on the plane grating *XY* as shown in Fig. 2.34. If *MN* is the field of view of telescope and here P_1 is the *n*th primary maximum of spectral line of wavelength λ at an angle of diffraction θ_n. P_2 is the n^{th} primary

maximum of a second spectral line of wavelength $\lambda + d\lambda$ at a diffracting angle $\theta_n + d\theta$. According to Rayleigh's criterion, these two lines can be just distinguished as separate in any order of the spectrum, if the principal maximum of one coincides with the first minimum of the other.

The direction of the nth primary maximum for a wave length λ is given by,

$$(e + d)\sin\theta_n = n\lambda \qquad ...(1)$$

and the direction of the nth primary maximum for a wavelength $(\lambda + d\lambda)$ is given by

$$(e + d)\sin(\theta_n + d\theta) = n(\lambda + d\lambda) \qquad ...(2)$$

where $(e + d)$ is the grating element.

Let the first minimum adjacent to the nth maximum be obtained in the direction $(\theta_n + d\theta)$.

The grating equation for the minima is,

$$N(e + d)\sin\theta = m\lambda \qquad ...(3)$$

where N is the total number of lines on the grating and m = all integral values except $0, N, 2N, \ldots nN$, because at these values 0th, Ist, 2nd, ... nth principal maximum respectively will appear. Clearly the first minimum adjacent to the nth principal maximum, in the direction of θ increasing, will be obtained for $m = (nN + 1)$. Therefore, if this minimum is obtained in the direction $((\theta_n + d\theta)$, then equation (3) becomes,

$$N(e + d)\sin(\theta_n + d\theta) = (nN + 1)\lambda$$

or
$$(e + d)\sin(\theta_n + d\theta) = \frac{(nN + 1)\lambda}{N} \qquad ...(4)$$

Equating right hand side of equation (2) and (4),

$$n(\lambda + d\lambda) = \frac{(nN + 1)\lambda}{N} = n\lambda + \frac{\lambda}{N}$$

or
$$nd\lambda = \lambda / N$$

or
$$\frac{\lambda}{d\lambda} = \mathbf{nN}$$

But the quantity $\lambda/d\lambda$ is the resolving power R of the grating.

Therefore,
$$\mathbf{R = nN}$$

Thus, it is quite obvious that the resolving power of grating is independent of the grating element but depends on the (i) total number of lines on the grating and (ii) the order of the spectrum.

Again, from equation (1)
$$n = \frac{(e + d)\sin\theta_n}{\lambda}$$

So,
$$R = Nn = \frac{N(e + d)\sin\theta_n}{\lambda}$$

where $N(e + d)$ = total width of the grating.

Hence, at a particular angle of diffraction θ_n, the resolving power of the grating is directly proportional to the total width of the grating. Since, $\sin\theta_n = 1$ (maximum)

Therefore, $$R_{max} = N(e + d)/\lambda$$

2.22 Experimental Determination of the Resolving Power of a Grating

The resolving power of a grating may be determined experimentally by finding the minimum width of the grating for just resolution of two close lines D_1 and D_2 of sodium. For this, the grating is mounted on the prism table of an adjusted spectrometer. An adjustable slit is placed in between the grating an telescope with its edges parallel to the ruling of the grating [Fig. 2.35]. The slit of the spectrometer is illuminated by sodium light and in the first order D_1, D_2 lines are observed through the telescope. Now, the adjustable slit is closed gradually until D_1 and D_2 lines just merge together. In this condition the slit width is measured with the help of a travelling microscope. Let it be w, then w is the minimum width of the grating for resolving D_1 and D_2.

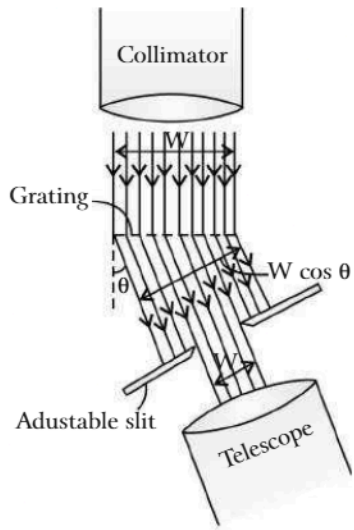


Fig. 2.35

We know that, wavelengths of D_1 and D_2 are 5896 Å and 5890 Å. Thus, the wavelength difference ($d\lambda$) is 6 Å at a mean wavelength (λ) 5893 Å. Hence, the necessary resolving power to resolve the D_1 and D_2 lines is,

$$\frac{\lambda}{d\lambda} = \frac{5893}{6} = 982$$

Clearly, the grating width w measured above gives the resolving power equal to 982. Since, the resolving power of the grating is proportional to its total width. So, the resolving power for the whole grating will be,

$$R = 982 \times \frac{W \cos\theta}{w}$$

where W is the total width of the grating surface and θ is the mean angle of diffraction of D_1 and D_2 lines.

Example 16: What would be the minimum number of lines in a gratings which will just resolve in the second order the lines whose wave lengths are 5890 Å and 5896 Å ?

Solution: The resolving power of a grating is given by,

$$\frac{\lambda}{d\lambda} = nN \quad \text{or} \quad N = \frac{\lambda}{nd\lambda}$$

Here, n = number of order of the spectrum = 2, N = total number of lines on the grating

$d\lambda = 5896\ \text{Å} - 5890\ \text{Å} = 6\ \text{Å}$ and $\lambda = 5890\ \text{Å}$

Putting the values, we get $$N = \frac{5890 \times 10^{-8}}{2 \times 6 \times 10^{-8}}$$

$$= \frac{5890}{12} \approx 491$$

Example 17: A grating has 1000 lines ruled on it. In the region of wavelength $\lambda = 6000$ Å, find (i) the difference between two wavelengths that just appear separated in the first order (ii) the resolving power in the second order.

Solution: The resolving power of a grating is given by,

$$\frac{\lambda}{d\lambda} = nN \quad \text{or} \quad d\lambda = \frac{\lambda}{nN}$$

Here, $\lambda = 6000\ \text{Å}$, $n = 1, N = 1000$

(i) Putting the values, we get $$d\lambda = \frac{6000}{1 \times 1000} = \mathbf{6\ Å}$$

(ii) Again, the resolving power of a grating,

$$R = \frac{\lambda}{d\lambda} = nN$$

Here, $n = 2, N = 1000$

So, $$R = 2 \times 1000 = \mathbf{2000.}$$

Example 18: A plane grating has 15000 lines per inch. Find the angle of separation of the 5048 Å and 5016 Å lines of helium in the second order spectrum.

Solution: Let θ_1 and θ_2 be the angles of diffraction for the second order, for the wavelengths λ_1 and λ_2 respectively.

$\therefore$ $$(e + d)\sin\theta_1 = n\lambda_1$$

and $$(e + d)\sin\theta_2 = n\lambda_2$$

Here given that, $e + d = \frac{2.54}{15000}$ cm, $n = 2$

$$\lambda_1 = 5048\ \text{Å} = 5048 \times 10^{-8}\ \text{cm} \quad \text{and} \quad \lambda_2 = 5016\ \text{Å} = 5016 \times 10^{-8}\ \text{cm}$$

Putting the values, we get $$\sin\theta_1 = \frac{2 \times 5048 \times 10^{-8} \times 15000}{2.54} = 0.5962 \text{ or } \theta_1 = 36°36'$$

and $$\sin\theta_2 = \frac{2 \times 5016 \times 10^{-8} \times 15000}{2.54} = 0.5924 \text{ or } \theta_2 = 36°20'$$

$\therefore$ Angle of separation $= \theta_1 - \theta_2 = \mathbf{16'}$

Example 19: A plane transmission grating has 40000 lines in all, with grating element 12.5×10^{-5} cm. Calculate the maximum resolving power for which it can be used in the range of wavelength 5000 Å.

Solution: The resolving power of a grating is given by, $R = nN$ (n is order of spectrum)

or $$R_{max} = n_{max}N$$

Also, we have $$(e+d)\sin\theta = n\lambda$$

For getting highest order $\theta = 90°$ or $\sin\theta = 1$

$\therefore$ $$n_{max} = \frac{e+d}{\lambda}$$

Here given that, $(e+d) = 12.5\times10^{-5}$ cm and $\lambda = 5000$ Å

So, $$n_{max} = \frac{12.5\times10^{-5}}{5000\times10^{-8}} = 2.5$$

Hence, the second order is the highest order which is observed with this grating. Therefore, maximum resolving power,

$$R_{max} = 40000\times2 = \mathbf{80000}$$

Example 20: A source containing a mixture of hydrogen and deuterium atoms emits a red doublet at $\lambda = 6563$ Å whose separation is 1.8 Å. Find the minimum number of lines required in a plane grating which can resolve the doublet in the first order.

Solution: The smallest wavelength difference $d\lambda$ that can be resolved by a grating at wavelength λ is given by,

$$\frac{\lambda}{d\lambda} = Nn$$

where N is the total number of lines on the grating and n is the order of the spectrum.

$\therefore$ $$N = \frac{1}{n}\frac{\lambda}{d\lambda}$$

Here given that, $n = 1, \lambda = 6563$ Å and $d\lambda = 1.8$ Å

$\therefore$ $$N = \frac{6563\text{ Å}}{1\times1.8\text{ Å}} = \mathbf{3646.}$$

2.23 Resolving Power of a Prism

Ability of a prism to separate two close spectral lines is known as resolving power or chromatic resolving power of a prism. It is measured by $\lambda/d\lambda$, where $d\lambda$ is the smallest wavelength difference that can be just resolved by the prism at the wavelength λ.

Let a plane wavefront BP of light of wavelengths λ and $\lambda + d\lambda$ be incident on the prism ABC which is placed in the position of minimum deviation. Let CQ and CQ' be the emergent wavefronts for wavelengths λ and $\lambda + d\lambda$ as shown in Fig. 2.36. Let δ and $(\delta - d\delta)$ be the angle of deviation, μ and $(\mu - d\mu)$ the refractive indices corresponding to wavelength λ and $\lambda + d\lambda$ respectively.

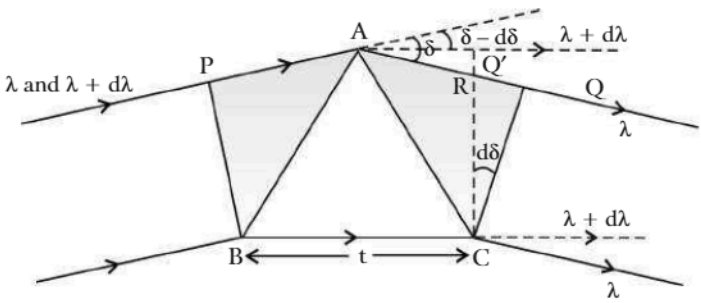


Fig. 2.36

According to Fermat's principle, the optical path between the incident and emergent wavefronts for any wavelength must be the same. Hence, for the wavelength λ,

$$PA + AQ = \mu BC \qquad ...(1)$$

Similarly, for the wavelength $\lambda + d\lambda$,

$$PA + AQ' = (\mu - d\mu)BC \qquad ...(2)$$

From equation (1) and (2), we have

$$AQ - AQ' = d\mu BC$$

But from Fig. 2.36, $AQ' \approx AR$, so that $AQ - AQ' \approx AQ - AR = RQ$

$\therefore$ $$RQ = d\mu BC$$

Again from [Fig. 2.36], $$\angle Q'CQ = d\delta$$

and $$RQ = CQ d\delta$$

$\therefore$ $$CQ d\delta = d\mu BC$$

The emergent beam has rectangular section. The prism may also be considered as a rectangular aperture of width e which is the width of the emergent beam CQ.

$\therefore$ $$e d\delta = t d\mu$$

or $$d\delta = \frac{t\, d\mu}{e} \qquad ...(3)$$

where t is the width of the base of the prism

Hence, the diffraction pattern for each wavelength is equivalent to that due to a rectangular aperture (single slit) of width e. According to Rayleigh criterion, if two spectral lines are just resolved, then principal maximum of one should fall on the first minimum of the other. It means that the angular separation $d\delta$ between the principle maxima of λ and $\lambda + d\lambda$ must be equal to the angle $d\theta$ between the principal maximum and the first minimum of λ. The angle $d\theta$ is given by the Fraunhofer diffraction at a single slit,

$$e \sin d\theta = \lambda \quad \text{if } d\theta \text{ is the angle of diffraction}$$

or $$ed\theta = \lambda \qquad (\because d\theta \text{ is very small}, \therefore \sin d\theta \approx d\theta)$$

or $$d\theta = \lambda / e \qquad \ldots(4)$$

For just resolution $d\delta = d\theta$. So, from equation (3) and (4), we get

$$\frac{t\, d\mu}{e} = \frac{\lambda}{e}$$

or $$td\mu = \lambda$$

Hence, the resolving power of the prism,

$$\lambda / d\lambda = t \frac{d\mu}{d\lambda}$$

It shows that the resolving power of a prism is directly proportional to the width of the base of the prism and also proportional to the rate of change of refractive index with wavelength. As we have $\frac{d\mu}{d\lambda}$ = dispersive power

So, **Resolving power = $t \times$ Dispersive power**

The spectral image for each wavelength is more sharp in a grating spectra than in prism spectra. Hence, grating can show lines of much smaller $\Delta\lambda$ separately as compared with prism. In other words, resolution of grating is much than prism.

Example 21: A prism is just able to resolve the sodium yellow lines 5890 Å and 5896 Å. Calculate its resolving power. How close spectral lines can a prism resolve in this range if the base width as well as dispersive power is doubled ?

Solution: The resolving power of a prism is given by, $\lambda / d\lambda$

Here given that, mean wave length, $\lambda = 5893$ Å $\qquad \left(\because \lambda = \frac{5890 + 5896}{2}\right)$

and $$d\lambda = 5896 - 5890 = 6 \text{ Å}$$

$$\therefore \quad \frac{\lambda}{d\lambda} = \frac{5893\ \text{Å}}{6\ \text{Å}} = 982$$

Also, we have $\frac{\lambda}{d\lambda} = t\left(\frac{d\mu}{d\lambda}\right) = t \times \text{Dispersive power}$, where t = base width of prism

So, the smallest resolvable wavelength difference is,

$$d\lambda = \frac{\lambda}{t \times \text{dispersive power}}$$

Now, if t and $d\mu/d\lambda$, both are doubled, then $d\lambda$ becomes one fourth of the original. Thus, the new prism can resolve a wavelength difference of $(6\text{Å}/4) = \mathbf{1.5\ Å}$

Example 22: A prism spectrometer uses a prism of base width 5 cm and of material whose dispersion $d\mu/d\lambda$ is 200 cm^{-1} in the range $\lambda = 5000$ Å.What is the smallest difference of wavelength in this range which this spectrometer can resolve ?

Solution: The smallest wavelength difference $d\lambda$ that can be resolved in the wavelength range λ by a prism spectrometer is given by,

$$\frac{\lambda}{d\lambda} = t\frac{d\mu}{d\lambda} = t \times \text{Dispersive power}$$

where t is the base width of the prism

Here given that, $t = 5$ cm, $\frac{d\mu}{d\lambda} = 200\ \text{cm}^{-1}$ and $\lambda = 5000\ \text{Å} = 5000 \times 10^{-8}$ cm

Putting all the values, we get $\frac{5000 \times 10^{-8}}{d\lambda} = 5 \times 200$

or $$d\lambda = \frac{5000 \times 10^{-8}}{1000} = 5 \times 10^{-8}\ \text{cm} = 5\ \text{Å}$$

Miscellaneous Problems

Problem 1: The diameter of the first ring of a zone plate is 1 mm. If plane waves ($\lambda = 5000\ \text{Å}$) falls on the plate, where should the screen be placed so that light is focussed to a brightest spot ?

Solution: The plane wave falling on the plate will be focussed to a brightest spot at the first focus of the plate. The focal length of the zone plate is given by,

$$f = \frac{r_n^2}{n\lambda}, \text{ where } r_n \text{ is the radius of the } n\text{th zone.}$$

For brightest spot $n = 1$, So, $f = \dfrac{r^2}{\lambda}$

Here given that, $r_1 = 0.5 \text{ mm} = 0.05 \text{ cm}$ and $\lambda = 5000 \times 10^{-8} \text{ cm}$

$\therefore$ $$f = \frac{(0.05)^2}{5 \times 10^{-5}} = \mathbf{50\ cm}$$

Hence, the screen should be placed **50 cm distant from the plate.**

Problem 2: Find the first three focal lengths of a zone plate for which the radius of zone is 0.3 mm for light of wavelength 5000 Å.

Solution: The focal length of the zone plate is given by,

$$f = \frac{r_n^2}{n\lambda\,(2m-1)}, \qquad \text{where } r_n \text{ is radius of } n\text{th zone.}$$

Given that, radius of first zone *i.e.*, $r_1 = 0.3 \text{ mm} = 0.03 \text{ cm}$, $n = 1$ and $\lambda = 5000 \times 10^{-8} \text{ cm}$

So, $$f = \frac{(0.03)^2}{5 \times 10^{-5}\,(2m-1)}$$

To get first three focal lengths, we put $m = 1, 2, 3$

So, $$f_1 = \frac{(0.03)^2}{5 \times 10^{-5} \times 1} = \frac{3 \times 3 \times 10^{-4}}{5 \times 10^{-5}} = \mathbf{18\ cm}$$

$$f_2 = \frac{9 \times 10^{-4}}{5 \times 10^{-5} \times 3} = \mathbf{6\ cm}$$

and $$f_3 = \frac{9 \times 10^{-4}}{5 \times 10^{-5} \times 5} = \mathbf{3.6\ cm}$$

Problem 3: Find the radii of the first three transparent zones of a zone plate whose focal length is one meter for $\lambda = 5890\,Å$.

Solution: The first focal length of the zone plate is given by,

$$f = \frac{r_n^2}{n\lambda}, \qquad \text{where } r_n \text{ is radius of the } n\text{th zone.}$$

or $$r_n^2 = f\,n\lambda$$

So, $$r_1^2 = f\lambda \qquad (\text{putting } n = 1)$$

$$= 1 \times 5890 \times 10^{-10}\,\text{m} \qquad [\because\ f = 1\text{m and } \lambda = 5890\ Å]$$

or $$r_1 = \sqrt{(5890 \times 10^{-10}\ \text{m})} = 76.7 \times 10^{-5}\,\text{m} = \mathbf{0.0767\ cm}$$

and $r_3 = \sqrt{(3f\lambda)}$ (since even numbers zone are opaque *i.e.*, not transparent zone)

So, $r_3 = \sqrt{3 \times 5890 \times 10^{-8}\,\text{cm}} = \mathbf{0.133\ cm}$

and $r_5 = \sqrt{5 \times 5890 \times 10^{-8}\,\text{cm}} = \mathbf{0.172\ cm}$

Problem 4: An object placed at 20 cm from a zone plate and the brightest image is situated at 20 cm from the zone plate on the other side with light of wavelength $= 4000$ Å. Calculate the number of Fresnel's zone in a radius of 1 cm of that plate.

Solution: For a zone plate, $$f = \frac{r_n^2}{n\lambda} \qquad \ldots(1)$$

and also, we have that the zone plate behaves like a convex lens. So, we use lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Given, that $v = 20$ cm, $u = -20$ cm

So, $$\frac{1}{20} + \frac{1}{20} = \frac{1}{f} = \frac{1}{10}$$

or $$\boldsymbol{f = 10\ cm}$$

Also given that, radius of the nth zone $r_n = 1\,cm$ *i.e.*, then

$$n = \frac{r_n^2}{f\lambda} = \frac{(1)^2}{10 \times 4000 \times 10^{-8}} = \frac{1}{10 \times 4 \times 10^{-5}} = \mathbf{2500}$$

Problem 5: A zone plate is made by arranging the circles of the radii which define the zone such that they are the same as the radii of Newton's ring formed between a plane surface and the surface having radius of curvature 200 cm. Find the primary focal length of the zone plate.

Solution: For Newton's ring, radius of the nth ring,

$$r_n = \sqrt{(n\lambda R)} \qquad \text{(for dark rings and for air film } \mu = 1\text{)}$$

where R is radius of curvature.

So, $$r_1 = \sqrt{(\lambda R)} \qquad \ldots(1)$$

For a zone plate, the primary focal length is given by,

$$f_1 = \frac{r_1^2}{\lambda} \qquad \ldots(2)$$

Putting the value of r_1 from equation (1), we get

$$f_1 = \frac{\lambda R}{\lambda} = R$$

Here given that, $R = 200$ cm

So, $$f_1 = \mathbf{200\ cm}$$

Problem 6: A fine aperture is placed at a distance of 12 cm from a sharp razor blade edge held vertically at a distance of 25 cm from the screen. If the aperture is illuminated by light ($\lambda = 5890$ Å), calculate the height of the 6th bright band above the line joining the edge to the aperture.

Solution: The distance of the nth maximum from the line joining the edge to the aperture is given by

$$x_n = \sqrt{\frac{b(a+b)(2n+1)\lambda}{a}}$$

where a is the distance between slit and straight edge and b is the distance between straight edge and screen.

Here given that, $a = 12$ cm, $b = 25$ cm, $\lambda = 5890 \times 10^{-8}$ cm and $n = 5$ (for 6^{th} bright band)

So, putting the values, we get

$$x_6 = \sqrt{\frac{25(12+25)5890 \times 10^{-8} \times 11}{12}}$$

$$= \sqrt{\frac{25 \times 37 \times 11 \times 5890}{12}} \times 10^{-4} = \mathbf{0.223\ cm}$$

Problem 7: A single slit of width 0.14 mm is illuminated normally by monochromatic light and diffraction bands are observed on a screen 2 m away. If the centre of the second dark band is 1.6 cm from the central bright band, deduce the wavelength of light used.

Solution: In Fraunhofer diffraction due to a narrow slit, the condition for dark band is given by,

$$e\sin\theta = n\lambda \quad \text{or} \quad \lambda = \frac{e\sin\theta}{n}$$

Here given that, e = width of slit $= 0.14$ mm $= 0.014$ cm, n = order of dark band $= 2$

and $$\sin\theta = \frac{x}{D} = \frac{1.6\text{ cm}}{200\text{ cm}} = \frac{1.6}{200}$$

Putting the values, we get,

$$\lambda = \frac{0.014 \times 1.6}{200 \times 2} = 5.6 \times 10^{-5}\text{ cm} = \mathbf{5600\ \text{Å}}$$

Problem 8: A linear aperture whose width is 0.02 cm is placed in front of a lens of focal length 60 cm. This aperture is illuminated normally by a parallel beam whose wavelength is 5×10^{-5} cm. What will be the distance between the centre and the first dark band of the diffraction pattern on a screen placed 60 cm from the lens.

Solution: It is case of Fraunhofer diffraction due to a narrow single slit. The direction of minima is given y by, $e\sin\theta = \pm n\lambda$

Here given that, e = width of slit $= 0.02$ cm, $f = 60$ cm

and $n = 1$ (first dark band on either side of central maxima)

$\therefore$ $$e\sin\theta = \lambda$$

or $$\sin\theta = \lambda / e$$

or $$\theta \approx \lambda / e \text{ radian} \qquad \ldots(1) \quad [\because \ \theta \text{ is very small}]$$

Also, if x be the linear distance of the first minimum from central maxima and f the focal length of the lens, then the angular distance between the central maxima and the first dark band is given by,

$$\theta = (x/f) \text{ radian} \qquad \ldots(2) \quad [\because \text{ angle} = \text{arc/radius}]$$

From equation (1) and (2), we have

$$\frac{\lambda}{e} = \frac{x}{f} \quad \text{or} \quad x = \frac{f\lambda}{e}$$

Putting all the values, we get $$x = \frac{60 \times 5 \times 10^{-5}}{0.02} = \mathbf{0.15\ cm}$$

Problem 9: Calculate the angles at which the first dark band and the next bright band are formed in the Fraunhofer diffraction pattern of a 0.3 mm wide slit ($\lambda = 5890$ Å).

Solution: In the Fraunhofer diffraction due to narrow single slit, the direction of minima is given by,

$$e\sin\theta = \pm n\lambda$$

For the first dark band, $n = 1$

$$\therefore \quad e\sin\theta = \lambda$$

or
$$\sin\theta = \frac{\lambda}{e} = \frac{5890\times10^{-8}}{0.03} = 0.00196$$

$$\therefore \quad \theta = 7'$$

Let the angle of diffraction θ' corresponds to the first bright band on either side of the central maxima, then approximate formula is given by,

$$e\sin\theta' = \frac{3\lambda}{2} \qquad [\because\ e\sin\theta = (2n+1)\lambda/2]$$

$$\therefore \quad \sin\theta' = \frac{3\lambda}{2e} = \frac{3}{2}\times0.00196 = 0.00294, \theta' = \mathbf{10'}$$

Problem 10: Find the half angular width of the central bright maximum in the Fraunhofer diffraction pattern of a slit o width 12×10^{-5} ***cm when the slit is illuminated by monochromatic light of wavelength 6000*** Å.

Solution: In Fraunhofer diffraction, if θ is the half angular width of the central maximum, then

$$\sin\theta = \frac{\lambda}{e}, \qquad \text{where } e \text{ is width of slit.}$$

Here given that, $e = 12\times10^{-5}$ cm and $\lambda = 6000$ Å

Putting all the values, we get,

$$\sin\theta = \frac{6000\times10^{-8}}{12\times10^{-5}} = 0.5 = \frac{1}{2} = \sin 30^\circ \qquad \text{or} \quad \theta = \mathbf{30^\circ}$$

We can also determine the angular width of central maximum,

$$2\theta = 60^\circ$$

Problem 11: In a plane transmission grating the angle of diffraction for the second order principal maximum is 30°. ***Calculate the number of lines in one cm of the grating surface. (The wavelength of used light is 5000*** Å)

Solution: The direction of maximum intensity in the diffracted light due to plane grating is given by,

$$(e+d)\sin\theta = n\lambda$$

where $(e+d)$ is grating element and $\frac{1}{e+d}$ is the number of lines per cm on the grating surface and n is order of maxima.

$$\therefore \quad \frac{1}{e+d}=\frac{\sin\theta}{n\lambda}$$

Here given that, $n=2, \theta=30°$ and $\lambda=5\times10^{-5}$ cm

Putting all the values, we get

$$\frac{1}{e+d}=\frac{\sin 30°}{2\times5\times10^{-5}}=\frac{1\times10^{5}}{2\times2\times5}=\frac{100000}{20}=\mathbf{5000}$$

So, the number of lines/cm on the grating surface will be **5000.**

Problem 12: Light of wavelength 5000 Å falls normally on a plane transmission grating having 15000 lines in 3cm. Find the angle of diffraction for maximum intensity in first order. (Given that, $\sin^{-1}0.25=14°29'$***)***

Solution: The direction of maximum intensity due to a diffraction grating is given by,

$$(e+d)\sin\theta=n\lambda$$

Here given that, $e+d=\frac{3\text{ cm}}{15000}, n=1$ and $\lambda=5000$ Å

Putting all the values, we ge $\sin\theta=\frac{1\times5000\times10^{-8}\times15000}{3}=0.25$

or $$\theta=\sin^{-1}(0.25)=\mathbf{14°29'}$$

Problem 13: The limits of the visible spectrum are 4000-7000 Å. Find the angular breadth of the first-order visible spectrum produced by a plane diffraction grating having 15000 lines per inch when light is incident normally on the grating. Given, $\sin13°40'=0.236$ ***and*** $\sin24°24'=0.413$

Solution: The grating equation is given by,

$$(e+d)\sin\theta=n\lambda$$

or $$\sin\theta=\frac{n\lambda}{(e+d)}$$

Here given that, $(e+d)=\frac{2.54}{15000}$ cm $\quad (\because$ 1 inch = 2.54 cm)

$n=1$ (first order) and $\lambda=4\times10^{-5}$ cm

$$\therefore \quad \sin\theta=\frac{1\times4\times10^{-5}\text{ cm}}{(2.54/15000)\text{ cm}}=0.236$$

or $$\theta = \sin^{-1}(0.236) = \mathbf{13°40'}.$$

Again, for $n = 1$ and $\lambda = 7 \times 10^{-5}$ cm, we have

$$\sin\theta = \frac{1 \times 7 \times 10^{-5}\ \text{cm}}{(2.54/15000)\ \text{cm}} = 0.413$$

or $$\theta = \sin^{-1}(0.413) = \mathbf{24°24'}.$$

Thus the angular positions for the 4000 Å and 7000 Å lines in the first order are 13°40′ and 24°24′ respectively. Therefore, the angular breadth of the first-order visible spectrum is,

$$= 24°24' - 13°40' = \mathbf{10°44'}.$$

Problem 14: In a grating spectrum, which spectral line in 4th order will overlap with 3rd order line of 5461 Å ?

Solution: The grating equation is,

$$(e + d)\sin\theta = n\lambda$$

If the nth order of wavelength λ_1 (say) coincides with the $(n+1)$th order of λ_2, then

$$(e + d)\sin\theta = n\lambda_1 = (n+1)\lambda_2. \quad \text{or} \quad \lambda_2 = \frac{n\lambda_1}{n+1}$$

Here given that, $n = 3, \lambda_1 = 5461\,\text{Å}, (n+1) = 4,$

$\therefore$ $$\lambda_2 = \frac{3 \times 5461\,\text{Å}}{4} = \mathbf{4096\ Å}.$$

Problem 15: A diffraction grating used at normal incidence gives a green line, $\lambda = 5400$ Å, in a certain order n superimposed on a violet line, $\lambda = 4050$ Å, of the next higher order $(n+1)$. If the angle of diffraction is 30°, calculate the spacing between the grating lines. Also, find how many lines are three cm in the grating ?

Solution: The grating equation is,

$$(e + d)\sin\theta = n\lambda$$

If the nth order of wavelength λ_1 (say) coincides with $(n+1)$th order of λ_2, then

$$(e + d)\sin\theta = n\lambda_1 = (n+1)\lambda_2 \qquad \ldots(1)$$

or $$\lambda_1 = \frac{n+1}{n}\lambda_2$$

Here given that, $\lambda_1 = 5400\,\text{Å}, \lambda_2 = 4050\,\text{Å}.$ and $\theta = 30°$

So, $$5400\,\text{Å} = \frac{n+1}{n}4050 \quad \text{or} \quad n = 3$$

Putting this value of n and $\theta = 30°$ in equation (1) we get,

$$(e+d)\sin 30° = 3 \times (5400 \times 10^{-8}\ \text{cm})$$

$$\therefore \quad (e+d) = \frac{3 \times (5400 \times 10^{-8}\,\text{cm})}{\sin 30°} = \mathbf{3.24 \times 10^{-4}\,cm}$$

$$\therefore \quad \text{Number of lines per cm} = \frac{1}{3.24 \times 10^{-4}} = \mathbf{3086}$$

Problem 16: How many orders will be visible if the wavelength of light, falling normally on a grating having 25400 lines per inch, is 5000 Å ?

Solution: The grating equation is,

$$(e+d)\sin\theta = n\lambda$$

The maximum value of $\sin\theta = 1$, so the highest order visible with this grating will be given by,

$$n_{max} = \frac{e+d}{\lambda}$$

Here given that, $e+d = \frac{2.54}{25400}$ cm and $\lambda = 5000\ \text{Å} = 5000 \times 10^{-8}$ cm [$\because$ 1 inch $= 2.54$ cm]

$$\therefore \quad n_{max} = \frac{2.54}{25400 \times 5000 \times 10^{-8}} = \frac{10}{5} = \mathbf{2}$$

Problem 17: What is the highest order spectrum which may be seen with light of wavelength 5×10^{-5}cm by means of a grating with 3000 lines/cm? Calculate the wavelengths in the visible spectrum (3.5×10^{-5}cm, 7.0×10^{-5}cm) which coincide with the fifth order spectrum of this light.

Solution: The grating equation is;

$$(e+d)\sin\theta = n\lambda$$

The maximum value of sin θ si 1. Therefore, the highest order visible with the grating will be given by,

$$n_{max} = \frac{e+d}{\lambda}.$$

Here given that, $e+d = \frac{1}{3000}$ cm and $\lambda = 5 \times 10^{-5}$ cm

$$\therefore \quad n_{max} = \frac{(1/3000)\,\text{cm}}{(5 \times 10^{-5}\ \text{cm})} = \mathbf{6.6}$$

Thus, the highest order which may be seen is 6.

The direction θ of the fifth order ($n=5$) spectrum of wavelength 5×10^{-5}cm is given by

$$(e+d)\sin\theta = 5 \times (5 \times 10^{-5}) \qquad \ldots(1)$$

Let λ' be the other wavelength falling in the same direction θ. Then, we may write,

$$(e+d)\sin\theta=(5\pm m)\lambda', \qquad \ldots(2) \quad \text{where } m=1,2,3,\ldots$$

From equations (1) and (2), we get

$$5\times(5\times10^{-5})=(5\pm m)\lambda'$$

$$\therefore \qquad \lambda'=\frac{25}{5\pm m}\times10^{-5}\text{cm}$$

Putting $m=1,2,3,\ldots$

$$\lambda'=(4.1,6.2,3.6,8.3,3.1,12.5,\ldots)\times10^{-5}\text{ cm}$$

Of these, the wavelengths in the visible region ($3.5\times10^{-5}-7.0\times10^{-5}$ cm) are,

$$= \mathbf{3.6\times10^{-5}, 4.1\times10^{-5}, 6.2\times10^{-5}\ cm}$$

Problem 18: A plane transmission grating having 5000 lines per cm is being used under normal incidence of light. (i) What is the maximum wavelength of light whose spectrum can be seen in the fourth order ? (ii) What is the highest order spectrum that can be seen for the light of wavelength 6000 Å. (iii) If the width of the opaque parts is double than that of transparent parts of the grating then which orders of spectra will be absent ?

Solution:

(i) The grating equation is,

$$(e+d)\sin\theta=n\lambda$$

The maximum value of $\sin\theta$ is 1 and $n=4$. Therefore, the maximum wavelength that can be seen is given by,

$$\lambda_{max}=\frac{(e+d)}{n}=\left(\frac{1}{5000}\text{ cm}\right)\times\frac{1}{4}=5\times10^{-5}\text{ cm}=\mathbf{5000\ Å}$$

(ii) Again, for $\lambda=6000$ Å

$$\therefore \qquad n_{max}=\frac{e+d}{\lambda}=\frac{(1/5000)\text{ cm}}{6\times10^{-5}\text{ cm}}=3.3$$

Thus, the highest visible order is **3.**

(iii) The condition of the absence of spectrum of order n is,

$$\frac{e+d}{e}=\frac{n}{m},\ \text{where } m=1,2,3,\ldots$$

If $d=2e$, then

$$n=3m=3,6,9,\ldots$$

Hence, the **3rd, 6th, 9th ..... order spectra would be absent.**

Problem 19: A plane transmission grating produces an angular separation of 0.01 radian between two wavelengths observed at an angle of 30°. If the mean value of the wavelength is 5000Å and the spectrum is observed in the second order, calculate the difference in the two wavelengths.

Solution: The grating equation is,

$$(e+d)\sin\theta = n\lambda \qquad \ldots(1)$$

Differentiating this equation w.r. to λ

$$(e+d)\cos\theta\, d\theta = nd\lambda$$

or
$$d\theta = \frac{n\,d\,\lambda}{(e+d)\cos\theta}. \qquad \ldots(2)$$

This is the angular separation between two wavelengths differing by $d\lambda$ observed at an angle θ.

Putting the value of $(e+d)$ in equation (2) from equation (1) we get

$$d\theta = \frac{n\,d\,\lambda}{\frac{n\lambda}{\sin\theta}\cos\theta} = \frac{d\,\lambda}{\lambda\cot\theta}$$

or
$$d\lambda = (\lambda\cot\theta)d\theta$$

Here given that, $\lambda = 5000$ Å, $d\theta = 0.01$ radian and $\theta = 30°$

Putting the values, we get

$$d\lambda = 5000\text{ Å} \times \cot 30° \times 0.01$$
$$= 5000 \times 1.732 \times 0.01\text{ Å}$$
$$= \mathbf{86.6\ Å}$$

Problem 20: What will be the diameter of a telescope objective required to resolve two stars separated by an angle of 10^{-3} degree ? Assume $\lambda = 5000$ Å.

Solution: The smallest angular separation $d\theta$ resolvable by a telescope objective of diameter D is given by,

$$d\theta = \frac{1.22\,\lambda}{D}\text{ radian}$$

or
$$D = \frac{1.22\,\lambda}{d\theta}.$$

Here given that, $\lambda = 5000\text{ Å} = 5\times10^{-5}$ cm and $d\theta = 10^{-3}$ degree $= \left(\frac{\pi}{180}\times10^{-3}\right)$ rad.

$\therefore$
$$D = \frac{1.22\times(5\times10^{-5}\text{ cm})}{\left(\frac{3.14}{180}\times10^{-3}\right)\text{rad}} = \mathbf{3.5\ cm}$$

Problem 21: Calculate the diameter of the image of a distant object formed by a lens of focal length 10 cm and diameter 2 cm with light of wavelength 5000 Å.

Solution: The angular separation $d\theta$ that can be resolved by a telescope objective of diameter D is given by,

$$d\theta = \frac{1.22\,\lambda}{D} \text{ radian}$$

Here given that, $\lambda = 5000 \times 10^{-8}$ cm, $D = 2$ cm

$$\therefore \quad d\theta = \frac{1.22 \times 5000 \times 10^{-8}}{2}$$

$$= 3.05 \times 10^{-5} \text{ radian}$$

The focal length of the lens is 10 cm. Therefore, the radius of the central image in the focal plane of the lens is,

$$r = f\,d\theta \qquad \text{(angle = radius/arc)}$$

$$= 10 \times 3.05 \times 10^{-5} = 3.05 \times 10^{-4} \text{ cm}$$

Hence, the diameter of the image $= 2r = \mathbf{6.1 \times 10^{-4}\ cm}$

Problem 22: What is the smallest angular separation between two point stars which a telescope of 10 cm diameter objective can resolve ? **($\lambda = 6000$ Å)**

Solution: The smallest angular separation $d\theta$ that can be resolved by a telescope objective of diameter D is given by,

$$d\theta = \frac{1.22\,\lambda}{D}$$

Here given that, $\lambda = 6000\ \text{Å} = 6 \times 10^{-5}$ cm and $D = 10$ cm

$$\therefore \quad d\theta = \frac{1.22 \times (6 \times 10^{-5}\ \text{cm})}{10\ \text{cm}} = \mathbf{7.32 \times 10^{-6}\ radian}$$

$$= (7.32 \times 10^{-6} \text{ radian}) \times \frac{180^\circ}{\pi} \text{ degree of arc}$$

$$= 7.32 \times 10^{-6} \times \frac{180^\circ}{3.14} \times 3600 \text{ second of arc}$$

$$= \mathbf{1.5\ second\ of\ arc.}$$

Problem 23: A spectrometer, having a circular scale reading upto 1 second, is fitted with a telescope of 1 inch objective. Estimate the resolution of the instrument and suggest any improvement to it for optimum use **($\lambda = 6000$ Å).**

Solution: The smallest angular separation $d\theta$ that can be resolved by a telescope objective of diameter D is given by,

$$d\theta = \frac{1.22\,\lambda}{D}$$

Here given that, $D = 1 \text{ inch} = 2.54 \text{ cm}$ and $\lambda = 6000 \text{ Å} = 6000 \times 10^{-8} \text{ cm}$

$$\therefore \quad d\theta = \frac{1.22 \times (6000 \times 10^{-8} \text{ cm})}{2.54 \text{ cm}}$$

$$= 3 \times 10^{-5} \text{ radian}$$

$$= 3 \times 10^{-5} \times \frac{180^\circ}{\pi}$$

$$= (172 \times 10^{-5}) \text{ degree of arc} \qquad (\pi = 3.14)$$

$$= 172 \times 10^{-5} \times 60 \times 60 \text{ second of arc}$$

$$= 6'' \text{ (seconds of arc).}$$

It is therefore, no use of having a scale reading upto 1''.

In order that the reading upto 1'' may be useful, the aperture of the telescope should be correspondingly increased so that the value of $d\theta$ may be 1''. That is, the diameter of the telescope objective should be increased to,

$$1 \times 6 = \textbf{6 inches}.$$

Problem 24: A Rowland grating has 6000 lines per cm. Find the angular separation between two yellow lines of neon 5882 Å and 5852 Å in the first order spectrum observed normally.

Solution: For a Rowland grating, we have that the condition on of maximum is,

$$(e + d) \sin i = n\lambda$$

For the first wavelength,

$$(e + d) \sin i_1 = n\lambda_1 \qquad \ldots(1)$$

Similarly, for the second wavelength

$$(e + d) \sin i_2 = n\lambda_2 \qquad \ldots(2)$$

Give that, $n = 1, e + d = \frac{1}{6000} \text{ cm}, \lambda_1 = 5882 \times 10^{-8} \text{ cm}$ and $\lambda_2 = 5852 \times 10^{-8} \text{ cm}$

From equation (1), we have

$$\sin i_1 = \frac{n\lambda_1}{e + d} = \frac{1 \times 5882 \times 10^{-8}}{1/6000} = 0.3529$$

$$i_1 = 20°36'$$

From equation (2), we have

$$\sin i_2 = \frac{n\lambda_2}{e + d} = \frac{1 \times 5882 \times 10^{-8}}{1/6000} = 0.3511$$

$$i_2 = 20°30'$$

Angular separation $= i_1 - i_2 = 20°36' - 20°30' = 6'$

$$= \textbf{1.74} \times \textbf{10}^{-3} \textbf{ rad.}$$

Problem 25: In a concave grating with Rowland mounting, two lines of known wavelengths 4886.24 Å and 4887.37 Å falls at linear positions 12.372 mm and 14.268 mm. Calculate the wavelength of the line whose linear position is 11.638 mm.

Solution: We have that Rowland mounting gives normal spectrum in which the distance between the spectral lines are proportional to the differences of their wavelengths, *i.e.*, $dx \propto d\lambda$

or $$\frac{d\lambda}{dx} = \text{constant}$$

Given that, $$d\lambda = (4887.37 - 4886.24)\,\text{Å} = 1.13\,\text{Å}$$

and $$dx = (14.268 - 12.372)\,\text{mm} = 1.896\,\text{mm}$$

Thus,

$$\frac{d\lambda}{dx} = \frac{1.13}{1.896} = 0.596\,\text{Å/mm} \qquad \ldots(1)$$

Let λ be the wavelength of the line whose linear position is 11.638 mm. Now we find $d\lambda$ and dx corresponding to the wavelength 4886.24 Å whose linear position is 12.732 mm.

$\therefore$ $$d\lambda = (4886.24 - \lambda)\text{Å}$$

and $$dx = (12.732 - 11.638)\,\text{mm} = 0.734\,\text{mm}$$

Thus, $$\frac{d\lambda}{dx} = \frac{4886.24 - \lambda}{0.734}\,\text{Å/mm} \qquad \ldots(2)$$

But, we have $$\frac{d\lambda}{dx} = \frac{1.13}{1.896} = \text{constant} \qquad \text{[Eq. (1)]}$$

So, equation (1) and (2) should be equal. Therefore,

or $$\frac{4886.24 - \lambda}{0.734} = 0.596$$

or $$4886.24 - \lambda = 0.596 \times 0.734 = 0.437$$

or $$\lambda = 4886.24 - 0.437 = \mathbf{4885.80\ Å}$$

Problem 26: Sodium yellow doublet has wavelengths 5890° and 5896 Å. What should be the resolving power of a grating to resolve these lines.

Solution: The resolving power of a grating is given by,

$$R = \frac{\lambda}{d\lambda}$$

Here given that, $$\lambda = \text{mean wavelength} = \frac{5890 + 5896}{2} = 5893\,\text{Å}$$

and $$d\lambda = 5896\,\text{Å} - 5890\,\text{Å} = 6\,\text{Å}$$

$\therefore$ $$R = \frac{5893\,\text{Å}}{6\,\text{Å}} = \mathbf{982}$$

Problem 27: A plane transmission diffraction grating has 40000 lines. Determine its resolving power in the second order ($n=2$) for a wavelength of 5000 Å.

Solution: The resolving power of the grating is given by,

$$R = Nn$$

Here, N = Total number of lines on the grating $=40000$ and $n=2$

$$\therefore \quad R = 40000 \times 2 = \mathbf{80000}$$

Problem 28: Show that for a transmission grating with one inch ruled space, the resolving power cannot exceed 5×10^4 at normal incidence for $\lambda = 5080$ Å.

Solution: The resolving power of a transmission grating is given by.

$$R = Nn$$

Using the grating equation, $(e+d)\sin\theta = n\lambda$, we can write above equation as,

$$R = \frac{N(e+d)\sin\theta}{\lambda}.$$

The maximum possible R will be for $\theta = 90°$.

Thus, $$R_{max} = \frac{N(e+d)}{\lambda}.$$

Where $N(e+d)$ is the total ruled width.

Here given that, $N(e+d) = 1$ inch $= 2.54$ cm and $\lambda = 5080$ Å $= 5.080\times10^{-5}$ cm

$$\therefore \quad R_{max} = \frac{2.54 \text{ cm}}{5.080\times10^{-5} \text{ cm}} = 5\times10^4$$

Problem 29: Light of wavelength = 5000 Å falls on a grating normally. Two adjacent principal maxima occur at $\sin\theta = 0.2$ and $\sin\theta = 0.3$ respectively. Calculate the grating element. If the width of the grating surface is 2.5 cm, calculate its resolving power in the second order.

Solution: For normal incidence, the position of a principal maximum of the order n is given by,

$$(e+d)\sin\theta = n\lambda$$

Two adjacent maxima of the orders (say) n and $n+1$ occur at $\sin\theta = 0.2$ and $\sin\theta = 0.3$.

Thus, $$(e+d)\times0.2 = n\lambda$$

and $$(e+d)\times0.3 = (n+1)\lambda$$

on subtraction, we get

$$(e+d)\times0.1 = \lambda$$

or $$(e+d) = \frac{\lambda}{0.1} = \frac{5000 \text{ Å}}{0.1} = \frac{5\times10^{-5}}{0.1} \text{ cm} \qquad [\because \lambda = 5000 \text{Å}]$$

$$= 5 \times 10^{-4} \text{ cm}$$

The width of the ruled surface is 2.5 cm. Therefore, the total number of rulings is,

$$N = \frac{2.5 \text{ cm}}{(e+d)} = \frac{2.5 \text{ cm}}{5 \times 10^{-4} \text{ cm}} = 5000.$$

The resolving power of the grating in the second order is,

$$R = Nn$$

Here given that, $N = 5000$ and $n = 2$

$\therefore$ $$R = 5000 \times 2 = \mathbf{10{,}000}$$

Problem 30: Find the minimum number of lines in a plane grating required to just resolve the sodium doublet (5890 Å and 5896 Å) in the (i) first order (ii) second order.

Sodium: The resolving power of a grating is given by,

$$R = \frac{\lambda}{d\lambda} = Nn \qquad \text{or} \qquad N = \frac{\lambda}{nd\lambda}$$

(i) Here given that, λ = mean wavelength = 5893 Å, $d\lambda = 6$ Å and for the first order, $n = 1$,

$\therefore$ $$N = \frac{5893}{1 \times 6} = \mathbf{982}$$

(ii) For second order $n = 2$

$\therefore$ $$N = \frac{5893}{2 \times 6} = \mathbf{491}$$

Problem 31: Calculate the least width of a plane diffraction grating having 500 lines per cm to resolve two sodium lines D_1 and D_2 (5896 Å and 5890 Å) in the second order.

Solution: The resolving power of a grating is given by,

$$R = \frac{\lambda}{d\lambda} = Nn$$

or $$N = \frac{1}{n}\frac{\lambda}{d\lambda}.$$

Here given that, $n = 2$; mean $\lambda = \dfrac{5890 + 5896}{2} = 5893$ Å and $d\lambda = 5896 - 5890 = 6$ Å.

$\therefore$ $$N = \frac{1}{2} \times \frac{5893}{6} = 491.$$

The grating has 500 lines per cm, therefore the least ruled width of the grating required for the given resolution,

$$= \frac{491}{500} \simeq \mathbf{1.0 \text{ cm}}$$

Problem 32: A plane transmission grating has 16000 lines to an inch over a length of 5 inches. In the wavelength region of 6000 Å, in the second order find, (i) the resolving power of the grating and (ii) the smallest wavelength difference that can be resolved.

Solution: (i) The resolving power of a grating is given by,

$$R = \frac{\lambda}{d\lambda} = nN$$

Here given that, $N = 16000 \times 5 = 80000, \lambda = 6000$ Å and $n = 2$

$\therefore$ resolving power $= 80000 \times 2 = \mathbf{160000}$

(ii) Smallest wavelength difference $d\lambda$ is given by,

$$\frac{\lambda}{d\lambda} = Nn = 160000$$

$\therefore$ $$d\lambda = \frac{\lambda}{160{,}000} = \frac{6000\text{ Å}}{160{,}000} = \mathbf{3.75 \times 10^{-2}\ Å}$$

Problem 33: Find, the possibility, If a 2 cm grating having 425 lines/cm can resolve the sodium D-lines (5890 Å 5896 Å) in the (i) second order spectrum. (ii) first order spectrum.

Solution: The resolving power of a grating is given by,

$$R = Nn$$

Here given that, $N = 2 \times 425 = 850$ (total lines on the grating surface)

and $n = 2$ (second order)

Putting the value, we get

$$R = 2 \times 850 = 1700$$

For the case of first order spectrum, the resolving power of the grating,

$$R = 1 \times 850 = 850$$

Now for *D*-lines, the resolving power $= \frac{\lambda}{d\lambda}$

Here given that, $\lambda = 5893$ Å, $d\lambda = 6$ Å

$\therefore$ $$\frac{\lambda}{d\lambda} = \frac{5893}{6} = 982$$

So, **second order spectrum can be resolved by this grating.**

Now, the **resolving power of grating is not sufficient to resolve *D*-line in the first order.**

Problem 34: In a grating the sodium doublet (5890 Å, 5896 Å) is viewed in third order at 30° to the normal and is barely resolved. Find the grating spacing and the total width of the ruling.

Solution: For normal incidence the grating equation is given by,

$$(e+d)\sin\theta = n\lambda$$

Here given that, $\theta = 30°, n = 3$ and $\lambda_{mean} = 5893$ Å

$$\therefore \quad (e+d) = \frac{n\lambda}{\sin\theta} = \frac{3\times 5893\times 10^{-8}}{\sin 30°} = 3\times 5893\times 10^{-8}\times 2$$

$$= 35358\times 10^{-8} \text{ cm}$$

So, the grating spacing = $\mathbf{3.536\times 10^{-4}}$ **cm**

Let N be the total number of rulings required to just resolve the doublet in the third order, then

$$\frac{\lambda}{d\lambda} = Nn$$

or

$$N = \frac{\lambda}{nd\lambda} = \frac{5893 \text{ Å}}{3\times 6 \text{ Å}} = \mathbf{327}$$

Total width of the ruled surface $= N(e+d)$

$$= 327\times 3.536\times 10^{-4}$$

$$= \mathbf{0.1156 \text{ cm}}$$

Problem 35: What must be the thickness of the base of a flint glass prism which would resolve the sodium D-lines ? Given that, $\frac{d\mu}{d\lambda} = 952 \text{ cm}^{-1}$, $d\lambda = 6$ Å ***and*** $\lambda = 5893$ Å.

Solution: The resolving power of a prism is given by,

$$\frac{\lambda}{d\lambda} = t\left(\frac{d\mu}{d\lambda}\right)$$

where t is the thickness of the base of the prism.

Given that, $\lambda = 5893 \text{ Å} = 5893\times 10^{-8}$ cm, $d\lambda = 6\times 10^{-8}$ cm

and $$\left(\frac{d\mu}{d\lambda}\right) = 952 \text{ cm}^{-1}$$

Putting all the values, we get

$$\frac{5893\times 10^{-8}}{6\times 10^{-8}} = t\times 952$$

or

$$t = \frac{5893}{6\times 952} = \mathbf{1.03 \text{ cm}}$$

Problem 36: Calculate the width of the base of a 60° prism made of thin glass which is just capable of resoling sodium D_1 and D_2 lines of wavelength 5896 Å and 5890 Å. The refractive index of the prism glass is 1.6545 and 1.6635 for wavelength 6563 Å and 5270 Å respectively.

Sodium: The resolving power of a prism is given by,

$$\frac{\lambda}{d\lambda} = t \times \frac{d\mu}{d\lambda}$$

where t is width of the base of the prism and is given by,

$$t = \frac{\lambda / d\lambda}{d\mu / d\lambda}$$

Given that,

$$\lambda = \frac{5896 + 5890}{2} = 5893 \text{ Å}$$

and $$d\lambda = 5896 - 5890 = 6 \text{ Å}$$

and $$\frac{d\mu}{d\lambda} = \frac{1.6635 - 1.6545}{(6563 - 5270)\text{ Å}} = \frac{0.0090}{1293 \text{ Å}}$$

$$= \frac{0.0090}{1293 \times 10^{-8} \text{ cm}} = 696 \text{ cm}^{-1}$$

Putting all the values, we get

$$t = \frac{5893 \times 10^{-8} / 6 \times 10^{-8}}{696} = \frac{5893}{6 \times 696}$$

$$= \mathbf{1.41 \text{ cm}}$$

Problem 37: A diffraction grating just resolve two lines $\lambda = 5140.35$ Å and 5140.85 Å in the first order. Will it resolve the lines 8037.20 Å and 8037.50 Å in the second order ?

[Meerut 2000B]

Solution: The smallest wavelength difference $d\lambda$ that can be resolved by a grating at wavelength λ in order n is given by,

$$\frac{\lambda}{d\lambda} = nN$$

Where N is the total number of lines on the grating

Here given that, $$\lambda = \frac{5140.35 + 5140.85}{2} = 5140.60 \text{ Å}$$

$$d\lambda = 5140.85 - 5140.35 = 0.50 \text{Å and } n = 1$$

$\therefore$ $$\frac{5140.60}{0.50} = N \times 1$$

or $$N = 10280$$

Now, for resolving 8037.20 Å and 8037.50 Å in the second order, the number of lines on the grating should be,

$$N = \frac{\lambda}{nd\lambda} = \frac{8037.35\ \text{Å}}{2 \times 0.30} = 13395$$

$$[\because \lambda = \frac{8037.20 + 8037.50}{2} = 8037.35 \text{ and } d\lambda = 8037.50 - 8037.20 = 0.30\ \text{Å}]$$

But these number of lines are much higher than the number of lines 10280 on the given grating.

Hence, **the gratings cannot resolve these wavelengths.**

Exercise

(A) Descriptive Type Questions

1. What do you understand by Fresnel's half period zones ? Prove that the area of a half period zone of a plane wavefront is independent of the order of the zone. Also prove that the amplitude due to a large wavefront at a point in front of it is just half of that due to the first half period zone acting alone. **[Meerut 2012, 10, 09, 07]**

2. Explain the term half period zone. Calculate the area of a Fresnel zone. **[Meerut 2009]**

3. Obtain intensity due to cylindrical wavefront by Fresnel half period zone method at an external point.

4. What is a zone plate and how is it made ? Explain how a zone plate acts like a convergent lens having multiple foci. Derive an expression for its focal length. **[Meerut 2011, 08B, 07, 05, 01]**

5. What is a zone plate ? Give its theory. Show that a zone plate has multiple foci. Compare the zone plate with a convex lens. What is meant by phase reversal zone plate ?

6. Describe with necessary theory, the Fresnel type of diffraction due to a straight edge. Show the intensity distribution in the diffraction pattern. **[Meerut 2009B, 08B, 08, 07B, 06B, 05, 04B, 03, 02, 01, 00]**

7. A sharp razor blade is held vertically in front of a beam of monochromatic light coming from a narrow slit parallel to the edge of the blade. Discuss the positions of diffraction bands and intensities observed on a screen placed behind the blade.

8. (i) Describe with necessary theory the Fresnel class of diffraction due to sharp razor blade. Draw its intensity distribution. **[Meerut 2006B]**

   (ii) How would you use it to determine the wavelength of light ? **[Meerut 2006B]**

9. Describe Fraunhofer diffraction pattern obtained with a narrow single slit illuminated by a parallel beam of monochromatic light. Also deduce the positions of maxima and minima and find the relative intensities of successive maxima. Plot the intensity distribution curve.

[Meerut 2011B, 10, 07B, 06, 05B, 04B, 03B, 02B, 00B]

10. Describe Fraunhofer diffraction due to a single slit and deduce the position of the maxima and minima. Show that the relative intensities of successive maxima are nearly 1: 1/22 : 1/61.

11. Describe phasor diagram method and integral calculus method to obtain Fraunhofer diffraction at a slit.

12. Discuss the fraunhofer diffraction at a circular aperture.

13. Define resolving power of an optical instrument. What do you mean by Rayleigh's criterion for the resolution of two images ?

[Meerut 2009B, 09, 08, 07, 05B, 05, 03B, 02B, 02, 01, 00B]

14. What do you understand by the resolving power of a telescope ? How would you measure it experimentally ?

[Meerut 2009, 08, 05B, 05]

15. Explain clearly what do you understand by the limit of resolution of a telescope and obtain an expression for it? Why do good telescopes have objectives of large apertures.

16. Describe resolving power of a microscope.

17. Explain Phase Contrast Microscope.

18. Explain Fraunhofer diffraction pattern by narrow N-parallel slits illuminated by a parallel beam of monochromatic light and obtain its intensity distribution.

19. Explain the construction and theory of a plane diffracting grating.

[Meerut 2012, 08B, 05B, 02, 01, 00]

20. Give the Construction and theory of a plane diffraction grating of transmission type and show how would you use it to find the wavelength of light.

[Meerut 2008B]

21. What do you mean by plane reflection grating and blazed grating ?

22. Describe the theory of a concave grating.

23. Describe different methods of mounting of concave grating. How Rowland mounting is used to determine the wavelength of light. Also, write advantages and disadvantages of them.

24. Give the theory of concave reflection grating. Describe the working of Rowland's mounting. Why are concave grating preferable to plane grating?

[Meerut 2006]

25. Deduce an expression for the resolving power of a plane transmission grating.

[Meerut 2011B, 09B, 07, 05, 04B, 01B]

26. Give a method for the experimental determination of the resolving power of a grating.

[Meerut 2005]

27. Obtain an expression for the resolving power of a prism.

28. Derive an expression for the chromatic resolving power of a prism placed in the position of minimum deviation and also, show that the chromatic resolving power of a prism is given by,

resolving power = prism's base width × dispersive power of prism.

(B) Short Answer Type Questions

1. What is difference between interference and diffraction ? **[Meerut 2008B]**

2. What is the difference between fresnal and fraunhofer's diffraction?

[Meerut 2008B, 02]

3. Compare a zone plate and a convex lens.

4. Explain phase reversal zone plate.

5. How we can determine the wavelength of light using Fresnel type of diffraction due to a straight edge ?

6. How would you use a plane diffraction grating to determine the wavelength of light ?

[Meerut 2005B]

7. Explain and obtain the condition of absent spectra in a plane transmission diffraction grating. If the width of slit e (transparent space) is equal to the width of opaque space d, which orders will be absent ? Which order will be absent if $d = 2e$?

[Meerut 2008, 01B]

8. Find the maximum number of orders available with a grating. Show that if the width of the grating element is less than twice the wave length of light then only first order is available.

[Meerut 2009B, 08, 03]

9. How may orders will be visible if the wavelength of incident radiation is 5000 Å and number of lines on the grating is 2000 per cm ?

10. Define dispersive power of a grating and obtain an expression for it.

[Meerut 2007B, 03B, 03, 02B, 02, 01B, 01]

11. Why a concave grating is preferable then a plane grating for spectroscopic work ?

12. What is a normal spectrum ?

13. How is the resolving power of a telescope determined in the laboratory using a rectangular slit ? Derive the formula used in it.

14. Explain dispersive power and resolving power of a grating and distinguish between both.

15. Deduce an expression for the half angular width of a principal maximum in the diffraction pattern of a grating having N slits. **[Meerut 2006]**

16. Explain the effect of the increase in number of lines on the grating (closeness of ruling) and the width of the ruled space on the grating spectrum. **[Meerut 2002B]**

17. Two plane diffraction grating A and B have same width of ruled surface but A has more number of lines than B. Compare intensity of fringes, width of principal maximum and dispersive powers.

18. What do you mean by Echelon grating ?

19. Why do good telescopes have objectives of large apertures ?

(C) Very Short Answer Type Questions

1. Define the phenomenon of diffraction. **[Meerut 2009, 08B, 06B, 05, 03, 02B]**

2. Define Fresnel's half period zones.

3. What is a zone plate ? **[Meerut 2011B]**

4. What do you mean by a plane transmission diffraction grating ?

5. Explain the resolving power of an optical instrument. **[Meerut 2012, 11, 10]**

6. Define the resolving power of a grating.

7. What are the advantages of increasing the number of rulings in a grating?

8. A plane diffraction grating is used with normal incidence. It has N lines, with spacing e each. Write down the expression for the direction of nth order principal maximum for wavelength λ.

(D) Multiple Choice Questions

1. A convex lens is used for getting :
   (a) Fresnel's diffraction
   (b) Fraunhofer's diffraction
   (c) Both types of diffraction
   (d) None of the above.

2. The intensity due to whole rectilinear wavefront at any external point O is equal to :
   (a) One fourth of the first half period zone
   (b) One third of the first half period zone
   (c) One sixth of the first half period zone
   (d) None of the above.

3. A zone plate behaves like :

(a) A convex lens (b) A concave lens

(c) A concave mirror (d) A convex

4. The focal length of a zone plate is given by :

(a) $\frac{r_n}{n\lambda}$ (b) $\frac{r_n^2\lambda}{n}$

(c) $\frac{r_n^2}{n\lambda}$ (d) $\frac{r_n \cdot n}{\lambda}$

5. The position of the nth minima in the diffraction band (due to straight edge) is given by :

(a) $x_n = k\sqrt{(2n+1)}$ (b) $x_n = k\sqrt{(2n)}$

(c) $x_n = k\sqrt{n}$ (d) $x_n = k\sqrt{(n-1)}$

6. Two plane diffraction gratings A and B have same width of ruled surface but A has more number of lines than B, then intensity of fringes of :

(a) A is greater than B (b) B is greater than A

(c) A and B are the same (d) None o the above.

(E) Fill in the Blank

1. In Fraunhofer diffraction pattern due to narrow N-parallel slits, the direction of principal maxima is given by ..... .

2. In Fraunhofer diffraction pattern due to narrow N-parallel slits, the direction of minima is given by $N(e+d)\sin\theta = \pm m\lambda$, where N is total number of slits, then the three value of m which are not possible are given ..... .

3. In a grating, let e be the width of each transparent and d be the width of each opaque portion, the $(e+d)$ is known as ..... .

4. For a rectangular slit, the minimum angle between two point sources whose images can be resolved by a telescope is ..... .

5. The expression for resolving power of a telescope is ..... .

6. The resolving power of a telescope is ..... to its objectives diameter.

7. The formula for the resolving power of a prism in terms its dispersive power is ..... .

8. The most compact mounting of concave grating is ..... .

(F) True/False

1. For a zone plate, focal length of red colour is less than that of violet.
2. In Fresnel diffraction due to a straight edge, the intensity due to entire half wavefront at a distant point is equal to the intensity due to first half period strip.
3. The position of the nth maxima in the diffraction band (due to a straight edge) is given by $x_n = k\sqrt{(2n+1)}$, where k is a constant.
4. In Fraunhofer diffraction at a single slit, the direction of minimum is given by, $e\sin\theta = \pm 2n\lambda$.
5. In Fraunhofer diffraction at a single slit, the direction of central principal maximum is the direction of the incident light.
6. The dispersive power of a grating is directly proportional to the grating element.
7. The number of lines on a grating is represented by, $\frac{1}{(e+d)}$.
8. The dispersive power of a grating is represented by, $\frac{\lambda}{d\lambda}$.
9. The resolving power of a grating is represented by $\frac{\lambda}{d\lambda}$.
10. The limit of resolution of a telescope is given by $1.22\,\frac{\lambda}{D}$, where D is the diameter of the telescope objective.
11. In Eagle mounting of a concave grating, the slit, grating and photographic plate always remain at the corners of a right angle triangle.

Answers

(B) Short Answer Type Questions

1. **Difference between interference and diffraction:**

   (i) Interference phenomenon is the result of the interaction of light waves from two different wavefronts originating from the same source, while diffraction is the interaction of light waves (secondary wavelets) of the same wave front.

   (ii) In an interference pattern the fringes may or may not be of the same width while in a diffraction pattern they are never of the same width.

   (iii) In an interference pattern all the bright fringes are of uniform intensity while in a diffraction pattern they are of varying intensity.

   (iv) In an interference pattern the regions of minimum intensity are perfectly dark while in a diffraction pattern they are not so.

4. **Phase reversal zone plate:** Instead of blocking the alternate zones of the zone plate, Wood in 1898, introduced an additional optical path of $\lambda/2$ between the waves from successive zones. Then the amplitude from successive zones helped each other. Such plates are known as "*phase reversal zone plates*". These plates are prepared in the following manner,

   A chemically cleaned glass plate is coated with a thin layer of gelatine and dried. It is, then sensitivitised by immersing it in a weak solution of potassium dichromate for a few seconds. Again it is dried but in the dark. Now, it is placed in contact of an ordinary zone plate and exposed to sunlight. Light passing through transparent zones acts on gelatine and makes it insoluble in water, while the gelatine in contact with opaque zones remains soluble in water. By immersing the glass plate in water the gelatine of unexposed parts is dissolved to such a depth that the optical paths from successive zones on the glass plate increases additionally by $\lambda/2$. This plate is known as phase reversal zone plate. Such plates produce four times more intense image than an ordinary zone plate.

5. The position of the nth maxima in this type of diffraction pattern is given by,

$$x_n = \sqrt{\frac{b(a+b)(2n+1)}{a}\lambda}$$

   where a is the distance between the source slit and straight edge, b is distance between screen and straight edge and λ is the wavelength of monochromatic light.

   The separation between the first and the nth maxima will be,

$$x_n - x_1 = \sqrt{\frac{b(a+b)(2n+1)}{a}\lambda} - \sqrt{\frac{b(a+b)}{a}\lambda} \qquad \text{[for first maximum } n=0\text{]}$$

$$= \sqrt{\frac{b(a+b)}{a}\lambda[(2n+1)-1]}$$

$$x_n - x_1 = \sqrt{\frac{b(a+b)\lambda}{a}.2n}$$

   Thus, by measuring the separation between the first and nth maxima $(x_n - x_1)$ and by noting the value of a and b, the wavelength of light λ can be determined.

6. **Determination of wavelength of light using plane diffraction grating:** The grating spectrum of a given source of light is obtained by using a spectrometer. The spectrometer is adjusted for parallel rays by Schuster's method. The eye-piece of telescope is focused on cross wires. The grating is adjusted for normal incidence of light. Then, the principal maxima is given by,

$$(e+d)\sin\theta = n\lambda$$

   where $(e+d)$ is the grating element, n is order of principal maxima and θ is the diffraction angle for the particular order of principal maxima. So, by noting the diffraction angle for particular order of principal maxima, we can determine the wavelength *i.e.*,

$$\lambda = \frac{(e+d)\sin\theta}{n}$$

Also, we can determine the value of grating element if we know the wavelength of light, *i.e.*,

$$(e+d) = \frac{n\lambda}{\sin\theta}$$

7. **Condition for Absent Spectra:** The principal maxima in the grating spectrum is given by,

$$(e+d)\sin\theta = n\lambda \qquad \ldots(1)$$

where n is order of the maxima

The minima in a single slit pattern are obtained at the position,

$$e\sin\theta = m\lambda \qquad \ldots(2)$$

where $m = 1,2,3\ldots.$,

If both equation (1) and (2) are simultaneously satisfied, a particular maximum of order n will be missing in grating spectrum. Thus, *if principal maximum of any order coincides with any minimum, then that particular maximum will be absent. This is known as condition of absent spectra*. To obtain this condition, we divide equation (1) by equation (2),

$$\frac{(e+d)\sin\theta}{e\sin\theta} = \frac{n\lambda}{m\lambda}$$

or

$$\frac{\mathbf{e+d}}{\mathbf{e}} = \frac{\mathbf{n}}{\mathbf{m}} \qquad \ldots(3)$$

This is the condition for the spectrum of the order n to be absent.

(i) If $d = e$, then by using equation (3),

$$\frac{e+e}{e} = \frac{2}{1} = \frac{n}{m}$$

or $\qquad n = 2m = 2,4,6,\ldots\ldots \qquad (m = 1,2,3)$

i.e., **the 2nd, 4th, 6th, ... order spectra will be absent.**

(ii) If $d = 2e$, then $\qquad \frac{e+2e}{e} = \frac{3}{1}\frac{n}{m}$

or $\qquad n = 3\,m \qquad (m = 1,2,3,\ldots\ldots)$

$n = 3,6,9,\ldots..$

i.e., **the 3rd, 6th, 9th, ... order spectra will be absent.**

8. **Maximum number of orders:** In the diffraction pattern of a diffraction grating illuminated normally by monochromatic light of wavelength λ, the principal maxima are obtained and the position is given by, $(e+d)\sin\theta = n\lambda$

where $(e+d)$ is the grating element, θ is the angle of diffraction and n is the order of maximum.

We can write also, $\qquad n = \frac{(e+d)\sin\theta}{\lambda}$

For a given grating element $(e+d)$ is constant and for constant λ, value of n will depend on $\sin\theta$. Now, the maximum possible value of the angel of diffraction is $90°$ and $\sin 90° = 1$. Therefore, the maximum possible order is given by $\mathbf{n = \frac{e+d}{\lambda}}$

This expression shows that if $(e+d) < 2\lambda$, than $n_{max} < 2\lambda/\lambda$ or $n_{max} < 2$

i.e., **only first order spectrum will be obtained.**

9. For a grating the condition of maxima is given by,

$$(e+d)\sin\theta = n\lambda$$

For maximum possible order, $\theta = 90°$

$\therefore$ $$(e+d) = n\lambda \text{ or } n = \left(\frac{e+d}{\lambda}\right)$$

Given that, $\lambda = 5000 \times 10^{-8}$ cm and $(e+d) = \frac{1}{2000}$ cm^{-1}

So, $$n = \frac{1}{2000 \times 5000 \times 10^{-8}} = \frac{10^8}{10^7} = \mathbf{10}$$

10. Dispersive power of a grating: The dispersive power of a grating is defined as the ratio of the difference in the angle of diffraction of any two neighbouring spectral lines to the difference in wavelength between the corresponding spectral lines. It is represented as $d\theta/d\lambda$. In other words we can say that dispersive power of a grating is the difference in the angle of diffraction per unit change in wavelength. The angle of diffraction of the nth order principal maximum for a wavelength λ, is given by,

$$(e+d)\sin\theta = n\lambda$$

Differentiating this equation with respect to λ, we get

$$(e+d)\cos\theta \frac{d\theta}{d\lambda} = n$$

or $$\frac{d\theta}{d\lambda} = \frac{n}{(e+d)\cos\theta} \quad \ldots(1)$$

This is the expression for the dispersive power of a grating. From above equation it is clear that the dispersive power of the grating is:

(i) Directly proportional to the order of the spectrum.

(ii) Inversely proportional to $\cos\theta$. Larger the value of θ, higher will be dispersive power. (Since the angle of diffraction for red colour is greater than the violet in the spectrum, so the dispersion in the red region is more than in the violet region).

(iii) Inversely proportional to the grating element $(e+d)$.

11. For spectroscopic work we prefer concave grating over the plane grating due to following reasons:

(i) Additional lenses are not required for focusing the spectrum and thus chromatic aberration is not developed in the spectrum.

(ii) Concave grating is used at grazing incidence. Thus, the spectrum obtained by it is more intense and its photograph can be taken.

(iii) It can be used in ultraviolet.

(iv) Wavelength can be measured more accurately with concave grating.

12. A normal spectrum is one, in which the distance between the spectral lines is proportional to the difference of their wavelengths. Thus, if dx be the change in the linear positions of spectral lines (or $d\theta$ be the angular change) corresponding to the change $d\lambda$ in wavelength, then $\frac{dx}{d\lambda}$ or $\frac{d\theta}{d\lambda}$ must be constant for normal spectrum.

13. Experimental determination of resolving power of a telescope: To determine the resolving power of a telescope, we require an adjustable slit and a pair of close and parallel illuminated slits. The adjustable slit is fitted just in front of the objective of the telescope whose resolving power is to be determined. The pair of slits is prepared by cutting to fine parallel lines on a glass plate coated with black colour.

The telescope is placed at sufficient distance from the illuminated slits and directed towards them. The adjustable slit is kept widely opened. The eyepiece of the telescope is focussed on the cross wires. The distance between the cross wires and the objective is varied until well defined images of the slits are seen. Now, the width of the adjustable slit is gradually decreased until the limit of resolution is reached. This happens when, even a slight further decrease in the width of the slit results in the blending of the two images into one. The width of the slit 'a' is measured by means of a microscope. The resolving power of telescope is, then calculated by the following formula,

$$R = \frac{\lambda}{a}$$

where λ is the wavelength of light illuminating the slits.

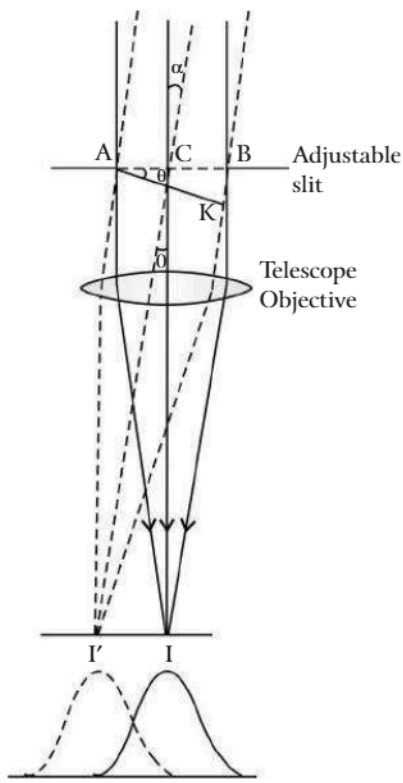


Fig. 2.37

Derivation of the formula: Suppose a parallel beam of monochromatic light starting form a distant object O is falling normally on the adjustable slit AB [Fig. 2.37].

According to Huygen's principle, each point in *AB* is sending secondary wavelets in all directions. Suppose the wavelets travelling normally to the slit. These wavelets meet at the focus *I* of the objective in the same phase, because they start from *AB* in the same phase and cover equal distances from *AB* to *I*. Hence, *I* is the central maximum of the diffraction pattern.

Similarly, we can assume the wavelets diffracted in a direction θ to the normal at the slit are focused at a point *I′* in the focal plane of the objective. Now, we draw a line *AK* normal to the rays. The path difference between wavelets starting from the extreme points *A* and *B* and moving the direction θ is *BK*. If *BK* is equal to λ, then the wavelets from *A* and *C* (the mid point of *AB*) shall reach *I′* with a path difference λ/2. Further, corresponding to each point in *AC* there will be point in *CB* such that the wavelets from them reach *I′* with a path difference of λ/2. The point *I′* will therefore, be the first minimum of the pattern. Thus, for the first minimum, we must have

$$BK = a\sin\theta = \lambda$$

where a is the width of the slit.

or $$\sin\theta = \lambda/a$$

If $a >> \lambda$, so $\sin\theta$ is small, then $\sin\theta \approx \theta$. Thus,

$$\theta = \lambda/a$$

Since, the central maximum is obtained in the direction, $\theta = 0$, So, λ/a is the angular separation between the central maximum and the first minimum.

Now, suppose there is another object *O′* situated near *O*. Its central maximum will be obtained in the focal plane of the objective on the line joining *O′* with the optical centre of objective.

According to Rayleigh's criterion, *O* and *O′* will be just resolved when the central maximum of *O′* falls at *I′*, the first minimum of *O*. Hence, at the limit of resolution, the angular separation between the central maxima of *O* and *O′* is

$$\theta = \frac{\lambda}{a}$$

From the Fig.2.37, it is clearly equal to α, the angle subtended by the objects *O* and *O′* at the objective. If *d* is the linear separation between the objects *O* and *O′* and *D* is their distance from the telescope objective, then

$$\alpha = \frac{d}{D}$$

Then, d/D is called the practical resolving power of a telescope, while λ/a is known as the theoretical resolving of the telescope.

15. **Angular half width of a principal maximum for a grating:** For a grating, the nth order principal maximum obtained in the direction θ_n is given by,

$$(e+d)\sin\theta_n = n\lambda \qquad ...(1)$$

Let the first minimum adjacent to the nth maximum be obtained in the direction $(\theta_n + d\theta_n)$. Then $d\theta_n$ is called the angular half width of the nth maximum.

The minima are obtained in the direction given by,

$$N(e+d)\sin\theta = m\lambda \qquad ...(2)$$

where N is the total number of ruling in the grating and m = any integer except $0, N, 2N \ldots nN$, because at these values of m, we get 0th , Ist, 2nd .... nth order principal maxima.

So, for getting adjacent minimum to nth order principal maxima, the value of $m = nN + 1$.

Hence, from equations (2), we get

$$N(e+d)\sin(\theta_n + d\theta_n) = (nN+1)\lambda$$

or $\quad N(e+d)[\sin\theta_n \cos d\theta_n + \cos\theta_n \sin d\theta_n = (nN+1)\lambda$

$\because$ $\quad d\theta_n$ is very small, $\quad \therefore \quad \cos d\theta_n = 1$ and $\sin d\theta_n = d\theta_n$

So, $\quad N(e+d)\sin\theta_n + N(e+d)\cos\theta_n d\theta_n = (nN+1)\lambda$

But from equation (1), $\quad (e+d)\sin\theta_n = n\lambda$

$\therefore \quad nN\lambda + N(e+d)\cos\theta_n\, d\theta_n = (nN+1)\lambda$

or $\quad N(e+d)\cos\theta_n\, d\theta_n = \lambda$

or $\quad d\theta_n = \dfrac{\lambda}{N(e+d)\cos\theta_n}$

This is the required expression, by Using equation (1), it can be written as,

$$d\theta_n = \frac{1}{\mathbf{Nn\cot\theta_n}}$$

16. If the number of lines on the grating increases, then $(e+d)$ decreases. Hence the dispersive power, $\dfrac{d\theta}{d\lambda} = \dfrac{n}{(e+d)\cos\theta}$ will be large. Therefore, the angular spacing between the maxima of two wavelengths will become large.

Now, the angular half width of principal maxima, $d\theta_n = \frac{\lambda}{N(e+d)\cos\theta_n}$, where $N(e+d)$ is the total of ruled surface.

If width of ruled surface increases, then the angular half width $d\theta_n$ decreases *i.e.*, sharper the maxima.

17. Since, the intensity of principal maxima is proportional to the square of the number of lines (N^2) on the grating. Therefore, principal maximum of grating *A* has greater intensity.

The angular half width of principal maxima is inversely proportional to the total width of the ruled surface. Hence, it is same for both grating *A* and *B*.

The dispersive power is inversely proportional to the grating element. Since grating *A* has greater number of lines on the same width of ruled surface than *B*. So, *A* has smaller grating element and hence a larger dispersive power than *B*.

18. The resolving power of a grating is given by,

$$= Nn$$

where N is the total number of lines on the grating and n is the order of spectra. Naturally to increase the resolving power, we have to increase N and n. The limit of N is 100000 but at these lines we can not get the spectrum higher than third order because the intensity is very poor in the higher orders.

Michelson, in 1898, devised a special type of grating in which N is kept small (20 to 40) whereas the order of spectrum n is made very high with no loss in intensity. This type of grating is known as *Echelon grating*.

19. The limit of resolution of the telescope $= 1.22\frac{\lambda}{D}$ radian, where D is the diameter of telescope objective.

Hence, larger is the diameter of apertures of telescope, smaller is the limit of resolution *i.e.*, higher the resolving power. Therefore, telescope with larger objective form sharper images and are capable of resolving the objects of smaller angular separation. The images are also brighter because lens with larger aperture collects more light.

(D) Multiple Choice Questions

1.	(b)	**2.**	(a)	**3.**	(a)	**4.**	(c)	**5.**	(b)	**6.**	(a)

(E) Fill in the Blank

1.	$(e+d)\sin\theta = \pm n\lambda$, where n is the order of spectrum	**2.**	$(0, N, 2N)$
3.	grating element or grating constant	**4.**	λ/D
5.	$\lambda/d\lambda$ or nN	**6.**	directly proportional
7.	resolving power of a prism = prism's base width × dispersive power of the prism		
8.	eagle mounting		

(F) True/False

1.	T	**2.**	T	**3.**	T	**4.**	F	**5.**	T	**6.**	T	**7.**	T
8.	F	**9.**	T	**10.**	T	**11.**	F						

○○○

POLARISATION OF LIGHT

- Transverse Nature of Light
- Polarisation
- Distinction Between Polarised and Unpolarised Light
- Representation of Light Waves
- Brewster's Law
- Polarisation by Reflection (Pile of Plates)
- Double Refraction and Doubly Refracting Crystals
- Nicol's Prism
- Huygen's Theory of Double Refraction in Uniaxial Crystals
- Polaroids
- Phase and Retardation Plate
- Quarter Wave Plate
- Half Wave Plate
- Babinet Compensator
- Plane, Circularly and Elliptically Polarised Light
- Production of Elliptically and Circularly Polarised Light
- Linearly Polarised Light

- Production of Polarization Preserved Light
- Production of Coaxial Overlay Preserved Light
- Production of Electrically Polarized Light
- Double Image Prism
- Optical Rotation
- Specification Sheet
- Big Quiz Preview

UNIT III

POLARISATION OF LIGHT

3.1 Transverse Nature of Light Wave

Experiments based on interference and diffraction phenomena have shown that light is some form of wave motion. But these phenomena do not tell about the type of wave *i.e.*, waves are longitudinal or transverse. The phenomenon of polarisation shows that the light waves are transverse waves.

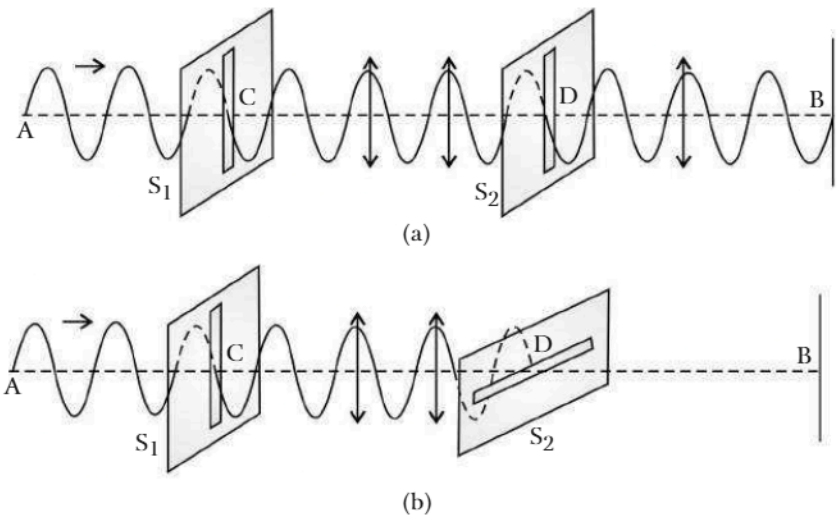


Fig. 3.1

To understand the transverse nature of waves, let a rope AB be passed through two parallel slits S_1 and S_2. The one end of the rope is fixed at point B. Hold the end A and move the rope up and down perpendicular to AB [Fig. 3.1(a)]. A wave emerges along CD due to transverse vibrations parallel to the slit S_1. If the slit S_2 is parallel to S_1 then the wave passes through it with undiminished amplitude. If, however, S_2 is rotated to become perpendicular to S_1, the slit S_2 does not allow the wave to pass through it [Fig. 3.1(b)].

If the end A is moved in a circular manner the rope will show circular motion up to the slit S_1. Beyond S_1, it will show only linear vibrations parallel to slit S_1 because the slit S_1 will stop the other components. Again if S_2 is perpendicular to S_1 then, the rope will not show any vibration beyond S_2.

If the longitudinal waves are set up by moving the rope AB forward and backward, the waves will pass through S_1 and S_2 irrespective of their position. A similar phenomenon has been observed in light when it passes through a tourmaline crystal. Thus, light waves are transverse in nature.

3.2 Polarisation of Light

When an ordinary light is made incident normally on a tourmaline crystal A which is cut parallel to its axis, the crystal A will act as the slit S_1 Fig.3.2(a). On rotating the crystal A, no remarkable change is noticed in the intensity which we are getting. Let another crystal B be placed parallel to A Fig.3.2(b). Now

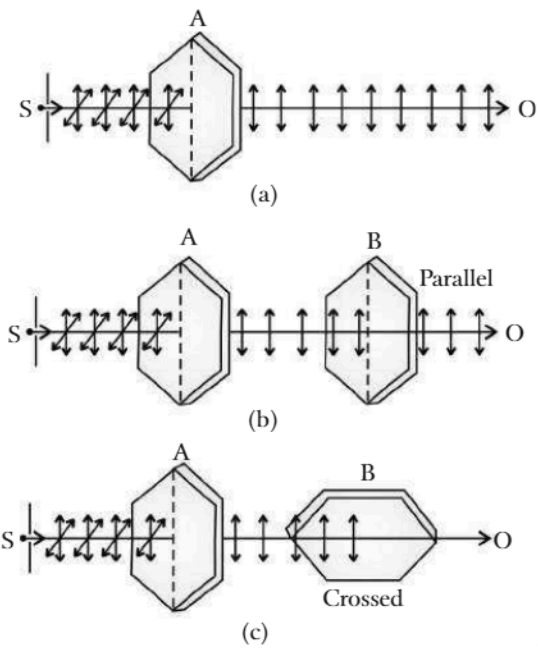


Fig. 3.2

1. Rotate both the crystal together so that their axes are always parallel. No change in the intensity of the light coming out of B is observed.

2. Keep the crystal A fixed and rotate the crystal B [Fig.3.2(c)]. The light transmitted through B gradually decreases and no light comes out of it when B is perpendicular to A. If the crystal B is further rotated, the intensity of light coming out of it gradually increases and is maximum again when the two crystals are parallel.

   This experiment shows that light does not propagate as longitudinal wave but it propagates as transverse waves.

Thus, it is clear that *on passing the light through a tourmaline crystal, the light waves are confined to a particular direction in a plane perpendicular to the direction of propagation of light*. [Fig. 3.2(a)] *The light which acquires the property of one-sideness is called polarised light and this phenomenon is called polarisation of light*. According to the electromagnetic theory of light, a light wave consists of electric and magnetic vectors vibrating in mutually perpendicular planes and both the planes are perpendicular to the direction of propagation of light. The electric vector acts as the light vector. In an unpolarised light, the light vector takes on all possible directions of vibration in a plane perpendicular to the direction of propagation. *But if the light vector vibrates only along a straight line in the plane, perpendicular to direction of propagation the light is said to be plane polarised or linearly plane polarised light.*

3.3 Distinction Between Polarised and Unpolarised Light

When an ordinary light is passed through a single rotating tourmaline crystal, there is no change in the intensity of the outcoming light from the crystal. It means that ordinary beam of light is symmetrical about its direction of propagation, *i.e.*, it consists of large number of light waves vibrating in all possible directions with equal probability in a plane perpendicular to the propagation (Fig.3.3). It is therefore said to be *unpolarised light*. Hence, the intensity of light emerging from the crystal is same in all the directions of the crystal.

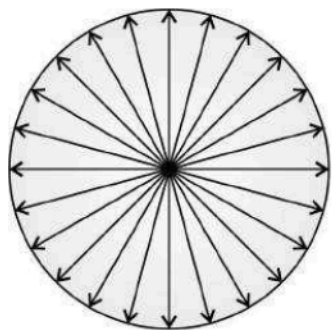


Fig. 3.3

But the polarised light, (obtained by passing it through tourmaline crystal), is not symmetrical about the direction of propagation of the light *i.e.*, vibrations are confined in a particular direction. Now, if this light is seen again through another rotating crystal, there will be a change in the intensity of light and it will be minimum when the axes of both the crystals are perpendicular to each other.

3.4 Representation of Light Vibrations

In the case of an unpolarised light beam, vibrations in all possible direction perpendicular to that of propagation of light are possible. Hence, it is represented by a star on a line Fig. 3.4(a). Also we know that a beam of ordinary light (unpolarised light) consists of two sets of waves vibrating in a plane perpendicular to the direction of propagation and at right angles to each other. Fig. 3.4(b) shows the direction of propagation of the beam (in the plane of the paper) and two sets of waves vibrating at right angles to each other, one in the plane of paper represented by *arrows* and second in the plane perpendicular to plane of the paper represented by *dots*.

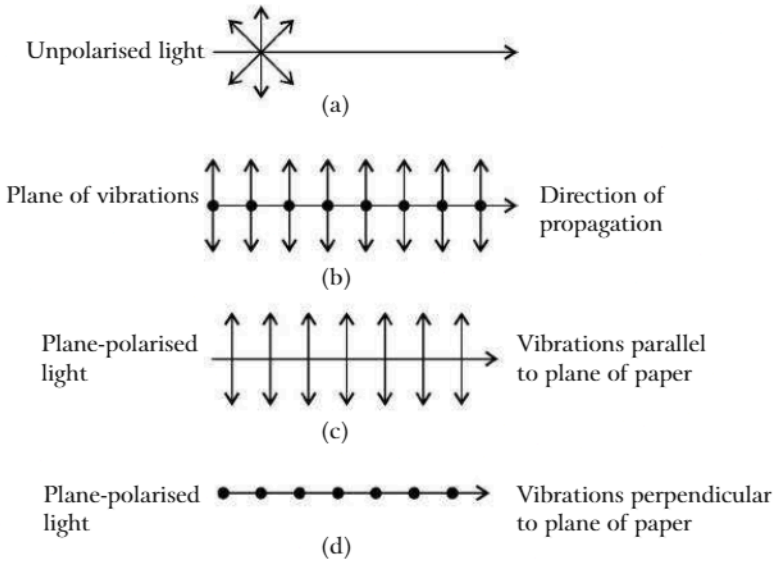


Fig. 3.4

In a plane polarised beam of light, the vibrations are in a single direction. If the vibrations are parallel to the plane of the paper, they are represented by arrows Fig. 3.4(c). If the vibrations are along a straight line perpendicular to the plane of the paper, they are represented by dots Fig. 3.4(d).

3.5 Brewster's Law

In 1808, Malus discovered that when an unpolarised light falls on the surface of any transparent substance, (*e.g.*, glass) the reflected and refracted rays are partially plane polarised and the degree of polarisation varies with the angle of incidence.

In 1811, Brewster found that ordinary light is completely polarised in the plane of incidence when it is reflected from a transparent substance at a particular angle of incidence (also the angle of reflection), called polarising angle. He discovered that there is a relation between the polarising angle p and the refractive index μ of the transparent substance with respect to surrounding medium. This is known as Brewster's law, and is given by,

$$\mu = \tan p$$

Now, let a beam AB of unpolarised light be incident on the surface of transparent substance at the polarising angle p. This beam is reflected along BC and refracted along BD (Fig. 3.5). Let r be the angle of refraction, then from Snell's law,

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin p}{\sin r} \qquad ...(1)$$

And from Brewster's law,

$$\mu = \tan p = \frac{\sin p}{\cos p} \qquad ...(2)$$

From equation (1) and (2), we have

$$\frac{\sin p}{\sin r} = \frac{\sin p}{\cos p}$$

or $$\cos p = \sin r$$

$$\sin(90° - p) = \sin r$$

or $$90° - p = r$$

or $$r + p = 90°$$

As $r + i = r + p = 90°$, $\angle CBD$ is also equal to 90°.

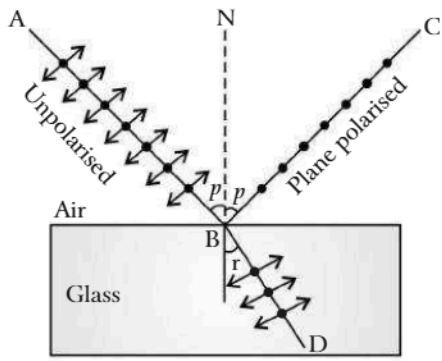


Fig. 3.5

Therefore, the reflected and the refracted rays are at right angle to each other.

From Brewster's law, also it is clear that for crown glass of refractive index 1.52, the polarising angle is given by,

$\mu = \tan p$ or $\tan p = 1.52$ or $p = \tan^{-1}(1.52)$ $p = 56.7°$

As the refractive index of a substance varies with the wavelength of the incident light, the polarising angle will be different for light of different wavelengths.

Example 1: A ray of light is incident on the surface of a glass plate of refractive index 1.5 at the polarising angle. What is the angle of refraction ?

Solution: Let p be the angle of polarisation and r the corresponding angle of refraction.

From Brewster's law, we have

$$\mu = \tan p$$

Here given that, $\mu = 1.5 \Rightarrow \tan p = 1.5$

or $$p = \tan^{-1}(1.5) = 56.3°$$

Also we have $$r + p = 90°$$

$\therefore$ $$r = 90° - p = 90° - 56.3° = \mathbf{33.7°}$$

Example 2: The refractive index for water is 1.33. Calculate the polarising angle for water.

Solution: According to Brewster's law, the polarising angle p is given by, $\mu = \tan p$

Here given that, $\mu = 1.33$ (For air to water)

$\therefore$ $\tan p = 1.33$ or $p = \tan^{-1}(1.33) = 53°$

3.6 Polarisation by Refraction (Pile of Plates)

When an ordinary light is incident on the glass surface at the polarising angle p Fig. 3.6 a small fraction is reflected which is completely plane polarised with vibrations perpendicular to the plane of incidence, while the major fraction is refracted and is partially polarised having vibrations both in the plane of incidence as well as perpendicular to the plane of incidence. Now, this refracted light is incident at the lower face at an angle r, where r is the angle of refraction at the upper face. From this face again the reflected light is completely plane polarised having vibrations perpendicular to the plane of incidence, while that refracted into the air is partly plane polarised.

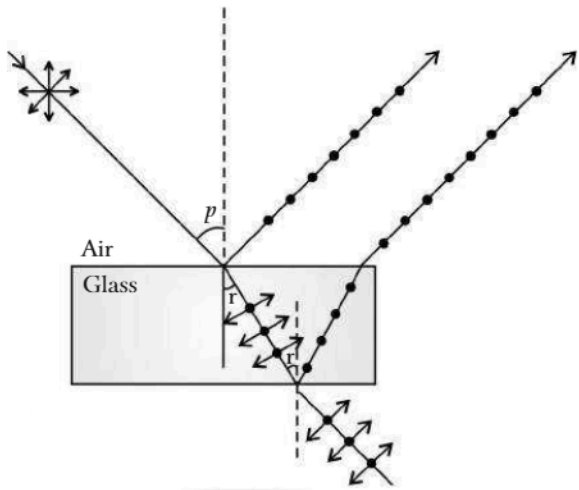


Fig. 3.6

Hence, if a beam of unpolarised light be incident on the *pile of plates* at the polarising angle Fig. 3.7 some of the vibrations perpendicular to the plane of incidence are reflected at each surface and all those parallel to it are refracted. The net result is that the refracted beam becomes poorer and poorer in the perpendicular components. The process continues and for a good number of plates (15 to 20) the transmitted light is completely free from the vibrations perpendicular to the plane of incidence and is having vibrations only in the plane of incidence *i.e.*, it will be almost plane polarised. Thus, we get plane polarised light by refraction using a pile of plates.

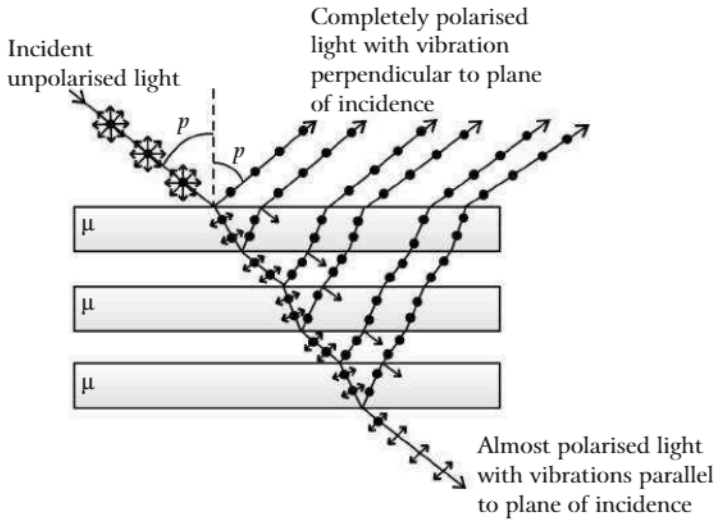


Fig. 3.7

3.7 Double Refracting Crystal and Double Refraction

When a beam of unpolarised light is incident on a calcite or quartz crystal, there are two refracted beams in place of the usual one as in the case of glass. Such crystals are said to be double refracting crystals and this phenomenon is called double refraction. It was first of all discovered by Erasmus Bartho Lines in 1669.

Normally calcite *i.e.*, calcium carbonate ($CaCO_3$) is used to get double refraction. Calcite, also known as Iceland spar (found in Iceland) is a colourless crystal, transparent to visible as well as ultraviolet light. The crystals are of two types, *uniaxial and biaxial. In uniaxial crystal, there is one direction called the optic axis along which the two refracted rays travel with the same velocity*. The examples are calcite, tourmaline and quartz. In biaxial crystals there are two optic axes. The examples are topaz and aragonite.

One of these two refracted rays is found to obey the laws of refraction, *i.e.*, it always lies in the plane of incidence and its velocity in the crystal is same in all the directions. This ray is known as the *ordinary ray* (*O-ray*). The other refracted ray does not obey the law of refraction. It travels in the crystal with different velocities in different directions. Hence, it is called the *extraordinary ray* (*E-ray*). The ordinary ray

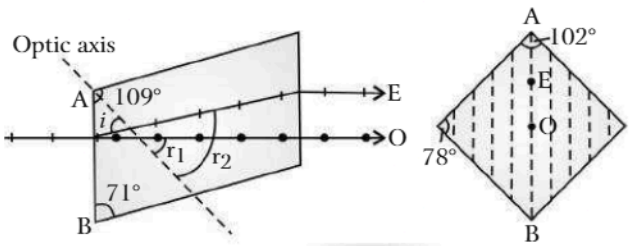


Fig. 3.8

has a refractive index, $\mu_0 = \frac{\sin i}{\sin r_1}$ and extraordinary ray has refractive index, $\mu_e = \frac{\sin i}{\sin r_2}$, where i is the angle of incidence, r_1 and r_2 are the angle of refractions for O and E-rays respectively.

In case of O-ray, refractive index is constant (as it obeys the laws of refraction) while for E-ray the refractive index varies with angle of incidence and is not fixed.

To see this phenomenon allow a narrow beam of light from point source to pass through a crystal of Iceland spar or calcite, two images will be obtained on the fig. 3.8.

3.7.1 Optic Axis of Uniaxial Crystals

Calcite crystal or Iceland spar is found in nature in different forms, all of which can be reduced by cleavage or breakage into a rhombohedron as shown in fig. 3.9. Each of the six faces of this crystal is a parallelogram having angles of 102° and 78° nearly. At two opposite corners A and G of the rhombohedron all the three angles are obtuse. These corners are known as *blunt corners* of the crystal. A line passing through any one of the blunt corners and making equal angles with the three faces which meet there, is the direction of the *optic axis* of the crystal. Optic axis is a direction not a line. In fact any line parallel to this line represents the optic axis.

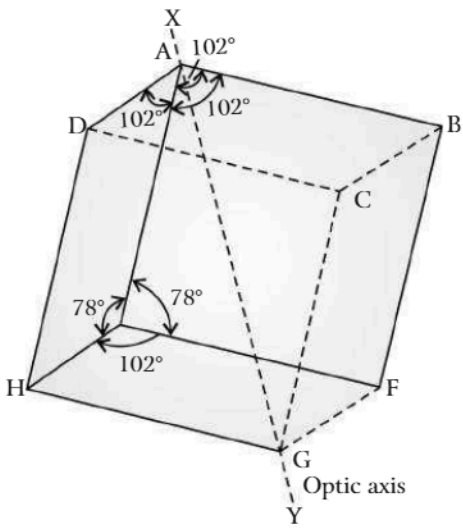


Fig. 3.9

If the rhombohedron is so cut that all of its three edges are equal as shown in fig. 3.9, then the line joining the opposite blunt corners (AG) or any line parallel to it gives the direction of the optic axis. If a ray of light is incident along the optic axis, it will not split into two rays, *i.e.*, their velocities remain the same along the optic axis. Thus, the phenomenon of double refraction is absent when light is allowed to enter the crystal along the optic axis. Hence, the crystal is symmetrical about the optic axis.

3.7.2 Principal Section and Principal Plane of the Crystal

A plane containing the optic axis of the crystal and perpendicular to a pair of opposite refracting faces is called a principal section of the crystal. As a crystal has six faces so for every point there are three principal sections. A principal section always cuts the surface of a calcite crystal in a parallelogram with angles 109° and 71°. A plane in the crystal drawn

through the optic axis and the ordinary ray is defined as the *principal plane of the ordinary ray*. Similarly, a plane in the crystal drawn through the optic axis and the extraordinary ray is defined as the *principal plane of the extraordinary ray*. In general, the two planes do not coincide.

It has been found that both ordinary and extraordinary rays are plane polarised. The vibrations of ordinary ray are perpendicular to the principal section of the crystal while the vibrations of E-ray are in the principal section of crystal. Hence, both the rays are plane polarised and their vibrations are at right angles to each other.

3.8 Nicol's Prism

The Nicol's prism is an optical device made from calcite crystal for producing and detecting plane polarised light. It was invented by William Nicol in 1828, who was an expert in cutting and polishing gems and crystals. The Nicol prism is made in such a way that it eliminates one of the two rays (produced by calcite crystal) by total internal reflection. Generally, the ordinary ray is eliminated and the *E*-ray is transmitted through the prism.

3.8.1 Construction

A calcite crystal *ABCD* whose length is three times of its breadth, is taken. Its end faces *AB* and *CD* are grounded in such a way that the angles in the principal section become 68° and 112° instead of 71° and 109°. The crystal is then cut into two parts along the plane A_1C_1 passing through the blunt corners and perpendicular to both the principal sections and the end faces, so that A_1C_1 makes an angle of 90° with ends C_1D and A_1B as shown in fig. 3.10. These two surfaces are grounded and polished to make then optically flat. They are then joined together by a thin layer of Canada balsam which is a clear transparent substance of refractive index 1.55 for sodium light.

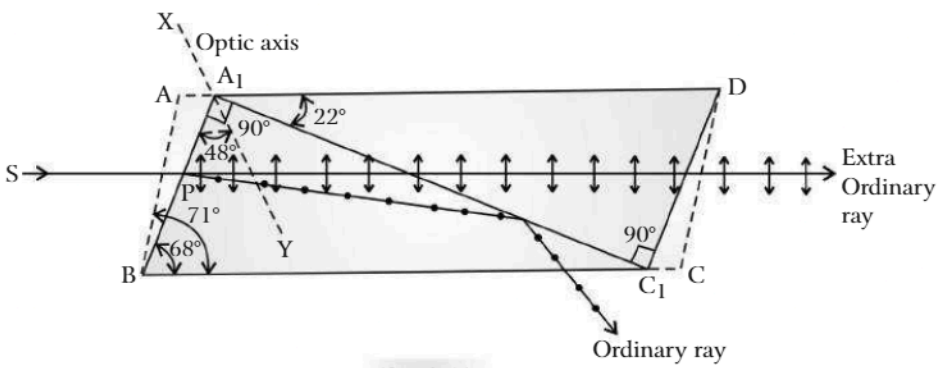


Fig. 3.10

3.8.2 Working

When an unpolarised light ray *SP* falls on the face A_1B, it splits up into refracted rays, *O*-ray and *E*-ray. Both the rays are plane polarised. The *O*-ray has vibrations perpendicular to the principle section and *E*-ray has vibrations parallel to the principal section of the crystal.

The refractive index of calcite for ordinary ray, $\mu_o = 1.658$ and for extraordinary ray is $\mu_e = 1.486$. As refractive index of Canada balsam for sodium light (=1.55) lies between μ_o and μ_e so, it is clear that Canada balsam acts as a rarer medium for an ordinary ray and it acts as a denser medium for extraordinary ray. Therefore, when *O*-ray passes from calcite crystal to Canada balsam layer it passes from denser to rarer medium. If the angle of incidence is greater than critical angle, the *O*-ray is totally internally reflected while the extraordinary ray is not affected and therefore transmitted through the prism. In this way, unpolorised light ray becomes plane polarised on passing through Nicol prism. The critical angle for ordinary ray =69°, as $\theta = \sin^{-1}[1/\mu] = 1.550/1.658$, where μ = refractive index for ordinary ray relative to Canada balsam. Since the length or crystal is large so, angle of incidence for *O*-ray becomes greater than critical angle and we obtain a beam of plane polarised light with vibrations parallel to principal section of the prism.

If the incident ray makes angle much smaller than *SPB* (with the face A_1B) the *O*-ray will strike the calcite balsam surface at an angle less than the critical angel (69°). Therefore, the *O*-ray will also be transmitted with *E*-ray. Now, if the incident ray makes an angle much greater than *SPB*, the *E*-ray will become more parallel to the optic axis XY so that its refractive index will be increased and become greater than that of the balsam. Then the *E*-ray will also be totally reflected from the calcite balsam surface and no light will emerge from the Nicol. Hence, to obtain a plane polarised light, the incident beam should not be too wide. This limits the angle of incidence of the *E*-ray about 14°. Hence, a Nicol's prism cannot be used for highly convergent or highly divergent light.

3.8.3 Nicol's Prism as an Analyser

The Nicol's prism can also be used for detection of plane polarised light, *i.e.* as an analyser. When two Nicol's prism P_1 and P_2 are placed coaxially as shown in fig. 3.11(a) they constitute a *polariscope*; the first prism acts as polariser and the second is analyser.

When an unpolarised ray of light is incident on a Nicol's prism P_1 the ray (*E*-ray) emerging from P_1 is plane polarised with vibration in the principal section of P_1. If this ray is incident on a second Nicol's prism P_2 as shown in fig 3.11(a), the *E*-ray will be completely transmitted through prism P_2 since the principal sections of both prisms are parallel to each other. The intensity of the emergent light ray will be maximum.

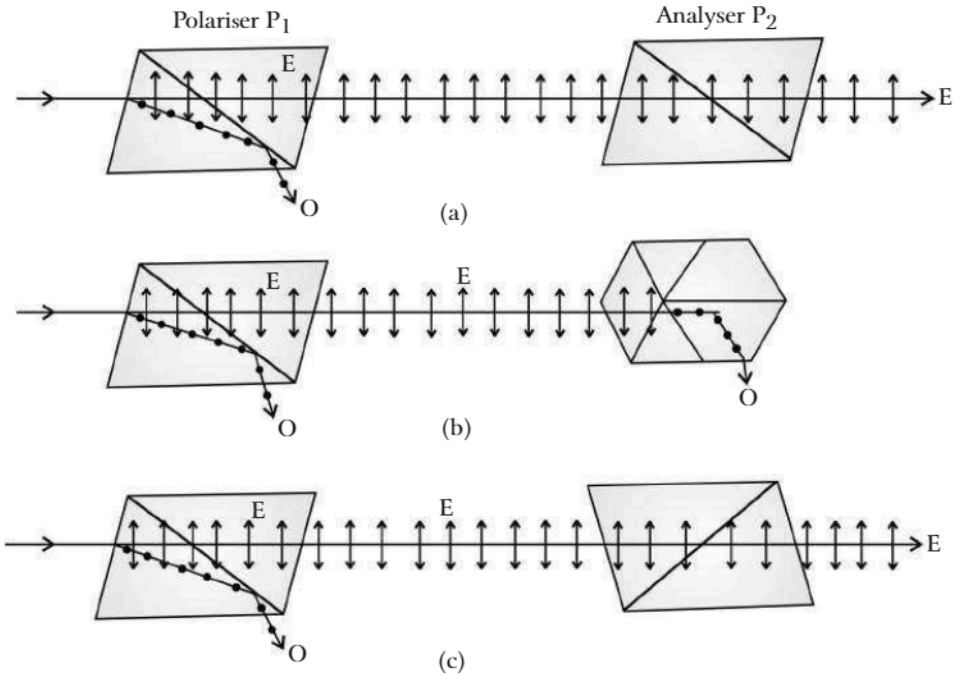


Fig. 3.11

Now if the second prism P_2 is gradually rotated, the intensity of the ray (*E*-ray) decreases in accordance with Malus's Law and when both the prism are crossed *i.e.*, they becomes perpendicular to each other [Fig. 3.11(b)], then no light ray comes out of the second prism P_2. It means that light coming out of P_1 is plane polarised and when this polarised extraordinary ray (*E*-ray) enter the prism P_2 in this position, it acts as an ordinary ray and is totally internally reflected by the Canada balsam layer and no light comes out of P_2. If the Nicol's prism P_2 be further rotated to have its principal section again parallel to that of P_1 Fig.3.11(c) the intensity of emergent light will again be a maximum. Thus, prism P_1 produces plane polarised light and is called *polariser*. Prism P_2 detects it and is called *analyser*.

Example 3: Plane polarised light is incident normally on a plate of doubly refracting uniaxial crystal with faces cut parallel to the optic axis. Compare the intensities of extraordinary and ordinary, rays, if the light is incident with vibration making an angle of 30° with the optic axis. Given $\lambda = 6000, A\ \mu_e = 1.5532, \mu_o = 1.5442$.

Solution: Given that, $\mu_e = 1.5532$ and $\mu_o = 1.5442$

i.e., $\mu_e > \mu_o$ it means that the given crystal is positive.

Now, let *A* be amplitude of the incident polarised light wave and θ be the angle made by incident beam with the optic axis. On entering the crystal, the light will split up into

two rays, an *E*-ray and an *O*-ray. The intensity of the *E*-ray will be $A^2\cos^2\theta$ having vibrations parallel to the optic axis and the intensity of the *O*-ray will be $A^2\sin^2\theta$ having vibrations perpendicular to the optic axis. Hence, the ratio of intensities of the *E*-ray and *O*-ray will be

$$\frac{I_E}{I_O}=\frac{A^2\cos^2\theta}{A^2\sin^2\theta}=\frac{\cos^2\theta}{\sin^2\theta}$$

Here given that, $\theta=30°$

$$\therefore \quad \frac{I_E}{I_O}=\frac{\cos^2 30°}{\sin^2 30°}=\frac{(\sqrt{3}/2)^2}{(1/2)^2}=\frac{3}{1}$$

Example 4: Unpolarised light falls on two polarising crystals placed one on the top of the other. What must be the angle between the characteristic directions of the crystal if the intensities of the transmitted light from the second crystal is (i) one third of the maximum intensity of the transmitted beam from the first crystal, (ii) one third the intensity of the incident beam? Assume that the crystal is ideal i.e., it reduces the intensity of unpolarised light by exactly 50% when beam is transmitted by the first crystal.

Solution: Let I_0 be the intensity of the unpolarised light (incident beam), then the intensity I of the transmitted (polarised) beam by the first crystal would be as,

$$I=\frac{I_0}{2} \qquad ...(1)$$

If both the crystals are parallel to each other, then I will be the maximum intensity transmitted by the second crystal. Let θ be the angle between the characteristic directions of the two crystals. By Malus's Law, the intensity transmitted by the second crystal is given by,

$$I_\theta=I\cos^2\theta \qquad ...(2)$$

(i) Given that, $I_\theta=\frac{1}{3}I$

$$\therefore \quad \frac{1}{3}I=I\cos^2\theta$$

or $\cos\theta=\pm\frac{1}{\sqrt{3}}$ or $\theta=\cos^{-1}\left(\sqrt{\pm\frac{1}{\sqrt{3}}}\right)$

or $\theta=\pm 55°$

(ii) Given that, $I_\theta=\frac{1}{3}I_0=\frac{1}{3}.2I=\frac{2}{3}I$ [from eq. (1)]

Using equation (2), we have $I\cos^2\theta=\frac{2}{3}I$

or $\cos^2\theta = \frac{2}{3}$ or $\theta = \cos^{-1}\left(\pm\sqrt{\frac{2}{3}}\right)$

or $\theta = \pm 35°$

3.9 Huygen's Theory of Double Refraction in a Uniaxial Crystal

When a beam of unpolarised light is incident on a calcite crystal, it splits up into two refracted rays. One of these ray obeys the law of refraction having same velocity in all directions. This ray is called ordinary ray (*O*-ray). The other ray behaves in an extraordinary way, having different velocities in different directions. This is called extraordinary ray (*E*-ray). This phenomenon is known as double refraction.

Huygen explained the phenomenon of double refraction with the help of secondary wavelets. To explain this, he gave the following postulates:

1. When a light ray is incident on the surface of a doubly refracting crystal, each point on the surface becomes the origin of two secondary wavelets, ordinary and extraordinary, which spread out into the crystal.

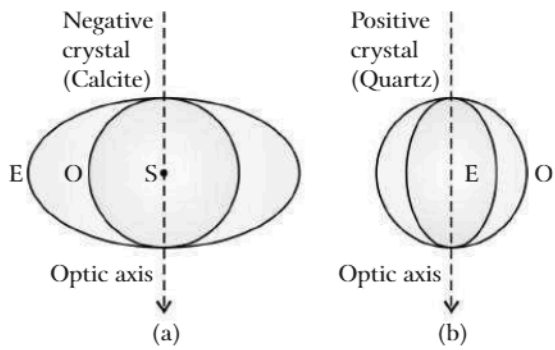


Fig. 3.12

2. For the ordinary ray, for which the velocity of light is the same in all directions, the wavefront is spherical.
3. For the extraordinary rays, the velocity varies with the direction (*i.e.* along the radius vector) and the wavefront is an ellipsoid of revolution.
4. The velocities of the ordinary and the extraordinary rays are the same along the optic axis *i.e.*, the sphere and the ellipsoid touch each other at points along the optic axis of the crystal as shown in fig 3.12.
5. For the negative uniaxial crystals ($\mu_O > \mu_e$), the sphere lies inside the ellipsoid [Fig. 3.12(a)], while for the positive uniaxial crystal ($\mu_O < \mu_e$), the ellipsoid lies inside the sphere [Fig. 3.12(b)].
6. For negative crystals, the velocity of extraordinary ray is least along optic axis but is maximum in perpendicular direction. For positive crystals, the velocity of extraordinary ray is maximum along optic axis and is least in perpendicular direction.

3.10 Polaroids

Polaroids are commercial polarising devices based on the principle of dichroism *i.e.* the property of certain doubly refracting crystals absorbing one of the two (*O* and *E*) rays completely and allowing the other ray to transmit. In polaroids dichroic crystals are embedded in a volatile viscous medium (cellulose acetate) and crystals are aligned with their optic axes parallel. But these crystals are not stable and are affected by slight strain. So, they are developed in form of large sheets. Thus, in a polaroid, a large sized polarising film mounted between two glass plates and is used to obtain plane polarised light for commercial purposes.

The film consists of a thin sheet of nitro cellulose packed with ultramicroscopic crystals of the organic compound iodosulphate quinine (herapathite) with their optic axis all parallel. These crystals are highly dichroic, *i.e.*, absorbing one of the doubly refracted beams completely. When a beam of unpolarised light passes through the polaroid film, the emerging light is plane polarised. Light is polarised or not it can be verified with a second polaroid. When the two polaroids are parallel to each other, the light transmitted by the first is also transmitted by the second [Fig. 3.13(a)]. But if the second is rotated through 90°, no light emerge out of it Fig. 3.13(b).

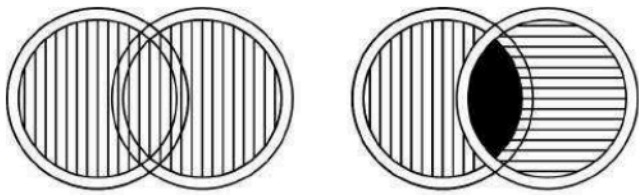


Fig. 3.13

3.10.1 Uses of Polaroids

1. Polaroids are used in the laboratory to produce and analyse plane polarised light. They are cheaper than the Nicols.

2. They are used in sun glasses to cut off the glare of light reflected from horizontal surfaces like wet roads, polished tables etc.

3. Polaroids are used to control the intensity of light entering trains and aeroplanes. A polaroid is fixed outside the window and other polaroid is fitted inside, which can be rotated. The intensity of light can be adjusted by rotating the inner polaroid.

4. Polaroid glasses are also used to see three dimensional pictures.

3.11 Phase Retardation Plate

When a plane polarised beam is incident normally on a double refracting crystal plate, its vibrations break at the first face into two components *i.e.* one vibration along the optic axis and the other vibration perpendicular to it. These components travel through the plate in the same direction (along the normal) but with different speeds v_o and v_e. If the thickness of the plate is d, the difference in time taken by two beams in traversing the plate is,

$$|\Delta t| = d\left(\frac{1}{v_o} - \frac{1}{v_e}\right)$$

In terms of equivalent path difference in vacuum, the relative retardation is,

$$\Delta x = c\Delta t = d|\mu_o - \mu_e|$$

where μ_o and μ_e are refractive indices of *O*-ray and *E*-ray respectively.

Thus, a relative phase retardation $(2\pi/\lambda)\Delta x$ is created between the two mutually perpendicular rays.

For this reason the crystal plate is called *phase retardation plate*. If we choose the thickness of plate in such a way that the path retardation is equal to $(\lambda/4)$ or $(\lambda/2)$, then the plate is called *quarter wave plate* or *half wave plate* respectively.

3.12 Quarter Wave Plate

It is an uniaxial double refracting crystal plate, having its optic axis parallel to the refracting face and thickness such as to produce a path difference of $(\lambda/4)$ *or phase difference of* $\pi/2$ *between the emergent ordinary and extraordinary rays.*

When a beam of plane polarised monochromatic light of wavelength λ is incident normally on double refracting crystal plate (Fig.3.14). The ray is broken up into *O* and *E*-rays, which travel in the same direction with different velocities (because refracting face is parallel to optic axis). If the crystal is of calcite (negative), the *E*-ray travels faster than the *O*-ray, so that $\mu_o > \mu_e$, where μ_o and μ_e are the principle refractive indices of the crystal for the *O* and *E*-rays.

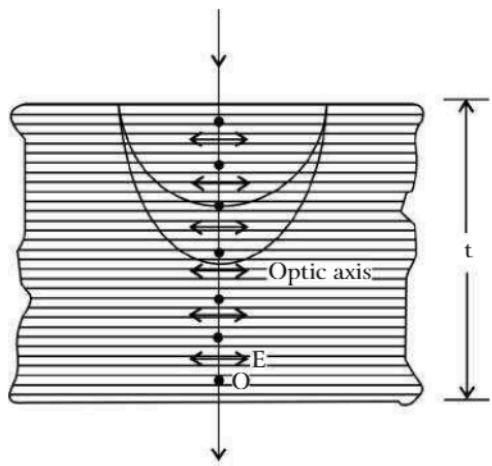


Fig. 3.14

If t is the thickness of the crystal plate, the optical path of the *O* and *E*-rays in the plate will be $\mu_o t$ and $\mu_e t$ respectively. Therefore, the path difference

between both the emergent rays will be $(\mu_o - \mu_e)t$. If the plate is to act as a quarter wave plate, this path difference should be equal to $\lambda/4$ *i.e.*,

$$(\mu_o - \mu_e)t = \lambda/4$$

or

$$t = \frac{\lambda}{4(\mu_o - \mu_e)}$$

Similarly, for a quartz crystal (positive) in which $\mu_e > \mu_o$

$$t = \frac{\lambda}{4(\mu_e - \mu_o)}$$

When plane polarised light is incident on a quarter wave plate, the emergent light is in general *elliptically polarised*; the axes of the ellipse are parallel and perpendicular to the optic axis and the ratio of the axes is given by $\tan\theta$, where θ is the angle between the plane of vibration of the incident beam and optic axis. When $\theta = 45°$, the light emerging from a quarter wave plate is *circularly polarised*.

3.13 Half Wave Plate

It is an uniaxial doubly refracting crystal plate, having its optic axis parallel to the refracting face and thickness such as to produce a path difference of $\lambda/2$ or phase difference of π between the emergent ordinary and extraordinary rays.

When a plane polarised monochromatic light beam of wavelength λ is incident normally on double refracting crystal plate (Fig. 3.15), the ray is broken up into *O* and *E*-rays, which travel in the same direction with different velocities (because refracting face is parallel to optic axis). If the crystal is of calcite (negative), the *E*-ray travels faster than *O*-ray, so that $\mu_o > \mu_e$, where μ_o and μ_e are the principle refractive indices of the crystal for *O*-ray and *E*-ray.

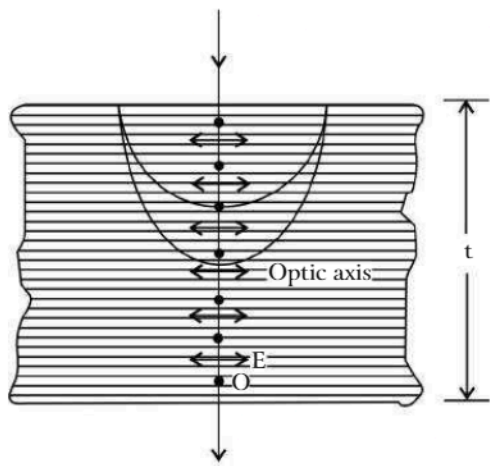


Fig. 3.15

If t is the thickness of the crystal plate, the optical path of the *O* and *E*-ray in the plate will be $\mu_o t$ and $\mu_e t$ respectively. Therefore, the path difference between the two emergent rays will be $(\mu_o - \mu_e)t$. If the plate is to act as a half wave plate, then this path difference should be equal to $\lambda/2$, *i.e.*,

$$(\mu_o - \mu_e)t = \lambda/2$$

or

$$t = \frac{\lambda}{2(\mu_o - \mu_e)}$$

Similarly, for a quartz crystal plate (positive) in which $\mu_e > \mu_o$

$$t = \frac{\lambda}{2(\mu_e - \mu_o)}$$

When a linearly polarised light is incident upon a half wave plate, the emergent light is also linearly polarised but its direction of vibration makes an angle 2θ with the incident light where θ is the angle between plane of vibration of incident beam and the optic axis. If $\theta = 45°$, then a *half wave plate rotates the azimuth of a beam of plane polarized light* by 90°. Hence, such a plate is used in polarimeters as half shade devices to divide the field of view into two halves presented side by side.

Example 5: Calculate the thickness of a quarter wave plate for a wavelength of 5890 Å. Given that, $\mu_o = 1.54$ ***and*** $\mu_e = 1.55$.

[Meerut 2005]

Solution: Since $\mu_e > \mu_o$ *i.e.*, plate crystal is positive. Then, the thickness of the quarter wave plate is given by,

$$t = \frac{\lambda}{4(\mu_e - \mu_o)}$$

Here given that, $\lambda = 5890\ \text{Å}, \mu_e = 1.55, \mu_o = 1.54$

Putting the values, we get

$$t = \frac{5890 \times 10^{-10}\ \text{m}}{4(1.55 - 1.54)} = \frac{5890 \times 10^{-10}}{4 \times 0.01}$$

$$= 1472.5 \times 10^{-8}\ \text{m} = 1.472 \times 10^{-5}\ \text{m}$$

$$= \mathbf{1.472 \times 10^{-2}\ mm}$$

Example 6: Calculate the thickness of a doubly refracting crystal required to introduce a path difference of $\lambda/2$ ***between O and E-rays when*** $\lambda = 6000\ \text{Å}, \mu_o = 1.65$ ***and*** $\mu_e = 1.48$

Solution: Since $\mu_o > \mu_e$ *i.e.*, plate crystal is negative. So, the thickness of the half wave plate is given by,

$$t = \frac{\lambda}{2(\mu_o - \mu_e)}$$

Here given that, $\lambda = 6000 \times 10^{-10}\ \text{m}, \mu_o = 1.65, \mu_e = 1.48$

Putting the values, we get $t = \dfrac{6000 \times 10^{-10}\ \text{m}}{2(1.65 - 1.48)} = \dfrac{6000 \times 10^{-10}}{2 \times 0.17}$

$$t = \frac{60}{34} \times 10^{-6} \text{ m} = 1.765 \times 10^{-6} \text{ m}$$

$$= 1.765 \times 10^{-3} \text{ mm}$$

Example 7: A beam of linearly polarised light is changed into circularly polarised light by passing it through a sliced crystal of thickness 0.003 cm. Calculate the difference in refractive indices of the two rays in the crystal, assuming this to be of minimum thickness. The wavelength of light used is 6×10^{-7} m.

Solution: We know that the linearly polarised light changes into the circularly polarised light with the help of quarter wave plate only. For quarter wave plate, we have

$$(\mu_e - \mu_O)t = \frac{\lambda}{4}$$

or

$$\mu_e - \mu_O = \frac{\lambda}{4t}$$

Here given that, $\lambda = 6 \times 10^{-7}$ m and $t = 0.003 \text{ cm} = 3 \times 10^{-5}$ m

$\therefore$

$$(\mu_e - \mu_O) = \frac{6 \times 10^{-7}}{4 \times 3 \times 10^{-5}} = 5 \times 10^{-3}$$

3.14 Babinet Compensator

For the analysis of plane polarised light it is required to have a device which can produce a phase-difference of any arbitrary value between two plane polarised waves in mutually perpendicular planes. A single crystal, such as a quarter wave plate can produce. Only particular phase-difference depending upon its thickness and wavelength of light used. Babinet devised an arrangement by means of which a measurable and variable path difference can be introduced. The device known as **Babinet compensator**, consists of two wedge shaped plates *ABC* and *DEF*, placed with their hypotenuse adjacent so as to form a small rectangular block. One of the wedge is fixed while the other can be moved in its own plane by means of a micrometer screw. The wedges are cut from crystal line quartz in such a way that the optic axis in one is parallel and in the other perpendicular to the two refracting faces.

If the plane polarised light is incident normally on the face *AB* of compensator with the plane of vibration making an angle θ with the optic axis, it will be broken up into ordinary and extraordinary components.

The *E* component parallel to the optic axis in the crystal travels slower than the *O*-component until it reaches *P* on *ED* of second crystal. At *P* point the *E* vibration becomes *O* vibration as now it is perpendicular to the optic axis and the *O*-vibration becomes *E* vibration in the second crystal. In other words the two vibrations exchange velocities in passing from one wedge to the other. The resultant effect is such that the one wedge tends to cancel the effect due to other.

If the two wedges are placed as shown in fig. 3.16 (a) and the ray of light is incident at the centre as at P so that the thickness of paths in the two is the same, then the cancellation is complete and the effect is that of *a* plate of zero thickness. If the wedge *DEF* is made to slide as shown in Fig 3.16 (b) so that the thickness traversed by the wave at P in the two wedges *ABC* and *DEF* is t_1 & t_2 respectively, then the path difference introduced by the first wedge will be $t_1(\mu_e - \mu_0)$ and that produced by the other will be $t_2(\mu_e - \mu_0)$.

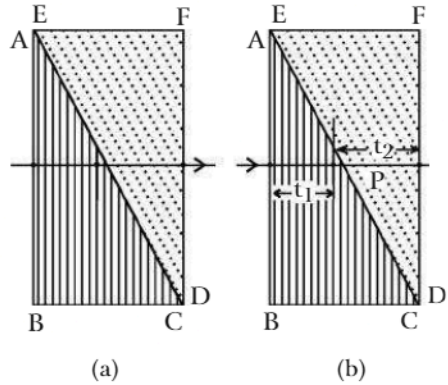


Fig. 3.16

Hence resultant path difference $= (t_1 - t_2)(\mu_e - \mu_0)$

Thus by adjusting the position of two wedges relative to each other, any desired path difference can be produced.

3.15 Plane, Circularly and Elliptically Polarised Light

According to the electromagnetic theory of light, a light wave consists of electric and magnetic vectors vibrating in mutually perpendicular planes and both the planes are perpendicular to the direction of propagation of light. The electric vector is responsible for the optical effects of the wave and is also called the "*light vector*". In unpolarised light, the light vector can vibrate in all possible directions in a plane perpendicular to the direction of propagation. But if, *the light vector vibrates along a fixed straight line in a plane perpendicular to the direction of propagation, the light is said to be plane polarised or linearly polarised light*.

If the light vector vibrates circularly or elliptically in the plane, the light is said to be circularly or elliptically polarised light. It is worth noting that circular or elliptical vibrations are the resultant of two linearly or (plane) polarised light waves superimposed on each other.

3.16 Production of Elliptically and Circularly Polarised Light from Linearly Polarised Light

Let us consider a monochromatic light beam falls on the Nicol's prism N_1. After passing through N_1 the light ray is plane polarised. This plane polarised light is incident normally on a uniaxial double refracting crystal (calcite) whose faces have been cut parallel to its optics axis, [Fig. 3.17(a)]. On entering the crystal the incident light will be splitted into two components, ordinary and extraordinary. In this case both the rays travel along the same direction but, with different velocities Fig. 3.17(b). Suppose the incident light is vibrating along PQ making an angle θ with the optic axis of the crystal Fig. 3.17(c). Let A be the amplitude of the vibrations of incident plane polarised light.

Therefore, the amplitude of the ordinary ray, vibrating along *PO* (perpendicular to optic axis) is $A \sin\theta$ and the amplitude of the extraordinary ray along *PE* axis parallel to optic is $A \cos\theta$. In a negative crystal (calcite) *E*-ray travels faster than *O*-ray. Hence, the two rays emerge out of the crystal having a phase-difference δ between them and it depends upon the thickness *d* of the crystal. Thus, the incident plane polarised light on leaving the crystal will consists of simple harmonic vibrations in two mutually perpendicular planes, having same period but different amplitudes and phases. They can be compounded into a single motion which can be **linear, circular** or **elliptical** depending upon the phase-difference and the value of θ.

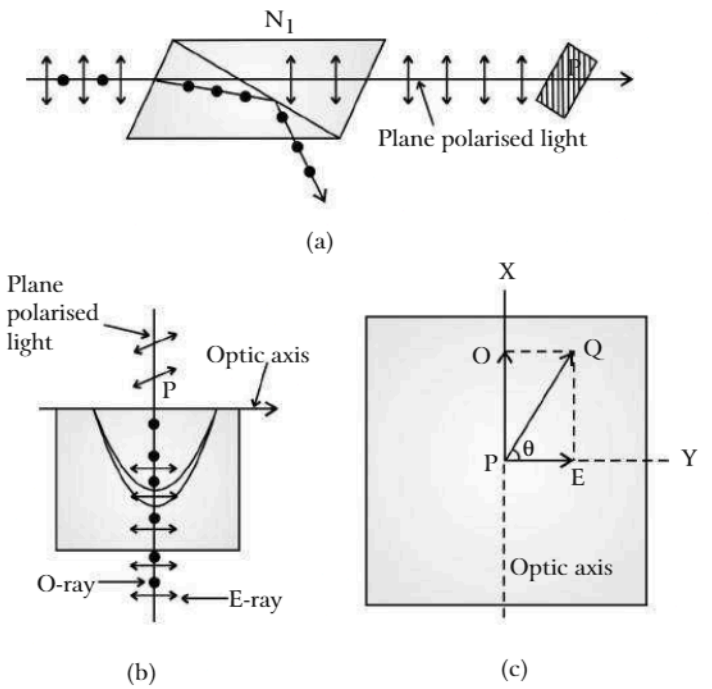


Fig. 3.17

If δ is the phase-difference introduced between the two (*E* and *O*-rays) rays in passing through the crystal, then *E*-ray be represented as,

$$x = A\cos\theta \sin(\omega t + \delta)$$

and *O*-ray will be as, $y = A\sin\theta \sin\omega t$

Let $A\cos\theta = a$ and $A\sin\theta = b$

then, $x = a\sin(\omega t + \delta)$ ...(1)

$y = b\sin\omega t$ ...(2)

From equation (2), $y/b = \sin\omega t$

and $$\cos\omega t = \sqrt{1-\sin^2\omega t} = \sqrt{1-\frac{y^2}{b^2}}$$

From equation (1), $x/a = \sin\omega t \cos\delta + \cos\omega t \sin\delta$

or $$\frac{x}{a} = \frac{y}{b}\cos\delta + \sqrt{1-\frac{y^2}{b^2}}\sin\delta \qquad \text{(using above value)}$$

or $$\frac{x}{a} - \frac{y}{b}\cos\delta = \sqrt{1-\frac{y^2}{b^2}}\sin\delta$$

or $$\left(\frac{x}{a} - \frac{y}{b}\cos\delta\right)^2 = \left(1-\frac{y^2}{b^2}\right)\sin^2\delta$$

or $$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab}\cos\delta = \sin^2\delta \qquad \ldots(3)$$

This is the general equation of an ellipse. Hence, the light emerging from the crystal is, in general, elliptically polarised.

Special Cases:

Case I: If the thickness of the crystal is such that $\delta = 0$ or $2n\pi$, then $\sin\delta = 0, \cos\delta = 1$ and equation (3) becomes,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} = 0$$

or $$\left(\frac{x}{a} - \frac{y}{b}\right)^2 = 0$$

or $$\pm\left(\frac{x}{a} - \frac{y}{b}\right) = 0$$

or $$\pm y = \pm\frac{b}{a}x$$

This equation represents a pair of co-incident straight lines passing through the origin and having a slope b/a. This indicates that the resultant (emergent) light is *linearly polarised* in the same plane of vibration as the incident light Fig. 3.18(a).

Case II: If $\delta = \frac{\pi}{2}$ or $(2n+1)\frac{\pi}{2}$, then, $\cos\delta = \cos\frac{\pi}{2} = 0$ and $\sin\frac{\pi}{2} = 1$

So, equation (3) becomes, $$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

This is the equation of a symmetrical ellipse. Thus, the emergent light is elliptically polarised Fig. 3.18(b), (c). The plane of the ellipse is normal to the direction of propagation.

Case III: If $\delta = (\pi/2)$ and $a = b$ $[\theta = 45°]$

This equation represents a circle of radius 'a'. Thus, the emergent light is *circularly polarised* Fig 3.18(d). In this case the incident plane polarised light makes an angle of 45° with the optic axis of the crystal.

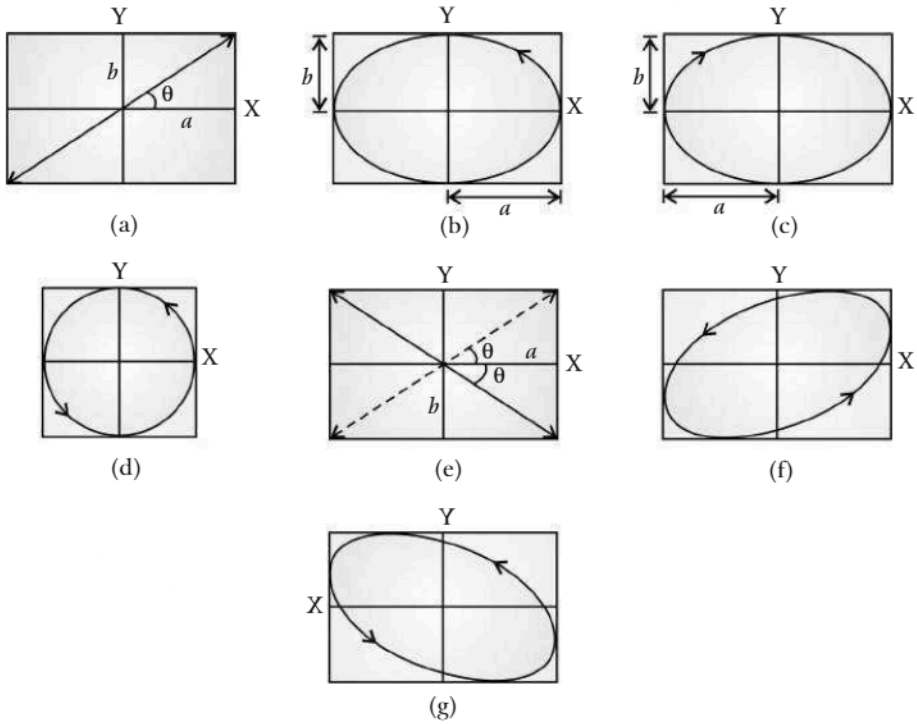


Fig. 3.18

Case IV: If $\delta = \pi$ or $(2n+1)\pi$, then equation (3) becomes,

$$\pm\left(\frac{x}{a}+\frac{y}{b}\right)=0 \quad \text{or} \quad \pm y=\left(\pm\frac{b}{a}\right)x$$

This is again a pair of co-incident straight lines through the origin with slope $(-b/a)$. In this case the emergent light is linearly polarised and the direction of vibration makes an angle equal to $2\tan^{-1}(b/a)=2\theta$ with that of the incident light Fig. 3.18(e).

Case V: If $\delta=\frac{\pi}{4}$ or $\frac{3\pi}{4}$, equation (3) becomes the equation of an oblique ellipse with major axis inclined in the positive direction of the x-axis, Fig. 3.18(f) or in the negative direction of x-axis Fig 3.18(g).

3.17 Production of Plane Polarised Light

To produce plane polarised light, a beam of monochromatic light is passed through a Nicol's prism. While passing through the Nicol's prism, the beam is split up into ordinary and extraordinary ray. The ordinary ray is totally internally reflected back at the Canada balsam layer and is absorbed, while the extraordinary ray emerges out of the Nicol's prism. This beam is plane polarised with its vibrations parallel to the principle section.

3.17.1 Detection of Plane Polarised Light

To detect plane polarised light, it is allowed to fall on another Nicol's prism. This prism is rotated about the direction of propagation of light. If the intensity of the emerging light varies with zero minimum, the light is plane polarised.

3.18 Production of Circularly Polarised Light

To produce circularly polarised light, the two waves vibrating at right angles to each other and having the same amplitude and time period should have phase-difference $\pi/2$ or path difference of $\lambda/4$. For this purpose, a parallel beam of monochromatic light is allowed to fall on Nicol's prism N_1, the beam after passing through the prism N_1, is plane polarised. This plane polarised light normally falls on a quarter wave plate in such a way that plane polarised light makes an angle 45° with the optic axis ($\theta = 45°$) of the quarter wave plate.

Inside the plate the plane polarised light is split up into two rectangular components having amplitude $A\cos 45°$ parallel to the optic axis (E-ray) and $A\sin 45°$ perpendicular to the optic axis (O-ray), where A is amplitude of incident polarised light. These components emerge from the plate with a phase-difference of $\pi/2$. Hence, the emerging light from a quarter wave plate is circularly polarised light.

3.18.1 Detection of Circularly Polarised Light

To detect the circularly polarised light, it is allowed to fall on another Nicol's prism N_2. This prism is rotated about the direction of propagation of light. For every position of Nicol's N_2 the circular polarised light is split up into two rectangular components of equal amplitudes. The direction of one vibration component will be parallel and the other will be perpendicular to the principle section of the prism. The parallel component will pass out while the perpendicular component will be totally reflected. So, on rotation of the Nicol's there is no variation in the intensity of the emergent light.

3.19 Production of Elliptically Polarised Light

To produce elliptically polarised light, the two waves vibrating at right angles to each other and having unequal amplitude should have a phase-difference of $\pi/2$ or path difference $\lambda/4$. For this purpose a parallel beam of monochromatic light is allowed to fall on Nicol's prism N_1, the beam after passing through the prism N_1, is plane polarised. This plane polarised light falls normally on a quarter wave plate in such a way that plane polarised light makes an angle other then 45° with the optic axis ($\theta \neq 45°$) of the quarter wave plate.

Inside the plate the plane polarised light is split up into two rectangular components having amplitude $A\cos\theta$ parallel to the optic axis (E-ray) and $A\sin\theta$ perpendicular to the optic axis (O-ray), where A is the amplitude of incident plane polarised light. These components emerge out of the plate with a phase-difference of $\pi/2$. Hence the emerging light from a quarter wave plate is elliptically polarised light.

3.19.1 Detection of Elliptically Polarised Light

To detect the elliptically polarised light, it is allowed to fall on another Nicol's prism N_2. This prism is rotated about the direction of propagation of light. For every position of Nicol's N_2 the elliptically polarised light is split up into two rectangular components of unequal amplitudes. The directions of the vibrations of the components will be parallel and perpendicular to the principle section of the prism. On rotating the Nicol's prism N_2 the intensity of emergent light varies from maximum to minimum according to the principle section of the Nicol becomes parallel to the major or the minor axis of the ellipse.

3.20 Double Image Prism

A Nicol's prism allows only one of the two plane polarised rays to transmit, the other being lost by total internal reflection. Also, Nicol's prism cannot be used in ultra-violet region, because Canada balsam absorbs ultra-violet light. It is sometimes necessary to retain both the rays so that two separate images are formed in the field of view which enable us to compare their intensities. An optical device used to form two separate images are known as double image prism. Double image prism is of two types: Rochon's prism and Wollaston's prism

3.20.1 Rochon's Prism

It consists of two right angled prism *ABC* and *BCD* of quartz cemented together with glycerine or castor oil as shown in Fig. 3.19(a). The optic axis of the prism *ABC* is parallel to its base *AB*, while that of prism *BCD* is perpendicular to the plane of paper.

A ray of light incident normally on the face *AC* of the prism *ABC* travels undeviated along the optic axis of the first prism. On entering the second prism *BCD*, it splits up into *O* and *E*-ray. The *O*-ray passes undeviated, thus remaining achromatic. The *E*-ray is deviated and dispersed. When only one plane polarised ray is desired, the *E*-ray can be cut off at sufficiently large distance from the prism.

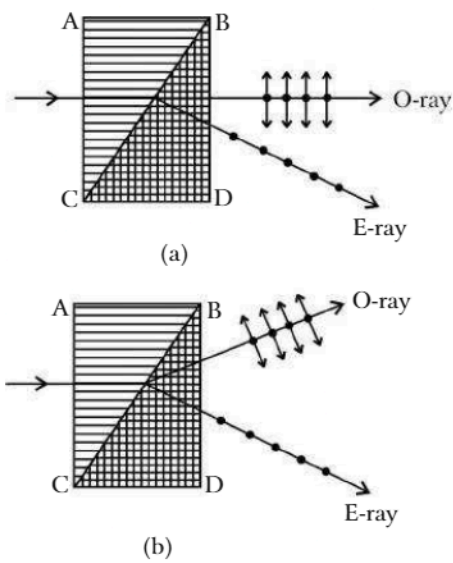


Fig. 3.19

3.20.2 Wollaston's Prism

It consists of two prism *ABC* and *BCD* of quartz cemented together with glycerine or castor oil as shown in fig. 3.19(b). The optic axis of the prism *ABC* is parallel to the face *AC*, while that of prism *BCD* is perpendicular to the first prism.

A ray of light incident normally on the face *AC* of the prism *ABC* travels perpendicular to the optic axis. Therefore, the *O*-ray and *E*-ray travel in the first prism in the same direction but with different velocities. Since, the second prism is cut with its optic axis perpendicular to that of the first prism, the *O*-ray in the first prism will become the *E*-ray in the second prism and *E*-ray becomes *O*-ray. Hence, both the rays are deviated and dispersed in the second prism resulting greater separation of two rays. This prism is specially used in experiments in which intensities of polarised lights are required.

Example 8: A double image prism is made of a material for which $\mu_O = 1.662$, $\mu_e = 1.474$. ***Deduce the angular separation of the emergent beams, if the angle of incidence is 45°.***

Solution: If the angle of incidence is 45°, then angle of refraction are given by,

$$\mu_o \sin i = \mu_e \sin r \qquad \text{or} \qquad \sin r = \frac{\mu_o}{\mu_e} \sin i$$

For prism *A*,
$$r_1 = \sin^{-1}\left(\sin i \times \frac{\mu_o}{\mu_e}\right)$$

For *B* prism μ_o and μ_e are interchanged.

So,
$$r_2 = \sin^{-1}\left(\sin i \times \frac{\mu_e}{\mu_o}\right)$$

Putting the values, we get
$$r_1 = \sin^{-1}\left(\frac{1}{\sqrt{2}} \times \frac{1.662}{1.474}\right) = 53°06'$$

and
$$r_2 = \sin^{-1}\left(\frac{1}{\sqrt{2}} \times \frac{1.474}{1.662}\right) = 38°50'$$

These rays falls on the opposite faces at the angles of incidence 8°6′ and −6°10′ respectively. So, the angles of emergence are, $\sin^{-1}(1.474 \sin 8°6') = 11°59'$ and $\sin^{-1}[1.662 \times \sin(-6°10')] = -10°17'$

Therefore, the angle between the emergent beams $= 11°59' - (-10°17') = 22°16'$.

3.21 Optical Rotation

When a polariser and an analyser are crossed, no light comes out of the analyser Fig. 3.20(a). When a quartz plate cut with its faces parallel to the optic axis, is introduced between N_1 and N_2 in such a way that light falls normally on the quartz plate, then the light comes out of N_2 as shown in Fig. 3.20(b). It means that plane of vibration of the plane polarised light changes (rotate) when it passes through an substance (quartz plate). The amount of rotation through which the plane of vibration is turned depends upon the thickness of the substance and the wavelength of light. *This property of rotating the plane of vibration by a substances is known as optical activity* and the substance is known as *optically active substance e.g.*, sugar solution, sodium chloride, turpentine etc.

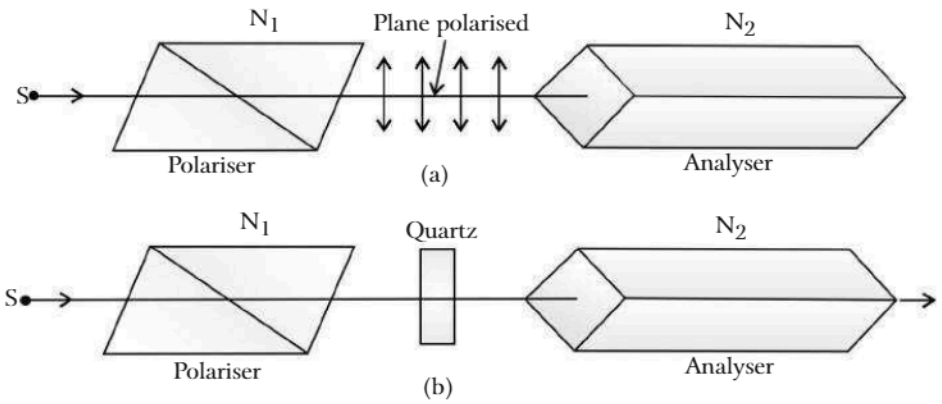


Fig. 3.20

There are two types of optically active substances (i) Right handed or dextro rotatory (ii) Left handed or laevo rotatory.

The substances, which rotate the plane of vibration in clockwise direction, are known as right handed or dextro rotatory substances and which rotate the plane of vibration in anticlockwise direction, are known as laevo rotatory or left handed substances.

It has been found that some quartz crystals are left handed and some are right handed.

3.21.1 Fresnel's Theory of Optical Rotation

A linearly polarised light may be taken as the resultant of two circularly polarised vibrations which are rotating in opposite directions with the same angular velocity. So, Fresnel assumed that:

1. The incident plane polarised light on entering a substance along the optic axis is resolved into two circularly polarised waves rotating in opposite directions, with the same angular velocity or frequency.

2. In an optically inactive substance, the two waves travel with the same velocity, but in an optically active substance they travel with different velocities. In a dextro rotatory substances, the clockwise wave travels faster, while in the laevo-rotatory substances the anticlockwise wave travel faster. Hence, a phase-difference is developed between them.

3. On emergence, the two circular components recombine to form a plane polarised light whose plane of vibration is rotated by an angle depending upon the phase-difference between them.

3.21.2 Rotation of Plane of Polarisation for Optically Inactive Crystal (Calcite)

Let, a plane polarised light beam incident normally on a crystal along the optic axis and its vibrations are represented by $y = 2a\sin\omega t$.

The plane polarised light, on entering the crystal is broken up into two equal and opposite circular vibrations. If the first face of crystal is in xy plane, then the circular components are represented as,

For clockwise circular vibrations,

$$x_1 = a\cos\omega t$$

$$y_1 = a\cos(\omega t + \pi/2) = a\sin\omega t$$

(As the circularly polarised light is the resultant of two rectangular components having a phase-difference of $\pi/2$).

and for anticlockwise circular vibrations,

$$x_2 = -a\cos\omega t$$

$$y_2 = a\sin\omega t$$

Therefore, the resultant vibrations, along X-axis,

$$x = x_1 + x_2 = a\cos\omega t - a\cos\omega t = 0$$

$$y = y_1 + y_2 = a\sin\omega t + a\sin\omega t$$

$$= 2a\sin\omega t$$

Thus, the resultant vibrations has an amplitude $2a$ and is plane polarised. The plane of vibration is along the original direction.

3.21.3 Rotation of Plane of Polarisation for Optically Active Crystal (Quartz)

Let the crystal be right handed optically active crystal. So, the clockwise vibration travels faster than anticlockwise vibration. Let the phase-difference between them $= \delta$

For clockwise vibrations,

$$x_1 = a\cos(\omega t + \delta)$$

$$y_1 = a\sin(\omega t + \delta)$$

and for anticlockwise vibrations,

$$x_2 = -a\cos\omega t$$

$$y_2 = a\sin\omega t$$

Therefore, the resultant displacements along the two axes are,

$$x = x_1 + x_2 = a\cos(\omega t + \delta) - a\cos\omega t$$

$$= 2a \sin\frac{\delta}{2} \sin\left(\omega t + \frac{\delta}{2}\right) \quad \ldots(1)$$

and $$y = y_1 + y_2 = a\sin(\omega t + \delta) + a\sin\omega t$$

$$= 2a\cos\frac{\delta}{2}\sin\left(\omega t + \frac{\delta}{2}\right) \quad \ldots(2)$$

The resultant vibrations along X-axis and Y-axis have the same phase. Therefore, the resultant vibration is plane polarised and it makes an angle $\delta/2$ with the original direction. Therefore, the plane of vibration has rotated through an angle $\delta/2$ on passing through the quartz crystal and this angle is given by $\tan\frac{\delta}{2} = \frac{x}{y}$ [from equation (1) and (2)].

If μ_R and μ_L are refractive indices of quartz for clockwise and anticlockwise circularly polarised light respectively and d is the thickness of the crystal, then optical path difference between two vibrations in passing through the crystal is given by

$$= (\mu_L - \mu_R)d \qquad \text{(for right handed optically active crystal)}$$

If the wavelength of light $= \lambda$, then

$$\text{Phase-difference, } \delta = \frac{2\pi}{\lambda} \times (\text{path difference})$$

or $$\delta = \frac{2\pi}{\lambda}(\mu_L - \mu_R)\, d$$

The plane of vibration is rotated through an angle $\delta/2$

So, $$\theta = \frac{\delta}{2} = \frac{\pi}{\lambda}(\mu_L - \mu_R)\, d$$

and for the left handed optically active crystal,

$$\theta = \frac{\delta}{2} = \frac{\pi}{\lambda}(\mu_R - \mu_L)\, d$$

3.21.4 Experimental Verification of Fresnel's Theory

To verify Fresnel's theory, a beam of plane polarised light is incident normally on the face AB of the rectangular block $ABFG$ (Fig. 3.21). The block consists of alternate prisms of right handed and left handed quartz crystal. The optic axis is parallel to the base of each prism. According to Fresnel's theory, when plane polarised light is incident normally on the first face AB, then it will break up into two opposite circularly polarised beam in opposite directions which travel along the same direction but with different speeds. On passing through the first oblique

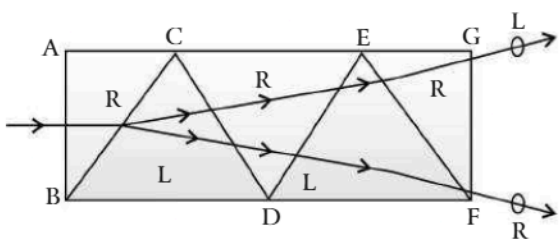


Fig. 3.21

boundary *BC*, the right beam which moves faster in first prism (right handed) becomes slower in the second prism (left handed) and also vice-versa is true. Therefore, one beam is bent away from the normal and the other is bent towards the normal. The two beams are separated apart, while they travel through the prism *L*. Again at the boundary *CD* of next prism *R*, the speeds are again interchanged and the beam that is bent towards the normal in *L* prism, is now bent away from the normal. The net result is that both beams are separated more and more while passing through the arrangement. Hence, when the two beams comes out, they are widely apart. When these beams are analysed by a quarter wave plate and a Nicol's prism, they are found to be circularly polarized in opposite direction. Therefore, if verify the Fresnel's theory.

Example 9: The rotation of the plane of polarisation of light $(\lambda = 5893\ \text{Å})$ ***in a material is*** $10°/\text{cm}$. ***What is the difference in refractive indices of the material for clockwise and anticlockwise circularly-polarised light ?***

Solution: The optical rotation is given by, $\theta = \dfrac{\pi d}{\lambda}(\mu_L - \mu_R)$

where μ_L and μ_R are the refractive indices of the material for anticlockwise and clockwise circularly polarised light.

Here given that, $\dfrac{\theta}{d} = 10°/\text{cm}$ and $\lambda = 5893\ \text{Å} = 5893 \times 10^{-8}\ \text{cm}$.

Putting the values, we get $(\mu_L - \mu_R) = \dfrac{10 \times 5893 \times 10^{-8}}{180}$ $[\because \pi = 180°]$

$$= 32.7 \times 10^{-7} = 3.27 \times 10^{-6}$$

3.22 Specific Rotation

The specific rotation S of a substance at a given temperature t and for a given wavelength of light λ *is defined as the rotation (in degrees) produced by one decimetre (10 cm) long column of the solution containing 1 gm of the active substance in 1 c.c. liquid i.e.,*

$$S = \frac{\theta}{l \times C}$$

where θ is the rotation in degree, l is the length of the solution in decimetre through which the plane polarised light passes and C is the concentration of the active substance in gm/cm^3 in the solution. So, we can write,

$$\text{Specific rotation} = \frac{\text{rotation in degrees}}{\text{length of the solution in decimetre} \times \text{solution's concentration in gm/cc.}}$$

The product of the specific rotation and the molecular weight of the active substance is called *molecular rotation*. Sugar is probably the most common of all optically active substances, which is used for the estimation of its strength in a solution. The instrument which is used to determine the optical rotation produced by a substance is called polarimeter.

3.22.1 Laurent's Half Shade Polarimeter

It is used to measure optical rotation of some substances. It consists of two Nicol prisms N_1 and N_2. N_1 is a polariser and N_2 is an analyser. After N_1 there is a half shade device *H*, called Laurent's plate. It is a combination of half wave plate *Q* of quartz and glass *G*. *Q* covers one half of the emergent beam from N_1, while the other half is covered by *G*. The glass plate *G* absorbs the same amount of light as the quartz plate *Q*. *T* is a hollow glass tube having a large diameter in its middle portion. This tube is filled with the solution of optically active substance and closed at the ends. There should be no air bubbles in the tube. The rotation can be measured with the help of a telescope Fig. 3.22.

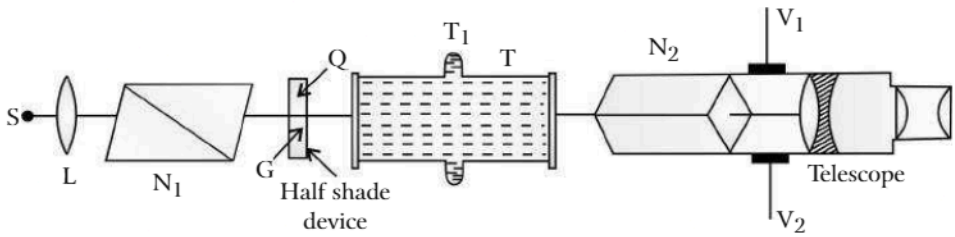


Fig. 3.22

3.22.2 Working

Light from a monochromatic source *S* is incident on the converging lens. *L*. After passing through N_1, the beam is plane polarised. One half of the beam passes through the quartz plate *Q* and the other half passes through the glass plate *G*. Let the plane of vibration of the plane polarised light incident on the half shade device be along *AB* Fig. 3.23. Here *AB* makes an angle θ with *YY'*, on passing through the quartz plate *Q*, the beam is split up into two components, one *E*-component (plane of vibration is parallel to the optic axis of the quartz) and the other *O*-component (plane of vibration is perpendicular to the optic axis). They travel along the same direction but with

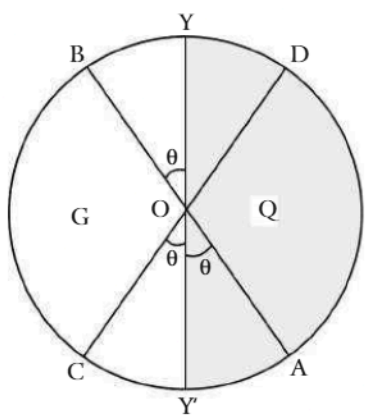


Fig. 3.23

different speeds. The *O*-component travels faster in quartz, and on emergence a phase difference of π or a path difference of λ/2 is introduced between them. Therefore, the vibrations of the beam emerging out of quartz will be along *CD* (where $\angle AOY' = \angle COY'$) whereas the vibrations of the beam emerging out of the glass plate will be along *AB*. Now, if the analyser N_2 has its principle section along *YY' i.e.*, along the direction which bisect the angle *AOC*, the amplitudes of light incident on the analyser N_2 the quartz half and glass half will be equal. Therefore, the field of view will be equally bright Fig. 3.24(a).

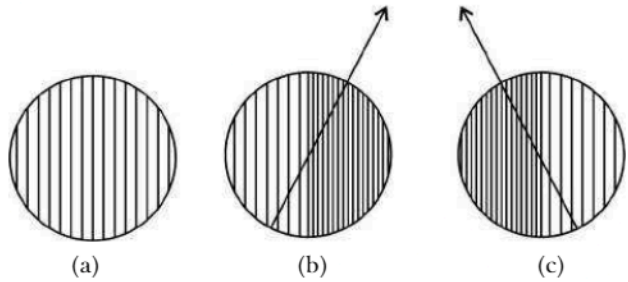


Fig. 3.24

If the analyser N_2 is rotated to from this position in the clockwise direction (right side of *YY'*), then the right half will be darker as compared to the left Fig. 3.24(b). On the other hand, if the analyser N_2 is rotated in the anticlockwise (left side of *YY'*), the left half is darker as compared to the right Fig. 3.24(c).

3.22.3 Determination of Specific Rotation of Sugar Solution

To find the specific rotation of sugar, the analyser Nicol N_2, is so adjusted that the two halves of the field are equally illuminated without the solution in the tube *T*. The reading of the verniers V_1 and V_2 are noted. Now the sugar solution of known concentration is filled in the tube. The solution rotates the vibrations from the quartz half and glass half. Therefore, the field of view is not equally bright. The analyser is rotated in the clockwise direction and is brought to a position so that the whole field of view is equally illuminated. The new positions of vernier V_1 and V_2 on the circular scale are noted. Thus, the angle through which the analyser has been rotated gives the angle through which the plane of vibration of the incident beam has been rotated by the sugar solution. Hence, the difference between the two positions of N_2 gives the angle of rotation θ. The length *l* of the tube (containing solution) is measured in decimetres. Then, the specific rotation of the sugar solution is determined by this formula,

$$S = \frac{\theta}{l \times C}$$

where *C* is the concentration of the solution.

3.23 Biquartz Plate

Instead of a half shade plate, a biquartz plate is used in this polarimeter. It is combination of two semicircular plates of quartz each of thickness 3.75 mm. One half consists of right handed optically active quartz, while the other is left handed optically active quartz, both cut perpendicular to the optic axis and cemented together so as to form a complete circular plate Fig. 3.25. Each half plate rotates the plane of polarisation of the yellow light through 90°, one anticlockwise and other clockwise

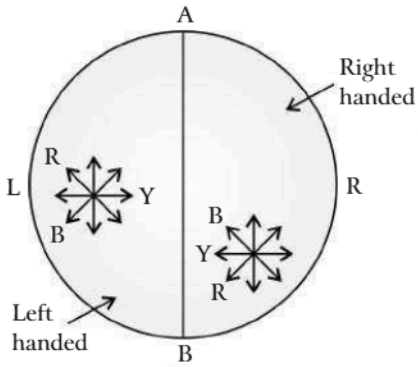


Fig. 3.25

3.23.1 Biquartz Polarimeter

It is an esay and accurate method to determine the specific rotation of optically active substances. It consists of a converging lens, a polarising nicol, biquartz plate (instead of half shad plate), a tube (containing the optically active solution) an analysing Nicol and a telescope arranged just as in a half shade polarimeter Fig. 3.26. In this polarimeter white light is used instead of sodium light.

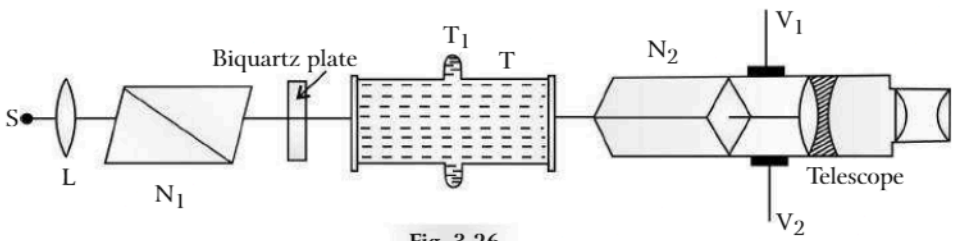


Fig. 3.26

3.23.2 Working

If white light is used, yellow light is quenched by the biquartz plate. When white light passes through polarising Nicol, it gives plane polarised light. This plane polarised light travels through the biquartz. Normally, it travels along the optic axis. Therefore, rotatory dispersion occurs in each half plate of the biquartz. If the vibrations in the incident polarised light are along *AB*, then, after passing through biquartz, the vibration of yellow light are along a direction perpendicular to *AB*. The vibrations of red and blue are along different directions as shown in fig. 3.26. If the principle section of analysing Nicol N_2 is parallel to *AB*, the yellow light will be eliminated in both halves of the field, while the red and blue light will be present in the same proportion in each half. Therefore, the two halves of the field will be equally illuminated with a reddish violet tint, called the **sensitive tint** or **tint of the passage**. The position of this tint is very sensitive because even a slight rotation of the analyser Nicol (one way or the other way) will make one half blue and the other half red.

3.23.3 Determination of Specific Rotation

By adjusting the particular position of the analyser, the field of view appears equally bright with **tint of passage** first without the optically active solution and then with optically active solution. The difference between the two positions of the Nicol gives the optical rotation θ produced by the optically active solution (sugar solution). If the length of the solution's tube is l decimetre and concentrations of solution is C gm/cc, then specific rotation S is determined by

$$S = \frac{\theta}{l \times C}$$

Example 10: Determine the specific rotation of the given sample of sugar solution if the plane of polarisation is turned through 13.2°. The length of the tube is 20 cm and concentration of sugar solution is 10%.

Solution: The specific rotation is given by,

$$S = \frac{\theta}{l \times C}$$

Here given that, $\theta = 13.2°, l = 20\text{ cm} = 2$ decimetre

and $C = 10\% = (10/100)\text{ gm/cm}^3 = 0.1\text{ g/cm}^3$

$\therefore$ $$S = \frac{13.2}{2 \times 0.1} = 66°$$

Example 11: A 20 cm long glass tube containing sugar solution rotates the plane of polarisation by 11°. If the specific rotation of sugar is 66°, calculate the concentration of the sugar solution.

Solution: Specific rotation is given by,

$$S = \frac{\theta}{l \times C}$$

Here given that, $S = 66°, \theta = 11°$ and $l = 20\text{ cm} = 2$ decimetre

Putting the values, we get

$$66 = \frac{11}{2 \times C}$$

or $$C = \frac{11}{2 \times 66} = \frac{1}{12} = \mathbf{0.0833\ g/cm^3}$$

Miscellaneous Problems

Problem 1: A beam of light is incident on a glass plate at angle of 58°6′ and the reflected beam is completely plane polarised. Find the refractive index of glass.

Solution: According to Brewster's law, $\mu = \tan p$

where μ is refractive index of glass and p is polarising angle.

Here given that, $p = 58°6'$ ($\because i = p$, polarising angle)

$\therefore$ $\mu = \tan 58°6' = 1.6$

Problem 2: A beam of light travelling in water strikes a glass plate which is also immersed in water. When the angle of incidence is 51°, the reflected beam is found to be plane polarised. Calculate the refractive index of glass.

Solution: By Brewster's law, $\mu = \tan p$

where μ = refractive index of glass with respect to water.

Here given that, $p = 51°$ ($i = p$)

$\therefore$ $\mu = \tan 51° = 1.235$

Problem 3: The refractive indices of glass and water are 1.54 and 1.33. Which will be greater, the polarising angle for a beam incident from water to glass or that for a beam incident from glass to water ?

Solution: According to Brewster's law, $\mu = \tan p$

where μ is the refractive index of the second medium with respect to the first and p is the polarising angle.

Here given that, for water to glass incidence,

$$\mu = \frac{1.54}{1.33} = 1.158$$

$\therefore$ $\tan p_1 = 1.158$ or $p = \tan^{-1}(1.158) = \mathbf{49.2°}$

And, for glass to water incidence,

$$\mu = \frac{1.33}{1.54} = 0.864$$

$\therefore$ $\tan p_2 = 0.864$ or $p_2 = \tan^{-1}(0.864) = \mathbf{40.8°}$

Thus, **the polarising angle p is larger for water to glass incidence.**

Problem 4: A ray of light is incident on the surface of a transparent plate of refractive index $\sqrt{3}$ at the polarising angle. Calculate the angle of refraction of the ray.

Solution: If the reflected light is completely plane polarised then the angle of incident will be equal to polarising angle p and by Brewster's law.

$$\mu = \tan p$$

Here given that, $\mu = \sqrt{3}$

$\therefore$ $\tan p = \sqrt{3}$ or $p = 60°$

Let r be the angle of refraction.

We also have that, $r + p = 90°$

$\therefore$ $r + 60° = 90°$ or $\boldsymbol{r = 30°}$

Problem 5: When sun light falls on the surface of water at an incidence of 53°, the reflected beam is found to be completely plane polarised. Find the angle of refraction and the refractive index of water.

Solution: As the reflected light is completely plane polarised, the angle of incidence is equal to polarising angle p. If r is the angle of refraction, then we have

$$r + p = 90°$$

or $r = 90 - p = 90° - 53° = \mathbf{37°}$

($\because$ angle of incidence = angle of polarisation)

And refractive index $\mu = \tan p$ (By Brewster's law)

$\therefore$ $\mu = \tan 53 = \mathbf{1.327}$

Problem 6: Estimate the angular position of the sun above the horizon when the light reflected from a calm water ($\mu = 1.33$) lake is completely plane polarised. Does this angle depend upon the wavelength of light ?

Solution: According to Brewster's law,

$$\mu = \tan p$$

Here given that, $\mu = 1.33$

$\therefore$ $\tan p = 1.33$ or $p = \tan^{-1}(1.33) = 53°$

So, angle of polarisation is 53° and this is equal to angle of incidence *i i.e.*, angle of the ray from the vertical. But we want to know the angle of sun above the horizon *i.e.*,

$$90° - 53° = \mathbf{37°}$$

Yes, this angle depend on the wavelength of light as the refractive index depends on the wavelength.

Problem 7: Calculate the velocities of ordinary and extraordinary rays in the calcite crystal in a plane perpendicular to the optic axis.

Given that, $\mu_o = 1.658, \mu_e = 1.486$ ***and*** $c = 3 \times 10^{10}$ **cm/sec.**

Solution: The velocity of the ordinary-ray in the crystal normal to the optic axis,

$$= \frac{c}{\mu_0} = \frac{3 \times 10^{10}}{1.658} = 1.81 \times 10^{10} \text{ cm/sec.}$$

Similarly, the velocity of the extraordinary-ray in the crystal normal to the optic axis,

$$= \frac{c}{\mu_e} = \frac{3 \times 10^{10}}{1.486} = 2.02 \times 10^{10} \text{ cm/sec.}$$

Problem 8: Two Nicols are crossed to each other. Now one of them is rotated through 60°. What percentage of incident unpolarised light will pass through the system ?

Solution: Initially Nicols are crossed *i.e.*, at 90°. When one is rotated through 60°, the angle between their principle planes become 30°.

Let I_o be the intensity of the incident wave. The incident light breaks up into an *O*-ray and *E*-ray when passes through first Nicol. The *O*-ray is lost by total internal reflection, while the *E*-ray is transmitted. Thus, the transmission of a Nicol for incident unpolarised light is 50% (neglecting absorption losses), *i.e.*, the intensity of the transmitted *E*-ray is $I_o/2$.

The plane polarised *E*-ray on entering the second Nicol again breaks up into *E'*-ray with intensity $(I_o/2)\cos^2\theta$ and *O'*-ray with intensity $(I_o/2)\sin^2\theta$ (By Malus law), where θ is the angle between the principle planes of the two Nicols. Again *O'*-ray is lost by total reflection and only *E'*-ray is transmitted. Thus, the intensity I_θ of the light emerging from the second Nicol is,

$$I_\theta = \frac{1}{2} I_o \cos^2\theta$$

Here given that, $\theta = 30°$

$$\therefore \quad \frac{I_\theta}{I_o} = \frac{1}{2}\cos^2 30° = \frac{(\sqrt{3}/2)^2}{2} = \frac{3}{8}$$

$$\therefore \quad \% \text{ transmission} = \frac{I_\theta}{I_o} \times 100 = \frac{3}{8} \times 100 = 37.5\%$$

Problem 9: Two Nicols have parallel polarising direction so that the intensity of transmitted light is maximum. Through what angle must either Nicol be turned if intensity is to drop by one fourth of its maximum value ?

Solution: Let θ be the angle through which the second Nicol is rotated. By Malus law, the transmitted intensity, through second Nicol is given by,

$$I_\theta = I\cos^2\theta$$

where I is the maximum value of intensity when both Nicol's are parallel.

Given that, $I_\theta = (3/4)I$

$\therefore$ $$(3/4)I = I\cos^2\theta$$

$$\cos\theta = \left(\pm\sqrt{\frac{3}{4}}\right) \quad \text{or} \quad \theta = \cos^{-1}\left(\pm\sqrt{\frac{3}{2}}\right)$$

$\therefore$ $$\theta = \pm 30°$$

Problem 10: A polariser and an analyser are set in such a way that the intensity of the emergent light is maximum. What percentage of the maximum intensity of light is transmitted from the analyser if either is rotated by 30°, 45°, 60° ?

Solution: If θ is the angle between polariser and analyser, then the transmitted intensity through analyser is given by,

$$I_\theta = I\cos^2\theta$$

where I is the maximum intensity of the emergent light.

Here given that,

1. If $\theta = 30°$

   $\therefore$ $$I_\theta = I\cos^2 30 = I\left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3I}{4}$$

   or Percentage of transmitted light from the analyser,

   $$\frac{I_\theta}{I}\times 100 = \frac{3}{4}\times 100 = 75\%$$

2. If $\theta = 45°$

   $\therefore$ $I_\theta = I\cos^2 45$ or $\frac{I_\theta}{I}\times 100 = \frac{1}{2}\times 100 = 50\%$

3. If $\theta = 60°$

$$\therefore \quad I_\theta = I\cos^2 60° \quad \text{or} \quad \frac{I_\theta}{I}\times 100 = \frac{1}{4}\times 100 = 25\%$$

Problem 11: An unpolarised beam of light is incident on a group of four polarising crystals which are lined up so that the characteristic direction of each is rotated by 30° clockwise with respect to the preceding crystal. What fraction of the incident intensity is transmitted ?

Solution: Let I_o be the intensity of the incident unpolarised light. The intensity I of the (polarised) transmitted beam by the first crystal is given by,

$$I = \frac{I_0}{2} \qquad ...(1)$$

Now, next crystal is at 30° with respect to first, then the intensity of transmitted beam by the second crystal is given by,

$$I_\theta = I\cos^2\theta = I\cos^2 30°$$

$$= (3/4)I \qquad ...(2)$$

Further, the third crystal is at 30° with respect to second, then the intensity of transmitted beam by the third crystal is given by,

$$I'_\theta = (3/4)\, I\cos^2\theta = (3/4)\, I\cos^2 30°$$

$$= \frac{3}{4}I\times\frac{3}{4} = \frac{9}{16}I$$

and again, the fourth crystal is at 30° with respect to third, then the intensity of transmitted beam by the fourth crystal is given by,

$$I''_\theta = \frac{9}{16}I\cos^2\theta = \frac{9}{16}I\cos^2 30°$$

$$= \frac{9}{16}I\times\frac{3}{4} = \frac{27}{64}I$$

or $$I''_\theta = \frac{27}{64}\cdot\frac{I_0}{2} = \frac{27}{128}I_0 \qquad \text{[(using eq. (1)]}$$

So, the fraction of incident beam finally transmitted will be $I''_\theta / I_o = 27/128$

***Problem 12:** An analysing Nicol examines two adjacent plane polarised beams A and B whose planes of polarisation are mutually perpendicular. In one position of the analyser, beam B shows zero intensity. From this position a rotation of 30° shows the two beams of equal intensity. Deduce the intensity ratio I_A/I_B of two beams.*

Solution: The intensity of the emergent polarised light by the analyser is given by,

$$I_\theta = I\cos^2\theta$$

where I is intensity of plane polarised light falling on analyser and θ is the angle between the plane of transmission of the analyser and the plane of vibration of the incident light. Given that, initially the beam B shows zero intensity. It means that, $\theta = 90°$ for beam B and $\theta = 0°$ for beam A because plane of polarisation are mutually perpendicular. Now, when the analyser is rotated through 30°, then we have

$\theta = 30°$ for beam A

and $\theta = 60°$ for beam B

Thus, if I' be the intensity of beam A and also of beam B as transmitted by the analyser, then we have

$$I' = I_A\cos^2 30° = I_B\cos^2 60°$$

$$\therefore \quad \frac{I_A}{I_B} = \frac{\cos^2 60°}{\cos^2 30°}\frac{(1/2)^2}{(\sqrt{3}/2)} = \frac{1}{3}$$

***Problem 13:** Plane polarised light passes through a quartz plate with its axis parallel to the faces. Calculate the least thickness of the plate for which the emergent beam will be circularly polarised. Given that,* $\mu_e = 1.5533, \mu_o = 1.5442$ *and* $\lambda = 5\times10^{-5}$ **cm.**

Solution: If a plane polarised light is converted into circularly polarised light on passing through a quartz plate, it means that quartz plate is quarter wave plate *i.e.*, it introduces a path difference of $\lambda/4$ to the rays. For quarter wave plate, we have

$$t = \frac{\lambda}{4(\mu_e - \mu_o)}$$

Here given that, $\lambda = 5\times10^{-5}$ cm, $\mu_e = 1.5533$, $\mu_o = 1.5442$

$$\therefore \quad t = \frac{5\times10^{-5}}{4\times(1.5533 - 1.5442)} = \frac{5\times10^{-5}}{4\times9.1\times10^{-3}}$$

$$= 1.37\times10^{-3}\text{cm}$$

Problem 14: Calculate the thickness of a calcite plate which would convert plane polarised light into circularly polarised light. Given that, $\mu_e = 1.658, \mu_o = 1.486$ ***at the wavelength of light used*** $= 5890\ \text{Å}$.

[Meerut 2005B]

Solution: The calcite plate must be quarter wave plate because it converts plane polarised light into circularly polarised light.

We have, $$t = \frac{\lambda}{4(\mu_e - \mu_o)}$$

Here given that, $\mu_e = 1.658,\ \mu_o = 1.486$ and $\lambda = 5890 \times 10^{-8}$ cm

$\therefore$ $$t = \frac{5890 \times 10^{-8}}{4(1.658 - 1.486)} = \frac{5.89 \times 10^{-5}}{6.88 \times 10^{-1}} = \mathbf{8.56 \times 10^{-5}\ cm}$$

Problem 15: Calculate the thickness of a doubly refracting crystal plate required to introduce a path difference of $\lambda/2$ ***between the ordinary and extraordinary ray, when*** $\lambda = 6000\ \text{Å}, \mu_o = 1.55$ ***and*** $\mu_e = 1.54$.

Solution: When a crystal plate introduces a path difference of $\lambda/2$ between the rays, then plate must be half wave plate. For half wave plate, we have

$$t = \frac{\lambda}{2(\mu_e - \mu_o)} \qquad [\because (\mu_e \sim \mu_o)]$$

Here given that, $\mu_e = 1.55$ and $\mu_o = 1.54$ and $\lambda = 6000 \times 10^{-8}$ cm

$\therefore$ $$t = \frac{6000 \times 10^{-8}}{2(1.55 - 1.54)} = \frac{6 \times 10^{-5}}{2 \times 0.01} = \mathbf{3 \times 10^{-3}\ cm}$$

Problem 16: Plane polarised light $(\lambda = 5 \times 10^{-7}\ \text{m})$ ***is incident on a quartz plate cut parallel to the optic axis. Find the least thickness of the plate for which the ordinary and extraordinary rays combine to form plane polarised light on emergence. What multiples of this thickness would give the same result. Given that*** $\mu_e = 1.5533$ ***and*** $\mu_o = 1.5442$

Solution: When a ray becomes plane polarised after passing through the quartz crystal plate, then it must be half wave plate. therefore, we have

$$t = \frac{\lambda}{2(\mu_e - \mu_o)}$$

Here given that, $\mu_e = 1.5533, \mu_o = 1.5442$ and $\lambda = 5 \times 10^{-7}$ m

$$\therefore \quad t = \frac{5 \times 10^{-7}\,\text{m}}{2\,(1.5533 - 1.5442)} = \frac{5 \times 10^{-7}\,\text{m}}{2 \times 0.0091}$$

$$= 2.75 \times 10^{-5}\ \text{m} = \mathbf{2.75 \times 10^{-3}\,cm}$$

The thickness which gives the same results are ***3t, 5t.***

Problem 17: A phase retardation plate of quartz has thickness 0.1436 mm. For what wavelength in the visible range (4500 Å - 8000 Å) will it acts as (i) quarter wave plate (ii) half wave plate ? Given that, $\mu_o = 1.5443$, $\mu_e = 1.5533$.

Solution:

(i) For a quarter wave plate,

$$t = \frac{n\lambda}{4\,(\mu_e - \mu_o)} \quad \text{or} \quad \lambda = \frac{4t\,(\mu_e - \mu_o)}{n}$$

where $n = 1, 5, 9, 13, \ldots$ etc.

Here given that, $\mu_e = 1.5533, \mu_o = 1.5443$ and $t = 0.1436\ \text{mm} = 1.436 \times 10^{-4}\ \text{m}$

$$\lambda = \frac{4 \times 1.436 \times 10^{-4} \times (1.5533 - 1.5443)}{n}$$

$$= \frac{4 \times 1.436 \times 10^{-4} \times 0.009}{n}$$

$$= \frac{51696 \times 10^{-10}}{n}\ \text{m} = \frac{51696}{n}\ \text{Å}$$

Putting $n = 1, 5, 9, 13 \ldots$, we get $\lambda = 51696$ Å, 10339 Å, 5745 Å, 3976 Å respectively. Thus, in visible range for **5745** Å, this plate will acts as a quarter wave plate.

(ii) For a half wave plate,

$$t = \frac{n\lambda'}{2(\mu_e - \mu_o)} \quad \text{or} \quad \lambda' = \frac{2t\,(\mu_e - \mu_o)}{n}$$

where $n = 1, 3, 5, 7 \ldots$

or $$\lambda' = \frac{2 \times 1.436 \times 10^{-4} \times 0.009}{n} = \frac{25848}{\text{n}}\ \text{Å}$$

Putting $n = 1, 3, 5, 7$, we get $\lambda' = 25848$ Å, 8616 Å, 5169 Å and 3639 Å respectively. Thus, in visible range for **5169 Å** this plate will acts as a half wave plate.

Problem 18: Calculate the required thickness of a quartz crystal plate whose faces are perpendicular to the optic axis in order to produce a rotation of 43.5° for sodium light of wavelength 5893 Å. The specific rotation in quartz for this light is 21.75°/mm

Solution: We know that the rotation is directly proportional to the thickness of quartz and Given that, 1 mm thick quartz produces 21.75° rotation, therefore to produce a rotation of 43.5°,

$$d = \frac{1\,\text{mm}}{21.75} \times 43.5° = \mathbf{2.00\ mm}$$

Problem 19: The specific rotation of quartz at 5086 Å is 29.73°/mm. Calculate the difference in refractive indices of the material for clockwise and anticlockwise circularly polarised light.

Solution: The optical rotation is given by,

$$\theta = \frac{\pi d}{\lambda}(\mu_A - \mu_C) \qquad [\because\ \mu_L = \mu_A \text{ and } \mu_R = \mu_C]$$

Here given that, $\theta/d = 29.73°/\text{mm}$ and $\lambda = 5086 \times 10^{-7}\,\text{mm}$

Putting the values, we get

$$(\mu_A - \mu_C) = \frac{29.73 \times 5086 \times 10^{-7}}{180} \qquad [\because\ \pi \text{ radian} = 180°]$$

$$= \mathbf{8.4 \times 10^{-5}}$$

Problem 20: The refractive indices of quartz for right handed and left handed circularly polarised sodium light propagating along the optic axis are 1.54420 and 1.54427 respectively. Calculate the specific rotation of quartz (given, $\lambda = 6 \times 10^{-5}$ cm).

Solution: We have the optical rotation,

$$\theta = \frac{\pi d}{\lambda}(\mu_L - \mu_R)$$

or specific rotation, $\frac{\theta}{d} = \frac{\pi}{\lambda}(\mu_L - \mu_R)$

Here given that, $\mu_L = 1.54427, \mu_R = 1.54420$ and $\lambda = 6 \times 10^{-5}$ cm

$$\therefore \quad \frac{\theta}{d} = \frac{180°(1.54427 - 1.54420)}{6 \times 10^{-5}} = \frac{180° \times 7 \times 10^{-5}}{6 \times 10^{-5}} \qquad [\because\ \pi \text{ radian} = 180°]$$

$$= \mathbf{210°/cm}$$

Problem 21: The indices of refraction of quartz for right handed and left handed circularly polarised light of wave length 7620 Å are 1.53914 and 1.53920 respectively. Calculate the rotation of the plane of polarisation of the light in degree produced by a plate 0.5 mm thick.

Solution: The optical rotation of plane of polarisation is given by,

$$\theta = \frac{\pi}{\lambda} d\,(\mu_L - \mu_R)$$

Here given that, $d = 0.5$ mm, $\mu_L = 1.53920$, $\mu_R = 1.53914$ and $\lambda = 7620 \times 10^{-7}$ mm

Putting the values, we get

$$\theta = \frac{180° \times 0.5 \times (1.53920 - 1.53914)}{7620 \times 10^{-7}} = \frac{180° \times 0.5 \times 6 \times 10^{-5}}{7620 \times 10^{-7}}$$

$$= 7.09°$$

Problem 22: A 20 cm long tube containing sugar solution is placed between crossed Nicols and illuminated by light of wavelength 6×10^{-5}cm. If the specific rotation of sugar is 60° and the optical rotation produced is 12°, what is the concentration of the solution ?

Solution: The specific rotation is given by,

$$S = \frac{\theta}{l \times C} \qquad \text{or} \qquad C = \frac{\theta}{l \times S}$$

Here given that, $\theta = 12°, S = 60°$ and $l = 20$ cm $= 2$ decimetre

Putting the values, we get,

$$C = \frac{12°}{60° \times 2} = \frac{1}{10} = 0.1 \text{gm/c.c.}$$

Problem 23: A solution of camphor in alcohol in a tube of 20 cm long is found to rotate the plane of vibration of light passing through it by 33°. What must be the density of camphor in gm/c.c. of solution ? The specific rotation of camphor is +54°.

Solution: The specific rotation is given by,

$$S = \frac{\theta}{l \times C} \qquad \text{or} \qquad C = \frac{\theta}{l \times S}$$

Here given that, $S = 54°, \theta = 33°$ and $l = 20\text{ cm} = 2$ decimetre

$$\therefore \quad C = \frac{33°}{54° \times 2} = \mathbf{0.306\,gm/c.c.}$$

So, the density of camphor is **0.306 gm/c.c.**

Problem 24: Calculate the specific rotation for sugar solution using the following data: Length of the tube =20 cm, Volume of the tube =120 cm^3, Quantity of sugar dissolved =6 gm and Angle of rotation of the analyser for restoring equal intensity 6.6°.

Solution: The specific rotation is given by,

$$S = \frac{\theta}{l \times C}$$

Here given that, $\theta = 6.6°, l = 20\text{ cm} = 2$ decimetre

and $C = (6/120)\,\text{gm}/\text{cm}^3 = 0.05\,\text{gm/c.c.}$

$$\therefore \quad S = \frac{6.6°}{2 \times 0.05} = \mathbf{66°}$$

Problem 25: Calculate the specific rotation if the plane of polarisation is turned through 26.4°, in traversing 20 cm length of 20% sugar solution.

Solution: The specific rotation is given by,

$$S = \frac{\theta}{l \times C}$$

Here given that, $\theta = 26.4°, l = 20\text{ cm} = 2$ decimetre,

and $C = 20\% = (20/100) = 0.2\,\text{gm/c.c.}$

$$\therefore \quad S = \frac{26.4°}{2 \times 0.2} = \mathbf{66°}$$

Problem 26: Calculate the specific rotation of sugar solution, if the plane of polarisation is rotated by 13.2°. The length of the tube containing sugar solution is 20 cm and 5 gm of sugar is dissolved in 50 c.c. of water.

Solution: The specific rotation is given by, $S = \frac{\theta}{l \times C}$

Here given that, $\theta = 13.2°, l = 20\text{ cm} = 2$ decimetre,

and $C = (5/50)\text{ gm/c.c.} = 0.1\text{ g/c.c.}$

$$\therefore \quad S = \frac{13.2°}{2 \times 0.1} = 66°$$

Problem 27: A 20 cm long tube is filled with a solution of 15 gm of cane sugar in 100 c.c. of water. Find the angle of rotation of the plane of polarisation of a beam of plane polarised light when it passes through the solution. Specific rotation for canesugar $= 66.5°/dmgmcm^3$.

Solution: The specific rotation is given by,

$$S = \frac{\theta}{l \times C} \qquad \text{or} \qquad \theta = S \times l \times C$$

Here given that, $S = 66.5°, l = 20\text{ cm} = 2$ decimetre

and $C = (15/100)\text{ gm/c.c.} = 0.15\text{ gm/c.c.}$

Putting the values, we get, $\theta = 66.5° \times 2 \times 0.15 = 19.95° \approx 20°$

Problem 28: On introducing a polarimeter tube 25 cm long and containing sugar solution of unknown strength, it is found that the plane of polarisation is rotated through 10°. Find the strength of the sugar solution in gm/cm^3. ***(Given that, the specific rotation of sugar solution is 60° per decimetre per unit concentration).***

Solution: The specific rotation is given by,

$$S = \frac{\theta}{l \times C}$$

Here given that, $S = 60°, \theta = 10°, l = 25\text{ cm} = 2.5$ decimetre

So, $$C = \frac{10°}{60° \times 2.5} = \frac{1}{15} = 0.067\text{ gm/c.c.}$$

Problem 29: A 200 mm long tube containing 48 cm^3 ***of sugar solution produces an optical rotation of 11° when placed in a saccharimetre. If the specific rotation is 66°, calculate the quantity of sugar contained in the form of a solution.***

Solution: We have that, $$S = \frac{\theta}{l \times C}$$

Here given that, $S = 66°, \theta = 11°, l = 200 \text{ mm} = 2$ decimetre.

$$\therefore \quad C = \frac{11}{66 \times 2} = \frac{1}{12} = 0.0833 \text{ gm/c.c.}$$

But Given that, volume of the tube, $V = 48 \text{ cm}^3$

So, mass of the sugar in the solution $= C \times V = 0.833 \times 48 = \mathbf{4 \text{ gm}}$

Problem 30: 80 gm of impure sugar when dissolved in a litre of water gives an optical rotation of 9.9° when placed in a tube of length 20 cm. If the specific rotation of sugar is 66°, find the % purity of the sugar sample.

Solution: The specific rotation is given by,

$$S = \frac{\theta}{l \times C} \quad \text{or} \quad C = \frac{\theta}{l \times S}$$

Here given that, $\theta = 9.9°, l = 20 \text{ cm} = 2 \text{decimetre and } S = 66°$

$$\therefore \quad C = \frac{9.9°}{2 \times 66°} = 0.075 \text{ gm/c.c.}$$

or $C = 75 \text{ gm/litre}$ [1 litre = 1000 c.c..]

But given that, 80 gm impure sugar is dissolved in 1 litre. So, purity of solution is,

$$\frac{75}{80} \times 100 = \mathbf{93.75\%}$$

Problem 31: A certain length of 5% solution causes the optical rotation of 20°. How much length of 10% solution of the same substance will cause 35° rotation ?

Solution: Let S be the specific rotation of solution and l_1 be the length of the tube for 5% and l_2 be the length of tube for 10% solution, then we have,

$$S = \frac{\theta_1}{l_1 \times C_1} = \frac{\theta_2}{l_2 \times C_2}$$

Here given that, $\theta_1 = 20°, \theta_2 = 35°, C_1 = 5\%, C_2 = 10\%$

Putting the values, we get, $\frac{20°}{l_1 \times 5} = \frac{35°}{l_2 \times 10}$ or $l_2 = \frac{35 \times 5}{20 \times 10} l_1 = \frac{7}{8} \boldsymbol{l_1}$

Problem 32: A sugar solution in a tube of length 20 cm produces optical rotation of 13°. The solution is then diluted to one third of its previous concentration. Find the optical rotation produced by 30 cm long tube containing the diluted solution.

Solution: Let S be the specific rotation of the sugar solution, then we have

$$S = \frac{\theta_1}{l_1 \times C_1} = \frac{\theta_2}{l_2 \times C_2}$$

Here given that, $\theta_1 = 13°, l_1 = 20 \text{ cm} = 2 \text{ decimetre}, C_1 = C \text{ (let)}$

and $l_2 = 30 \text{ cm} = 3 \text{ decimetre}, C_2 = C/3$

$$\therefore \quad \frac{13°}{2 \times C} = \frac{\theta_2}{3 \times C/3}$$

or $$\theta_2 = 13°/2 = \mathbf{6.5°}.$$

Exercise

(A) Descriptive Type Questions

1. What do you mean by polarisation of light ? Distinguish between polarised light and ordinary (unpolarised) light. How do we represent light vibrations ?
[Meerut 2003B, 03, 02B]

2. Polarisation requires that light waves are transverse. Comment on this.
[Meerut 2008B]

3. What is Brewster's law ? Show that when light is incident on a transparent substance at the polarising angle, the reflected and refracted rays are at right angles.
[Meerut 2003]

4. What is polarising angle and its value for air glass combination? **[Meerut 2007]**

5. Explain the action of a pile of plates in producing plane polarised light.

6. What do you mean by Double refracting crystal and explain the phenomenon of double refraction in uniaxial (calcite) crystal ?
[Meerut 2005, 03B, 03, 02]

7. Describe the optic axis, principal section and principal plane of the uniaxial crystals.

8. Describe the construction of a Nicol's prism and show how it can be used as a polariser or as an analyser ?
[Meerut 2012, 11, 07B, 06, 05B, 04B, 03B, 02B, 01B, 00B]

9. What do you understand by double refraction in crystals ? Explain Huygen's theory of double refraction in a uniaxial crystal. **[Meerut 2009, 06B, 01]**

10. Explain Polaroids and give their uses. **[Meerut 2004B, 03B, 02]**

11. Explain phase retardation plate.

12. What are quarter wave and half wave plates ? Explain how they are used in the analysis of different types of polarised light beams ? **[Meerut 2008, 05B, 05, 03B, 01B, 01]**

13. Describe Babinet's Compensator.

14. Explain plane polarised, circularly and elliptically polarised light. How we produce the circularly and elliptically polarised light from linear polarised light ? **[Meerut 2011B, 10, 04B, 03B, 02B, 02, 00]**

15. With the help of Nicol's prism and a quarter wave plate, how the plane polarised, circularly polarised and elliptically polarised light are produced and detected ? **[Meerut 2005, 00]**

16. Describe double image prism and how it is used in spectro-photometer?

17. What do you understand by optical rotation ? Give an outline of Fresnel's theory of optical rotation. Describe an experiment to verify it. **[Meerut 2009, 08B, 07, 05]**

18. What do you mean by optical rotation? Show how Fresnel's theory explain optical rotation? What is experimental evidence there in support of the theory? **[Meerut 2007]**

19. Define specific rotation. Describe the construction and working of a Laurent's half shade polarimeter. How will you use it to find the specific rotation of sugar (glucose) solution? **[Meerut 2012, 11, 09B, 08, 07B, 06B, 05, 03, 02B, 01B, 01]**

20. Explain the action of biquartz plate in a biquartz polarimeter. **[Meerut 2005]**

21. Explain the construction and working of a biquartz polarimeter. Describe the action of biquartz plate in a biquartz polarimeter. How you would use it to find the specific rotation of an optically-active substance ? **[Meerut 2011B, 07B, 05, 03B]**

22. Define specific rotation. Explain the working of a biquartz polarimeter. How would you use it to determine the specific rotation of sugar solution? **[Meerut 2007B, 06]**

23. Explain the working of Laurent half shade and biquartz plate in optical rotation. **[Meerut 2009B, 08B]**

(B) Short Answer Type Questions

1. Define plane of vibration and plane of polarisation. **[Meerut 2003]**

2. What do you mean by double refraction? How this phenomenon can be shown experimentally?

3. State and prove the law of Malus. **[Meerut 2007B, 02]**

4. If the plane of vibration of the incident beam makes an angle of 30° with the optic axis, compare the intensities of extraordinary and ordinary light.

5. What do you understand by double refraction in a crystal ? Are the ordinary and extraordinary rays plane polarised ? What happens when two such crystals are rotated relative to one another ?

6. Explain, what do you mean by dichroism ? **[Meerut 2009B, 08, 07, 03B, 02]**

7. What do you understand by the principle refractive indices of a double refracting crystal?

8. What will be the state of polarisation of the light, emerging from a quarter-wave plate when the incident light is circularly polarised, unpolarised?

9. How would you distinguish between unpolarised light and partially plane polarised light ?

10. How would you distinguish between circularly and elliptically polarised light ? **[Meerut 2003]**

11. If you have a quarter and half wave plate, how would you distinguish between them ?

12. How would you distinguish between circularly polarised light and a mixture of circularly polarised and unpolarised light? **[Meerut 2005, 01]**

13. How would you distinguish between circularly polarised and unpolarised light? **[Meerut 2003]**

14. How would you distinguish between elliptically polarised light and a mixture of upolarised and plane polarised light? **[Meerut 2001B]**

15. Draw a chart which shows the distinction between unpolarised and polarised (circularly, elliptically or plane polarised) light.

(C) Very Short Answer Type Questions

1. Define polarisation of light. **[Meerut 2011]**
2. What is Brewster's law? **[Meerut 2011]**
3. Define polarising angle. **[Meerut 2009]**
4. Define principle section of a crystal.
5. What do you mean by Nicol's prism?
6. State the law of Malus. **[Meerut 2011B, 10, 06B]**
7. Explain double refraction in a crystal. **[Meerut 2012, 09, 05]**
8. Define plane polarised light. **[Meerut 2005B]**
9. Define circularly polarised light. **[Meerut 2007B]**
10. Define elliptically polarised light. **[Meerut 2007B]**
11. What do you mean by quarter wave plate? **[Meerut 2008B, 05B]**
12. What do you mean by half wave plate ? **[Meerut 2008B, 06, 05B]**
13. Define optical activity and optically active substances.
14. Define specific rotation of a substance.
15. Write two types of optically active substances.

(D) Multiple Choice Questions

1. In a plane polarised beam of light if the vibrations are parallel to the plane of paper, then they are represented by :

   (a) Arrows (b) Dots

   (c) Both arrow and dots (d) None of the above

2. A ray of light is incident on the surface of a glass plate of refractive index 1.5 at the polarising angle. What is the angle of polarisation ?

   (a) 56.3° (b) 33.7°

   (c) 90° (d) None of the above

3. In an uniaxial double refracting crystal, the extraordinary ray has vibration:

   (a) In the principle plane

   (b) Perpendicular to the principle plane

   (c) In the direction of propagation of light ray

   (d) None of the above

4. In Nicol's prism the ray, eliminated by total internal reflection is :

(a) Ordinary ray (b) Extraordinary ray

(c) Both (d) None of the above

5. Nicol's prism is used as :

(a) Polariser (b) Analyser

(c) Polariser and analyser both (d) None of the above

6. When a polariser and an analyser are perpendicular then the intensity of emergent beam will be :

(a) Maximum (b) Minimum

(c) Zero (d) None of the above

7. A half wave plate introduces a further phase-difference of :

(a) $\pi/2$ (b) π

(c) $\pi/4$ (d) 2π

8. If the plane polarised light falls normally on a quarter wave plate making an angle 45° with the optic axis, then the emergent light will be :

(a) Plane polarised (b) Circularly polarised

(c) Elliptically polarised (d) All of the above

9. Half shade device consists of :

(a) Only glass

(b) Only quartz

(c) Half part of glass and half part of quartz

(d) None of the above

10. If the plane of polarisation is turned through 13.2°, then the specific rotation of sugar will be, (given that the length of tube is 20 cm and concentration of sugar solution is 10% :

(a) 66° (b) 33°

(c) 90° (d) 180°

(E) Fill in the Blank

1. In a plane polarised light, if the vibrations are along a straight line perpendicular to the plane of the paper then they are represented by .....

2. Double refracting crystal splits up the incident ray into ..... rays.

3. In Nicol's prism, one ray is eliminated by total ..... .

4. In double refracting crystal, the ray which does not obey the law of refraction, is called ..... .

5. In double refracting crystal, the ray which obeys the law of refraction, is called ..... .

6. The velocities of the ordinary and extraordinary rays are the ..... along optic axis.

7. A half wave plate introduces a further path difference of ..... .

8. The thickness of a quarter wave plate is given by ..... .

9. If the intensity of the emergent beam varies with zero minimum after passing through rotated Nicol, then the light is ..... .

10. If a plane polarised light falls normally on a quarter wave plate, then the emergent light will be circularly or ..... polarised.

(F) True/False

1. An unpolarised light is represented by a dot.

2. According to Brewster's law, $\mu = \tan r$, where μ is the refractive index and r is the angle of refraction.

3. In an uniaxial double refracting crystal, the ordinary ray has vibrations perpendicular to the principle section of the crystal.

4. In Nicol's prism, the ordinary ray is eliminated by total internal reflection.

5. In double refraction crystal, the ray which does not obey the law of refraction is called ordinary ray.

6. A quarter wave plate introduces a further path difference of $\lambda/2$.

7. The angle by which the plane of vibration is rotated after passing through right handed optically active substance is given by $(\pi d/\lambda)(\mu_L - \mu_R)$.

Answers

(B) Short Answer Type Questions

(Hint and Solutions)

1. When an ordinary light is passed through a tourmaline crystal, the light is polarized and vibrations of light waves are confined to only one direction perpendicular to the direction of propagation of light. The plane of polarization is that plane in which no vibrations occur and passing through the direction of propagation. The plane *ABCD* in fig. 3.27 is the plane of polarization. *The plane in which vibrations occur and passing through the direction of propagation is known as plane of vibration*. The plane *EFGH* in fig. 3.27 is the plane of vibration.

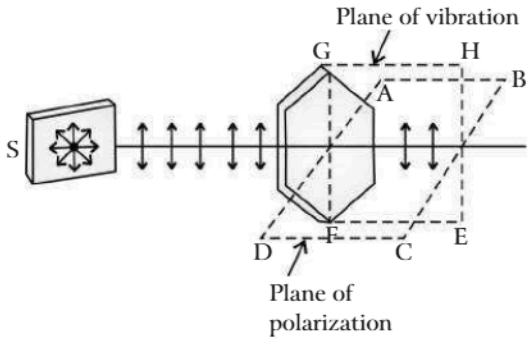


Fig. 3.27

3. **Malus's Law:** According to law of Malus, when a completely plane polarized light ray is incident on an analyser, the intensity of the emergent light through the analyser varies as the square of the cosine of the angle between the plane of transmission of the analyser and the plane of polarizer.

Let a be the amplitude of the light transmitted by the polarizer and θ is the angle between the plane of polarizer and the plane of analyser (Fig. 3.28). Resolving the amplitude into two components:

(i) $a\cos\theta$, along OA (parallel to the plane of transmission of the analyser)

(ii) $a\sin\theta$, along OB (perpendicular to the plane of transmission of the analyser).

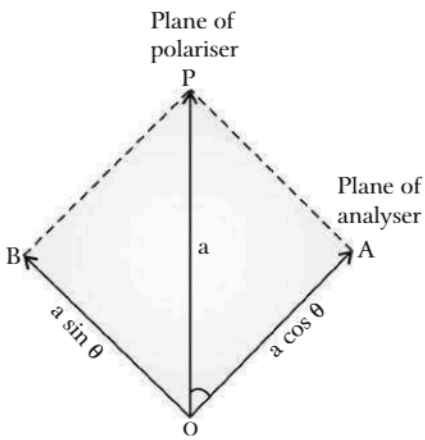


Fig. 3.28

Only the component $a\cos\theta$ is transmitted through the analyser. Therefore, the intensity of light through the analyser will be,

$$I_\theta = a^2 \cos^2\theta = I\cos^2\theta \qquad ...(1)$$

where $I = a^2$, the intensity of the incident plane polarized light.

So, $$I_\theta \propto \cos^2\theta$$

If the polarizer and analyser are parallel *i.e.*, $\theta = 0°$, then

$$I_\theta = I \quad (\cos 0° = 1)$$

It the polarizer and analyser are perpendicular, ($\theta = 90°$), then,

$$I_\theta = 0 \quad (\cos 90° = 0)$$

4. Intensity of the extraordinary ray is given by,

$$I_e = a^2 \cos^2\theta$$

Intensity of the ordinary ray, $I_o = a^2 \sin^2\theta$

$\therefore$ $$\frac{I_e}{I_o} = \frac{a^2\cos^2\theta}{a^2\sin^2\theta} = \frac{\cos^2\theta}{\sin^2\theta}$$

Here, $\theta = 30°$

$\therefore$ $$\frac{(\sqrt{3}/2)^2}{(1/2)^2} = 3$$

5. **Double Refraction:** When a ray of light is passed through a crystal (calcite), the ray is split up into two refracted rays. This phenomenon is called double refraction. One of these two rays obeys the refraction laws, called ordinary ray (*O*-ray) and other ray behaves in an extraordinary way and is, therefore called the extraordinary ray (*E*-ray).

Yes it is found that both the ordinary and extraordinary rays are plane polarized having vibrations perpendicular to each other. The polarization of light by double refraction crystal can be demonstrated by using a pair of such crystals.

If the two crystals are placed in the path of the beam with their principal planes parallel to each other, the two resultant images O_1 and E_1 are seen Fig. 3.29(a) separated by a distance equal to the sum of the displacements produced separately.

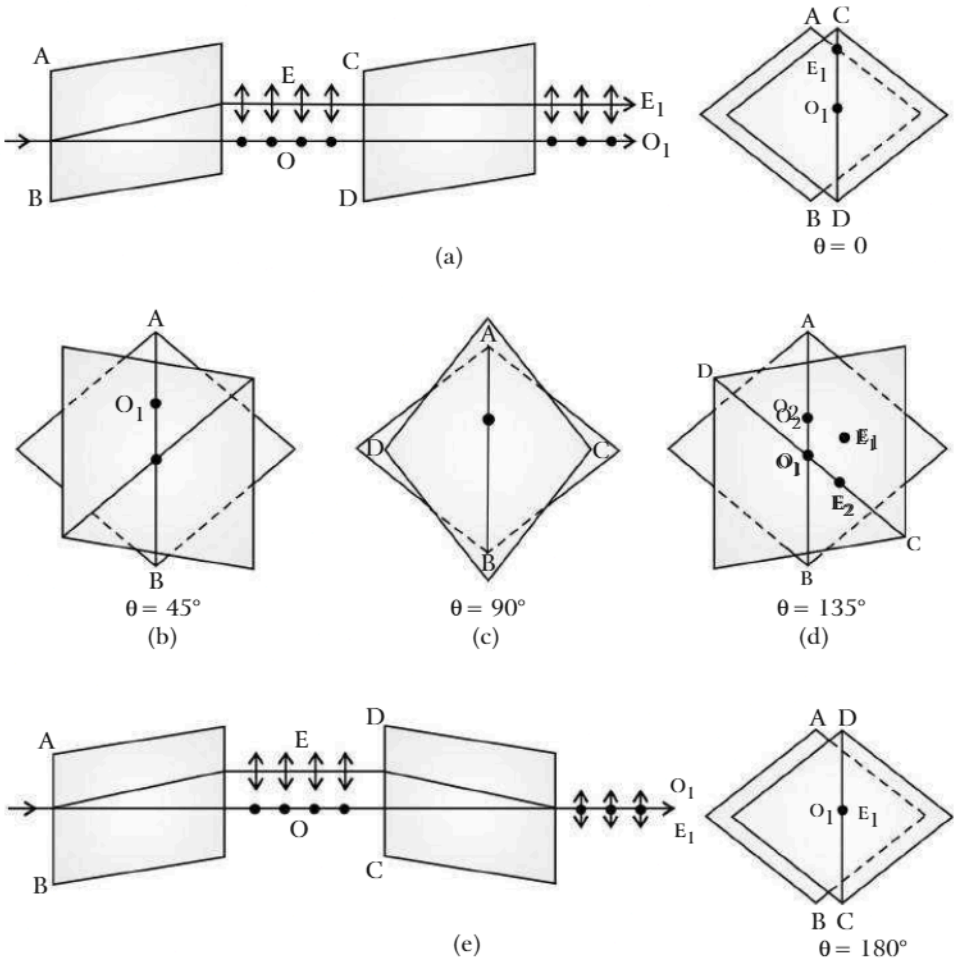


Fig. 3.29

When the second crystal is rotated, each of the ray *O* and *E* from the first crystal suffers double refraction in the second crystal, giving rise to four image. As the rotation is continued O_1 and O_2 remain fixed while E_1 and E_2 increases Fig.3.29(b). At 90° rotation, we again have two beams but their image changes the ordinary of the first becomes the extraordinary of the second and vice-versa. Fig. 3.29(c). On further rotation O_1 and E_1 reappear and start gaining intensity, while O_2 and E_2 start losing intensity so, again four image appears Fig. 3.29(d). At 180°, the principal sections of the crystals are parallel having optic axis in opposite directions. The images O_2 and E_2 are vanished, while O_1 and E_1 combine into single beam in the centre Fig. 3.29(e), if the crystals are of equal thickness.

6. **Dichrosm:** Some double refracting crystals have the property of absorbing one of the two refracting rays completely while allowing the other to emerge. The selective absorption by the crystal is known as dichroism. Tourmaline is the best example of such type crystal.

   When a beam of unpolarized light is incident on a tourmaline crystal of about 1mm thickness.

   The beam is split up into two plane polarized rays, the ordinary ray (*O*-ray) and the extraordinary ray (*E*-ray), vibrating in mutually perpendicular planes. The *O*-ray is completely absorbed while the *E*-ray is transmitted and it is plane polarized ray. In this way dichroic crystal produce plane polarized light.

7. **Principal refractive indices of a crystal:** The refractive index of a medium is defined as the ratio of the velocity of light in vacuum to the velocity of light in the medium. In the case of double refracting crystals, the *O*-ray travels with the same velocity in all directions but *E*-ray has different velocities in different directions. We have that along the optic axis the velocity of *E*-ray is same as that of *O*-ray and hence in the direction perpendicular to the optic axis the velocity of the rays have the maximum difference. So there are two principal refractive indices of double refracting crystal, one corresponding to the *O*-ray, and the other corresponding to the *E*-ray travelling normal to the optic axis.

   The principal refractive index of the *O*-ray is,

$$\mu_0 = \frac{\text{Velocity of light in vacuum}}{\text{Velocity of O-ray in the crystal}}$$

   and the principal refractive index of the *E*-ray is,

$$\mu_e = \frac{\text{Velocity of light in vacuum}}{\text{Velocity of E-ray in the crystal normal to the optic axis}}$$

8. If the light incident on quarter wave plate is circularly polarized, the incident wave can be resolved into two plane polarized rays having a phase difference of $\pi/2$. The quarter wave plate introduces another phase difference of $\pi/2$. So, the emergent light will be plane polarized light.

   If the incident light is unpolarized, then the emergent light is also unpolarized.

9. To distinguish between unpolarized and partially plane polarized light, the given light is allowed to pass through a rotating nicol. On rotation of nicol, the unpolarized light shows no variation in the intensity of the emergent light while the partially plane polarized shows the intensity variation with non-zero minimum.

10. To distinguish between circularly polarized light and elliptically polarized light, the given light is allowed to pass through a rotating nicol prism. On rotation of nicol, the circularly polarized light shows no variation in the intensity of the emergent light while in the case of elliptically polarized light, the emergent light shows intensity variation with non-zero minimum.

11. To distinguish between a quarter wave plate and half wave plate, plane polarized light is made to fall normally on each of them. The emergent light is then examined with a nicol prism rotating about the direction of light.

    In the case of quarter wave plate the emergent light may be elliptically, circularly or plane polarized depending upon the orientation of plate. Hence, on examining through the rotating nicol we shall observe (i) variation in intensity with non-zero minimum, when the emergent light is elliptically polarized. (ii) no variation in intensity when the emergent light is circularly polarized. (iii) variation in intensity with zero minimum when the emergent light is plane polarized.

    In the case of half wave plate, the emergent light is plane polarized light for all orientations of the plate. Hence, rotation of the nicol will always give variation in intensity with zero minimum.

12. To distinguish between circularly polarized light and a mixture of circularly polarized and unpolarized light, the given light is first allowed to pass through a quarter wave plate and then through a rotating nicol. The quarter wave plate introduces another $\pi/2$ phase difference to the circularly polarized light and emergent light becomes plane polarized. On rotating the nicol prism it shows the variation of intensity with zero minimum. While if the light is mixture of circularly polarized light and unpolarized light, again the quarter wave plate introduces phase difference of $\pi/2$ and the circularly polarized light becomes plane polarized but unpolarized light still remains unpolarized. So, on rotating the nicol, it shows the variation of intensity with non-zero minimum.

13. **Distinction between circularly polarized light and unpolarized light:** When a beam of light is allowed to fall on a nicol prism and prism and there is no variation in the intensity of the light on rotating the nicol, then the beam is either circularly polarized or unpolarized light.

    To distinguish between circularly polarized and unpolarized light, the light is first allowed to fall normally on a quarter wave plate and then on a rotating nicol prism. If the light is circularly polarized then it consists of ordinary ray and extraordinary ray having equal amplitudes and phase difference $\pi/2$ (path difference $\pi/4$). On passing through the quarter wave plate, it introduces a further phase difference of $+\pi/2$ (path difference $\pi/4$) to the rays. So, the phase difference between the two rays on emergence will be 0 or π. Hence, the emergent light will now be plane polarized. This plane polarized light is made to fall on rotating nicol prism. The intensity of light will vary with zero minimum, and shows given light is circularly polarized Fig. 3.30(a).

If the given light is unpolarized, then on passing through the quarter wave plate, there is no effect on the light and after passing through a rotating nicol prism, the emergent light will show no variation in the intensity Fig. 3.30(b).

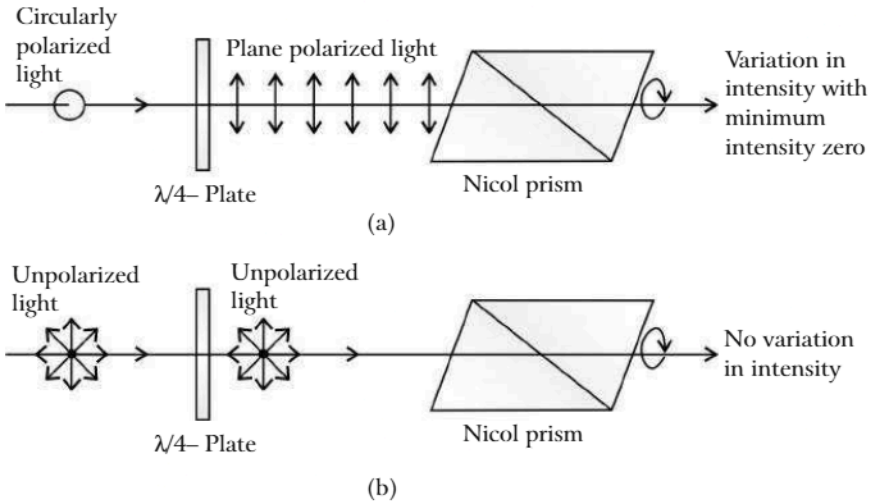
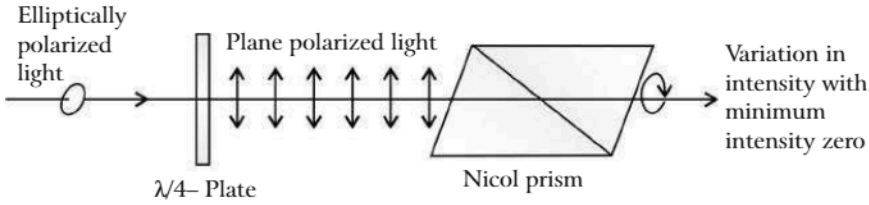


Fig. 3.30

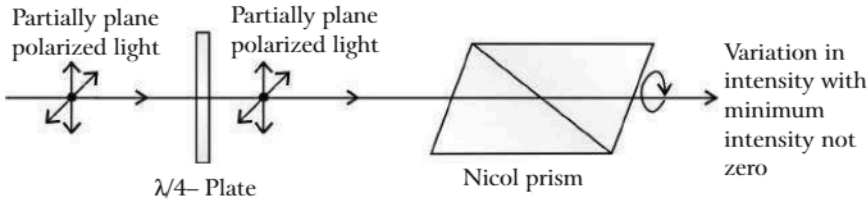
14. Distinction between elliptically polarized light and mixture of unpolarized and plane polarized light: When a beam of light is allowed to fall on a prism and the intensity of light varies from maximum to minimum (non-zero) value on rotating the nicol prism, then the beam is either elliptically polarized or a mixture of unpolarized and polarized light.

To distinguish between the two, the light is allowed to fall normally on a quarter wave plate and then on a rotating nicol prism. If the beam is elliptically polarized, it consists of the ordinary ray and extraordinary ray having unequal amplitudes and phase difference of $\pi/2$. On passing through it, the quarter wave plate introduces a further phase difference of $+\pi/2$ to the rays. So, the Phase difference between two rays on emergence will be 0 or π. Hence, the emergent light will be plane polarized. This plane polarized light is made to fall on rotating nicol prism. The intensity of light will vary with zero minimum and shows that given light is elliptically polarized Fig. 3.31(a).

If the given light is a mixture of unpolarized and polarized light,it will remain as such on passing through the quarter wave plane and after passing through a rotating nicol, prism, the intensity of light will vary with a non-zero minimum Fig.3.31(b).


Elliptically polarized light
Plane polarized light
Variation in intensity with minimum intensity zero
λ/4– Plate
Nicol prism

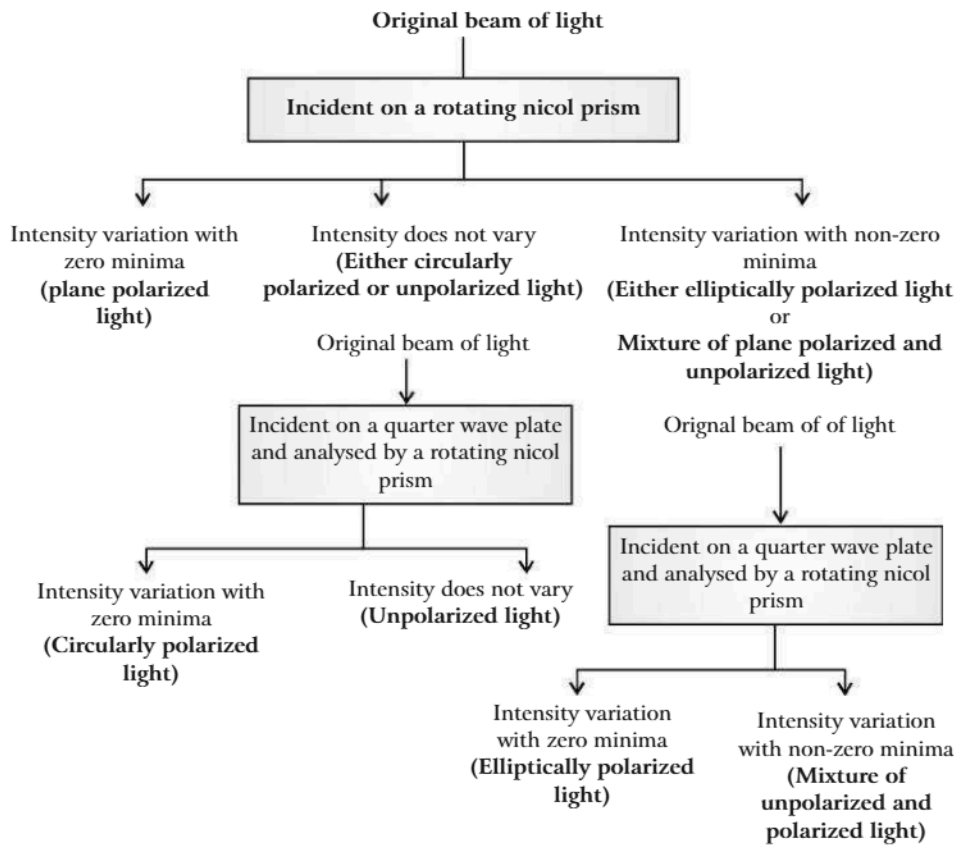
(a)


Partially plane polarized light
Partially plane polarized light
Variation in intensity with minimum intensity not zero
λ/4– Plate
Nicol prism

(b)

Fig. 3.31

15.


Original beam of light
Incident on a rotating nicol prism
Intensity variation with zero minima (plane polarized light)
Intensity does not vary (Either circularly polarized or unpolarized light)
Intensity variation with non-zero minima (Either elliptically polarized light or Mixture of plane polarized and unpolarized light)
Original beam of light
Incident on a quarter wave plate and analysed by a rotating nicol prism
Intensity variation with zero minima (Circularly polarized light)
Intensity does not vary (Unpolarized light)
Orignal beam of of light
Incident on a quarter wave plate and analysed by a rotating nicol prism
Intensity variation with zero minima (Elliptically polarized light)
Intensity variation with non-zero minima (Mixture of unpolarized and polarized light)

(D) Multiple Choice Questions

1.	(a)	**2.**	(a)	**3.**	(a)	**4.**	(a)	**5.**	(c)
6.	(c)	**7.**	(b)	**8.**	(b)	**9.**	(c)	**10.**	(a)

(E) Fill in the Blank

1.	arrows	**2.**	two	**3.**	internal reflection
4.	extraordinary ray	**5.**	ordinary ray	**6.**	same
7.	$\lambda/2$	**8.**	$t = \frac{\lambda}{4(\mu_e - \mu_o)}$	**9.**	plane polarised
10.	elliptically				

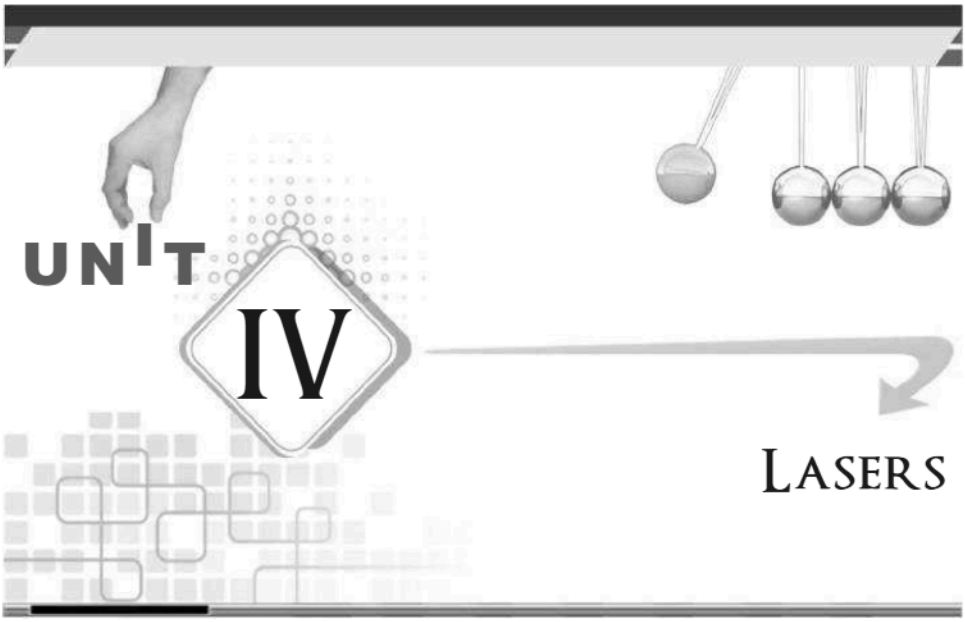
(F) True/False

1.	F	**2.**	F	**3.**	T	**4.**	T	**5.**	F
6.	F	**7.**	T						

○○○

LASERS

- Purity of a Spectral Line
- Coherence, Coherence Time and Coherence Length
- Spatial Coherence
- Temporal Coherence
- Spontaneous Emission and Stimulated Emission of Radiation
- Einstein's A and B Coefficients
- Conditions for Laser Action
- Population Inversion
- Pumping Process
- Three and Four Level Laser Systems
- Pulsed and Tunable Lasers
- Ruby Laser
- He-Ne Laser
- Applications of Lasers
- Spatial Coherence and Directionality
- Estimation of Beam Intensity



Laser is an abbreviation of Light amplification by stimulated emission of radiation. It is a quantum electronic device which generate coherent electromagnetic radiation in the infrared, visible, ultraviolet and X-ray regions. Laser began to be developed in 1958 but first laser was fabricated by Maiman in 1960. Towner, Basov and Prokhorov were awarded Noble Prize in 1964 for the invention of laser. Now a days lasers are widely used in science, industry, technology, communications, medicines etc.

4.1 Purity of a Spectral Line

The laser light has some special properties that make it different from other light sources. The most important property is its extra ordinary mono chromaticity (*i.e.*, single wavelength) as compared to any source of light. Practically no light source including laser is capable of producing absolute monochromatic light. One can only make better approximations to ideally mono chromatic light. It is observed experimentally and can be explained on the basis of quantum theory that spectral lines have finite purity (*i.e.* mono chromaticity of light). Which is denoted by *Q*. It is defined as

$$Q = \frac{\lambda}{\Delta\lambda}$$

Where λ is the central wave length of spectral line and $\Delta\lambda$ is spread in wave length of a line called wave length width. Every spectral line has a finite width which means that it has a finite purity.

(*i.e.* it corresponds to a continuous distribution of wavelengths in some narrow interval $\Delta\lambda$). Thus, *purity of a spectral line gives the degree of mono chromaticity of a light.*

4.2 Coherence, Coherence Time and Coherence Length

The other important property of laser light is coherence. A wave that appear to be a pure sine wave for infinitely Fig. 4.1(a) long period of time or at infinite distances in space is said to be a perfectly coherent wave. But, no light emitted by an actual source produces an ideal sinusoidal field for all values of time or at all distances from source. This is because when an excited atom returns to its initial state, it emits light pulses of short durations (e.g. of the order of 10^{-10} sec for sodium atom). Thus field remains sinusoidal only for time interval of the order of 10^{-10} sec after which the phase changes as shown in fig. 4.1 (b).

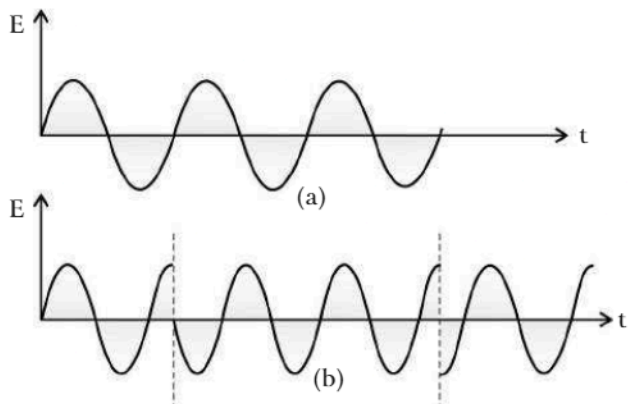


Fig. 4.1

The average time interval for which the field remains sinusoidal is known **as coherence time of the light beam and is denoted by** τ. The distance travelled by light beam in time τ is defined as **coherence length and is denoted by *L*.** Coherence time and length are related as

$$L = C\,\tau. \qquad \text{where } C \text{ is the velocity of light.}$$

In a coherent wave; a definite relationship exists between phase of a wave at a given time and at a certain time later, or at a given point and at a certain distance away. No actual light source emits perfectly coherent wave. Practically the light sources emit pure sine waves only for a limited period of time or at a limited distance in space. So light sources are partially coherent.

There are two different criteria of coherence, the criteria of space and criteria of time. This give rise to concept of spatial coherence and temporal coherence.

4.3 Spatial Coherence

Spatial coherence is the phase relationship between the radiation fields at different points in space. If a constant phase difference (or same phase) exists at different locations in a wavefront at the same instant, then it is said to be spatially coherent. In other words, spatial coherence measures the area of a wavefront over which the light is coherent. The directional property of a laser beam is due to its spatial coherence.

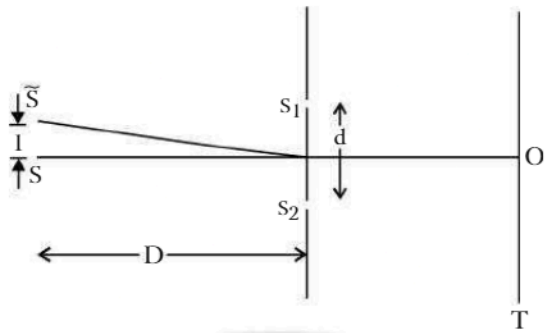


Fig. 4.2

The concept of spatial coherence can be understood by the double slit experiment as shown in fig. 4.2. It so be an extented source of light of width l and S_1 & S_2 represent two pin holes (*i.e.*, slits) with separation 'd'. For coherent light beam good interference fringes will be observed on the screen when $l << \frac{\lambda D}{d}$ where D is the distance of source so and pin holes S_1 & S_2 and λ is the wavelength of light used, But λ/D is the angle subtended by the source at the slits; (say θ), then $d << \frac{\lambda}{\theta}$.

The distance $\frac{\lambda}{\theta} = l_\omega$ is termed as lateral coherence width. It is the distance over which the beam is spatially coherent. Hence, to obtain a good interference pattern with young's double slit, the separation between the slits S_1 & S_2 should be kept much less than the coherence width. Using ordinary extended light source the light is passed through a pin hole to obtain a spatially coherent light beam.

4.4 Temporal Coherence

The temporal coherence of a source is the phase relationship between the radiation fields at different times (but at the same point). For example in a source like sodium lamp, two radiations of slightly different wavelengths are given out which have slightly different coherence times. If a definite phase relationship exists between two times of coherence, then the light beam of sodium lamp is temporally coherent. The temporal coherence measures the duration over which this phase relationship is maintained. The monochromaticity property of laser beam is due to temporal coherence.

To understand the concept of temporal coherence, the path difference between two interfering beams in Michelson interferometer is measured as shown in fig. 4.3.

The interference pattern observed in telescope would disappear when path difference between the two close wavelengths λ_1 & λ_2 producing interference is

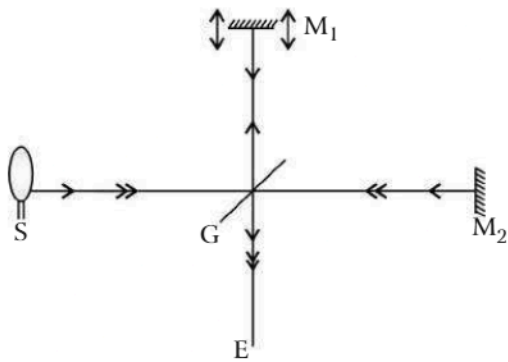


Fig. 4.3

$$\text{path difference} = \frac{\lambda^2}{2(\lambda_1 - \lambda_2)}$$

where λ is the mean wavelength of λ_1 & λ_2.

If instead of two discrete wavelengths, we consider a beam having wavelengths lying between λ and $\lambda + \Delta\lambda$ then fringes would disappear if path difference between the beams become equal to coherence length *L*.

$$\therefore \qquad L = \frac{\lambda^2}{\Delta\lambda}$$

(Here 2 does not appear because it has been assumed that beam contains all wavelength lying between λ and $\lambda + \Delta\lambda$)

It means that spectral line width of mean wavelength λ is given by

$$\Delta\lambda = \frac{\lambda^2}{L}$$

Further since $\nu = c/\lambda$, the frequency spread $\Delta\nu$ of a line would be

$$\Delta\nu = \frac{c}{\lambda^2}\Delta\lambda \approx \frac{c}{L} \qquad \text{[using equation (2)]}$$

But, we know that $\tau = \frac{L}{c}$, where τ is coherence time.

$$\therefore \qquad \Delta\nu = \frac{1}{\tau}$$

Thus, longer the coherence time, the smaller is the frequency spread. For a perfect monochromatic line $\Delta\nu = 0$ or $\tau = \infty$

For an ordinary source $\tau = 10^{-10}$ sec, and so $\Delta\nu = 10^{10}$ Hz.

For sodium yellow line ($\lambda = 5893$ Å),

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^{10}}{5893 \times 10^{-8}} = 5 \times 10^{14} \text{ Hz.}$$

Thus spectral purity $\frac{\Delta\nu}{\nu} = \frac{10^{10}}{5 \times 10^{14}} = 2 \times 10^{-5}$ $\left[\text{Spectral purity} = \frac{\Delta\nu}{\nu} = \frac{\lambda}{\Delta\lambda}\right]$

It represents the monochromaticity of the source.

Example 1: For a source radiating at mean wavelength λ = 6000 Å the coherence time τ is $\approx 2 \times 10^{-10}$ sec

Deduce the order of magnitude of

(i) ***Coherence length L***

(ii) ***The spectral width of the line***

(iii) ***Purity factor Q***

Solution:

(i) Coherence length $L = \tau c$

$$L = 2 \times 10^{-10} \times 3 \times 10^{8} = \mathbf{6 \times 10^{-2}} \text{ meter}$$

(ii) The spectral width $\Delta\lambda \approx \frac{\lambda^2}{2L} = \frac{(6000 \times 10^{-10})^2}{2 \times 6 \times 10^{-2}} = \mathbf{3 \times 10^{-12}}$ meter

$$= 0.03 \times 10^{-10} \text{ m} = \mathbf{0.03 \text{ Å}}$$

(iii) Purity factor, $Q \approx \frac{\lambda}{\Delta\lambda} = \frac{6000 \times 10^{-10}}{0.03 \times 10^{-10}}$

$$Q = \mathbf{2 \times 10^{5}}$$

Example 2: The coherence length of sodium D_2 line is 2.5 cm. Calculate

(i) Coherence time

(ii) Spectral width of line

(iii) Purity factor Q

Solution:

(i) We know $L = \tau c$

or $\tau = \frac{L}{c} = \frac{2.5 \times 10^{-2}}{3 \times 10^{8}} = \mathbf{0.83 \times 10^{-10}}$ sec

(ii) Spectral width of line $\Delta\lambda = \frac{\lambda^2}{2L} = \frac{(5896\times10^{-10})^2}{2\times2.5\times10^{-10}} = 7\times10^{-12}$ meter = **0.07 Å**

(iii) $$Q = \frac{\lambda}{\Delta\lambda} = \frac{5896\times10^{-10}}{0.07\times10^{-10}} = \mathbf{8.4\times10^4}$$

4.5 Spontaneous Emission and Stimulated Emission of Radiation

The interaction of radiation (photon) with matter may occur in three ways:

1. Stimulated or Induced Absorption
2. Spontaneous Emission
3. Stimulated or Induced Emission

4.5.1 Stimulated or Induced Absorption

The atomic energy levels are quantised and they are characterised by integral numbers. Consider a simple two levels system having energy levels E_1 and E_2 in any atom such that $(E_2 > E_1)$.

At ordinary temperature most of the atoms are in the ground state E_1. If a photon of frequency ν is incident on the system(Fig. 4.4) it will be absorbed by an atom in ground state E_1 and will therefore goes to the excited state E_2, Such that

$$h\nu = E_1 - E_2$$

This process is known as stimulated (or induced) absorption. *The absorbed photon is called stimulating photon. The process of absorption is necessarily stimulated.* Once the atom is in excited state. The Probability of absorption depends on the rate at which photons fall on the atom *i.e.* energy density $u(\nu)$ of incident radiation. It can decay after a short time ($\simeq 10^{-8}$ sec) by two different processes.

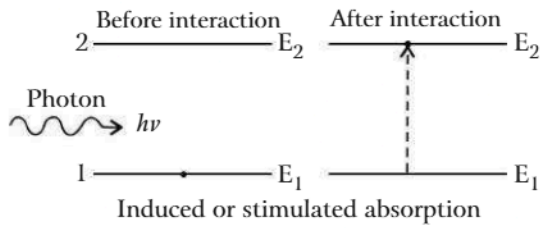


Fig. 4.4

1. Spontaneous emission
2. Stimulated emission

4.5.2 Spontaneous Emission

An atom cannot remain in an excited state for long time as the life of the excited state is very small ($\simeq 10^{-8}$ sec). Hence, it falls back to its ground. State E_1 (Fig. 4.5) by spontaneously emitting a photon of frequency v. where v is given by

$$hv = E_2 - E_1$$

$$v = \frac{E_2 - E_1}{h}$$

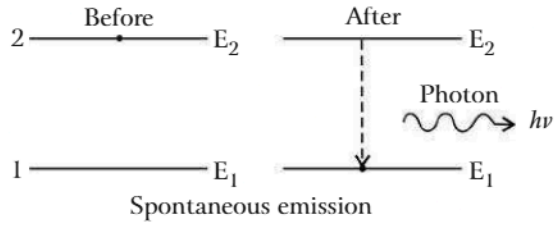


Fig. 4.5

This process is known as *spontaneous emission.* In a spontaneous emission of radiations it is noticed that:

1. The energy of emitted photon is hv and it can travel in any random direction.
2. The rate at which atoms fall from excited state E_2 to lower state E_1 at every instant is proportional to number of atoms in the excited state E_2.
3. In a substance, there are billions of atoms and when they give spontaneous emission, then the direction and time of emisson is different. So there is no phase relationship between them.

   Thus the radiation emitted in spontaneous emission will be incoherent.

4.5.3 Stimulated or Induced Emission

When a photon of frequency exactly equal to $v = (E_2 - E_1)/h$ is incident on the atom in the excited state E_2, then it stimulates the atom to move to ground state E_1 (Fig. 4.6) by emitting a photon of same frequency v. This process is known as *stimulated or induced emission of radiation* (or negative absorption of radiation). It was proposed by Einstein. The probability of this emission is proportional to the energy density of stimulating radiation. In stimulated emission, the important points to be noted are:

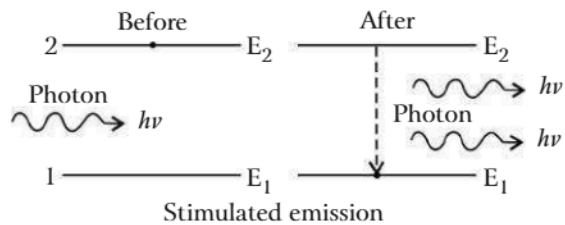


Fig. 4.6

1. For every incident photon, we have two emitted photons.
2. The directions of propagation of emitted photon and incident photon are same.
3. The energy of emitted photon and incident photon are same.
4. The state of polarisation of emitted photon is exactly same as that of incident photon.

In other words, the stimulated photon (emitted photon) have the same frequency and is in phase with stimulating photon (incident photon). In this way, we get a amplified and unidirectional coherent beam.

4.6 Einstein's A and B Coefficients

The phenomenon of stimulated emission was first predicted by Einstein in 1916. He considered the equilibrium between matter and electromagnetic radiation. The probability rate of a transition (*i.e.*, absorption, spontaneous emission, stimulated emission) are expressed in terms of three constants B_{12}, A_{21} & B_{21} which are known as Einstein co-efficients. The relation among Einstein co-efficients is derived as follows:

Let us consider an assembly of atoms (sample of matter) which is in thermal equilibrium with electromagnetic radiation of frequency $\nu \geq E_2 - E_1/h$ and energy density $u(\nu)$ at temperature T. Let N_1 and N_2 be the number of atoms per unit volume in energy state 1 and 2 with energies E_1 and E_2 respectively. The energy density is defined as the radiation energy per unit volume within frequency interval ν and $\nu + d\nu$. The atoms in the lower energy state E_1 can absorb radiation and get excited to state E_2. The probability rate of occurrence of the absorption transition $1 \rightarrow 2$ would be proportional to population of state 1 and energy density of radiation $u(\nu)$. Thus, the number of absorption transitions per unit time per unit volume will be

$$= N_1 B_{12} u(\nu) \qquad [\because P_{12} \alpha N_1 u(\nu)]$$

Where proportionality constant B_{12} is characteristic of energy states 1 and 2 and is known as **Einstein's co-efficient of absorption.**

When the atom de exits from energy state E_2 to E_1, then emission transitions at frequency ν takes place. It may be spontaneous emission or stimulated emission.

The probability rate of spontaneous emission $2 \rightarrow 1$ is determined by the properties of the states 2 and 1 and is proportional to the population (number of atoms) N_2 of energy state 2. Thus the number of spontaneous emission transitions per unit time per unit volume will be

$$= N_2 A_{21}$$

Where the proportionality constant A_{21} depends on the properties of state 2 and 1 and is known as Einstein co-efficient of spontaneous emission.

The probability rate of stimulated emission transition $2 \rightarrow 1$ is proportional to the population of state 2 and the energy density.

$u(\nu)$ of the stimulating radiation. Thus the number of stimulated emissions per unit time per unit volume $= N_2 B_{21} u(\nu)$.

Where B_{21} is Einstein co-efficient of stimulated emission of radiation.

At thermal equilibrium, the number of absorption transitions and emission transitions should be equal, *i.e.*,

$$N_1 B_{12}\, u(\nu) = N_2 [A_{21} + B_{21}\, u(\nu)] \quad \ldots(1)$$

$$u(\nu) = \frac{N_2 A_{21}}{N_1 B_{12} - N_2 B_{21}}$$

$$u(\nu) = \frac{A_{21}/B_{21}}{\left(\frac{N_1}{N_2}\right)\left(\frac{B_{12}}{B_{21}}\right) - 1} \quad \ldots(2)$$

The equilibrium distribution of atoms among different energy states is given by Boltzmann's Law. According to this law, the number of atoms N_1 and N_2 in energy states E_1 and E_2 at temperature T are given by,

$$N_2 = N_0\, e^{-E_2/kT}$$

$$N_1 = N_0\, e^{-E_1/kT}$$

$$\therefore \quad \frac{N_2}{N_1} = \frac{e^{-E_2/kT}}{e^{-E_1/kT}}$$

$$\frac{N_2}{N_1} = e^{-(E_2 - E_1)/kT}$$

where $E_2 - E_1 = h\nu$

So, $$\frac{N_2}{N_1} = e^{-h\nu/kT}$$

or $$\frac{N_1}{N_2} = e^{h\nu/kT} \quad \ldots(3)$$

Putting the value of N_1/N_2 from equation (3) in equation (2), we get

$$u(\nu) = \frac{A_{21}/B_{21}}{\left(\frac{B_{12}}{B_{21}}\right) e^{h\nu/kT} - 1} \quad \ldots(4)$$

Also, we have planck's radiation eqn.

$$u(\nu) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/kT} - 1} \quad \ldots(5)$$

Now, Compare equation (4) and (5)

we get $$\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} \quad \ldots(6)$$

and $$\frac{B_{12}}{B_{21}} = 1 \qquad ...(7)$$

or $$B_{12} = B_{21}$$

or *The probability of stimulated emission is equal to the probability of stimulated absorption.*

From equation (6) we can write

$$\frac{A_{21}}{B_{21}} \propto \nu^3$$

The *ratio of probability of spontaneous emission and stimulated emission is proportional to* ν^3. It means that *the probability of spontaneous emission increases rapidly when the energy difference between two states is large.*

$$(E_2 - E_1 = h\nu).$$

4.7 Conditions for Laser Action

A laser is a device to produce an intense highly concentrated, monochromatic, unidirectional and highly coherent beam of light. Let us consider a assembly of atoms distributed in energy state E_1 and E_2 $(E_1 < E_2)$. When a light beam of frequency $\nu = E_2 - E_1/h$ pass through this medium, then a light photon may be absorbed by the atom in energy state E_1 and atom is excited. In this case the intensity of light beam decreases gradually in passing through the medium.

Now, if a light photon of frequency ν $(\nu = E_2 - E_1/h)$ is incident on an atom which is already in excited state E_2 then atom falls back to lower state E_1, emitting a photon of same energy in the same direction (stimulated emission).

In order to have a unidirectional, coherent and intense beam we should have more stimulated emission than absorption.

At thermal equilibrium the population of upper energy state N_2 in an atomic system is considerably smaller than that of the lower state N_1. 1* A system of atoms in which more than half of atoms are in excited state is said to have undergone *population inversion. i.e.*

$$N_2 > N_1$$

or $$N_2 - N_1 > 0$$

1* $(N_2 < N_1)$ so that there is very little stimulated emission compared to absorption. If, some how the *number of atoms in higher energy state be made sufficiently larger than the number of atoms in the lower state then stimulated emission predominates over absorption.*

A material having population inversion is called an *active material.* A suitable photon (of energy $h\nu = E_2 - E_1$) in passing through such a excited material will stimulate these atoms to fall back to their ground state. The photons emitted by stimulated emission further knocks out the excited atoms in state E_2 and give rise to emission of more photons. This process continues and intensity of light beam increases exponentially. This increase in light intensity is known as light amplification.

The population inversion alone is sufficient to produce light amplification by the stimulated emission of radiation. To sustain this amplification process, the amplifier should work as an oscillator.

The amplifier can be made to behave as oscillator by using positive feedback. In lasers, positive feedback is achieved by placing the active material between two highly reflective mirrors. The radiation repeatedly travels back and forth between reflecting surfaces and gets amplified.

If one of the two mirrors is made partially transparent, then the light that escape out form the laser cavity form the strong laser beam. Theoretically a laser with a pair of totally reflective mirrors, like any positive feedback system could amplify itself to infinity, but in practice cavity losses limit the positive feedback and degree of amplification.

The laser oscillations build up when gain of the active material exceeds the sum of cavity losses and output mirror transmission losses. When population inversion $(N_2 - N_1)$ reaches a critical value known as initial inversion, threshold is reached. Once the initial inversion is reached, oscillations are build up from the spontaneous emission. The photons which are emitted spontaneously along the axis initiate the amplification process. This is the condition of laser action.

4.8 Population Inversion

The establishment of a situation in which number of atoms in higher energy state is greater than that in lower energy state is called the **population inversion.**

Suppose we have a large number of atoms (N_0) in thermal equilibrium, then the distribution in different energy states obey the Maxwell-Boltzmann statistics. Suppose at temperature T, the instantaneous populations in energy states E_1 and E_2 are N_1 and N_2 respectively. Now we can write

$$N_1 = N_0 e^{-E_1/kT} \quad \ldots(1)$$

$$N_2 = N_0 e^{-E_2/kT} \quad \ldots(2)$$

Hence, the relative population

$$\frac{N_2}{N_1} = e^{-(E_2 - E_1)/kT} \quad \ldots(3)$$

where k is Boltzmann constant. From above equation it is clear that $N_2 << N_1$ at equilibrium because $E_2 > E_1$. It implies that the number of atoms in the higher state (N_2) is less than the number of atoms in lower state (N_1).

The ratio of population densities at room temperature (17°C) is $\frac{N_2}{N_1} = 10^{-32}$

As invisible region with optical sources, $E_2 - E_1 >> kT$

In such situations, the probability of stimulated emission is much less than the probability of spontaneous emission. But in spontaneous emission, the photons emitted from various atoms have random direction and a random phase. Hence, light emitted from a source will be incoherent. On the other hand, in stimulated emission, the radiation emitted is coherent and amplified. Hence, *the essential condition of amplification is to have predominately stimulated emission*. For it two condition should be satisfied:

1. The excited state should have a longer mean life time or it should be **metastable state**.
2. The number atoms in the higher state E_2 should exceed that in lower state E_1.

$$N_2 > N_1$$

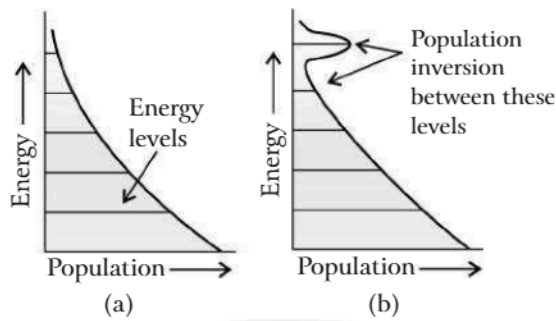


Fig. 4.7

The condition $N_2 > N_1$ is known as *population inversion*. It is an abnormal situation. It is never observed at thermal equilibrium as shown in Fig. 4.7. If however, by *any means a large number of atoms are made available in the higher energy state* Fig. 4.7(*b*) , *stimulated emission is promoted.*

The point at which the population of both states is equal is called the **inversion threshold.** When population inversion is achieved the emitted photons having same frequency and phase interact with other atoms and thus a sort of chain emission takes place in which photons have same frequency and phase and travel in same direction. As a result the number of identical photons goes on multiplying by repeated stimulated emission and we get a very intense monochromatic, coherent and collimated beam of light.

4.9 Pumping Process

The *process by which the population inversion is achieved is known as pumping process.* There are two types of pumping processes.

1. Optical Pumping (excitation by a strong source of light e.g. flash lamp, or lamp)
2. Electrical Pumping (excitation by electron impact)

But the most natural pumping is the optical pumping.

4.9.1 Optical Pumping

In this pumping process, the energy in the form of light radiation is absorbed by the active material and thereby that energy pumps atoms into the higher energy level. This method is mainly suited for solid state laser. For example in Ruby laser population inversion is achieved by optical pumping process as shown in Fig. 4.8.

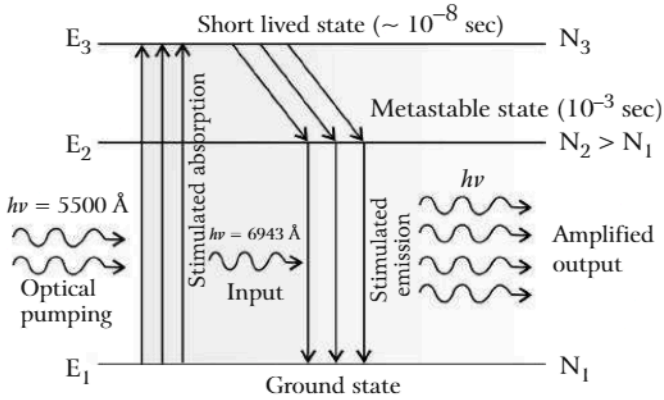


Fig. 4.8

In optical pumping, population inversion is brought about by three level scheme. Let us consider a material whose atoms when raised from the ground state E_1 to a excited state E_3 with a life time $\sim 10^{-8}$ second, do not come back to state E_1 by emitting radiation as the transition from state E_3 to state E_1 is not allowed by selection rules. But the transition from state E_3 is followed to an intermediate state E_2 (with a life time $\sim 10^{-3}$ second) known as metastable state. In other words, when atoms of the active material are irradiated with an specified energy $h\nu' = E_3 - E_1$ [frequency $\nu' = (E_3 - E_1)/h$], the atoms are excited to the state E_3 by the process of stimulated absorption. Some of the atoms decay spontaneously to metastable state E_2 where they can stay for a much longer time $\sim 10^{-3}$ second as compared to 10^{-8} second. Due to much longer life time of metastable state the number of atoms in the state E_2 becomes more than the number of atoms in the state E_1. This is known as *population inversion*.

After achieving *population inversion*, the atoms in metastable state E_2 are exposed to photons of energy $h\nu = E_2 - E_1$ [frequency $\nu = (E_2 - E_1)/h$]. Stimulated emission will start. In it a single photon produces a very large number of photons of the same frequency in the same direction, so that the radiation beam is now highly amplified. Thus, we get a coherent, and very intense beam in the direction of incident photons.

4.9.2 Electrical Pumping

Electrical pumping is mainly used for *gas and semiconductor lasers*. In this pumping process, the pumping is achieved by flowing a suitable current (electron) through the gas. Due to this current ions and electrons are produced. The electrons are accelerated by the electric field and gain enough K.E. energy to excite the neutral atoms through collisions.

In this impact excitation process, the movement of ions have less significance than of electrons. The electrons have larger velocity than the corresponding ion velocity. Electrical pumping takes place through following two processes:

1. **Excitation by electron impact:** In this; the excitation takes place by electron impact with atoms

$$e + A \rightarrow A^* + e$$

where A and A^* are the ground and excited state of the same atom. *This is known as collision of first kind.*

2. **Excitation by collision of atoms of different species:** When a gas have atoms of two species, the excitation occurs due to collision of atoms of the two species such as:

$$A^* + B \rightarrow A + B^*$$

This is known as collision of second kind. In this process it is assured that value of energy difference ΔE is less than kT.

4.10 Three and Four Level Laser Systems

It is seen that most of the laser system belong either to three level or to four level systems. The two level system of inversion is not appropriate for population inversion because it contains few difficulties. Two level system utilises the same two levels which are used in the process of amplification. But the inversion and amplification must be separated from one another in either space or time. Hence, multilevel system is employed. The most commonly the three level and four level systems are used.

4.10.1 Three Level Laser System

Initially most of the atoms of laser material are in lowest level 1. Pumping process raises the atoms in the level 3 which is known as *pump band.* Pump band is made up of large number of spectral levels. The excited atoms from the pump band are transferred to sharp level by *radionless transitions*. The energy lost by electron is absorbed by lattice. Electron come back to ground state after emitting photon. This is known as *laser transition*. Three level system is shown in Fig. 4.9.

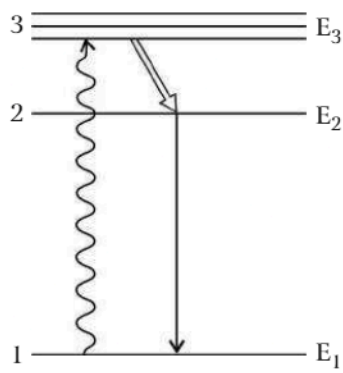


Fig. 4.9

If pumping intensity is below laser threshold, the atoms of level 2 predominantly come back to ground state by spontaneous emission in the form of fluorescence. But if the pump intensity is above the laser threshold, the excited atom in level 2 return to ground level both by spontaneous emission and stimulated emission. *Only stimulated emission constitutes laser output.*

The scheme of Fig. 4.9 is applicable only if there is a fast relaxation between level 1 and 3, it at once drops level 2. Then if an atom is pumped from level 1 to 3 it at once drops to level 2. Therefore the life time of level 2 state should be large in comparison to relaxation time of $3 \rightarrow 2$ transition.

or $$\tau_{21} >> \tau_{32}$$

It results into a situation in which N_3 is very small and we can assume that

$$N_1 + N_2 = N_{total}$$

Thus, one can continue pumping without equalizing the populations of level 1 and level 3 and can actually pump more than half of the atoms into level 2. Practically in three level system, we assume that atoms are effectively pumped from level 1 to metastable level 2 with a momentary pause in level 3. The laser action take place between level 2 and level 1. In three level system the terminal level is ground level and more than half of the atoms are to be pumped to level 2. For this high inversion, power is required. But as $N_2 - N_1$ is small, power required to maintain the population inversion is small. Due to the presence of large number of atoms in level 2, large number of spontaneous radiationless transition take place. The energy is usually carried by lattice photons. The three level laser system has got low efficiency.

4.10.2 Four Level Laser System

The four level system is available in many rare earth ions such as Neodymium. Suitable materials may be found in the form of gases or in condensed materials such as solids or liquids. In four level systems we have four levels ground level 1 sharply defined level 2, metastable level 3 and pump band 4. The four level system is shown in Fig. 4.10.

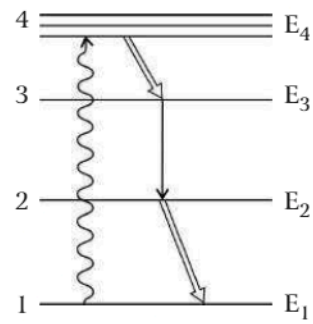


Fig. 4.10

The pump energy raises the atoms from ground level to pump band. The excited atoms proceed rapidly to metastable level 3 then the transition from level 3 to level 2 takes place. This is *laser transition*. From the level 2 atoms undergo rapid radiationless transition to ground level.

For a four level system, following condition should be satisfied.

$$\tau_{43} << \tau_{32}$$

and $$\tau_{21} << \tau_{32}$$

In four level laser system, the terminal level is almost empty, few atoms raised to level 3 from ground level are sufficient for population inversion. Thus to sustain laser action in four level system small pump power and small pump rate is required.

4.11 Pulsed and Tunable Lasers

Lasers can operate continuously for many years or generate ultra short pulses lasting only a for duration femto seconds ($1\ fs = 10^{-15}$ s). The differences reflect both the design of the lasers and the physics involved in them. It is to be noticed that the time scale of atomic physics is entirely different from that of the human race, being measured in nanoseconds or microsecond rather than seconds and minutes.

4.11.1 Pulsed Lasers

In the laser world, anything operating for a second or more at a time is called continuous wave. Most commercial wave lasers can emit for longer time, sometimes many years. But from the stand point of the physics, the laser emission that is steady for a second is to be *continuous-wave.* Such short term continuous wave operation is most likely for experimental lasers, specially high power ones used in military laser weapon research.

Many types of lasers can operate in *pulsed mode*, for a variety of reasons. For examples in solid state lasers, the main problem is heat dissipation. It takes time for the excess pump energy delivered to the laser rod to make its way out as heat, and continuous wave operation can cause heat to build up to laser damaging levels. Damage in this case is not physical damage to the laser rod but also disruption of the laser beam by thermally induced gradients.

In some other cases, the only way to produce a population inversion is by using input energy pulses with high power that cannot be sustained. Or we can say that population inversion is short lived because of rapid decay of excited state or rapid buildup of species in the ground state (as in ruby laser).

Pulse repetition rates also vary widely, Ruby lasers can generate at the most a few pulses per second, while excimer lasers can produce up to a thousand and copper vapour laser several thousand. Efforts are under way to increase repetition rates as a way to increase power from certain lasers particularly excimers.

4.11.2 Tunable Lasers

A laser which can have its output wavelength changed by some tuning process is known as tunable laser. The best-known tunable laser is *dye laser*. But many other types can be tuned over a limited range.

Fixed wavelength emission is good for many applications, but a choice of wavelength is required in some other applications.

By ordinary standard, laser emission is monochromatic, but there are considerable variation in line width of the emitted light. The range of wavelengths emitted by a laser depends on the choice of laser medium and optical cavity.

It is possible to insert some special elements into laser cavity to restrict oscillations to a narrower range of wavelength. Usual choice is a Fabry Perot etalon, a pair of glass plates aligned so they effectively form a resonator. Plates being closely spaced, the resonant modes are far apart and oscillation can be restricted to a single longitudinal mode. The placing can be adjusted by titling the etalon, by moving one of the plates or by changing the gas pressure inside or outside of the etalon cell. By such adjustments, it is possible to change the wavelength selected by etalon. The cavity mirrors can be coated to reflect light at only certain wavelengths, preventing oscillations from taking place at other wavelengths.

Where a broad range of wavelengths emission is possible, it is desirable to tune the laser wavelengths. Such tuning is done by inserting a wavelength dispersive element *i.e.* a prism or grating into optical cavity to serve as a wavelength selective element which serves as a cavity mirror. The dispersive element spreads out a spectrum, and only a narrow range of wavelengths in that spectrum is reflected along the axis of the laser cavity so that it can oscillate. Light of other wavelengths is scattered out of the laser medium hence would not oscillate or be amplified. *The net effect is to give optical cavity high losses at all other wavelengths except the desired wavelength.* The wavelength can be changed by turning the dispersive element so different wavelengths are aligned to resonate in the laser cavity. Dye laser is a very good example of *tunable laser*. An attraction of dye laser is *its tunable output wavelength and ability to produce ultra short pulses or ultra narrow line width.*

The active medium in dye laser is a fluorescent organic dye, dissolved in a liquid solvent. The excitation is done by optical pumping with the help of a flash tube. By employing a Fabry Perot interferometer inside a dye laser cavity, it is possible to select a particular portion of the band width so that laser not only produce very short pulses but also generate them at a particular wavelengths. By using different dyes almost complete visible spectrum is covered. Tuning is also used with CO_2 and chemical lasers.

For classification of lasers several criteria can be used such as type of active medium, excitation method, output wavelengths, output power, spectral band width, types of transitions etc. The first two criteria are basic to laser operation. Generally lasers are grouped in four classes:

1. Solid State Laser
2. Gas Laser
3. Semiconductor Laser
4. Liquid Laser

Each class of laser have so many different lasers that they are further sub-divided into many types.

4.12 Ruby Laser

Ruby laser is the first working laser developed by Maiman in 1960. *It is solid state laser and makes use of the three level scheme of population inversion.* It consists of three main parts:

1. **The Working Substance**
2. **The Resonant Cavity**
3. **The Optical Pumping System**

1. **The Working Substance:** The working substance is a ruby crystal. Ruby is a crystal of aluminium oxide (Al_2O_3) doped with 0.05%. Chromium oxide (Cr_2O_3). Aluminium atoms in the crystal are replaced by Cr^{+++}ions, These impurity (choramium) ions give a pink color to ruby and give rise to laser action.

2. **The Resonant Cavity:** Ruby crystal is grown in special furnaces and then shaped into rods of nearly 10 cm length and 0.8 cm in diameter. Flat end faces *A* and *B* of the rod are made parallel, plane and polished so that one of its ends is completely silvered to become fully reflecting while the other is only partially silvered (Fig. 4.11). An intense laser beam emerges out from partially reflecting end *B*.

   Ruby rod is surrounded by a cylindrical glass envelope through which liquid nitrogen or water is circulated to keep the rod cool.

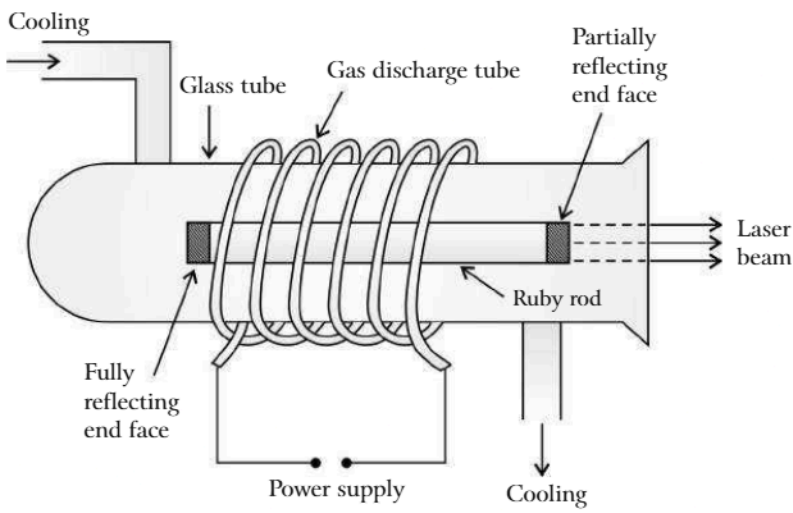


Fig. 4.11

3. **The Optical Pumping System:** Optical pumping system is in the form of helical xenon flash tube, optical pumping is done by this flash tube. It is wound round the glass envelope surrounding the ruby rod. The flash tube is provided with a suitable power supply.

4.12.1 Laser Operation

In ruby laser, energy levels are those of Cr^{+++}ions. Energy levels diagram is shown in Fig. 4.12.

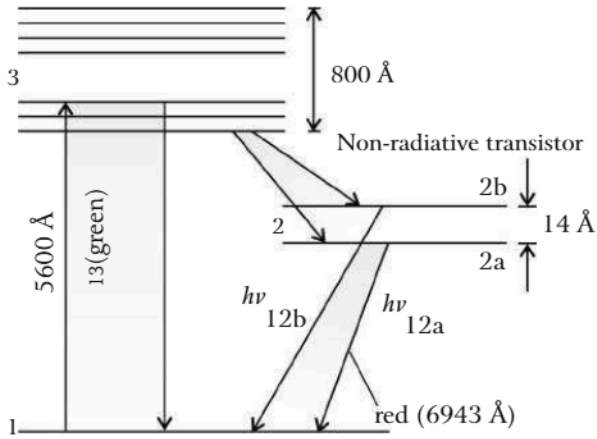


Fig. 4.12

The lines 1, 2, 3 shows the energy levels of chromium. the optical pumping results when a flash of light from xenon flash tube falls upon the ruby rod which last only for about a millisecond. The chromium atoms absorbs the component of wavelength $\lambda \approx 5600$ Å and get excited from energy level 1 to energy level 3. Which has a band width nearly equal to 800 Å. The chromium atoms has a double energy level 2 with the difference of about ≈ 14 Å. From this level 3 some excited atoms returns to the ground state as excited state has life time nearly equal to 10^{-8}sec. But some other atoms moves to level 2 which is a metastable state. The transitions $3 \rightarrow 2$ are non-radiative.

Since metastable state has a very long life ($\approx 3 \times 10^{-3}$sec) as compared to 10^{-8}sec of energy level 3. The probability of transition from level 3 to level 2 is much higher than that from 3 to 1. The probability of transition from 3 to 1 is very small in comparison to 2 to 1. Hence, the number of atoms in the energy level 2 keep on increasing and ultimately become more populated than 1. Hence, a **population inversion** is established between level 2 and level 1.

Such a system with population inversion is very unstable because one spontaneous transition may cause an induced transition resulting in two photons. This photon travels through the ruby rod and it moves parallel to the axis of the crystal, is reflected back and forth between the two ends of the crystal. The two ends of the rods act as resonator. It stimulates an excited atom to emit a photon in phase with the stimulating photon. The process build up, helped by multiple reflections at both ends (cavity). Thus, *avalanche effect* is produced till the beam becomes sufficiently intense to emerge out of partially silvered end of the crystal.

A stage is reached where the population inversion caused by one flash of xenon tube is used up. The laser beam then closes, till the next flash of Xenon tube repeat the process. Hence, *Ruby laser is called pulsed laser.* The duration of laser beam is less than a millisecond. The level 2 in ruby is a doublet level. The doublet separation is 14 Å. The

spontaneous transition would give two week lines at 6943 Å and 6929 Å each of width 6 Å. But under lasting conditions the line of wavelength 6943 Å dominates over the other line. The power output is 0.1 watt at 1.5 kW input.

In this way, a *highly intense, coherent, monochromatic* and *unidirectional* beam is obtained.

4.12.2 Uses of Ruby Laser

Pulsed ruby laser is used successfully in many ways such as:

1. Precision welding and drilling in metals.
2. Drilling of industrial diamonds.
3. Repairing of detached retinas as in opthamalogy.
4. Holography and photography of moving objects.

4.13 He-Ne Laser

Many laser applications require a continuous wave. Javan, Benett and Herriott in 1961 discovered a gas laser which emits light continuously, rather than in pulses. He – Ne gas laser device is one of the most common continuous wave laser source now-a-days with power ranging from 0.5 milliwatt to 60 milliwatt. They are available in different dimensions ranging from few centimeters to few meters.

4.13.1 Construction

The active medium in He-Ne gas laser is a mixture of helium and neon in the ratio 9 : 1 at the pressure of about 1 mm of Hg. A schematic diagram of the experimental arrangement for this gas laser is shown in Fig. 4.13.

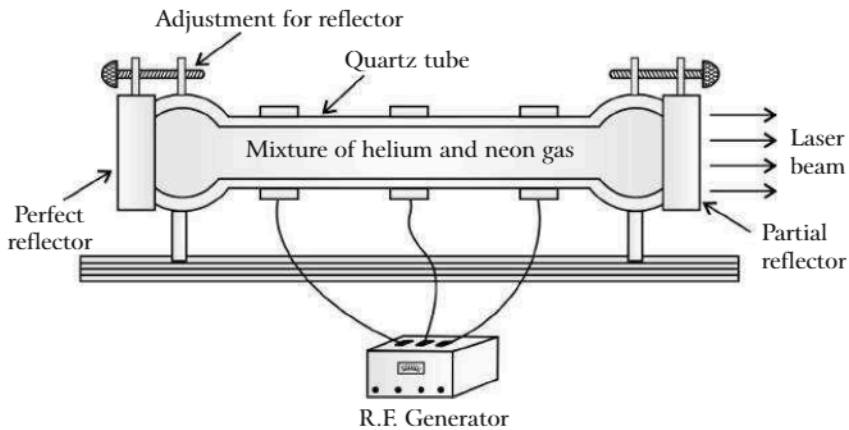


Fig. 4.13

The mixture is contained in quartz tube. The one end of the tube is perfect reflector while the other end is partial reflector. The spacing of the mirrors is equal to an integral number of half wavelengths of the laser light. He-Ne gas laser is electrically pumped continuously with d.c. power supply. But a large amount of noise is generated in d.c. excitation. The other way of excitation is high-frequency discharge. In this excitation method noise is not present.

4.13.2 Working

When electromagnetic energy is injected into the tube, helium atoms are excited to metastable state by means of collision with the electrons in tube. The excited helium atoms collide with the unexcited neon atoms. The population inversion takes place in **Ne** atoms. Since **He** atom has metastable levels at almost the same energies as the 2s and 3s levels of Ne, it allows a resonant exchange of energy and a resultant increase in population of the corresponding Ne levels. Thus **He** atoms help in achieving a popualtion inversion in the **Ne** atoms. The *laser action takes place in neon atom and helium serves the only purpose to increase the excitation process.*

The laser mirrors are reflecting mirrors of high reflecting power at a particular wavelength. This confines the laser action to one wavelength. The most popular wavelength is 6328 Å. This wavelength is emitted due to the transition from 3s–2p level. The transition from one of the 3s states to 3p state give a strong line at 33900 Å (3.39 μm). The transition from 2s to 2p gives line at 11523 Å (1.15 μm). Transitions from 2s to 2p states also produce laser lines at 1.12, 1.16, 1.29 and 1.21 μm. Dominant transition in neon showing laser action are given in table 4.1:

Table 4.1

Dominate transitions in Ne showing laser actions

S. No.	Wavelength		Lasing Transitions
	(Å)	(μ)	
1.	5435	0.5435	—
2.	6328	0.6328	$3s_2 \rightarrow 2p_4$
3.	11177	1.1177	$2s_5 \rightarrow 2p_9$
4.	11523	1.1523	$2s_2 \rightarrow 2p_4$
5.	11614	1.1614	$3s_3 \rightarrow 2p_5$
6.	11985	1.1985	$2s_3 \rightarrow 2p_2$
7.	12066	1.2066	$3s_5 \rightarrow 2p_6$
8.	33913	3.3913	$3s_2 \rightarrow 3p_4$

The transition 6238 Å lies in the visible region while other transitions lie in the infrared region. These transitions are shown in Fig. 4.14.

Light waves emitted parallel to the tube axis reflected back and forth by the mirror ends and stimulate emission of the same frequency from other excited neon atoms. Thus, the photons get multiplied. This process is continued and a beam of coherent radiation builds up in the tube. At last, an intense coherent, parallel beam emerges out through the partially silvered end of the tube.

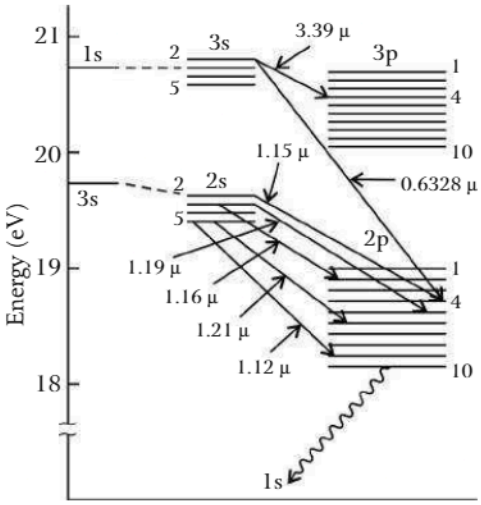


Fig. 4.14

He-Ne gas laser device upto a power of 60 mW with operating life time exceeding 20000 hours has been manufactured by **Spectra Physics** which is the world's largest laser manufacturer.

He-Ne gas laser has following advantages over a ruby laser

1. Output beam of ruby laser is not continuous. It is in the form of pulse while in He-Ne gas laser we get a continuous beam.
2. Ruby laser requires a large amount of optical energy for population inversion.
3. In three level ruby laser, ions in the ground state absorb the photon of wavelength 6943 Å from the beam as it build-up. This difficulty has been removed in four-level He-Ne gas laser.

4.14 Applications of Lasers

Laser is a very intense, highly directional, monochromatic and coherent visible light. It has narrow band width and narrow angular spread which is not available in light from ordinary source. Due to these unique features laser has got very wide applications in different branches of science, technology, etc. A special class of application is of non-linear optics. Some of its important uses are given below:

1. **In Communication:** The use of coherent light for communication is an important application. Laser light being highly coherent, can be modulated to transmit lot of messages at a time on radio and television.

2. **In Surgery:** In medical field, the laser beam is used in delicate surgery like cornea grafting. Intense and powerful laser beam has also been used to remove eye tumour. It is also used in the treatment of kidney stone, cancer and in cutting and sealing the small blood vessels in brain operation.

3. **In Military Purposes:** Laser beam is being used for effective and automatic control and guidance of rockets and satellites. During war-time lasers are used to detect and destroy the enemy missiles. Now-a-days, laser-rifles, laser pistols and laser bombs are also made.

4. **In Scientific Research:** Laser is very useful in science and research. Laser beam provides a new source of exploring the molecular structure and nature of chemical reactions. It is very useful in the study of Raman spectroscopy. It can be used to determine the temperature of plasma and the density of electrons.

5. **In Holography:** Lasers are used to obtain very sharp images of moving objects. Hence, lasers are used for three dimensional photography.

6. **In Industry:** Lasers has wide applications in industries. It is used for cutting fabric for clothing and steel sheets. It can drill extremely fine holes in hard materials including diamonds. Laser beam can also initiate certain chemical reactions. The laser beam is used to vaporise unwanted material during the manufacture of electronic circuits on semiconductor chips.

7. **In Computers:** The use of lasers in computers is under solid investigations. It is being tried to transmit an entire memory bank from one computer to another by the use of laser beam.

8. **Miscellaneous Uses:**

   (i) A laser beam attached to the wallstick of a blind man can enable him to see provided his ears have got a hearing mechanism.

   (ii) Since laser rays are parallel, so these are used in measuring long distances such as the distance between earth and moon has been measured by laser rays.

   (iii) It is used as a tool for mining and tunnelling through rocks.

   (iv) We have listed above only some uses of lasers Field is expanding very fast. These applications are likely to grow until they become many times more important and economical.

4.15 Spatial Coherence and Directionality

The ordinary light source emits radiations in all directions. Apertures are used to produce narrow beams of light from such sources, but lasers emit radiations only in one direction *i.e.*, they have highly directional beam. The directionality is expressed in terms of *full angle beam divergence.*It is twice the angle that the outer edge of the beam makes with the axis of beam. The outer edge of the beam is that point at which the strength of the beam drops to 1/*e* times its value at centre.

A beam of light from a plane wave front coming from an aperture of diameter '*d*' propagates as a parallel beam for a distance d^2/λ called "Rayleigh range". After that it begins to spread out because of diffraction effects. The angular spread $\Delta\theta$ of the beam is given by

$$\Delta\theta = \frac{\lambda}{d}$$

For a typical laser (small), the beam divergence is about 1 milliradian. It means that beam spreads less than 0.01 mm for every meter. Greater is the beam divergence, lesser is the directionally of the beam.

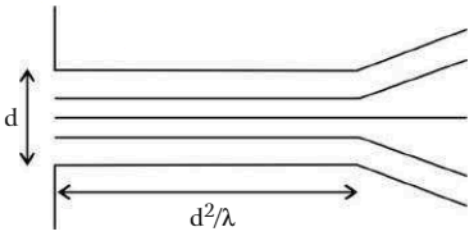


Fig. 4.15

More directional laser beam is obtained from a spatially coherent laser source. The reason is that the spatial coherence (also known as transverse or lateral coherence) indicates how far apart from each other, the two sources of light can be located in a direction transverse to the direction of observation. *i.e.*, the value of '*d* ' will be larger for spatially coherent laser source, and the beam divergence will be smaller giving out more directional laser beam.

4.16 Estimation of Beam Intensity

Light from a lamp is issued out almost uniformly in all directions. Laser gives out light into a narrow beam and its energy is concentrated in a small region and therefore its intensity is very high. Even a one watt laser appears many thousand times more intense than a 100 watt filament lamp. The photon output of a source is the measure of its intensity. Higher is the photon output of a laser, higher is its beam intensity.

The power output from a small gas laser is of the order of 10^{-3}watt where as a pulse solid state laser may give power output as high as 10^{9}watts. One photon of visible light represents nearly 10^{-19} Joules of energy. Therefore

$$\text{Photon output of a laser} = \frac{\text{power output of a laser}}{\text{photon energy}}$$

(No. of laser photons per sec)

$$= \frac{P}{h\nu} = \frac{10^{-3} \text{ to } 10^{9}}{10^{-19}} = 10^{16} \text{ to } 10^{28}$$

In lasers due to stimulated emission, all the atoms in laser medium emit photons having harmonious level of same frequency, phase, direction and polarization. The beam intensity is larger for a temporally coherent source as the number of laser photons emitted per second will be greater.

Miscellaneous Problems

Problem 1: The coherence time of a laser beam of wavelength 7400 Å is 4×10^{-5} second. Calculate coherence length and half spectral width.

[Meerut 2011]

Solution: Coherence length is given by, $L = \tau c$

Here, given that, $\tau = 4 \times 10^{-5}$ sec

$\therefore$ $L = 4 \times 10^{-5} \times 3 \times 10^{8}$ [C is velocity of light]

$= 1.2 \times 10^{4}$ meter $= \mathbf{12\ km}$

Half spectral width, $= \frac{\lambda^2}{L}$

$$= \frac{(7400 \times 10^{-10})^2}{1.2 \times 10^{4}}$$

$= 0.45 \times 10^{-10}$ meter $= \mathbf{0.45\ Å}$

Problem 2: With a He-Ne laser the fringes in Michelson interferometer remained visible when the path difference was increased upto 8 meter. Deduce the lower limit for
(i) the coherence length (ii) coherence time
(iii) spectral half width (iv) purity factor Q
(Given $\lambda = 11.5 \times 10^{-7}$ meter)

Solution:

(i) Coherence length, $L = 8$ **meter**

(ii) Coherence time, $\tau = \frac{L}{c} = \frac{8}{3 \times 10^{8}} = \mathbf{2.7 \times 10^{-8}}$ **sec**

(iii) Spectral half width, $\Delta\lambda \approx \frac{\lambda^2}{L} = \frac{(11.5 \times 10^{-7})^2}{8}$

$$\Delta\lambda = 1.6 \times 10^{-13} \text{ meter}$$

(iv) purity factor, $$Q = \frac{\lambda}{\Delta\lambda} = \frac{\lambda}{\lambda^2/L}$$

$$Q = L/\lambda$$

$$Q \approx \frac{8}{11.5 \times 10^{-7}} = 7 \times 10^6$$

Exercise

(A) Descriptive Type Questions

1. Define purity of a spectral line. **[Meerut 2009, 06]**
2. Explain how the purity of a spectral line is related with the concept of temporal coherence. Show that frequency spread of a spectral line is of the order of inverse of the coherence time.
3. Explain the properties of laser light.
   (i) Spatial coherence (ii) Temporal coherence.
   Define coherence length and coherence time. **[Meerut 2009, 07B]**
4. Explain stimulated absorption, Spontaneous emission and Stimulated emission of radiation. **[Meerut 2006B, 06, 05]**
5. Define Einstein's A and B coefficients and discuss their physical significance. Derive a relation between these co-efficients. **[Meerut 2011, 09B, 06]**
6. What is laser ? Write down the condition for laser action. **[Meerut 2003]**
7. Discuss the essential requirements for producing laser action. **[Meerut 2003]**
8. What are important features of stimulated emission ? Explain the principal of population inversion and hence discuss the essential requirement for producing laser action. **[Meerut 2007]**
9. Explain the principle of population inversion. **[Meerut 2012, 08, 07]**
10. What is pumping process ? Define optical pumping. Describe the construction and working of the three energy level laser. **[Meerut 2012, 08, 07]**
11. Explain population inversion in three and four level laser systems.
12. How four level laser system is better than three level laser system ? Explain.
13. Describe pulsed lasers and tunable lasers.
14. Write down the types of lasers. Describe the construction and working of a three level Ruby laser. Also write its application. **[Meerut 2008, 04,01]**
15. Describe the principle and working of He-Ne laser. Explain, how a helium-neon Laser is superior to Ruby Laser ? **[Meerut 2012, 10, 09B, 08B, 07B, 06B, 02]**
16. Describe the principle of laser. Explain how a helium-neon is superior to a Ruby laser ?

17. (i) Describe a He-Ne gas laser. How is population inversion achieved in this type of laser ?
    (ii) Explain in what respects He-Ne laser is superior to solid state Ruby laser
    [Meerut 2010]
18. Explain the construction and working of helium neon gas laser. How the population inversion can be achieved in it ? Why it is superior than the three energy level laser ? **[Meerut 2006B]**
19. What are characteristic properties of laser beam and its applications. **[Meerut 2006B]**
20. Write down applications of lasers in various fields of modern science. **[Meerut 2010]**
21. Explain spatial coherence and directionality.
22. Explain temporal coherence and spectral energy density.

(B) Short Answer Type Questions

1. Define laser and maser. **[Meerut 2009B]**
2. What is stimulated emission ? Also write important features of stimulated emission. **[Meerut 2012, 06B]**
3. Define pumping process.
4. What are semiconductor and P-N junction lasers ?
5. Write down the importance of metastable state.
6. What is stimulated absorption ?
7. Write down the characteristic properties of laser beam. **[Meerut 2012, 10, 08]**
8. Explain directionality of laser beam.
9. Write the name of different parts used to construct a laser.
10. How lasers are used in radio communication and space exploration ?
11. How lasers are used in Raman spectroscopy ?
12. Why lasers are known generators of light ?
13. Define coherence length and coherence time. **[Meerut 2012, 10, 06]**

(C) Very Short Answer Type Questions

1. What is laser ?
2. What is maser ?
3. Write down the name of first working laser.
4. Define Population inversion. **[Meerut 2009]**
5. Which type of emission is used in laser ?
6. What is pumping ? **[Meerut 2009]**
7. Give the name of a gas laser.
8. What is the active material in ruby laser ?
9. Write down the relation between coherence length and coherence time.
10. Write down a relation between transition probabilities of spontaneous and stimulated emission. **[Meerut 2003]**

(D) Multiple Choice Questions

1. If N_1 and N_2 are the number of atoms in ground state and excited state respectively then in population inversion:
   (a) $N_1 = N_2$ (b) $N_1 < N_2$
   (c) $N_1 > N_2$ (d) none of these
2. Ratio of probability of spontaneous emission and stimulated emission is proportional to:
   (a) ν (b) ν^2
   (c) ν^3 (d) None of these
3. The time interval for which field remains sinusodial is called:
   (a) Life time (b) Coherence time
   (c) Inversion time (d) None of these
4. Due to high directionality, high monochromaticity and coherence, a laser beam is used in:
   (a) Diffraction experiments (b) Polarization
   (c) Interference experiments (d) None of these

(E) Fill in the Blank

1. ..... is the phase relationship between the radiation field at different point in space.
2. The laser beam is used in ..... surgery.
3. ..... results in high temporal coherence of a laser beam.
4. Narrow angular spread results in high ..... in a laser beam which is not available in light from ordinary sources.
5. ..... of a laser beam give rise uses of lasers typically in communication system and computers.
6. In laser ..... can be achieved by placing the active material between two highly reflective mirrors.
7. Laser output is ..... until threshold is passed.
8. The threshold condition is established when photon density after reflection at two ends is equal to initial ..... .
9. The oscillation build up if feedback is sufficiently large to ..... for internal and external losses of the system.
10. The amount of feedback is determined by the ..... of the mirrors.

(F) True/False

1. Spontaneous emission is defined as the rise of an atom from ground state to higher excited state.
2. Ruby laser is a three level laser.
3. The life time of a metastable state is of the order of 10^{-8} second.
4. In stimulated emission the photons are coherent.
5. The important feature of stimulated emission is that the emitted photons are in the different directions.

6. He-Ne laser is a four level laser.
7. Stimulated emission was postulated by Einstein in 1917.
8. Ruby laser operates in pulses.
9. He-Ne gas laser is a continuous laser.
10. The lifetime of metastable states is less than the lifetime of excited state.
11. The laser beam is narrow, intense, parallel polychromatic and highly coherent.

[Meerut 2003]

Answers

(B) Short Answer Type Questions

1. Laser is a quantum electronic device which generate coherent electromagnetic radiation in the infrared, visible, ultraviolet and *X*-ray region. The word laser means *light amplification by stimulated emission of radiation*.

   Maser is a quantum electronic device which generate coherent electromagnetic radiation in the micro wave region. The word maser means microwave amplification by stimulated emission of radiation.

3. The process by which population inversion is achieved is known as pumping process. The population inversion can be achieved by two type of pumping process.

   (i) Optical pumping (ii) Electrical pumping

4. These lasers differ from solid, liquid and gas lasers in several manner. The size of these semiconductor lasers is very small and have high efficiency. These lasers operate at low power. Thus they provide a portable and easily controlled source of low power coherent radiation. For communication through optical fibres, semiconductor lasers have found to be of considerable use.

5. The presence of metastable state is of great importance for laser action to occur. The life time of metastable state is approximately 10^5 times the life time of any excited state ($\approx 10^{-8}$ sec). Hence relatively long life time provides mechanism helpful to achieve *population inversion*.

8. When the active material placed between plane parallel reflecting surfaces, only electromagnetic wave which propagates along cavity direction or very close to it is sustained in the cavity and other leaps out. Thus high directionality is obtained. Due to its high energy density and high directionality a laser beam can produce temperatures of the order of 10^4 °C at a focussed point.

9. There are mainly three parts of a laser.

   (i) Active material

   (ii) Resonant cavity

   (iii) Optical pumping system

10. Laser light being highly coherent, can be modulated to transmit hundred of messages at at a time on radio and television. It can accommodate in a single frequency band larger number of channels of carrier wave with enough band

width. Narrow angular spread of the laser beam is being utilized for communications with earth satellites and rockets to the moon and other planets.

11. Using laser beam Raman spectrum can be obtained from much smaller samples and faster too. High directionality of laser beam enables laser Raman spectra to show finer details of energy level, separations of molecules. The high intensity gives us additional information. Hence laser light is useful for detailed Raman spectroscopy since its direction, polarization and frequency are well defined.
12. *A laser is an amplifier of light in real sense*. When the laser is suitably excited by optical or electrical pumping process, the light of proper frequency entering the laser cavity is amplified in such a manner that output wave is in phase with input. Laser are seldom used as amplifiers because their internal noise is very high. There are practical difficulties in making laser gain very high and also there are difficulties in avoiding oscillations. Hence practically, laser is used as an oscillator which is known as *generator of coherent* light. Thus lasers are also known as *generator of light*.

(D) Multiple Choice Questions

1.	(b)	**2.**	(c)	**3.**	(b)	**4.**	(c)		

(E) Fill in the Blank

1.	Spatial coherence	**2.**	blood less	**3.**	Narrow band width
4.	directionality	**5.**	Narrow band width	**6.**	positive feedback
7.	zero	**8.**	photon density	**9.**	compensate
10.	reflectivity				

(F) True/False

1.	F	**2.**	T	**3.**	F	**4.**	T	**5.**	F	**6.**	T
7.	T	**8.**	T	**9.**	T	**10.**	T	**11.**	F		

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